Supplement to “Eclipses by Appointment: 104 Uncatalogued Eclipsing Systems, Catalogue Corrections and a Measured Null Planet Search from Public TESS and Gaia DR3 Data”

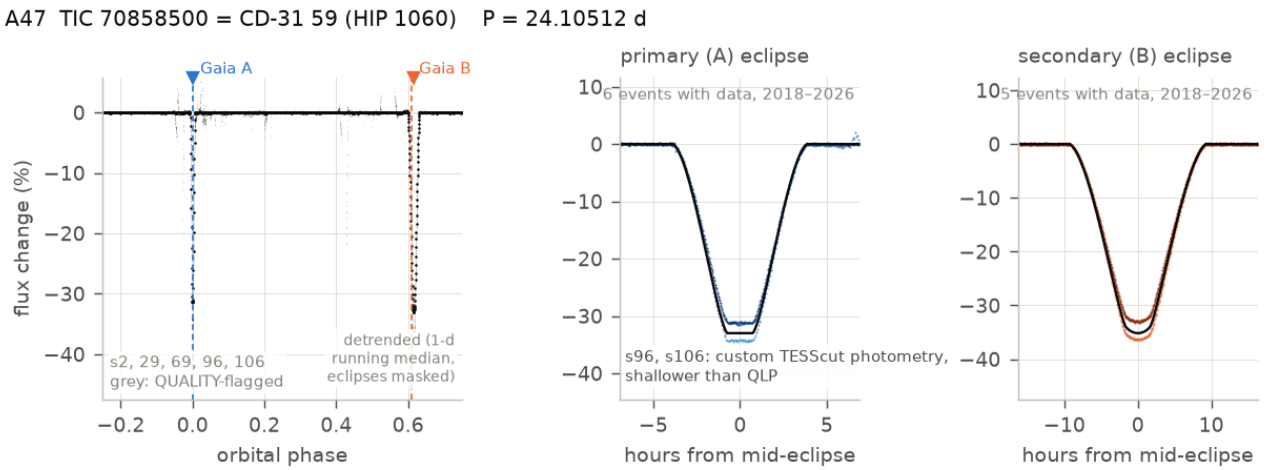
Submitted October 14, 2026

About this supplement

This supplement holds the per-object entries of the paper, one for each eclipsing system, heartbeat star, planet candidate and withdrawn object, with their figures. They are grouped under the section of the paper that discusses them (S3.1.2 holds the entries of the paper's Section 3.1.2, and so on) and are in the paper's order. Each entry gives the identifiers, the eclipse measurements, the ephemeris, the Gaia orbit, the independence checks, the catalogue status and the status word of the paper's Section 1 (Conventions), which also defines the terms used. Figure numbers follow the entry labels (Figure A47 belongs to entry A47). Depths are fractional losses of the system's light in the named pipeline's aperture. BTJD = BJD_{TDB} − 2457000. The analysis file behind each entry is named in the paper's Appendix B deposit (`tables/objects.csv`, column `source_path`).

S3.1.2 Confirmed and probable eclipsing binaries

A47. CD-31 59 = HIP 1060 (TIC 70858500)



A47. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 24.10512 d, Gaia's predicted conjunctions dashed. Middle: the six 34%, 6.9-h total eclipses at A, flat-bottomed during totality (sectors 2, 29, 69, 96, 106 ×2; the sector-96 event, in flagged cadences, is from custom TESScut photometry and shallower). Right: the five 36%, 16-h annular transits at B, with limb-darkened rounded bottoms.

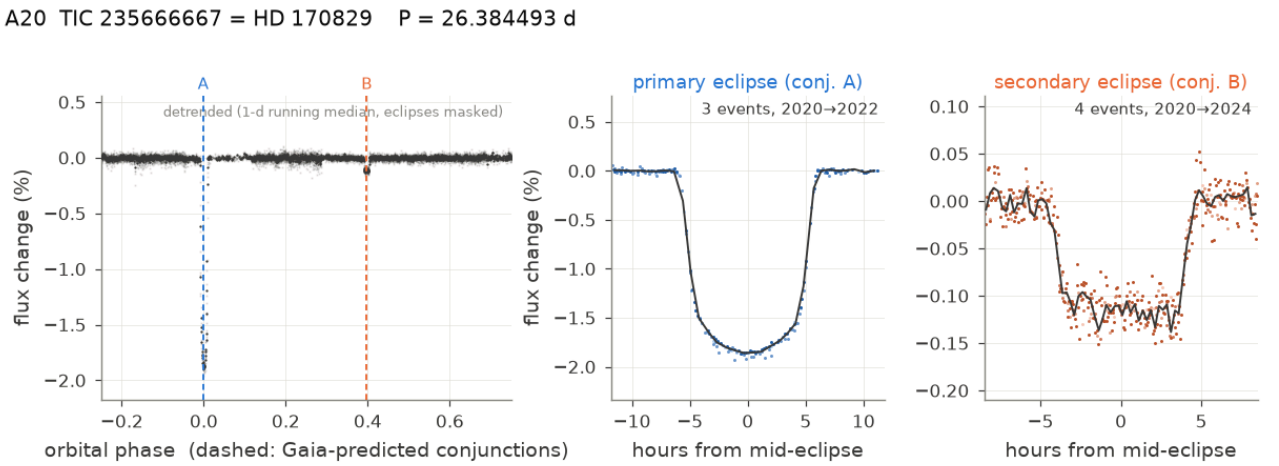
Identifiers	TIC 70858500; Gaia DR3 2320220880389321856; CD-31 59 = HIP 1060 = HE 0010-3037 = SAO 192405 = TYC 6990-585-1 = CPD-31 25 = PPM 275814; WDS 00132-3021 (HDS 31)
Position (J2000)	00:13:12.0 −30:20:59.5
Brightness	*G* 10.39; Tmag 10.01
Stars	Gaia SB2: star 1 (Gaia's “primary”, K_1 = 57.1 km/s) M_1 = 1.10 $M_\odot$, R_1 = 1.26 $R_\odot$, log *g* 4.28; star 2 M_2 = 1.22 $M_\odot$, R_2 = 1.83 $R_\odot$, log *g* 4.00, 67% of the TESS light (more evolved). SIMBAD type G2. Total luminosity 6.1 $L_\odot$, agreeing with Gaia's absolute magnitude. The WDS B component (*G* 13.24) lies 11.6″ away
Period	24.10512 ± 0.00001 d (Gaia 24.11059 ± 0.00424 d; −1.3σ)
Epochs	T_0(A) = BTJD 1367.0102 ± 0.0005 = BJD_{TDB} 2458367.0102; T_0(B) = BTJD 1357.7451 ± 0.0008 = BJD_{TDB} 2458357.7451. The two eclipses are nearly equal in depth (B deeper by 1.7 points). Phase 0 is set at A, the total eclipse, in the figure and the timing analysis; the VSX submission and Appendix A use B, the deeper, as the epoch, following VSX's convention
Eclipse at A	34.2% deep (PDCSAP 33.0%), 0.289 d (6.9 h) with a 0.069-d (1.7-h) flat bottom: total eclipse of star 1 behind star 2, near periastron; six events (s2, s29, s69, s96, s106 ×2) (the s96 A event lies entirely in QLP-flagged cadences and is measured in TARS and in custom photometry, this work)

Eclipse at B	35.9% deep (PDCSAP 34.7%; s096 diluted to 28.8% in QLP, 34.0% in TARS), 0.681 d (16 h) with a 0.128-d bottom: annular transit of star 1 across star 2, near apastron; five events (s2, s29, s69, s96, s106). B at phase 0.61564 after A
Orbit	Photometric $e \cos \omega = 0.1663$, $e \sin \omega = 0.4142$: $e = 0.446$, $\omega = 68.1°$; Gaia 0.4417 ± 0.0127, 69.2 ± 1.7° (within 1σ). Model: $i = 89.87°$, $k = 1.455$, χ^2 914/920
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s2, s29, s69); TASOC (s2); TGLC (s2); TARS (all sectors); custom TESScut photometry of s96 and s106 (this work)
Pixel attribution	TESScut difference images of all eclipses: flux-lost centroid 0.03–0.21 px from the target; the flux-lost image is 0.25× the direct image with 2% residuals; the *G* 13.24 neighbour 0.5 px away (would need about 470%) is excluded
Catalogue status	Not listed in: VSX, GCVS, NSV, ASAS-3, ASAS-SN, ATLAS, ZTF, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4 (including the unvetted table 2), Justesen & Albrecht 2021, Fetherolf et al. 2023, Gaia DR3 variability tables, SB9, ExoFOP, general web search; SIMBAD's 13 references include none on variability. Hipparcos: scatter 0.032 mag, no variability flag. A TESS-SPOC data-validation report for sector 69 carries a threshold-crossing event at 9.266 d, the B-to-A spacing, at very high signal-to-noise, never promoted to a TOI (Table A1)
Open issues	Fitting A and B separately shrinks the A-to-B separation by about 8.6 min over 120 cycles (3.5σ), the same sign as relativistic apsidal motion (which would give about 2.7 min); reported as tentative only
Next eclipses	A: BTJD 4331.940 (2026-10-18 10:26 UTC), 4356.045, 4380.150 (allow ±5 min); B: BTJD 4346.780 (2026-11-02 06:37 UTC), 4370.885
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-31 59 (HIP 1060) is a Hipparcos star with 35% eclipses that is not catalogued as eclipsing in any archive we searched. It is a double-lined pair in an eccentric ($e = 0.446$) orbit seen within 0.13° of edge-on. Its components are a 1.10 $M_\odot$ star and a slightly heavier, more evolved 1.22 $M_\odot$ companion that is 1.45 times larger and gives two-thirds of the light. Near periastron, where the stars are 26.5 $R_\odot$ apart and move at 154 km/s across the line of sight, the smaller star passes wholly behind the larger one. The result is a 6.9-h total eclipse whose flat bottom is totality and whose depth is the smaller star's share of the light. Near apastron (63.7 $R_\odot$, 64 km/s) the smaller star crosses the larger one, giving an annular transit 2.4 times longer, with a rounded bottom from limb darkening. One physical model reproduces the depths, shapes, durations, Gaia's *e* and ω, and the SB2 masses, and gives both radii.

The ephemeris was tested out of sample: three eclipses in sector 106, predicted before those data were examined, fall at O−C = +1.4, −1.5 and +1.4 min. Ground-based photometry of one eclipse, a joint fit with Gaia DR4 epoch radial velocities (masses and radii to about 1%), and continued timing to test the apsidal-motion hint are warranted. The system is the first of these eclipsing binaries to be submitted to the AAVSO VSX, and the submission is pending moderation.

A20. HD 170829 = HR 6950 = NSV 24480 (TIC 235666667)



A20. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 26.384493 d, Gaia's predicted conjunctions dashed. Middle: the three 1.65%, 11.8-h primary transits of the M dwarf across the subgiant (the fourth, in sector 80, lies in flagged cadences and is not drawn; see Open issues). Right: the four 0.106%, flat-bottomed secondary occultations.

Identifiers	TIC 235666667; Gaia DR3 4525943082344640256; HD 170829 = HR 6950 = HIP 90729 = NSV 24480
Position (J2000)	18:30:41.7 +20:48:54.2
Brightness	G 6.29; Tmag 5.79
Stars	G8 IV subgiant with an asteroseismic mass (1.11 $M_\odot$, Malla et al. 2024), R_1 = 1.76–1.86 $R_\odot$ + M dwarf of 0.19–0.21 $M_\odot$, R_2 = 0.21–0.23 $R_\odot$, about 3050 K; mass function 0.00534 $M_\odot$. SB9 orbit #1066; Gaia DR3 astrometric orbit
Period	26.384493 ± 0.000040 d (SB9 26.3847 ± 0.0001 d, −1.9σ; Gaia 26.384663 d, −0.2σ)
Epoch of primary	BTJD 2403.9798 = BJD_{TDB} 2459403.9798 (the epoch proposed for VSX)
Primary eclipse	1.65% (SPOC) / 1.76% (QLP) deep, 11.8 h, rounded U (full transit of the M dwarf across the subgiant); three events, sectors 26, 40, 53
Secondary eclipse	0.106% deep, 8.5 h, flat-bottomed (the M dwarf fully hidden); four events, sectors 26, 40, 53, 80; at phase 0.39677 after the primary
Orbit	Eclipse spacing gives e cos ω = −0.161 (SB9 −0.1607); duration ratio 1.39 gives e sin ω = −0.145 to −0.16 (SB9 −0.154). SB9: e 0.2223 ± 0.0002, ω 223.69 ± 0.06°. Gaia astrometric orbit: i = 89.7°; eclipse solution: impact parameter about 0, $i \approx 90°$
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP; TASOC raw (times within 3 min, depths within about 6%). TARS s053 fails on the saturated star
Pixel attribution	Centroids uninformative (saturated star, bleed columns). The brightest Gaia neighbour within 60″ (G 14.82) would need a 4,500% primary or a 290% secondary; no visual companion in WDS or CCDM (SIMBAD's "**" is the Gaia astrometric orbit). A Gaia E-type eclipsing binary 12″ away (P 0.92 d) contaminates the SPOC aperture (TCE_3, 75 ppm); it cannot produce the 1.65% or 0.106% eclipses
Catalogue status	**VSX type revision.** VSX: NSV 24480, type "VAR:", amplitude 0.08 V, no period. ASAS-SN "VAR", no period (saturated). Nothing in Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables; Hipparcos "constant" (the eclipses cover about 1.9% of the time); no TOI or CTOI; no eclipse in any of the 21 SIMBAD references or in a general web search. SPOC DV runs s14–s55 and s14–s86 carry both eclipses as TCEs
Open issues	Every event is 0.7–1.0 h early against the SB9 orbit. A full limb-darkened model joined to the SB9 orbit and the asteroseismic mass has not yet been made. A fourth primary falls in sector 80, in flagged cadences of QLP, SPOC and TESS-SPOC: a 1.7–1.9% dip centred near BTJD 3485.74 in flagged QLP cadences, with the data gap beginning at 3486.46, after egress
Next eclipses	Primary: BTJD 4330.048 (2026-10-16 13:07 UT), 4356.432, 4382.817; secondary: 4340.516, 4366.901 (about ±5 min formal; allow ±1 h given the SB9 tension)
Discovery	

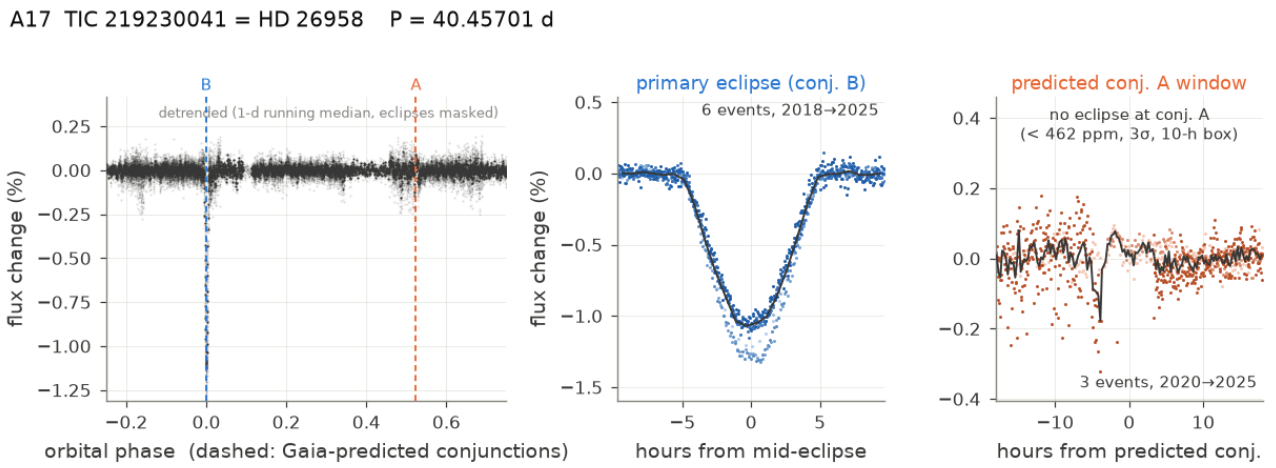
	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision, not a new star. HD 170829 is in VSX as NSV 24480 with type “VAR:” and no period; the eclipses are reported here for the first time. We propose revising it to type EA, P = 26.38449 d, epoch 2459403.9798.

HD 170829 is a bright G8 subgiant with an eclipsing M dwarf. At the primary eclipse the M dwarf crosses the subgiant, giving a limb-darkened U. At the secondary it is fully hidden behind the subgiant, giving a flat bottom. One geometry fits both. Taking the radius ratio from the primary depth and solving both durations gives R_2 = 0.21–0.23 $R_\odot$, as expected for a 0.2 $M_\odot$ dwarf, and an inclination that matches Gaia's independent astrometric orbit. The secondary depth implies a companion of about 3,050 K. Every eclipse falls within 1 h of the published SB9 spectroscopic orbit. A bright subgiant with asteroseismology, an eclipsing M dwarf and an astrometric orbit make an unusually complete combination. However, every event is 0.7–1.0 h early against the SB9 orbit, and no full model yet exists.

Both eclipses are observed in sector 26, and the sector-40 primary is complete in the raw SAP flux.

A17. HD 26958 (TIC 219230041)



A17. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 40.45701 d, Gaia's predicted conjunctions dashed. Middle: the six events at conjunction B, 2018–2025: five full 1.2%, 9.7-h eclipses and the partial egress of sector 5. Right: the data at conjunction A, where no eclipse is expected or seen.

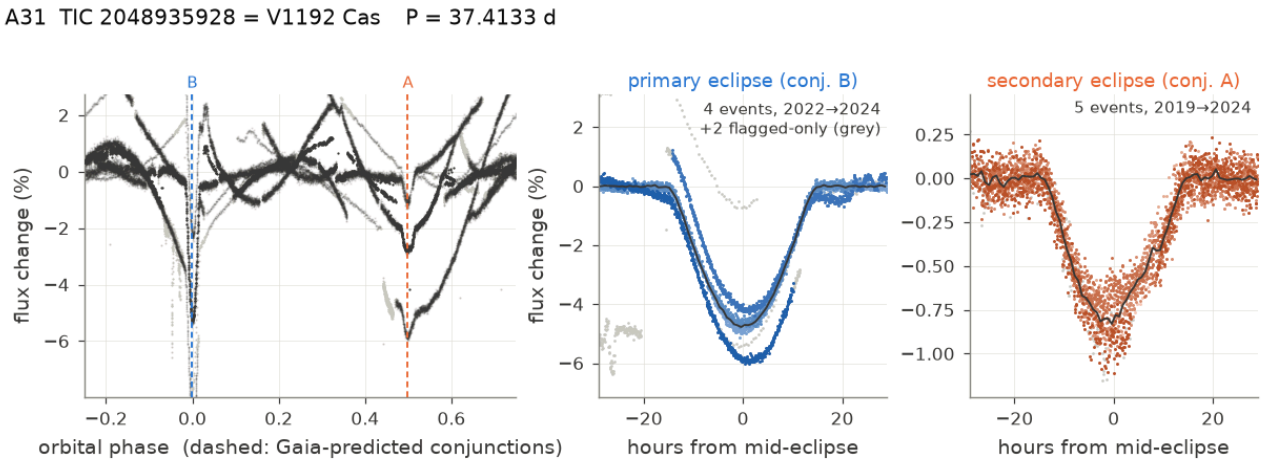
Identifiers	TIC 219230041; Gaia DR3 4830714373974009600; HD 26958
Position (J2000)	04:13:28.4 −49:07:18.6
Brightness	G 8.15; Tmag 7.51
Stars	K1 III giant (SIMBAD) + companion of at least 0.90–1.30 $M_\odot$ (minimum mass for a 1.0–2.0 $M_\odot$ giant), plausibly a ~1 $M_\odot$ dwarf or a fainter object of that mass such as an inner pair; mass function 0.2005 $M_\odot$; 163 pc. Gaia SB1, significance 617, all 18 RV epochs used
Period	40.45701 ± 0.00002 d (Gaia 40.454393 ± 0.001757 d; the eclipse period is longer by 1.49σ)
Epoch of eclipse	Ephemeris BTJD 386.0846 + 40.45701 E (BJD_{TDB} 2457386.0846); observed e.g. BTJD 3946.3012 ± 0.0009 (E88) = BJD_{TDB} 2460946.3012
Eclipse (B)	0.99–1.28% deep in QLP SAP (1.19–1.32% in the large-aperture flux), 9.4–9.8 h, rounded V (a pure V is worse by $\Delta\chi^2$ 21–175); five full events, sectors 3, 30, 31 and 97 (two), residuals −0.3 to +1.0 min; a partial egress in sector 5 also on the ephemeris
Other eclipse (A)	None: 113 ± 116 ppm in s031 (3σ limit 462 ppm for a 10-h box), below 561 ppm in s032; less than 4% of the B depth. The geometry predicts none (impact parameter at A exceeds 1 + k)
Orbit	Gaia e 0.1913, ω 258.54° (Gaia NSS solution); not refitted from the eclipses (only one kind seen). An inclination near 85° is estimated

Pipelines agreeing	QLP SAP and large-aperture flux; TGLC PSF and aperture, TASOC raw, eleanor-lite, T16 (s003); TARS (s003, s030, s031). No SPOC or TESS-SPOC light curve exists
Pixel attribution	No difference image; no centroid shift at mid-eclipse (TASOC, eleanor); no Gaia source within 42″, the two within 60″ would need 5944% and 67542% eclipses. TGLC's PSF photometry, which models this Gaia source and its neighbours separately, shows the same dip
Catalogue status	Not listed in: VSX, ASAS-SN, Kostov et al. 2025 tables 2–4, IJspeert et al. 2024, Prša et al. 2022, Gaia DR3 variability tables (`phot_variable_flag` NOT_AVAILABLE), Hipparcos (no variability flag), SIMBAD (SB*, K1III, 13 references, none on variability), general web search
Open issues	Radius ratio and impact parameter are degenerate (k 0.10–0.20 at impact 0.98–1.08); the companion's nature is inferred, not observed
Next eclipse	BTJD 4350.871 ± ~1.5 min (E = 98)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 26958 is a K giant at 163 pc with one eclipse per 40.457-d orbit. Five full eclipses from 2018 to 2025 fit a single linear ephemeris within 1.0 min, and five independent pipelines show the same dip at the same time and depth. The eclipse falls at the B conjunction. Under the standard radial-velocity convention the giant is the near body there, so the eclipse is most likely a grazing occultation of the companion rather than a transit of the companion across the giant. A total occultation is ruled out ($\Delta\chi^2 \approx 2100$ against a flat bottom), and the observed 9.7 h is half the central duration, so the eclipse must be grazing. The companion would then supply about 2–6% of the TESS light. A 1.0–1.16 $M_\odot$ main-sequence star would supply 7–10% in a blackbody estimate, within a factor of about 1.5 of the fitted value. K and M dwarfs fall below the minimum mass and are excluded.

The astrometric excess (RUWE 1.77 in DR3) is explained by the 40-d orbit's own photocentre motion, and no third body is required. Further constraints require spectroscopy to find the companion's lines, multi-colour photometry of the next eclipse (an occultation of a hotter dwarf should be deeper in the blue), and a joint RV and light-curve model.

A31. V1192 Cas = NSVS 3545309 (TIC 2048935928 = TIC 271237873)



A31. Left: QLP SAP flux folded on P = 37.4133 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 4.7%, 29-h grazing primary eclipses (the hotter companion occulted by the giant). Right: the 0.9%, 29-h secondary eclipses (the companion transiting the giant).

Identifiers	TIC 2048935928 (duplicate entry TIC 271237873, under which the TESS-SPOC files exist); Gaia DR3 2016683927990598784; V1192 Cas; NSVS 3545309
Position (J2000)	23:36:30.5 +63:27:28.7
Brightness	G 10.66; Tmag 9.79; VSX range 10.8–11.0
Stars	

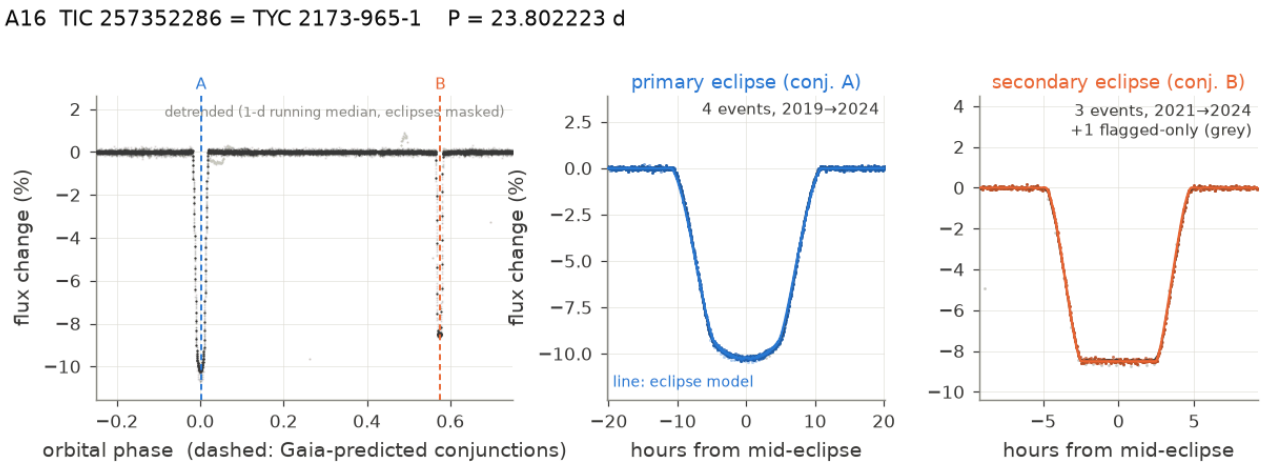
	K giant (Gaia: T_{eff} 5440 K, log *g* 2.74, R = 11.2 $R_\odot$, L = 98 $L_\odot$, FLAME mass 3.1 $M_\odot$) + hotter main-sequence companion of 1.34–1.97 $M_\odot$ (A/F type, about 5–10% of the TESS light); mass function 0.296 $M_\odot$; Gaia SB1. The composite light explains the hotter TIC T_{eff} (6850 K)
Period	37.4133 ± 0.0005 d from both eclipse kinds independently (Gaia 37.4195 ± 0.0041 d; −1.5σ)
Epoch of primary	T_0 = BTJD 1764.906 ± 0.018 = BJD_{TDB} 2458764.906 (*P* = 37.41329 ± 0.00047); secondary T_0 = BTJD 1783.585 ± 0.011
Primary eclipse (B)	4.3–4.9% deep, 28.4–29.2 h, V-shaped: grazing occultation of the hotter companion by the giant (Gaia's B conjunction, giant in front); six events, 2019–2025 (two in QLP-flagged cadences, clean in TEQUILA; two partial)
Secondary eclipse (A)	0.7–1.1% deep, 28–30 h: grazing transit of the companion across the giant; five events (the s085 one inflated by about 1 ppt of common mode)
Orbit	Secondary at phase 0.4993 ($e \cos \omega \approx -0.001$), duration ratio 1.0 ($e \sin \omega \approx 0$), in agreement with Gaia's e = 0.005 ± 0.006: circular. A central eclipse would last 46–56 h, so the geometry is grazing
Pipelines agreeing	QLP SAP; TESS-SPOC (as TIC 271237873, every post-s056 event); TEQUILA (s017, s018)
Pixel attribution	TESScut s058 difference images: largest flux-loss pixel is the target pixel for both kinds, offsets 1.8″ and 1.3″. Brightest star within 60″ (*G* 15.9 at 44″) would need 630%; the *G* 12.5 star at 109″ lies outside the difference signal
Catalogue status	**VSX type revision SRA → EA.** VSX/GCVS: V1192 Cas, SRA, *P* 37.4 d, epoch JD 2451512 (discovery: Kuzmin 2008, “New semiregular pulsating stars”); SIMBAD Mi*; ASAS-SN ROT, *P* 2.0518 d (probably spurious); Kostov table 2. Not listed in: Gaia DR3 variability tables, SB9, WDS, the TESS-era EB catalogues, general web search. ExoFOP was not queried
Open issues	Spectra for the companion's lines; multi-colour depths (the occultation should be deeper in the blue). Low-level intrinsic variability of the giant is possible but not a 37.4-d pulsation
Next eclipses	Primary: BTJD 4346.42, 4383.84 (±0.04); secondary: BTJD 4327.69, 4365.10 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision, not a new star. V1192 Cas is catalogued as a semi-regular pulsator (SRA) with *P* 37.4 d; that period is the orbital period found from the eclipses. We propose reclassification of the SRA (VSX) and Mi* (SIMBAD) types as EA.

V1192 Cas is a K giant of about 11 $R_\odot$ with a hotter main-sequence companion on a circular 37.4-d orbit, eclipsing at grazing incidence. The deeper eclipse falls where the giant is in front, so it is the occultation of the hotter companion. The shallow one is the companion's transit, which at a grazing chord reaches about 1%. Both last about 29 h, against 46–56 h for a central eclipse.

The catalogued pulsation was tested directly. ASAS-SN V photometry from 2014–2017 has a periodogram peak at 35.9 d, within the frequency resolution of 37.4 d. Folded on the TESS ephemeris extrapolated back about 25 cycles, it shows the primary eclipse at phase 0.00. Out of eclipse the rms is only 0.027 mag, not a 0.2-mag semi-regular range. The 3–5% slow variation in TESS is not locked to orbital phase and is partly QLP level jumps. Using Gaia's full parameter covariance (corr(ω, T_p) = 0.99999 for this near-circular orbit) shrank the conjunction-time error to 0.15–0.38 d. This makes the phase agreement a significant test, and the eclipses pass it.

A16. TYC 2173-965-1 (TIC 257352286)



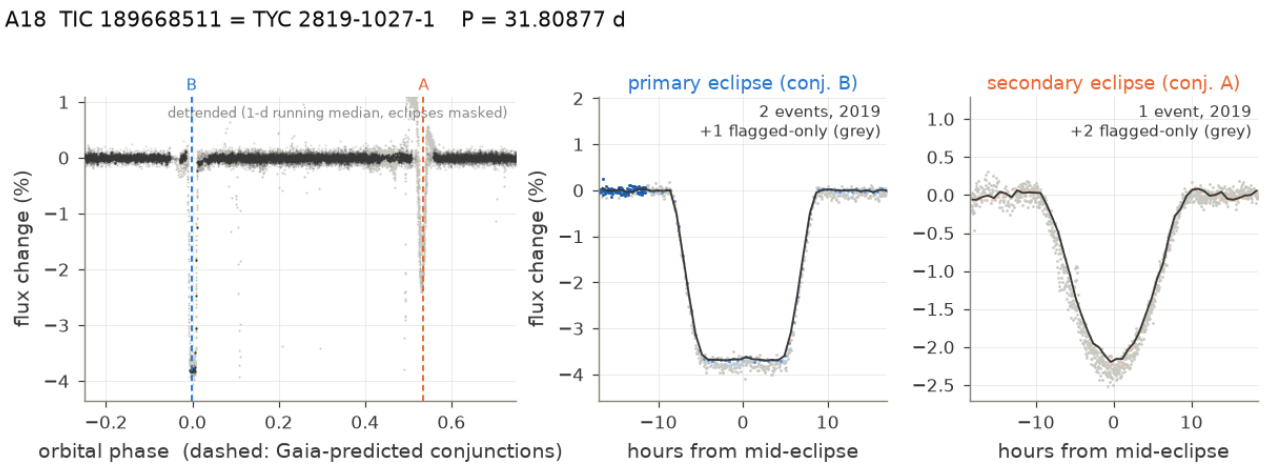
A16. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.802223 d, Gaia's predicted conjunctions dashed. Middle: the four round-bottomed 10% eclipses at A (sectors 15, 41, 55, 82). Right: the three flat-bottomed 8.45% eclipses at B (sectors 41, 55, 82).

Identifiers	TIC 257352286; Gaia DR3 1840217568772936448; TYC 2173-965-1
Position (J2000)	21:09:36.6 +23:45:30.2
Brightness	G 9.87; Tmag 9.42
Stars	Gaia SB1 (17 RV epochs, no secondary RVs). About 1.5 $M_\odot$ F/G subgiant (91.5% of the light) + about 1 $M_\odot$ G dwarf ($R \approx 0.94\ R_\odot$, about 5900 K); mass function 0.1473 $M_\odot$
Period	23.802223 ± 0.000022 d (Gaia 23.800734 ± 0.001153 d; difference +1.29σ)
Epoch of primary	T_0(A) = BTJD 2804.28519 ± 0.00089 = BJD_{TDB} 2459804.28519
Primary eclipse (A)	9.90–10.23% deep, 20.2 h, round-bottomed (limb-darkened transit of the subgiant by the companion), ingress 5.5 h; four events, sectors 15, 41, 55, 82
Secondary eclipse (B)	8.40–8.51% deep, 9.2 h, flat bottom 5.3 h (total occultation of the companion); three events, sectors 41, 55, 82; B − A = 13.65454 ± 0.00111 d (phase 0.57367)
Orbit	Eclipse model (sector 82): e = 0.4198, ω = 284.62°; Gaia 0.4239 ± 0.0073 (−0.56σ), 284.57 ± 0.87° (+0.05σ). i = 88.96°, $k = R_2/R_1$ = 0.317
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s41, s55); TESS-SPOC (s82); TARS (s41)
Pixel attribution	Difference images (SPOC s55 target pixel file; TESScut s41) have the target's PSF shape, centroid within 0.1 px of the target; no neighbour brighter than G 15 within 60″ could produce the dip (each would need a drop of more than 100%)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, ExoFOP (no TOI or CTOI), general web search; SIMBAD lists it as SB* with one reference (the Gaia DR3 multiplicity paper). Prior partial detection: a SPOC s14–s55 TCE at P = 13.5987 d, an alias folding both eclipses together, which never became a TOI
Open issues	None; spectroscopy at quadrature would give absolute masses and radii
Next eclipses	A: BTJD 4327.6275, 4351.4297; B: BTJD 4341.2820, 4365.0842 (1σ about 2.5–3 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 2173-965-1 is a detached, eccentric, totally eclipsing binary. At conjunction A the smaller companion transits the subgiant that Gaia's radial velocities track, giving the 20-h, round-bottomed eclipse. At B the subgiant totally hides the companion, giving the flat 8.45% eclipse, whose depth equals the companion's share of the light. The A eclipse lasts 2.2 times as long because it happens at a separation of $1.39a$ against $0.58a$ at B. One Keplerian eclipse model, fitted to sector 82 alone, reproduces all seven eclipses with residual rms 0.5–0.9 ppt, and its e and ω agree with Gaia's independent radial-velocity orbit. With the primary radius taken from the TIC and corrected for the companion's light, the solution is a self-consistent pair of a 1.5 $M_\odot$ F/G subgiant and a 1 $M_\odot$ G dwarf.

Seven eclipses are observed, including the sector-15 primary. The secondary's 5.3-h flat bottom is distinct from its 9.2-h total duration. The SPOC TCE at the alias period and the Gaia SB1 solution are prior partial detections of the system. QLP's detrended flux removes the 20-h eclipse in sectors 15 and 41; only SAP keeps it.

A18. TYC 2819-1027-1 (TIC 189668511)



A18. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 31.80877 d, Gaia's predicted conjunctions dashed. Middle: the three flat-bottomed 3.75% total occultations at B (sectors 17, 18, 85). Right: the three rounded 2.2% transits at A (sectors 17, 58, 85).

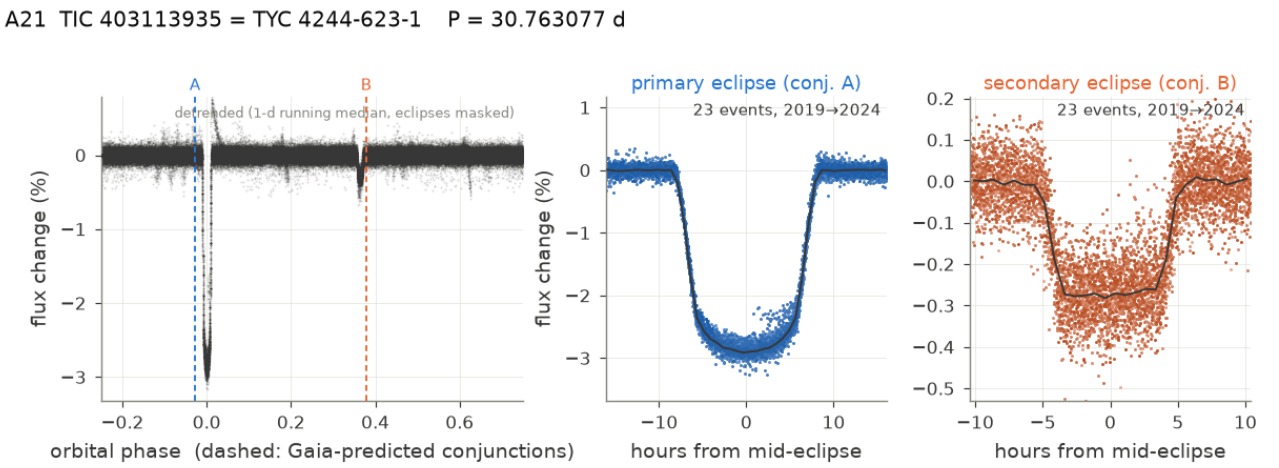
Identifiers	TIC 189668511; Gaia DR3 3468957854207709 44; TYC 2819-1027-1
Position (J2000)	01:42:18.1 +39:53:21.1
Brightness	G 9.84; Tmag 9.22
Stars	K giant of about 1.3–1.6 $M_\odot$ (R_1 fixed at 4.78 $R_\odot$) + roughly solar-type dwarf, 0.86–0.96 $M_\odot$, 0.67–0.75 $R_\odot$, 3.75% of the light; mass function 0.1364 $M_\odot$; Gaia SB1
Period	31.80877 ± 0.00001 d from B; 31.80862 ± 0.00003 d from A (Gaia 31.799234 ± 0.003676 d; the eclipse period is longer by 2.6σ)
Epoch of primary	T_0(B) = BTJD 1766.5861 = BJD_{TDB} 2458766.5861; each A follows its B by 16.894 d
Primary eclipse (B)	3.73–3.81% deep, 16.75–16.99 h, flat-bottomed (flat part 10.22 h): total occultation of the dwarf by the giant; sectors 17, 18, 85; O–C −0.9, +2.3, 0.0 min
Secondary eclipse (A)	2.12–2.25% deep, 18.0–18.6 h, short rounded bottom: transit of the dwarf across the limb-darkened giant; sectors 17, 58, 85; O–C +1.0, −0.7, +1.1 min. Predicted from the B geometry (19.3 h, about 2.2%) before being measured
Orbit	A follows B by phase 0.5311 (Gaia's elements predict 0.5379); e cos ω = −0.048 against Gaia's −0.059 ± ~0.009 (about 1.2σ). i ≈ 86.1–86.65°, off-centre chord
Pipelines agreeing	QLP SAP; TARS (s017, s018, s058, s085); TESS-SPOC SAP (s085, s058). No SPOC or TESS-SPOC product exists for s017–s018
Pixel attribution	TESScut difference images of the B eclipses in s017, s018 and s085 and the A eclipses in s017 and s058 (made after the pre-submission review; `appointments/vet2pix/`): in every clean event the deficit centroid lies within 0.16 px (3.4″) of the catalogue position and 0.2–2.8″ from the light centroid, and the deficit is a scaled copy of the target's image (residual 0.08–0.19); s058 A needs a 0.02-px pointing shift fitted out, and s085 A can be tested only on stray-light-flagged cadences, where it agrees too. The B depth is 3.7% in 1×1, 3×3 and 5×5 apertures. The only Gaia source within 60″ brighter than G 15 (G 13.31 at 58.7″, a rotating variable) would need a 95% eclipse, lies 2.9 px from the target where the loss is 0.2–12 ppt against 37 ppt on the target, and varies by 0.03 mag in Gaia; all others would need more than 1000%. QLP and TESS-SPOC give the same depths; TARS runs shallower
Catalogue status	Not listed in: VSX (only the rotating variable 58″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SIMBAD (SB* only), general web search
Open issues	

	Three of the six events lie in QLP-flagged scattered-light stretches (their depths, shapes and times match the clean ones, and TESS-SPOC SAP reproduces them); radial velocities or a spectrum would fix the giant's mass and the companion's light
Next eclipses	B at 1766.5861 + n × 31.80877; A 16.894 d after each B
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed: six eclipses of two kinds in four sectors at Gaia's period, the deficit on the target in difference images of five of them (added after the pre-submission review), and depth agreement between QLP and TESS-SPOC (not with TARS (s085 B 2.71% against QLP's 3.74%; s058 A 1.79% against 2.25%); the B and A periods, 31.80877 ± 0.00001 and 31.80862 ± 0.00003 d, differ by 4.7σ formally.

TYC 2819-1027-1 is a K giant with a roughly solar-type dwarf companion on a 31.8-d orbit. At B the giant is in front and totally hides the dwarf, giving flat-bottomed occultations. Their depth makes the companion 3.75% of the light, about 1.5 times its area ratio, so in the TESS band the companion is brighter per unit area than the giant, as a G dwarf of about 0.9–1.0 $M_\odot$ would be. With the giant's radius fixed, the B eclipse yields the companion's size and the impact geometry, and these predict the A transit (19.3 h, about 2.2% after limb darkening). The transit is observed three times at 18.0–18.6 h and 2.06–2.25%. Both kinds of eclipse give nearly the same period (4.7σ apart formally).

No other dip of 1% or more occurs in the four sectors.

A21. TYC 4244-623-1 (TIC 403113935)



A21. Left: QLP SAP flux over 28 sectors, detrended with a 1-d running median with the eclipses masked, folded on P = 30.763077 d, Gaia's predicted conjunctions dashed. Middle: the 2.77%, 16.1-h primary transits. Right: the 0.27%, 10.4-h flat secondary occultations, one of which was reported as a planet candidate.

Identifiers	TIC 403113935; Gaia DR3 2245031052604901120; TYC 4244-623-1; CTOI 403113935.01
Position (J2000)	20:14:39.7 +66:08:00.4 (TESS northern continuous viewing zone)
Brightness	G 9.84; Tmag 9.41
Stars	F/G subgiant, R_1 = 2.50–2.66 $R_\odot$ (assuming 1.2–1.4 $M_\odot$) + early-M dwarf of 0.39–0.43 $M_\odot$, 0.39–0.41 $R_\odot$, about 3530 K; mass function 0.0239 $M_\odot$. Gaia SB1, significance 53, RUWE 34.9
Period	30.763077 ± 0.000013 d (Gaia 30.785932 ± 0.010666 d; the eclipse period is 2.1σ shorter)
Epoch of primary	BTJD 2849.89772 = BJD_{TDB} 2459849.89772; secondary BTJD 2830.31820
Primary eclipse	2.77 ± 0.09% deep, 16.1 h, rounded U (transit); 20 clean events (23 fitted), rms 1.3 min, no curvature
Secondary eclipse	0.266 ± 0.014% deep, 10.4 h, flat-bottomed (occultation); 20 clean events (23 fitted), rms 3.6 min
Orbit	

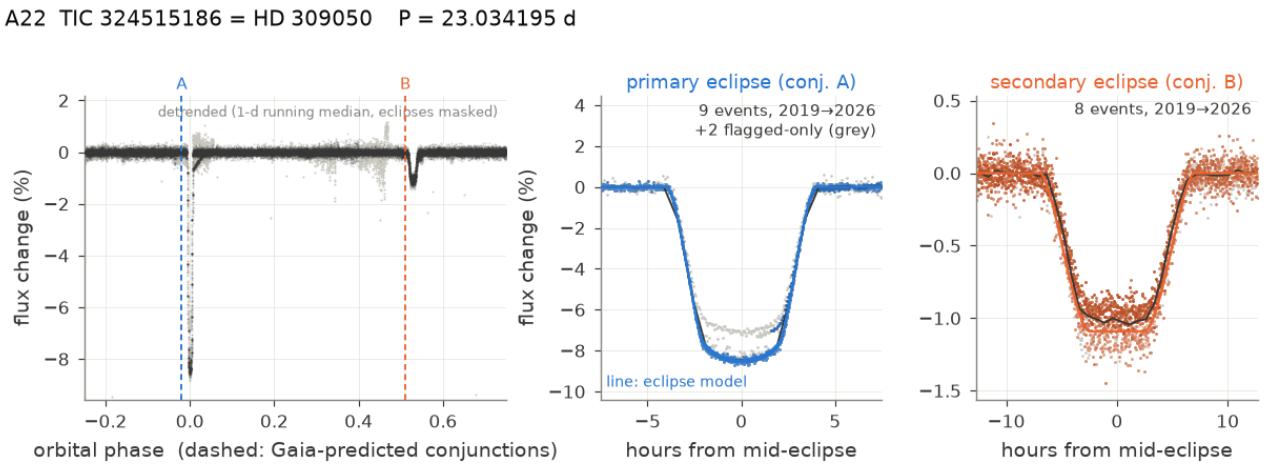
	Eclipse spacing 0.36354 of an orbit (Gaia predicts 0.40655): $e \cos \omega = -0.209$ against Gaia's -0.143 ± 0.018 (3.7σ); duration ratio gives $e \sin \omega = -0.23$ (Gaia -0.248). Eclipse-based $e \approx 0.32$, $\omega \approx 230°$ (Gaia 0.286, 240.06°). $i = 88.7°$, $k = 0.155$. No apsidal motion (primary and secondary periods agree to $(0.8 \pm 2.8) \times 10^{-5}$ d)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s056, s075); TASOC (s015); times within 6 min
Pixel attribution	No centroid shift in TESS-SPOC s056 during the primary. The only Gaia neighbour brighter than G 15 (G 13.93 at 49.9″) would need a 119% primary. RUWE 35 implies a probable star within about 1″
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, general web search; Kostov table 2 flag only. **A previous report misidentifies the signal:** ExoFOP CTOI 403113935.01 (submitted 2025-07-01 as a “single transit event”, epoch BJD 2459891.8453, 2,483 ppm, 9.9 h, 14.9 $R_\oplus$) is this star's secondary eclipse in sector 58. In the TESS-SPOC DV report (s56–s69), TCE 2 corresponds to the secondary eclipses
Open issues	Gaia's period and $e \cos \omega$ are inconsistent with the eclipse timing; an RV refit would constrain the true e and ω. High-resolution imaging for the RUWE-35 companion (a possible triple)
Next eclipses	Primary: BTJD 4357.2885 (2026-11-12 18:52 UTC), 4388.0516, 4418.8147; secondary: 4337.7090 (2026-10-24 04:57 UTC), 4368.4721 (about ±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with caveats. No difference image (only a null centroid shift in TESS-SPOC s056); RUWE 35 implies a star within about 1″; Gaia's period and $e \cos \omega$ are 2.1σ and 3.7σ off, so the eclipses follow Gaia's orbit only approximately.

Note: the star has an ExoFOP community planet candidate, CTOI 403113935.01, which is a misreading of the secondary eclipse of this 30.763-d eccentric eclipsing binary as a single planetary transit.

TYC 4244-623-1 is an F/G subgiant with an early-M companion on an eccentric 30.76-d orbit, observed in 28 TESS sectors. Twenty clean primaries and twenty clean secondaries give a period with 1.3-min timing rms. The primary depth is $1.14 \times k^2$, as limb darkening predicts. The ~1-d offset from Gaia's predicted times is a linear drift of −0.0229 d per orbit for both kinds of eclipse, which means that Gaia's period is too long. Extrapolated back to Gaia's epoch, the primaries and secondaries are offset in opposite directions, so Gaia's $e \cos \omega$ is also biased. Because TESS rules out apsidal motion, we attribute the discrepancy to Gaia's spectroscopic solution (RUWE 35, low significance).

RUWE 35 cannot come from this orbit, so a further star probably lies within about 1″, making the system a possible triple. The eclipse period is within 2.1σ of Gaia's SB1 period for the target, and the eclipses fall near its conjunctions. The secondary is phase-locked to the primary.

A22. HD 309050 (TIC 324515186)



A22. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.034195 d, Gaia's predicted conjunctions dashed. Middle: the 8.3%, 7.5-h flat-bottomed transits at A. Right: the 1.1%, 12.7-h total occultations at B.

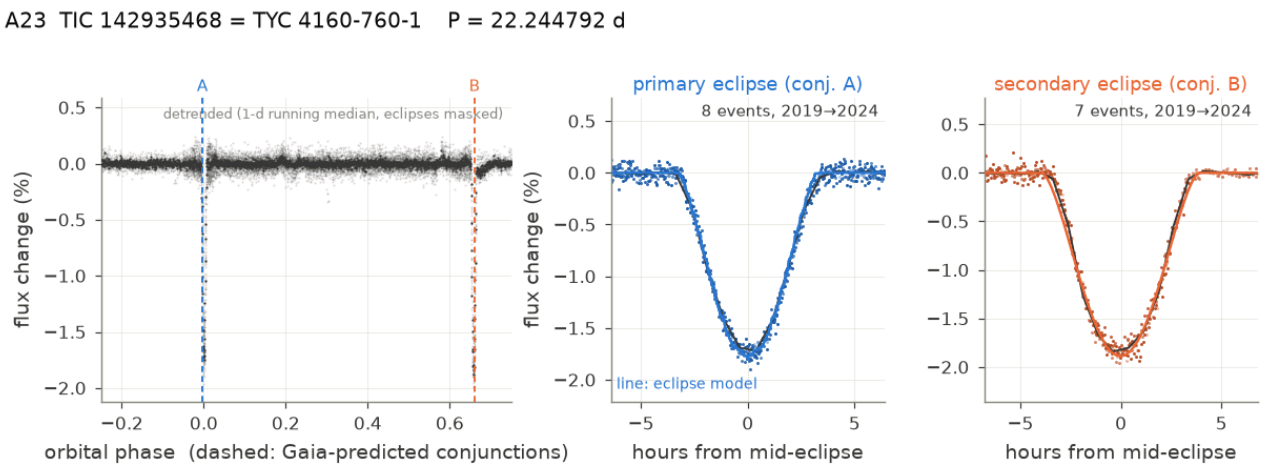
Identifiers	TIC 324515186; Gaia DR3 5333119169896513536; HD 309050
Position (J2000)	11:49:05.6 −63:46:58.3
Brightness	*G* 9.85; Tmag 9.44; V 9.97 (SIMBAD)
Stars	F2 subgiant, R_1 = 2.12 $R_\odot$ (TIC 2.21), M_1 = 1.39 $M_\odot$ (TIC) + M0 dwarf, m_2 = 0.572 $M_\odot$, R_2 = 0.59 $R_\odot$, about 3900 K, 1.09% of the light; mass function 0.0487 $M_\odot$; Gaia SB1
Period	23.034195 ± 0.000010 d (Gaia 23.0619 ± 0.0129 d; the eclipse period is 2.15σ shorter)
Epoch of primary	T_0(A) = BTJD 2657.89250 ± 0.00031 = BJD_{TDB} 2459657.89250
Primary eclipse (A)	7.9–8.33% deep, 7.5 h (flat part about 3.9 h), nearly flat-bottomed transit of the companion across the F star (b = 0.32); nine events, sectors 10–99, two in QLP-flagged cadences measured in other pipelines; O−C within ±0.3 min over 1500 d
Secondary eclipse (B)	0.92–1.14% deep, 12.4–13.8 h (flat part about 6 h), total occultation of the companion (b = 0.58); eight events; B − A = 12.17047 ± 0.00079 d (phase 0.52837)
Orbit	Joint physical model, e and ω free: e = 0.316, ω = 82.3° against Gaia 0.300 ± 0.017, 81.5 ± 3.7° (+0.9σ, +0.2σ); timing e cos ω = 0.0425 (Gaia 0.0443 ± 0.019). i = 88.72°, k = 0.277
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TGLC PSF; eleanor-lite; TARS (mid-times within 0–3 min for A, 2–11 min for B)
Pixel attribution	TESScut difference images (s37, s64): largest loss at the target pixel for both kinds, centroid of the loss within 0.04–0.1 px of the target; no loss at the *G* 14.1 star 18″ or the *G* 15.5 star 15″ away
Catalogue status	Not listed in: VSX (in a 120″ search the nearest entry, at 72″, is unrelated), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, SB9, WDS, Hipparcos, Gaia DR3 variability tables, CCDM, ExoFOP (no TOI/CTOI), 5 SIMBAD references, general web search. Kostov table 2 flags 16 faint stars 30–67″ away but not the star (probably contamination by the target's eclipses). A TESS-SPOC multi-sector DV report (s56–s69) detected the A eclipse at SNR 659; it never became a TOI
Open issues	None for the binary; two RV spectra would give the companion's absolute mass, an infrared spectral type would test T_2 ≈ 3900 K
Next eclipses	A: BTJD 4339.389, 4362.423; B: BTJD 4328.525, 4351.559 (±1.2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 309050 is an F2 subgiant with an early-M companion. At A the faint companion crosses the F star whose radial velocities Gaia measured. At B it is completely hidden, so the 1.09% depth is the companion's light directly. One geometry (k = 0.277, surface-brightness ratio 0.144, i = 88.7°) fits all six

modelled eclipses to the noise. Forcing 8% third light (SPOC's crowding) only moves k to 0.288. The light-curve e and ω, fitted freely, match Gaia's within 1σ. There is no sign of apsidal motion or timing variations.

The ~2-d offset from Gaia's predicted times is not constant. It grows from about −1 d to −4 d because Gaia's period is too long, and extrapolated back to Gaia's epoch the eclipses agree with Gaia's conjunctions (+1.3σ and +0.8σ). A Monte-Carlo conjunction predictor can jump a whole cycle when a sampled ω crosses 90° (conjunction A falls just after periastron here), inflating the window. The search's predictor re-centres the draws (Section 2.2) and was not affected.

A23. TYC 4160-760-1 (TIC 142935468)



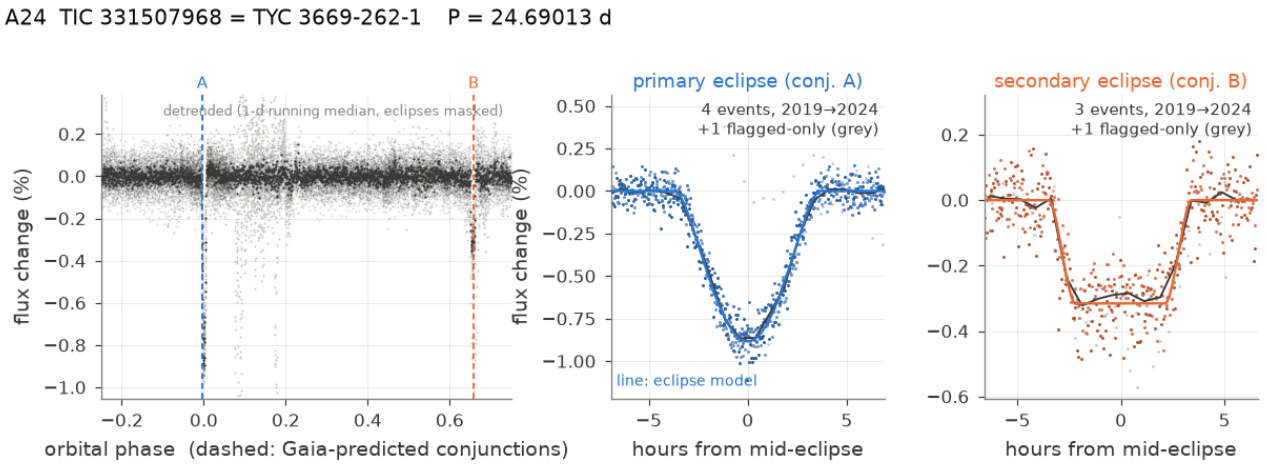
A23. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.244792 d, Gaia's predicted conjunctions dashed. Middle: the eight 1.70%, 6.4-h grazing transits at A. Right: the 1.8%, 6.8-h near-total occultations at B.

Identifiers	TIC 142935468; Gaia DR3 864679171266740736; TYC 4160-760-1
Position (J2000)	11:54:54.3 +66:03:48.2
Brightness	G 10.00; Tmag 9.45
Stars	Subgiant, R_1 = 1.97–2.15 $R_\odot$ (TIC 2.06), M_1 = 1.1–1.5 $M_\odot$ + M dwarf, m_2 = 0.57–0.68 $M_\odot$, R_2 = 0.59–0.65 $R_\odot$, about 3600 K, 1.9% of the light; mass function 0.0651 $M_\odot$; Gaia SB1
Period	22.244792 ± 0.000017 d (Gaia 22.247626 ± 0.00183 d; −1.55σ)
Epoch of primary	T_0(A) = BTJD 2228.68805 ± 0.00056 = BJD_{TDB} 2459228.68805
Primary eclipse (A)	1.66–1.76% deep, 6.0–6.7 h (about 6.4 h), rounded V/U: grazing transit (b = 1.10); eight events, sectors 14–75, O−C within ±3.5 min
Secondary eclipse (B)	1.73–1.91% deep, 6.6–7.0 h (about 6.8 h): near-total occultation of the companion (b = 0.75); five full and two partial events; B − A = 14.62903 ± 0.00091 d (phase 0.65764)
Orbit	Timing e cos ω = 0.2455 against Gaia's 0.2546 ± 0.0065 (−1.4σ); a model grid prefers Gaia's own e sin ω (−0.193). i = 87.17°, k = 0.30 (grazing eclipses let k, i and e sin ω trade off)
Pipelines agreeing	QLP SAP; SPOC 2-min s75; TESS-SPOC s75; TASOC s14/s15; TARS (13 events, within ±10 min)
Pixel attribution	SPOC s75 pixel file and TESScut s41/s48: the loss has the star's image shape, centroid within 0.1 px of the target; no Gaia source brighter than G 16 within 100″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 (all tables), WDS, Hipparcos, SB9, Gaia DR3 variability tables, CCDM, ExoFOP, 2 SIMBAD references (SB* from Gaia DR3), general web search. No SPOC DV products
Open issues	The k–i–e sin ω trade-off; radial velocities, multi-colour photometry of B and higher-cadence ingress/egress would break it
Next eclipses	A: BTJD 4341.943, 4364.188; B: BTJD 4334.328, 4356.572 (±2.5 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4160-760-1 is a subgiant with a cool dwarf companion of about 0.6 $M_\odot$. Its two eclipses are nearly equal in depth. At A the companion just clips the primary's limb and blocks about 1.7% of the light. At B, the closer conjunction, most of the companion is hidden, and the depth of about 1.8% is roughly the companion's whole light. The near-equal depths are therefore a coincidence of geometry. A companion carrying 2% of the light would not show in Gaia's spectra, so a single-lined solution is what one expects.

The eclipse durations are about 6.4 h and 6.8 h, and the offset from Gaia's predicted times is again a slow drift caused by the error in Gaia's period. Because both eclipses graze, the duration ratio does not constrain e sin ω well. The best grid point is Gaia's own value, with k = 0.30 (rms 410 ppm over eight modelled eclipses).

A24. TYC 3669-262-1 (TIC 331507968)



A24. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.69013 d, Gaia's predicted conjunctions dashed. Middle: the 0.86%, 6.9-h near-grazing V-shaped transits at A. Right: the 0.32%, 6.6-h flat-bottomed total occultations at B.

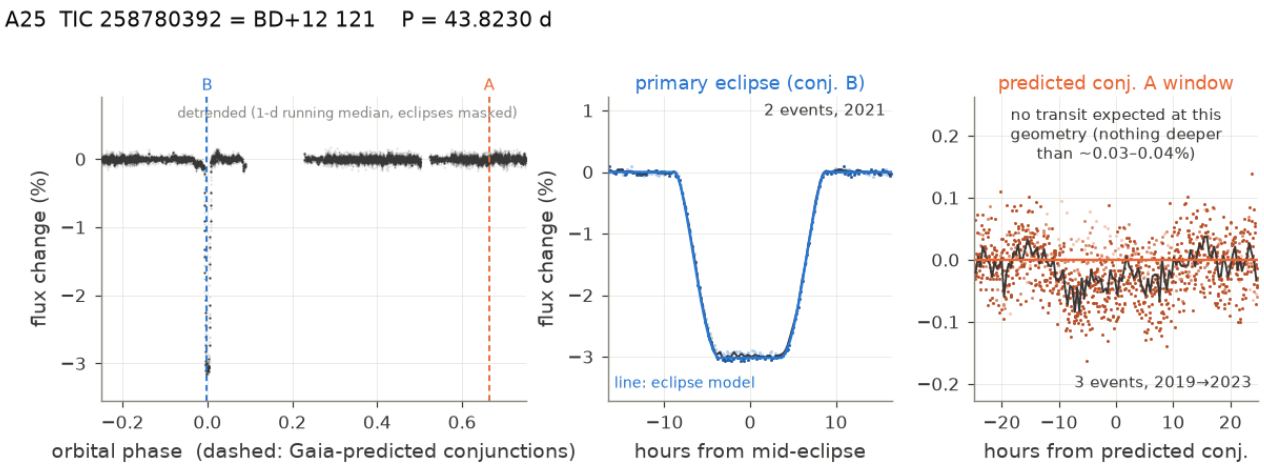
Identifiers	TIC 331507968; Gaia DR3 411152889920798336; TYC 3669-262-1
Position (J2000)	01:11:25.8 +54:21:26.3
Brightness	G 9.83; Tmag 9.28
Stars	G/K subgiant, R_1 = 1.89–2.11 $R_\odot$ (TIC 2.06), log g ≈ 3.85–3.9 + mid-M dwarf, M_2 = 0.29–0.36 $M_\odot$, R_2 = 0.29–0.33 $R_\odot$, roughly 3300 K, 0.32% of the light; mass function 0.01654 $M_\odot$; Gaia SB1
Period	24.69013 ± 0.00003 d (Gaia 24.68806 ± 0.00454 d; 0.46σ)
Epoch of primary	T_0(A) = BTJD 1797.4225 ± 0.0015 = BJD_{TDB} 2458797.4225; T_0(B) = BTJD 1813.6099 ± 0.0022
Primary eclipse (A)	About 0.86% deep, 6.9 h, nearly V-shaped (flat part 1.6 h): near-grazing transit of the subgiant (impact 0.997 R_1); four complete events (s18, s58 ×2, s85) and a partial in QLP-flagged s17 cadences; O−C −3.3 to +2.3 min
Secondary eclipse (B)	About 0.32% deep, 6.6 h, flat-bottomed (T_{23} ≈ 4.6 h): total occultation of the M dwarf (impact 0.535 R_1); three events (s18, s58, s85); phase 0.65562 ± 0.00011 after A
Orbit	One two-sphere Keplerian model fits all seven complete eclipses at χ^2/dof 1.055: e = 0.3824, ω = 307.96°, i = 87.67°, k = 0.155. Gaia: e 0.4019 ± 0.0059 (3.3σ apart), ω 307.30 ± 1.21° (within 0.5σ); e cos ω 0.2340 against 0.2435 ± 0.0076 (1.2σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP; TARS (mid-times within 0.002 d). No SPOC 2-min data
Pixel attribution	TESScut difference images (s58, s85 A): same shape as the target's PSF, peak in the target's own pixel, centroid offset 0.2–0.55 px (within the noise of a 0.8% signal). The VSX “RS” star 23″ away (G 17.89) would need a 1440% A dip; every other source within 60″ is G ≥ 17.07. No common mode (median of 110–150 stars −73 to +261 ppm)
Catalogue status	

	Not listed in: VSX (the "RS" entry 23″ away is a separate *G* 17.9 background star with no period), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI/CTOI), SIMBAD (SB*, one reference), general web search
Open issues	The photometric e is 3.3σ below Gaia's by Gaia's formal error; the tension is mild, since Gaia SB1 errors are commonly underestimated by a factor of about 1.5–2. A few RVs would refine e
Next eclipses	A: BTJD 4340.506, 4365.196 (±5 min); B: BTJD 4332.003, 4356.693 (±6 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3669-262-1 is a G/K subgiant of about 2.0 $R_\odot$ with an M dwarf of about 0.3 $M_\odot$. The A eclipse is a grazing transit of the subgiant, hence the V shape and a depth well below k^2. The B eclipse is a total occultation of the M dwarf, hence the flat bottom. The separation at A is 1.22a against 0.66a at B, so one inclination gives an impact parameter 1.9 times larger at A. The model's durations (7.3 and 6.8 h) match those measured. The occultation depth puts the companion at an absolute TESS magnitude of about 9.4, as expected for a dwarf of about 0.3 $M_\odot$.

The eclipse timing constrains e cos ω to about 10^{-4}. Fixing e and ω at Gaia's values raises χ^2 by more than 1600, so we adopt the photometric e and ω as the better determined. A further A eclipse, in fully flagged sector-17 cadences, is present in TESS-SPOC and TARS.

A25. BD+12 121 = ASAS J005930+1304.9 (TIC 258780392)



A25. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 43.8230 d, Gaia's predicted conjunctions dashed. Middle: the B windows with data; two identical 2.9%, 16.5-h flat-bottomed occultations (sectors 42 and 43) and a third window covering only the edge of an event. Right: the data at conjunction A, where the fitted geometry predicts no eclipse and none is seen (< 0.04%).

Identifiers	TIC 258780392; Gaia DR3 2776639323993879680; BD+12 121; ASAS J005930+1304.9
Position (J2000)	00:59:30.3 +13:04:56.1
Brightness	*G* 9.16; Tmag 8.52
Stars	G5/K giant (SIMBAD Sp G5), R_1 = 4.9–6.1 $R_\odot$ (TIC 6.07) + K0–K2 dwarf, $M_2 \approx 0.8\ M_\odot$ (0.73–1.12 allowed), R_2 = 0.89–1.11 $R_\odot$, 3.0% of the light; mass function 0.1270 $M_\odot$; Gaia SB1
Period	43.8230 ± 0.0025 d from two eclipses one cycle apart; cycle count fixed by Gaia (Gaia 43.8190 ± 0.0114 d; 0.33σ)
Epoch of eclipse	T_0(B) = BTJD 2451.287 ± 0.0013 = BJD_{TDB} 2459451.287
Eclipse (B)	2.89–2.92% deep, 16.5 h, flat-bottomed (T_{23} 8.8–8.9 h, ingress/egress about 3.9 h): total occultation of the dwarf by the giant; two events (s42, s43) matching to 0.03% in depth and 0.1 h in duration; both 0.24 d (0.4σ) after Gaia's predicted B times
Other eclipse (A)	None, as predicted: at the fitted i the impact parameter at A is 1.29–1.34 R_1, beyond 1 + k = 1.18 in all 11 Gaia draws. Box limits at A: 120–240 ppm (s043), 280–430 ppm (s070)

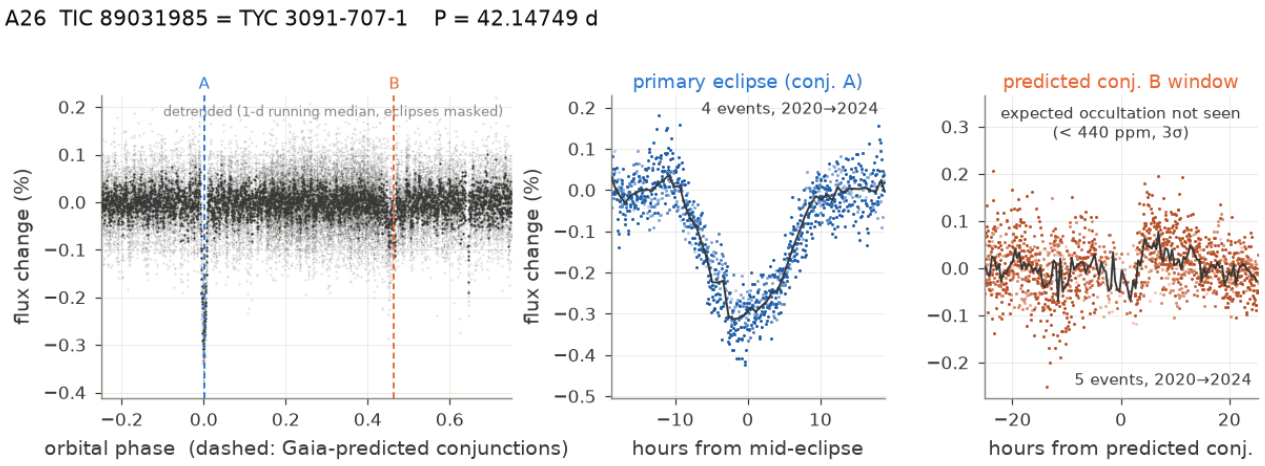
Orbit	e and ω held at Gaia draws (e 0.3841, ω 229.40°); one eclipse kind gives no independent $e \cos \omega$. i = 85.0–85.2°, k = 0.182
Pipelines agreeing	QLP SAP (quality-0 and all cadences); TARS (times within 0.003 d; TARS s43 depth 2.44%). TESS-SPOC exists only for s070
Pixel attribution	TESScut difference images (s42, s43): same shape as the target's PSF, offsets 1.0″ and 3.2″; the raw per-pixel depth is a flat ~2.8% across the whole PSF. Nearest Gaia neighbour G 19.58 at 17.7″. No common mode
Catalogue status	**VSX type revision.** VSX: ASAS J005930+1304.9 at the target, type MISC, P 91.5 d, amplitude 0.11 mag V; ASAS-SN "ROT:", no period. No eclipsing-binary catalogue, no Gaia DR3 variability row, nothing in SB9 or WDS, no TOI/CTOI, SIMBAD (SB*, 4 references); a general web search found nothing
Open issues	Only two eclipses; a third B eclipse and a spectrum (for M_1, R_1 and the companion's type) would improve the solution. An A transit, if ever seen, would force $i \geq 86°$
Next eclipses	B: BTJD 4335.68 ± 0.11, 4379.50 ± 0.11; A conjunction (no eclipse expected): 4364.83 ± 0.20
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision. The VSX entry ASAS J005930+1304.9 (MISC, 91.5 d) is this star, but it describes slow variability of the giant, not the eclipses. The ASAS-3 periodogram has power 0.35 at 91.2 d and 0.02 at twice the orbital period, and 0.03-mag eclipses filling 1.6% of the orbit cannot produce a 0.1–0.2-mag, 91-d signal. We propose reclassification from MISC/ROT to EA + ROT.

BD+12 121 is a G5/K giant with a K-dwarf companion. TESS observed two identical total eclipses one orbit apart, in which the giant hides the dwarf. Their depth, 3.0%, is the companion's share of the light. The same geometry predicts that the dwarf passes clear of the giant at the other conjunction. The two covered A windows are flat to 0.03–0.04%, so the data require the absence of an A eclipse, as Gaia's geometry predicts. The companion's absolute TESS magnitude (about 5.0) points to a K0–K2 dwarf of about 0.8 $M_\odot$. The light, mass function and radii are consistent.

The 91-d ASAS signal is attributed to slow, season-dependent modulation of the giant, most likely spots or activity, in agreement with ASAS-SN's independent "ROT:". TESS shows drifts of at most about 1.5% per sector and neither confirms nor contradicts that period.

A26. TYC 3091-707-1 (TIC 89031985)



A26. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 42.14749 d, Gaia's predicted conjunctions dashed. Middle: the four 0.29%, 18.8-h limb-darkened transits at A, 2019–2024. Right: the B window, where an occultation of about 0.2% was expected and none is seen (< 440 ppm, 3σ).

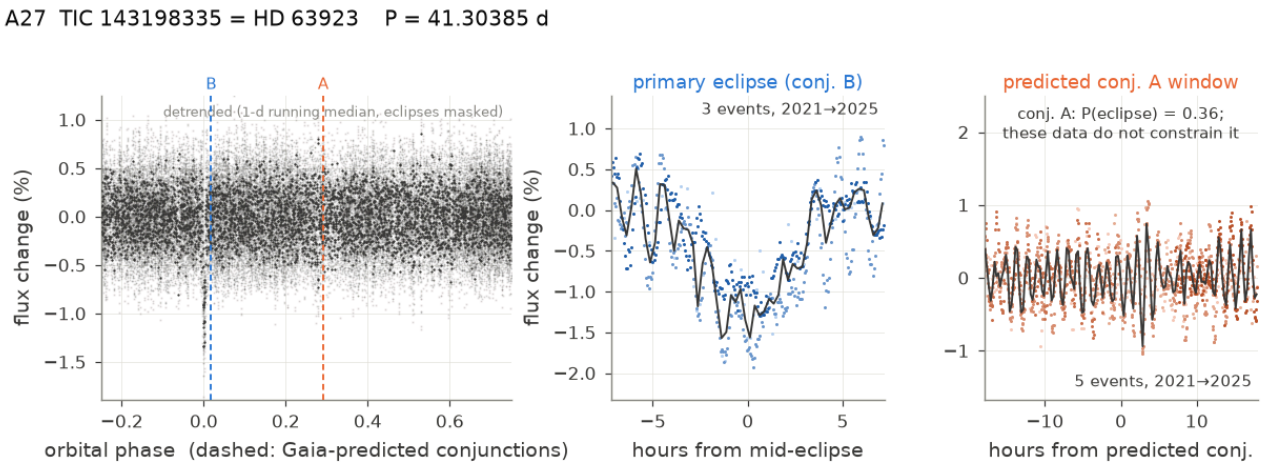
Identifiers	TIC 89031985; Gaia DR3 1344257629016319744; TYC 3091-707-1
Position (J2000)	17:36:17.8 +40:49:02.9

Brightness	*G* 9.46; Tmag 8.79
Stars	K giant, $R_1 \approx 7.5\ R_\odot$ (7.53 used) + a body of $R_2 \approx 0.54\ R_\odot$, minimum mass 0.50–0.67 $M_\odot$ for M_1 = 1.2–2.5 $M_\odot$; mass function 0.0341 $M_\odot$; systemic RV −57 km/s (SIMBAD); Gaia SB1
Period	42.14749 ± 0.00117 d (Gaia 42.144303 ± 0.003614 d; +0.8σ)
Epoch of eclipse	T_0(A) = BTJD 2645.4034 ± 0.0122 = BJD_{TDB} 2459645.4034 (cycle 15 of the fit)
Eclipse (A)	Stack: 2880 ppm deep (2535–3481 ppm per event), T_{14} = 18.8 h (17.0–21.7 h), flat fraction 0.2, limb-darkened U: a near-limb transit of the giant (impact ≈ 0.92 R_1); four events, sectors 26, 40, 52, 80; O−C −55 to +22 min (the outlier is the 30-min sector 26)
Other eclipse (B)	**Absent, though expected:** combined depth +29 ± 137 ppm over five windows, < 440 ppm at 3σ (< 917 ppm at any *e* cos ω); the physical Monte Carlo model predicts B with probability 0.999 and depth 2100 ppm [900, 4300]; only 4.6% of draws fall below 440 ppm
Orbit	With the TESS period and Gaia's t_{peri}, *e* (0.0631), ω (154.80°), the eclipses sit 0.0–0.1σ from Gaia's A time (±0.53 d) and about 35σ from B. *e*, ω not derivable from one eclipse kind
Pipelines agreeing	QLP SAP; TARS (all four); TASOC FLUX_RAW (s026); TESS-SPOC SAP and PDCSAP (s080)
Pixel attribution	TESScut difference images peak on the target pixel in all four sectors (offsets 0.28–0.35 px in the clean sectors); the *G* 15.65 star at 19.8″ would need 83% and is 10–24σ off; the *G* 11.56 star at 79″ is excluded by its own pixel. No common mode (target ranks 2nd–5th of 87–232 stars)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos (main, new reduction, variability annexes), SB9, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, one reference), general web search; not in the PTPS/JOTA SB1 sample (Jaros et al. 2026)
Open issues	The missing B eclipse: the companion's TESS-band surface brightness must be ≲ 0.13 of the giant's ($T_2 \lesssim 3050$ K), which contradicts a single main-sequence star of ≥ 0.5 $M_\odot$. The A depth is slightly shallow against the 4000 ppm predicted
Next eclipses	A: BTJD 4331.303 ± 0.048 (2026-10-17 19:15 UT), 4373.451 ± 0.049 (allow ±1 h); B predicted, not seen: 4350.85 ± 0.11
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3091-707-1 is a K giant of about 7.5 $R_\odot$ crossed every 42.15 d by a body of about 0.54 $R_\odot$ near the giant's limb, consistent with the mass function for a single K7–M0 dwarf. The four transits are on the target, reproduced by three other pipelines, and fall at Gaia's A conjunction.

The absence of an occultation at B is unexplained. The separation grows only 5.5% at B, so the companion should pass behind the giant and an occultation of about 0.2% was expected. The data limit it to below 440 ppm (injected 1000 and 2000 ppm signals were recovered exactly). To hide it, the companion must be fainter per unit area than a single main-sequence star of its mass. Two explanations remain viable. The companion may be a close M+M pair: two 0.3 $M_\odot$ stars at about 3400 K predict about 500 ppm, close to the limit. Alternatively, Gaia's *e* or ω may be substantially in error. A hot compact companion is excluded by GALEX's far-ultraviolet magnitude and by the radius. The missing occultation does not affect the detection. A spectrum, multi-band photometry of a transit, a TESS sector covering B, and an RV orbit would resolve it.

A27. HD 63923 (TIC 143198335)



A27. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 41.30385 d, Gaia's predicted conjunctions dashed. Middle: the three 1.07%, 7.2-h eclipses at conjunction B (sectors 35, 62, 88). Right: the data near conjunction A, which do not constrain an eclipse there.

Identifiers	TIC 143198335; Gaia DR3 5518561289173094400; HD 63923
Position (J2000)	07:49:14.9 −47:07:24.1
Brightness	G 9.94; Tmag 9.68
Stars	F0V (Michigan Spectral Survey), $R_1 \approx 2.7\ R_\odot$, with δ Sct-like variability; companion of m_2 = 0.63–0.71 $M_\odot$ for M_1 = 1.5–2.0 $M_\odot$ (a K dwarf on the main sequence); mass function 0.051 $M_\odot$. Gaia SB1, significance 20
Period	41.30385 ± 0.00030 d (Gaia 41.295873 ± 0.068698 d; +0.12σ)
Epoch of eclipse	T_0(B) = BTJD 2963.0484 ± 0.0044 = BJD_{TDB} 2459963.0484
Eclipse (B)	1.07% deep (1.03–1.17% per event), T_{14} = 7.1–7.4 h, flat fraction 0.3–0.4 (a V shape is worse by $\Delta\chi^2$ = 13, weak given the red noise); three events, sectors 35, 62, 88; O−C −5 to +9 min
Other eclipse (A)	Not constrained: P(eclipse) only 0.36 at A; the predicted centre falls on flagged data or a gap in s035 and s088
Orbit	With the TESS period and Gaia's t_{peri}, e, ω, the eclipses sit −0.6σ (±1.2 d) from B and 7.7σ from A; chance coincidence of an unrelated eclipser about 10^{-4}. e sin ω = −0.14 (Gaia), so B is near closest approach. Not derivable from one eclipse kind
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s035, s062); TARS (s035, s062, s088); CDIPS (s035); t_0 within 3–9 min
Pixel attribution	Difference images peak on the target in all three sectors; the fitted offset of the eclipse light excludes the 6.2″ G 14.38 neighbour at 8.7–16σ and the 35–80″ stars at 63–80σ (the target hypothesis is itself off by 2.2, 4.9 and 4.5σ). Only an unresolved source within about 2″ remains possible; TIC contamination ratio 0.243. No common mode
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP (contamination ratio 0.243), SIMBAD (SB*, nine references, none on eclipses), general web search; no GALEX match
Open issues	**Physical model unresolved.** An occultation of the mass-function companion cannot easily be this deep (at most 0.84%, range 0.62–1.13%, even if total) or this short (a total occultation would last ≳ 10 h); a ~0.3 $R_\odot$ transiting body fits the shape but not the mass
Next eclipses	B: BTJD 4367.379 ± 0.011 (2026-11-22 21:04 UTC), 4408.683 ± 0.011; A possible: 4337.45 ± 1.13
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	

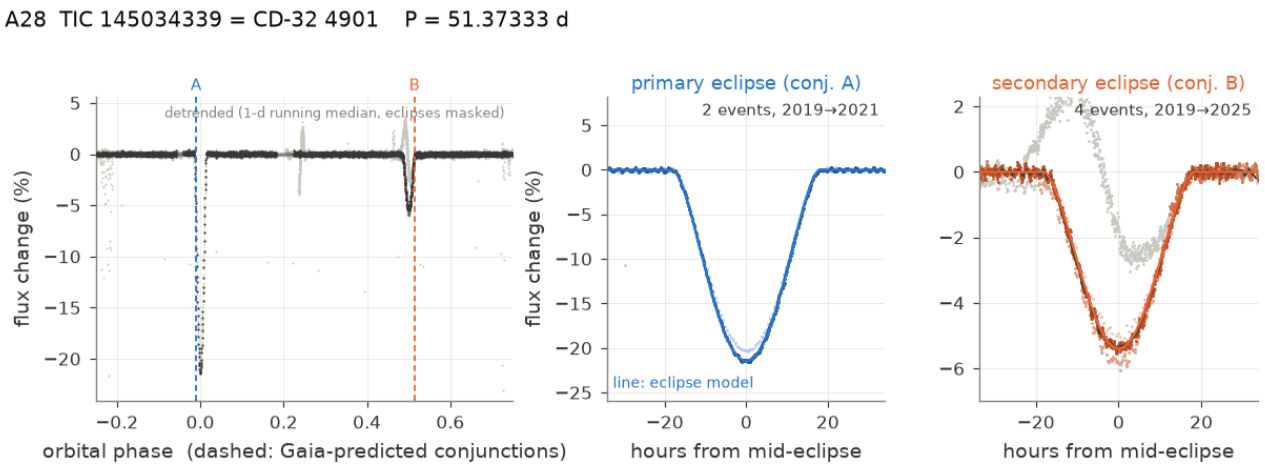
Confirmed, with caveats. The difference-image fit places the eclipse light on HD 63923 against the 6.2″ neighbour at 8.7–16σ, but the target hypothesis itself is off by 2.2, 4.9 and 4.5σ in the three sectors, so an unresolved source within about 2″ is not excluded; the 1.07% depth lies at the top of the 0.62–1.13% range allowed for an occultation of the mass-function companion (central value 0.84%); the physical model is unresolved.

Note: confirmed as an eclipsing system on the target (or on an unresolved source within about 2″), with its physical model unresolved. Either Gaia's orbit is in error (SB1 significance only 20, goodness-of-fit −1.5), or the companion is not a single main-sequence star.

HD 63923 is an F0 dwarf with three 1.07%, 7.2-h eclipses at Gaia's B conjunction. Three other pipelines reproduce the eclipses, and the difference images put the light on HD 63923 and not on its 6.2″ neighbour. The sector-62 eclipse (BTJD 3004.36) is distinct from a sector-wide systematic feature at BTJD 3007.

What the system is remains open. Under the convention validated on TIC 403113935 (A21), B means the companion passes behind the F star. However, an occultation of a 0.64 $R_\odot$, ~4300 K dwarf by a 2.7 $R_\odot$ F0 star can be at most 0.84% deep. The duration and flat fraction also imply R_1R_2 = 0.77–0.87 $R_\odot^2$ where about 1.7 is needed. A transit of the F star by a ~0.3 $R_\odot$ body ($k^2 \approx 1.2\%$) fits depth and shape. That reading, though, is the opposite of the convention at this conjunction, and no 0.67 $M_\odot$ object has that radius. An independent RV orbit is the decisive measurement. Further constraints would come from multi-colour eclipse depths (an occultation of a cool companion deepens to the red; a transit is grey), from high-resolution imaging within 2″, and from uncontaminated TESS coverage of an A window.

A28. CD-32 4901 (TIC 145034339)



A28. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 51.37333 d, Gaia's predicted conjunctions dashed; the secondary sits at phase 0.5000. Middle: the two 20–21%, 1.42-d primary eclipses (sectors 7 and 34). Right: the 5%, 1.4-d secondary eclipses.

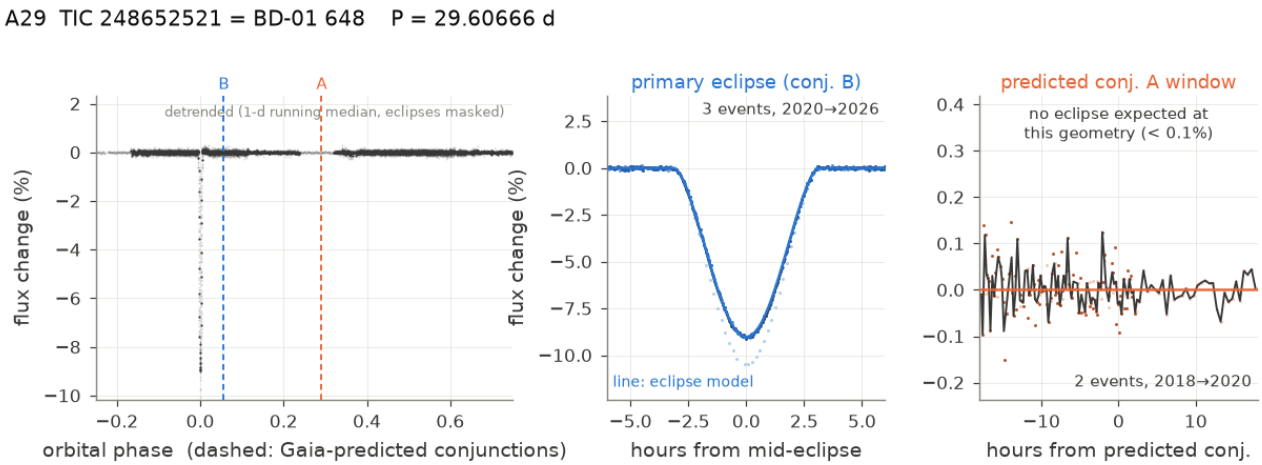
Identifiers	TIC 145034339; Gaia DR3 5547641825546896128; CD-32 4901 = CPD-32 2009 = TYC 7125-1800-1
Position (J2000)	08:08:46.9 −32:41:34.1
Brightness	*G* 9.74; Tmag 9.25
Stars	Gaia SB2 pair of large stars: $M_1 \sin^3 i$ = 1.147 $M_\odot$, $M_2 \sin^3 i$ = 0.577 $M_\odot$ (mass ratio 0.50 from Gaia's amplitudes); radii sum 8.5 $R_\odot$; companion 66% of the TESS light; TIC radius 8 $R_\odot$. X-ray source 1RXS J080846.6-324131 at 5.2″ and EUVE J0808-32.6 at 10″
Period	51.37333 ± 0.0002 d (Gaia 51.310621 ± 0.013853 d; the eclipse period is 0.063 d longer, 4.5σ)
Epoch of primary	T_0(A) = BTJD 1512.2307 ± 0.002 = BJD_{TDB} 2458512.2307

Primary eclipse (A)	20.0–21.2% deep, 1.42 d, partial V; two events (s7, s34)
Secondary eclipse (B)	4.9–5.5% deep, 1.38–1.42 d; four events (s8, s35, s61 on a scattered-light bump measured in TESS-SPOC, s88); O−C ≤ 0.0018 d; B − A = 25.6878 d (phase 0.50002)
Orbit	Circular: $e \cos \omega = 0.00004 \pm 0.0001$ against Gaia's 0.038 ± 0.007 (5.6σ); equal durations give $\lvert e \sin \omega \rvert \lesssim 0.01$ (Gaia -0.017 ± 0.006, not ruled out). Gaia $e = 0.042$, goodness-of-fit 14.9. Two-sphere fit: $i = 85.4°$
Pipelines agreeing	QLP SAP; TGLC PSF and aperture; GSFC-eleanor-lite; T16 difference imaging; TESS-SPOC (s61)
Pixel attribution	TESScut: the missing flux has the target's PSF shape, difference-centroid offset 0.026 px (A); neighbours within 60″ (G 14.75, 14.92) would need about 2100% and 2500%; RUWE 0.83
Catalogue status	Not listed under any name in: VSX, ASAS-SN, ASAS-3, GCVS, SB9, WDS, Hipparcos, Gaia DR3 variability tables (never tested by Gaia), Prša et al. 2022, Justesen & Albrecht 2021, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4 (three neighbours are in table 2, the target is not), ExoFOP, SIMBAD (SB*, one paper), general web search
Open issues	Gaia's e is probably spurious; Gaia's mass ratio would make the 0.58 $M_\odot$ star evolved, probably stripped. The weakest constraint is Gaia's mass ratio, not the eclipses. Partial eclipses leave the radius ratio free (the fit chose 2.3 and 6.2 $R_\odot$)
Next eclipses	A: BTJD 4337.764, 4389.137 (±0.011); B: BTJD 4363.450, 4414.823 (±0.005)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-32 4901 is a double-lined binary of two large stars on a 51.4-d orbit, with eclipses lasting 1.4 d. A 1.42-d eclipse needs the two radii to sum to at least 6 $R_\odot$. A circular two-sphere fit gives 8.5 $R_\odot$, partial V-shaped eclipses as observed, and a companion supplying 66% of the light. The depths need both stars large and bright, consistent with an SB2 and with the TIC radius. The nearby X-ray and EUV sources are consistent with an active giant binary. The photometric period is 4.5σ longer than Gaia's, but Gaia's conjunctions carried forward with it land within 0.3–0.5σ of the eclipses, and the deep eclipse falls at Gaia's A.

The secondary sits at phase 0.50002, so the orbit is circular, as expected for a giant at P = 51 d. Gaia's e = 0.042 disagrees by 5.6σ and is probably spurious given its goodness-of-fit of 15. If Gaia's velocity amplitudes are right, the less massive star must be evolved and probably stripped, as in a giant binary after mass transfer. Ground-based radial velocities and a spectrum of the cool component would resolve it.

A29. BD-01 648 (TIC 248652521)



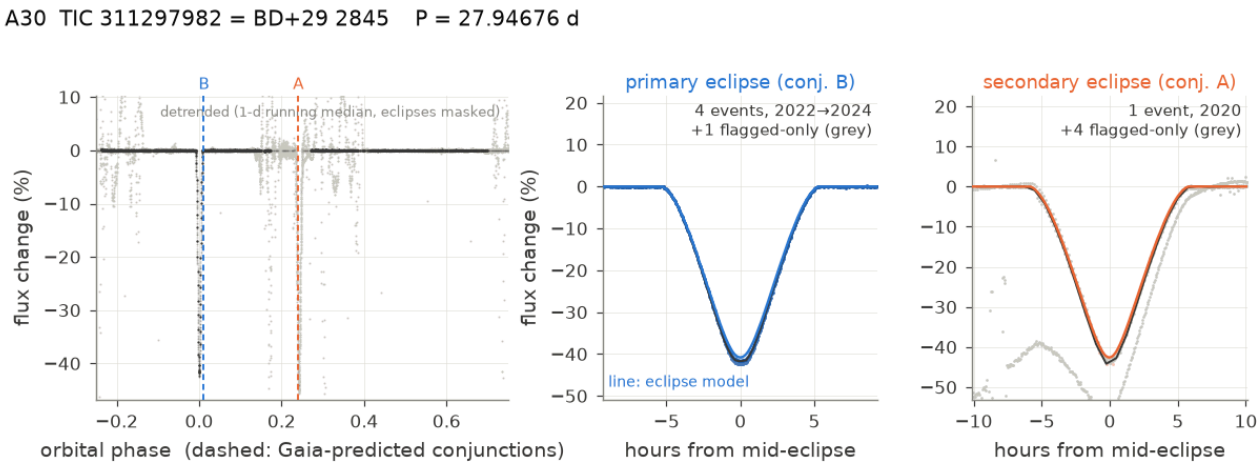
A29. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.60666 d, Gaia's predicted conjunctions dashed (the eclipse falls 1.67 d before Gaia's B). Middle: the three 9.5%, 5.8-h partial eclipses (sector 32, and two in sector 98; the sector-61 window, on a flagged scattered-light feature, is shown in a lighter tone and not used). Right: the A windows, flat to 0.1%, as the geometry predicts.

Identifiers	TIC 248652521; Gaia DR3 3206318533868790016; BD-01 648 = TYC 4727-450-1
Position (J2000)	04:28:42.2 −01:22:16.5
Brightness	*G* 9.99; Tmag 9.57
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 1.15 $M_\odot$, $M_2 \sin^3 i$ = 1.20 $M_\odot$ (mass ratio 1.04); twin model R = 1.69 $R_\odot$ each (radii free: 1.89 and 1.59), matching the luminosity (about 1.5 $R_\odot$ each at 6000 K)
Period	29.60666 ± 0.00002 d (Gaia 29.588805 ± 0.005297 d; the eclipse period is 0.0178 d longer, 3.4σ)
Epoch of eclipse	T_0 = BTJD 2177.8384 ± 0.0005 = BJD_{TDB} 2459177.8384
Eclipse (B, periastron side)	9.47–9.61% deep in a fixed aperture (QLP 8.8–10.2%), 0.24 d (5.8 h), rounded V (partial); three events (s32, s98 ×2); all O−C below 2×10^{-5} d (under 3 s)
Other eclipse (A)	Absent, not in a gap: all four A windows have data and are flat to 0.1%; the quality-0 data cover 99.7% of the orbit with no second eclipse deeper than 0.1% at any phase. At A the stars miss each other by 1.53 times the contact distance
Orbit	Gaia *e* 0.5486, ω 316.76° held fixed; *i* = 85.07°, B at 0.70 of contact. B is the conjunction nearest periastron (separation 0.51*a*, sky speed 150 km/s; A at 1.12*a*, 68 km/s)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s32); TARS (s32)
Pixel attribution	TESScut difference images: PSF-shaped deficit, centroid offset 0.03 px in all three events; the *G* 12.20 star at 90″ visibly does not dim; the *G* 14.30 star at 38″ would need 530%; RUWE 0.91
Catalogue status	Not listed under any name in: VSX, ASAS-SN, ASAS-3, GCVS, SB9, WDS, Hipparcos, Gaia DR3 variability tables, Prša et al. 2022, Justesen & Albrecht 2021, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP, SIMBAD (SB*, one paper), general web search
Open issues	The eclipse falls 1.669 d (0.056 in phase) before Gaia's B conjunction at Gaia's epoch: formally 25σ. Most likely Gaia's SB2 solution is distorted (goodness-of-fit 20); the alternative, that the eclipsing pair is not the Gaia pair, is much less likely but not excluded
Next eclipses	B: BTJD 4339.125, 4368.731 (±0.002); A (expected absent): 4346.09 ± 0.15
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with a caveat. The eclipse is 1.67 d (formally 25σ) from Gaia's B conjunction at Gaia's own epoch; we judge a distorted Gaia SB2 solution the likelier cause, but that the eclipsing pair is not the Gaia pair remains open.

BD-01 648 is a pair of near-twin stars of about 1.2 $M_\odot$ and 1.7 $R_\odot$ in an eccentric ($e \approx 0.55$) 29.6-d orbit. The stars partially eclipse each other near periastron, where they are close and fast. At the other conjunction they are more than twice as far apart and miss each other by 1.53 contact distances. The model predicts that A would eclipse only above about 87.5°, and B would then be at least 30% deep. The data show no second eclipse anywhere in the orbit. Three eclipses spanning five years agree to under three seconds.

The eclipse phase remains unresolved. Gaia's periastron time sits in the middle of its data span, so Gaia's period error cannot move the phase there. Yet the eclipse falls 1.67 d early. Shifting Gaia's orbit by that much would change the primary's velocity by 18 km/s rms, which Gaia's own velocities would show. The likelier explanation is that Gaia's SB2 solution is distorted (goodness-of-fit 20). Swapping the two components between epochs is the usual way near-twin SB2 fits go wrong. The much less likely alternative is that the eclipsing pair is not the Gaia pair. A few ground-based radial velocities near BTJD 4339 and around the following periastron would resolve it.

A30. BD+29 2845 (TIC 311297982)



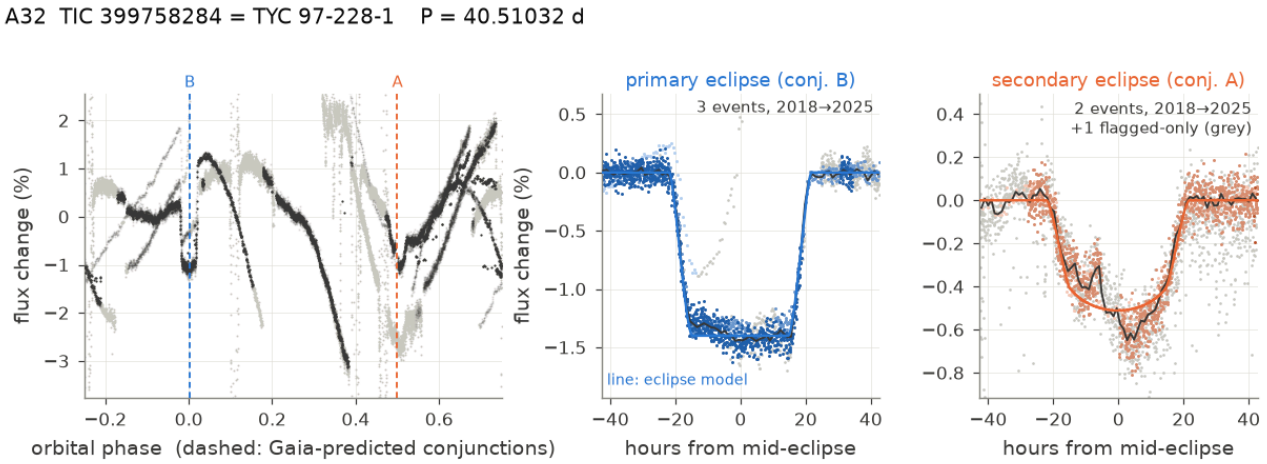
A30. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 27.94676 d, Gaia's predicted conjunctions dashed; flagged cadences, which hold half the eclipses, in a lighter tone. Middle: the five 40–42%, 9.3-h eclipses at B. Right: the five 42–45%, 10-h eclipses at A.

Identifiers	TIC 311297982; Gaia DR3 1305751357421836416; BD+29 2845 = TYC 2056-380-1 = SAO 84430 = PPM 104960 = AG+29 1593 = 2MASS J16314244+2901378
Position (J2000)	16:31:42.5 +29:01:37.9
Brightness	G 10.00; Tmag 9.60
Stars	Two slightly evolved F/G stars, Gaia SB2 (and APOGEE SB2, Kounkel et al. 2021): M $\sin^3 i$ = 1.12 + 1.09 $M_\odot$ with the TESS e (essentially the true masses at i = 89.5°); R_1 = 1.58, R_2 = 1.34 $R_\odot$; $L_2/L_1 \approx 0.71$; Gaia component temperatures 6110 and 5760 K; a = 50.4 $R_\odot$
Period	27.94676 ± 0.00001 d (Gaia 27.94708 ± 0.0013 d; −0.25σ)
Epoch of primary	T_0(B) = BTJD 1987.74538 ± 0.00042 = BJD_{TDB} 2458987.74538; T_0(A) = BTJD 1994.50441 ± 0.0018
Eclipse at A	42–45% deep, 9.9–10.3 h, near-V (nearly central, impact about 0.12–0.13 of R_1+R_2): star 1 (Gaia's primary) eclipsed by star 2; five events (s25, s51, s52, s78, s79), four in flagged cadences; O−C ±3 min
Eclipse at B	40–42% deep, 9.3–9.4 h: star 2 eclipsed; five events (s25 flagged in QLP, s51, s52, s78, s79); O−C ≤ 0.1 min. A→B phase 0.75815 (Gaia 0.77114 ± 0.00232)
Orbit	Physical fit to all ten eclipses: e = 0.420, ω = 353.6° against Gaia 0.4476 ± 0.0042, 348.67 ± 0.74° (roughly 6σ by Gaia's quoted errors; Gaia goodness-of-fit 23.7); i = 89.5°, k = 0.85, surface-brightness ratio 0.99. χ^2/n is 9.7 (red noise), so these values are indicative
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (all 10); TARS (6)
Pixel attribution	TESScut s079 B: largest flux-loss pixel is the target pixel, centroid offset 0.2″; nothing brighter than G 17.5 within 60″ (a 42% dip would need more than 1000%)
Catalogue status	Not listed under any name in: VSX, ASAS-SN, GCVS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 (all tables), WDS, Hipparcos and its variability annex, SB9, CCDM, Gaia DR3 variability tables, general web search; SIMBAD's one binary reference is the APOGEE SB2 list. ExoFOP was checked separately
Open issues	A joint fit of the APOGEE and Gaia velocities with the eclipses would give absolute radii and constrain e and ω
Next eclipses	B: BTJD 4335.273, 4363.220 (±0.001); A: BTJD 4342.028, 4369.975 (±0.004)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+29 2845 is a detached binary of two slightly evolved F/G stars of about 1.1 $M_\odot$ each, seen almost exactly edge-on, so the smaller star crosses the larger one nearly centrally and both eclipses are about 42% deep. Ten eclipses from 2020–2024 fit a single period. The radii (1.58 and 1.34 $R_\odot$, which add in quadrature to 2.07 $R_\odot$) match the TIC and Gaia single-star radius, and the Gaia multi-star temperatures are consistent.

Five of the ten eclipses lie in cadences flagged for scattered light, which may explain why the system has not been reported. Using Gaia's full covariance, the A eclipses fall within 0.23σ of prediction while the B eclipses are a constant 0.34–0.36 d early. Because the offset does not grow, it reflects the orbit shape rather than the period. The eclipses prefer $e \approx 0.42$, $\omega \approx 354°$, which amounts to a correction to Gaia's orbit rather than a problem with the eclipses. Gaia's errors are very probably understated. There is no apsidal motion within TESS.

A32. TYC 97-228-1 = ASASSN-V J045420.57+062810.8 (TIC 399758284)



A32. Left: QLP SAP flux folded on P = 40.51032 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 1.3–1.4%, 42-h flat total occultations of the G dwarf (sectors 32 and 98). Right: the 0.45% flat transits of the G dwarf across the giant (sectors 5 and 98).

Identifiers	TIC 399758284; Gaia DR3 3288089217923738112; TYC 97-228-1; ASASSN-V J045420.57+062810.8
Position (J2000)	04:54:20.6 +06:28:10.9
Brightness	G 10.66; Tmag 9.96
Stars	Chromospherically active K giant (T_1 = 4574 K), R_1 = 11.4–12.8 $R_\odot$ (TIC 10.8) + G dwarf, m_2 = 0.79–0.98 $M_\odot$, R_2 = 0.84–0.93 $R_\odot$, $T_2 \approx$ 5900–6000 K, 1.40% of the TESS light (for M_1 = 1.0–1.5 $M_\odot$); mass function 0.148 $M_\odot$; 886 pc; Gaia SB1
Period	40.51032 ± 0.00011 d (Gaia 40.501287 ± 0.007798 d; +1.16σ)
Epoch of primary	T_0(A) = BTJD 1443.6733 ± 0.0170 = BJD_{TDB} 2458443.6733; B at phase 0.49988 ± 0.00043
Primary eclipse (B)	1.32–1.44% deep, T_{14} 42.0–43.4 h, flat bottom 31.2–31.3 h: total occultation of the hotter G dwarf; two full events (s32, s98) and an ingress at the end of s5
Secondary eclipse (A)	0.45–0.46% deep (0.42–0.70% across pipelines), about 40 h, flat: full transit of the dwarf across the giant; two events (s5; s98, ingress lost, the weakest event)
Orbit	Circular: e cos ω = −0.0002 ± 0.0007 (Gaia +0.003 ± 0.006; Gaia's e = 0.015 ± 0.007 is a 2σ detection). Two-sphere fit: a/R_1 = 5.27, b = 0.78, i = 81.4°, k = 0.072–0.073
Pipelines agreeing	QLP SAP; ELEANOR-lite, T16, TASOC, TGLC PSF and aperture (s5 A: five pipelines)
Pixel attribution	TESScut s32 B difference image peaks on the target pixel (noisier); TGLC PSF photometry puts the signal on the target; nothing brighter than G 17.95 within 60″. Common mode: target ranks first or second of 219–284 stars
Catalogue status	**VSX type revision ROT → EA+ROT.** VSX: ASASSN-V J045420.57+062810.8, ROT, P 39.81 d, amplitude 0.25 mag in g. Not in ASAS-SN II/366, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables or any eclipsing-binary catalogue; no TOI/CTOI; SIMBAD (SB*, one paper); a general web search found nothing
Open issues	The s98 A eclipse is weakly measured and the A durations carry spot red noise; a B eclipse with both edges in clean data and a spectrum for the faint companion would constrain the system further. Masses depend on the assumed M_1
Next eclipses	B: BTJD 4340.157, 4380.667; A: BTJD 4360.417, 4400.927 (±0.02 d; T_{14} about 1.78 d)
Discovery	

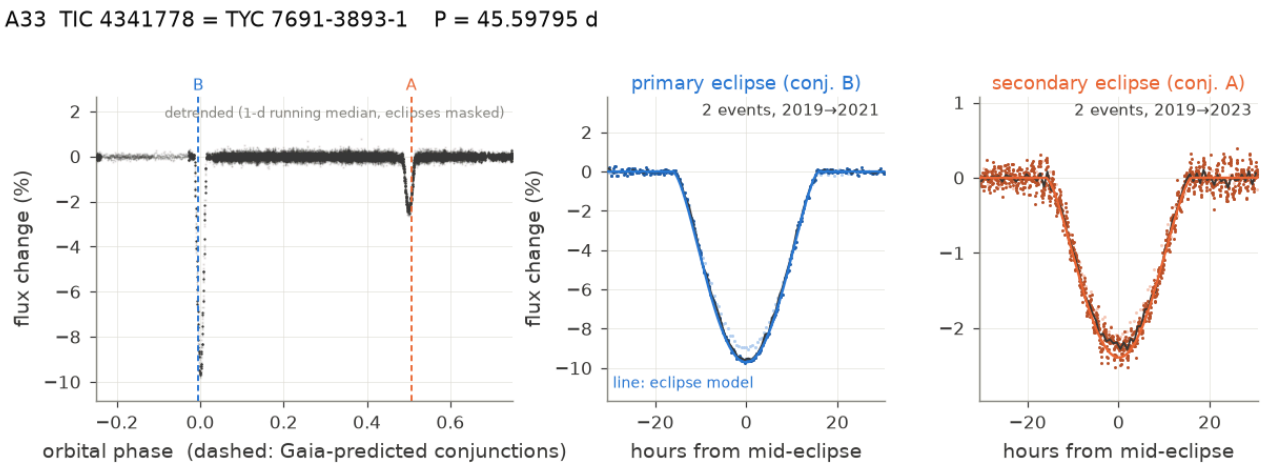
	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: a VSX type revision. The star is in VSX as a rotating variable (ROT, 39.81 d) from ASAS-SN. That signal is real but due to spots, not eclipses; the eclipses are what is new, and we propose EA with P = 40.5103 d, keeping ROT (EA+ROT/RS).

TYC 97-228-1 is a detached eclipsing binary of a K giant of about 12 $R_\odot$ and a G dwarf on a circular 40.5-d orbit. Both eclipses are flat-bottomed. At A the dwarf crosses the giant's disk, and at B the giant hides the dwarf completely. B is the deeper because the dwarf is hotter per unit area. The central duration would be 63 h; the observed 42–43 h follows from an impact parameter of 0.78. The dwarf's radius, temperature, light share and mass all agree with the mass function.

The ASAS-SN rotation signal has an amplitude of 0.25 mag in g, far more than the eclipses or an ellipsoidal effect could produce. The star is spectroscopically active and TESS shows 2–3% spot-like drifts in every sector. Its 39.81-d period is 0.983 of the orbital period, which indicates near-synchronous rotation typical of an RS CVn-type giant. The rotation period could not be measured independently.

A33. TYC 7691-3893-1 (TIC 4341778)



A33. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 45.59795 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the two 8.8–9.5%, 30.5-h occultations of the A star (sectors 8 and 35). Right: the two 2.1–2.3%, 30.5-h near-V transits of the A star across the giant (sectors 9 and 62).

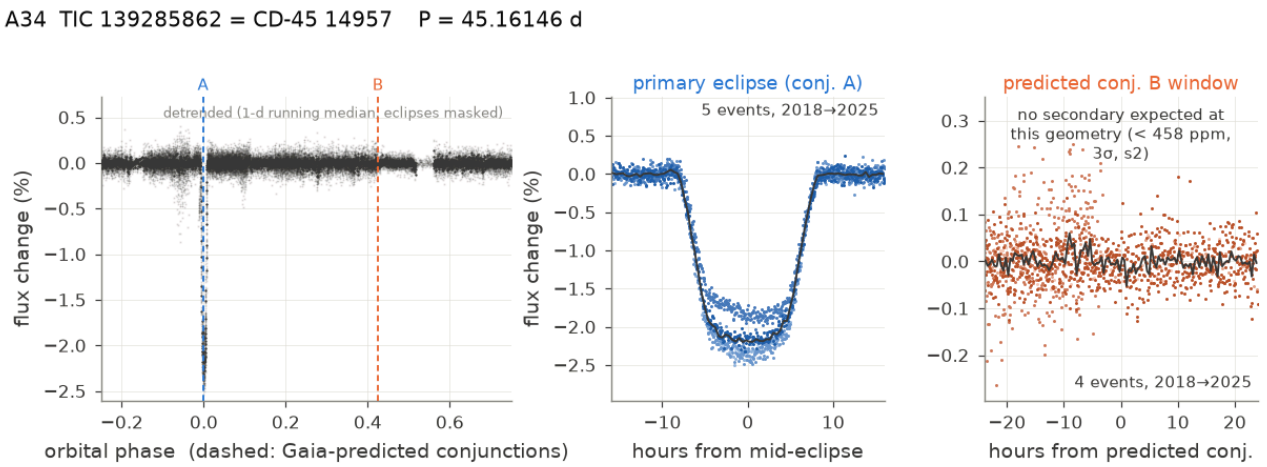
Identifiers	TIC 4341778; Gaia DR3 5430897742488880640; TYC 7691-3893-1
Position (J2000)	09:23:06.0 −37:32:12.4
Brightness	G 10.30; Tmag 9.83
Stars	Giant, R_1 = 10.2–10.7 $R_\odot$ (TIC 9.0) + A-type star, m_2 = 2.0–2.2 $M_\odot$, R_2 = 3.0–3.1 $R_\odot$, T_2 = 7200–9400 K, 23–24% of the TESS light (for M_1 = 2.5–3.0 $M_\odot$); mass function 0.373 $M_\odot$; Gaia SB1, spectral type F (Gaia); the TIC's 5957 K is a blend
Period	45.59795 ± 0.00008 d (Gaia 45.588324 ± 0.005264 d; +1.83σ)
Epoch of primary	T_0(A) = BTJD 1550.8760 ± 0.0030 = BJD_{TDB} 2458550.8760; B at phase 0.50010 ± 0.00006
Primary eclipse (B)	8.76–9.49% deep, T_{14} 30.5 h, 7-h bottom, ingress 11.7 h, near-V: partial occultation of the hot companion by the giant; two full events (s8, s35)
Secondary eclipse (A)	2.05–2.29% deep, 30.4–30.9 h, same near-V shape: partial transit across the giant's limb; two full events (s9, s62) plus an ingress (s35) and an egress (s89, after a gap, weaker)
Orbit	Circular: e cos ω = +0.00015 ± 0.00010, equal durations (e sin ω ≈ 0); Gaia's e cos ω = −0.016 ± 0.007 disagrees by 2.3σ, and Gaia's e (0.0174) is itself a 2.3σ detection and is biased upward. Joint fit: a/R_1 = 8.65, b = 1.02–1.03, k = 0.29, i = 83.2°
Pipelines agreeing	QLP SAP; ELEANOR, T16, TGLC PSF (s8 B); ELEANOR, T16, TEQUILA, TGLC PSF (s9 A); TESS-SPOC SAP and PDCSAP (s62 A)

Pixel attribution	TESScut s35 B: per-pixel depth peaks at 8.76% on the target pixel (5.0% at the *G* 13.3 star's pixel, the target's wings); the *G* 13.30 star at 45.7″ would need a 33% A dip and over 100% for B; TGLC PSF puts both signals on the target
Catalogue status	Not listed in: VSX (only a ROT star 51″ away, *G* 15.9), ASAS-SN II/366, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*; Turner 5 membership probability 0.00), general web search; Kostov table 2 only
Open issues	A correlated a/R_1–b–k valley in the fit; depths about 8% lower in s8/s9 (absorbed as extra dilution); masses depend on the assumed M_1. RVs or a spectrum to detect the A star, and multi-colour depths
Next eclipses	A: BTJD 4332.351, 4377.949; B: BTJD 4355.155, 4400.753 (±0.006 d; T_{14} about 1.27 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 7691-3893-1 is a detached, near-grazing eclipsing binary of a giant of about 10 $R_\odot$ and an A-type star of about 2 $M_\odot$ on a circular 45.6-d orbit. Both eclipses have the same duration and near-V shape, as a circular orbit requires, and cover the same overlap area. The occultation is about 4.3 times deeper because it hides the hot companion, which is 3.3–3.5 times brighter per unit area, while the transit covers the giant's darker limb. One geometry fits all four full eclipses to the noise, and the 2 $M_\odot$ A-type companion it requires is what the mass function demands.

All events fall within 0.17σ of Gaia's conjunctions.

A34. CD-45 14957 (TIC 139285862)



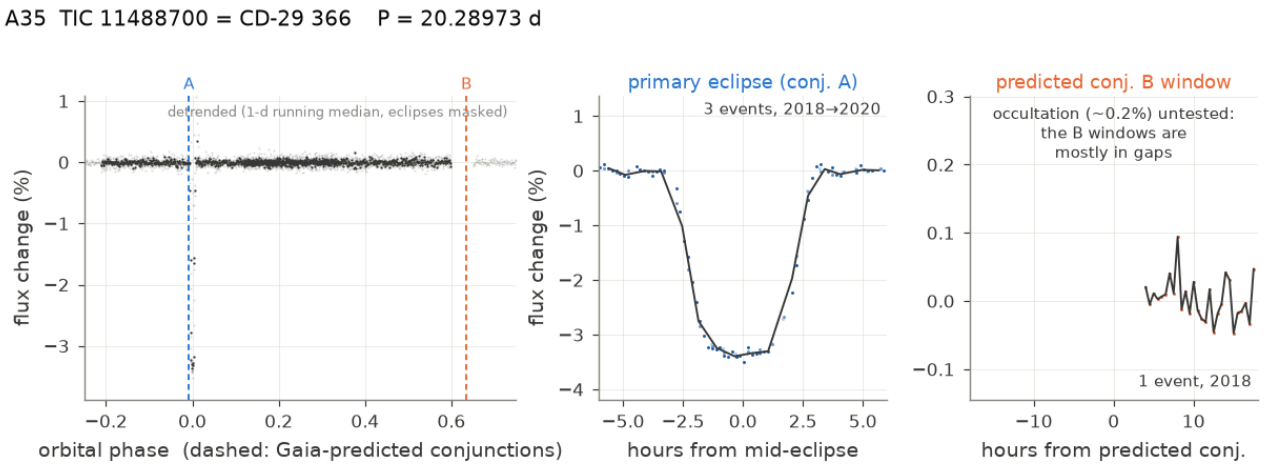
A34. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 45.16146 d, Gaia's predicted conjunctions dashed. Middle: the 2.3%, 16-h transits of the K dwarf across the K giant, 2018–2025. Right: the B window, where the geometry predicts no eclipse and none is seen.

Identifiers	TIC 139285862; Gaia DR3 6541487327863944960; CD-45 14957
Position (J2000)	23:12:03.2 −44:45:45.1
Brightness	*G* 10.51; Tmag 9.93
Stars	K giant, R_1 = 4.4–5.0 $R_\odot$ (Gaia FLAME 4.92 $R_\odot$, 1.63 $M_\odot$, log *g* 3.06) + K dwarf, R_2 = 0.67–0.76 $R_\odot$, $M_2(i = 90°)$ = 0.72/0.82/0.92 $M_\odot$ for M_1 = 1.3/1.63/2.0; mass function 0.0917 $M_\odot$; Gaia SB1
Period	45.16146 ± 0.00011 d (Gaia 45.1578 ± 0.0065 d; +0.6σ)
Epoch of eclipse	T_0(A) = BTJD 1354.3033 = BJD_{TDB} 2458354.3033
Eclipse (A)	2.11–2.31% deep (1.73% in s95, on a ramp), T_{14} 15.4–16.1 h, T_{23} 7.3–8.8 h, impact b = 0.67: transit of the K dwarf across the giant; five events (s2 egress only, s28, s68, s95, s96); O−C −5.5 to +16 min. A sixth, predicted at BTJD 4244.6367 ± 0.008 in s105 FFIs before those data were examined, was observed at 4244.6376 ± 0.001 (O−C +1.3 min) in custom TESScut photometry (this work)
Other eclipse (B)	

	None expected: the separation at B is 1.75 times larger, so $b_B = 1.17 > 1 + k$. Measured 3σ limits 570 and 458 ppm (s1, s2), < 1000 ppm (s95)
Orbit	Gaia e 0.294, ω 112.3°; e cos ω and e sin ω not measurable without B. Events +0.19 to +0.39 d after Gaia's A times (+0.73 to +0.83σ). $a/R_1 = 15.2$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s68); TGLC raw aperture (s2)
Pixel attribution	s28 difference image: loss on the target (offset +0.08, 0.00 px), core pixels lose 2.0–2.3% uniformly; the only Gaia neighbour within 60″ is G 19.9
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, general web search. SIMBAD's RAVE papers include it among RAVE + Gaia DR2 SB1 candidates (Birko et al. 2019); no eclipse
Open issues	RVs of the companion
Next eclipses	A: BTJD 4334.960, 4380.121, 4425.283 (±0.009)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-45 14957 is a K giant of about 5 $R_\odot$ crossed every 45.16 d by a K dwarf. Five transits from 2018 to 2025 define the period. A sixth transit, predicted for sector 105 (which has full-frame images but no QLP light curve), was recovered in TESScut photometry (this work) 1.3 min from the prediction, with the same depth and duration. Only one eclipse kind occurs: the transit impact parameter of 0.67 grows to 1.17 at the other conjunction, where the stars are 1.75 times farther apart, so the giant cannot occult the dwarf.

A35. CD-29 366 = Bok I 16B (TIC 11488700)



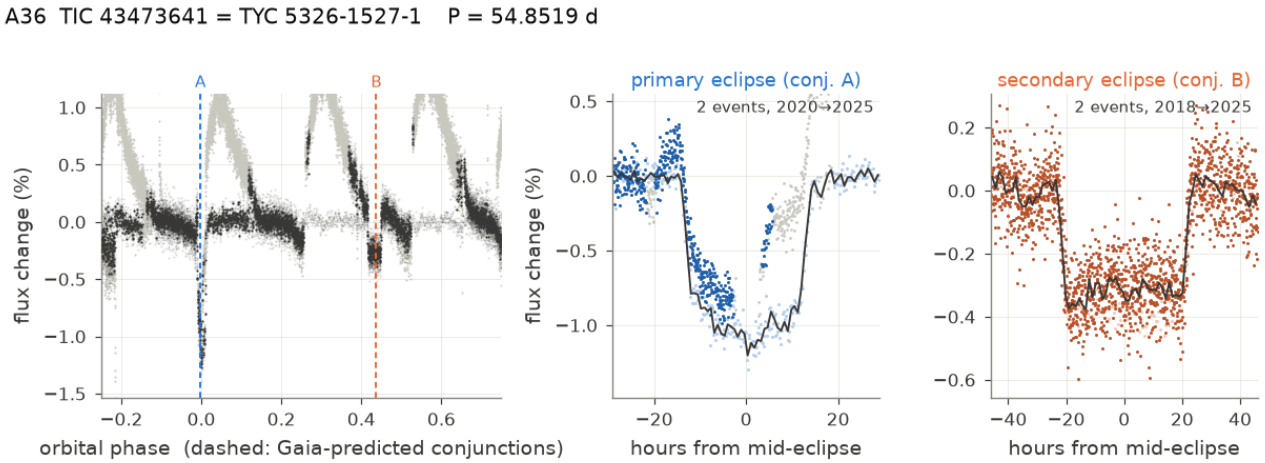
A35. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 20.28973 d, Gaia's predicted conjunctions dashed. Middle: the three 3.3%, 5.9-h U-shaped transits of the M dwarf across the F star (2018–2020). Right: the B window, mostly in data gaps; the expected 0.2% occultation is untested.

Identifiers	TIC 11488700; Gaia DR3 5032956687360117376; CD-29 366; Bok I 16B
Position (J2000)	01:11:28.1 −28:41:49.5
Brightness	G 10.64; Tmag 10.30
Stars	F star, R_1 = 1.41–1.48 $R_\odot$ (FLAME 1.65) + M dwarf, M_2 = 0.27–0.30 $M_\odot$, R_2 = 0.26 $R_\odot$; mass function 0.0090 $M_\odot$; Gaia SB1. A co-moving G 14.02 companion lies 1.65″ away (parallax 3.32 vs 3.29 mas, same proper motion)
Period	20.28973 ± 0.00016 d (Gaia 20.2758 ± 0.0109 d; +1.3σ)
Epoch of eclipse	BTJD 1404.1283 ± 0.0003 = BJD_{TDB} 2458404.1283
Eclipse (A)	3.28–3.35% deep, T_{14} 5.9 h, T_{23} 3.1 h, U-shaped, impact b = 0.62: transit of the M dwarf across the F star; three events (1383.8453 partial, 1404.1283, 2134.5620); O−C −4.9 to +4.8 min
Other eclipse (B)	Untested: expected occultation about 0.2%; only 16–21% of the predicted B windows have good data. B is therefore undetected rather than excluded

Orbit	Gaia e 0.228, ω 4°; events +0.85 to +1.37 d after Gaia's A times (+0.8 to +1.0σ). a/R_1 = 25
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s3, s30); TARS (s30); TGLC (s3); depths 3.1–3.7%, mid-times within 1 min
Pixel attribution	s30 difference image: loss lands on the pair (−0.02, +0.06 px); no pixel product can separate the two stars 1.65″ apart. The companion would need a 76% eclipse; the eclipse repeats at the period and phase of the F star's Gaia orbit
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS (nothing within 180″), Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Detecting B at about 0.2%; high-resolution imaging or RVs to rule out the 1.65″ companion completely. The s106 cutout is empty (target off-detector)
Next eclipses	A: BTJD 4346.142, 4366.432, 4386.722 (±0.024); B (about 0.2%): 4338.84 ± 0.54
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with a caveat. The co-moving G 14.02 companion 1.65″ away cannot be separated in any TESS product; the attribution rests on the period and phase of the F star's Gaia orbit (three events, one partial, 0.8–1.0σ from Gaia's A times) and on the physics.

CD-29 366 is an F star of about 1.45 $R_\odot$ transited every 20.29 d by an M dwarf of about 0.26 $R_\odot$. Two of the three dips lie in continuous clean data with flat background and centroids. The third is partly covered, after the flagged start of sector 3. No TESS pixel product can separate the co-moving companion 1.65″ away, but that companion would need a 76% eclipse to produce the dip. The eclipse also repeats at the period and phase of the F star's own Gaia orbit, and the physics is fully consistent with an M dwarf crossing the F star. The B windows are mostly in gaps, and the expected occultation is only about 0.2% deep.

A36. TYC 5326-1527-1 (TIC 43473641)



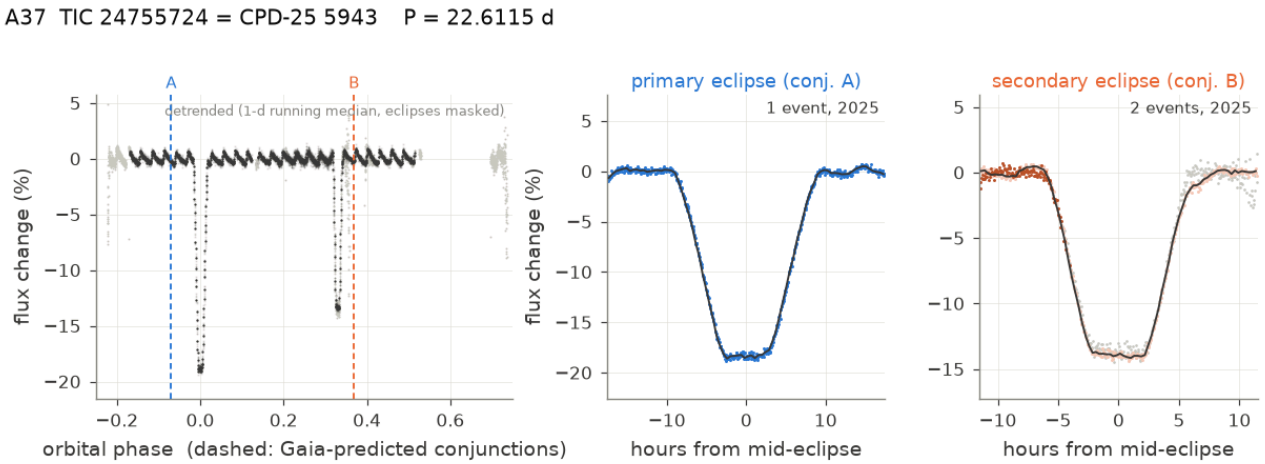
A36. Left: QLP SAP flux folded on P = 54.8519 d, Gaia's predicted conjunctions dashed. Middle: the 1.0%, 29-h flat transits at A (sectors 32 and 98). Right: the 0.39%, 46-h flat occultations at B (sectors 5 and 98).

Identifiers	TIC 43473641; Gaia DR3 3180643154951048064; TYC 5326-1527-1
Position (J2000)	04:59:10.9 −11:10:02.0
Brightness	G 11.00; Tmag 10.34
Stars	Giant, R_1 = 6.6/7.5/8.7 $R_\odot$ for M_1 = 1.0/1.5/2.5 $M_\odot$ (TIC 7.0, matching $M_1 \approx 1.1$) + K/M dwarf, R_2 = 0.67–0.88 $R_\odot$, about 3900 K, consistent with $M_2 \approx$ 0.55–0.65 $M_\odot$; mass function 0.0636 $M_\odot$; Gaia SB1
Period	54.8519 ± 0.0002 d (P_B = 54.85193 ± 0.00019, P_A = 54.8520 ± 0.0009; Gaia 54.8533 ± 0.0111 d, −0.1σ)
Epoch of primary	A: BTJD 2192.7826 ± 0.005 = BJD_{TDB} 2459192.7826; B: BTJD 1448.5445 ± 0.004

Primary eclipse (A)	1.03% deep, T_{14} 29.3 h, T_{23} 21.7 h, flat: transit of the dwarf across the giant (b = 0.56); two events (s32; s98, egress in flagged scattered light)
Secondary eclipse (B)	0.375–0.391% deep, T_{14} 45.8–47.2 h, T_{23} about 40 h, flat: total occultation of the dwarf at 1.64 times the A separation; two events (s5, s98)
Orbit	A→B phase 0.43189 and duration ratio 1.611 give e cos ω = −0.104, e sin ω = +0.241, e = 0.262, ω = 113°; Gaia −0.092 ± 0.008 and +0.241 ± 0.008 (1.6σ and 0.0σ; Gaia e 0.2579, ω 110.93°). a/R_1 = 10.6
Pipelines agreeing	QLP SAP; raw eleanor (s5 B 0.36%); raw TGLC aperture 0.46% and PSF 0.51%. Calibrated TGLC fluxes erase the two-day eclipse with a one-day filter
Pixel attribution	s32 difference image: A loss on the target pixel (+0.12, +0.18 px; 2.3 px from the G 11.85 neighbour at 58.5″, which would need 2.2%/0.9% eclipses); target pixel loses 1.3%, pixels toward the neighbour ≤ 0.3%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	B's ingress (2.9–3.7 h) is shorter than A's scaled by the velocity ratio, attributed to limb darkening biasing the A fit; this is not a refutation. A fully covered A eclipse at 200-s cadence fitted with limb darkening would resolve it
Next eclipses	A: BTJD 4332.01 (±0.04; 4332.008 ± 0.009 with P_B), 4386.86; B: BTJD 4355.703, 4410.555 (±0.05)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 5326-1527-1 is a giant of about 7 $R_\odot$ with a K/M dwarf companion on an eccentric 54.85-d orbit. The dwarf's transit lasts 29 h. Its occultation happens at 1.64 times the separation and so lasts almost two days. With both kinds of eclipse seen, the eclipse timing and durations give e cos ω and e sin ω independently of Gaia's radial velocities, and they reproduce Gaia's to 1.6σ and 0.0σ. The occultation depth (0.39% of the light) needs a companion about 0.4 times as bright per unit area as the giant, about 3900 K, consistent with the mass function.

A37. CPD-25 5943 (TIC 24755724)



A37. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.6115 d, Gaia's predicted conjunctions dashed. Middle: the single 18.7%, 17.5-h flat transit at A in sector 91. Right: the two 13.4%, 11.5-h flat total occultations at B (the second in flagged cadences).

Identifiers	TIC 24755724; Gaia DR3 4109331632663213440; CPD-25 5943
Position (J2000)	17:10:32.5 −26:00:49.1
Brightness	G 10.55; Tmag 10.21
Stars	A5 star (Wallenquist 1939), R_1 = 2.7–2.9 $R_\odot$ + companion, R_2 = 1.24–1.36 $R_\odot$, 13.4% of the light (about 6400 K and 1.3 $M_\odot$ needed); mass function 0.0926 $M_\odot$ gives $M_2(i = 90°)$ = 0.81–0.98 $M_\odot$ for M_1 = 1.6–2.2; Gaia SB1 (significance 11)

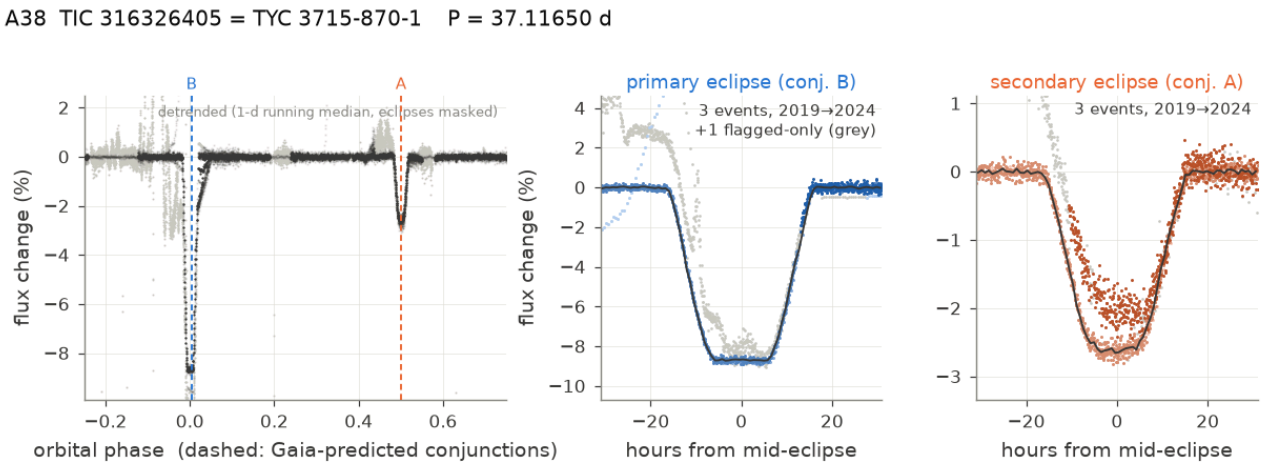
Period	P_B = 22.6115 ± 0.0029 d from two B eclipses (Gaia 22.5960 ± 0.0215 d; +0.7σ). The Harvard plates refine it (Section 5.6): the eclipse appears on 47 plates from 1890–1990 at P = 22.6131 ± 0.0002 d, inside the TESS error and twenty times tighter
Epoch of primary	A: BTJD 3793.2998 ± 0.0014 = BJD_{TDB} 2460793.2998
Primary eclipse (A)	18.73% deep, T_{14} 17.5 h, T_{23} 5.9 h, flat (b_A = 0.22); one event, sector 91
Secondary eclipse (B)	13.43–13.46% deep, 11.4–11.5 h, T_{23} 4.7–4.8 h, flat: total occultation; two events (3778.1317; 3800.7432 in flagged cadences)
Orbit	A→B phase 0.3292 and duration ratio 0.658 give e cos ω = −0.266, e sin ω = −0.204, e = 0.335, ω = 217°; Gaia −0.092 ± 0.058 (3.0σ off) and −0.220 ± 0.041 (0.4σ); Gaia e 0.246 ± 0.037, ω 247 ± 14°. a/R_1 = 17
Pipelines agreeing	QLP SAP (one sector, s91); no second pipeline was listed for this star
Pixel attribution	TESScut s91 difference images: loss centroid (−0.05, +0.16) px from the target at A and (−0.01, +0.04) px at B; peak-loss pixel is the target pixel (16% A, 11% B). The VSX "S" source 8.3″ away (G 18.47, probably a spurious detection in the target's halo) would need more than 26,000%; every Gaia star with G < 16 within 150″ would need more than 100%
Catalogue status	Not listed in: VSX (an "S" entry 8.3″ away, from the Gaia DR3 classifier, no period; an "RS" source at 53″, G 17.1), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Two tensions with Gaia: e cos ω 3.0σ off (plausibly an underestimated ω error in a weak orbit), and the 13.4% occultation needs a brighter companion than the mass function allows (Gaia's K1 is probably diluted). Neither moves the eclipse off the target. Needs a second A eclipse and ground spectroscopy. The A eclipse falls +3.88 d (+1.1σ) after Gaia's predicted A time. The Harvard plates (Section 5.6) supply a second, independent century of eclipses at this ephemeris (z = 11.9, false-alarm probability below 0.5%), so the period no longer rests on one sector and Gaia; the status is kept pending a second TESS epoch
Next eclipses	A: BTJD 4335.98, 4358.59; B: BTJD 4343.42, 4366.03 (±0.07–0.08)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable, see below
Status	Probable. One sector (s91) only: one transit and two occultations, the second in flagged cadences; no second pipeline; e cos ω 3.0σ from Gaia; the 13.4% total occultation needs a companion of about 1.3 $M_\odot$ against about 0.85 $M_\odot$ from the mass function. The A eclipse falls +3.88 d (+1.1σ) after Gaia's predicted A time. A second A eclipse measured in an independent pipeline would confirm it.

Note: probable, not confirmed. Both eclipse kinds were seen in a single sector (s91): one transit and two occultations, the second in cadences QLP flagged for scattered light, and no second pipeline was listed. The eclipse spacing puts e cos ω 3.0σ from Gaia's. The total occultation needs a companion of about 1.3 $M_\odot$ where the mass function gives about 0.85 $M_\odot$. The A eclipse falls 3.88 d (+1.1σ) after Gaia's predicted A time. A second A eclipse measured in an independent pipeline would confirm it.

CPD-25 5943 is an A5 star of about 2.8 $R_\odot$ with a companion of about 1.3 $R_\odot$, observed in a single TESS sector that contains one deep flat transit and two identical flat total occultations one orbit apart. The signal is confirmed on the target. The difference images put both losses on the A star, and the VSX short-timescale source 8.3″ away is faint, astrometrically poor and probably spurious. The ~1-day, ~1% sawtooth in the light curve tracks pointing resets and is not stellar.

Two results disagree with Gaia's orbit. The eclipse spacing gives e cos ω 3.0σ from Gaia's, while the duration ratio matches. Because B is a total occultation, its depth shows that the companion supplies 13.4% of the light. That needs a star of about 1.3 $M_\odot$, where the mass function gives about 0.85; a 0.85 $M_\odot$ dwarf would give about 7%. The likely explanation is that the companion's lines blend into the A star's spectrum and dilute Gaia's velocity amplitude. The alternative is a much more massive primary.

A38. TYC 3715-870-1 (TIC 316326405)



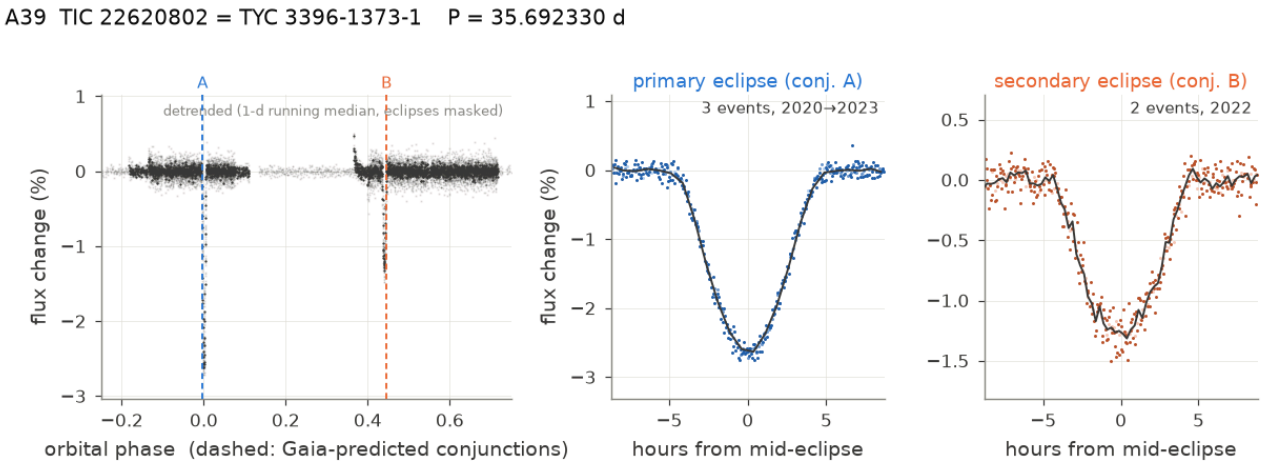
A38. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 37.11650 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 8.7%, 31-h flat total occultations of the hot companion at B. Right: the 2.5%, 31-h flat transits at A.

Identifiers	TIC 316326405; Gaia DR3 450207237998808064; TYC 3715-870-1
Position (J2000)	03:28:28.7 +59:17:36.2 (2° from the Galactic plane, 845 pc)
Brightness	G 10.63; Tmag 9.93
Stars	Giant, R_1 = 9.08 $R_\odot$ (TIC), probably near 5000 K + hot dwarf, $M_2(i = 90°)$ = 1.38/1.56/1.72 $M_\odot$ for M_1 = 2.0/2.5/3.0, R_2 ≈ 1.4–1.5 $R_\odot$, 8.7% of the light; mass function 0.229 $M_\odot$; Gaia SB1 (K_1 39.06 km/s)
Period	37.116502 ± 0.000308 d (Gaia 37.1120 ± 0.0055 d; +0.81σ)
Epoch of primary	T_0(B) = BTJD 1839.8511 ± 0.009 = BJD_{TDB} 2458839.8511; T_0(A) = BTJD 1821.2932 ± 0.011
Primary eclipse (B)	8.65–8.82% deep, T_{14} 31.4–31.6 h, T_{23} 14.2–14.3 h, flat: total occultation of the hot companion; two measured (s19, s59) plus a flagged ingress in s86
Secondary eclipse (A)	2.49–2.56% deep, T_{14} 30.3–31.3 h, T_{23} about 12 h, flat: transit of the companion across the giant (k ≈ 0.15–0.17); three events (s19, s59, s86 on a systematic feature, timing only)
Orbit	Circular: B–A phase 0.49999 ± 0.0004, e cos ω = 0.0000 ± 0.0006 (about ±0.001 including systematics; Gaia +0.0057 ± 0.0048); duration ratio 1.025, e sin ω ≈ +0.012 (Gaia +0.0045 ± 0.0049). b ≈ 0.74, i ≈ 85°
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s59); TEQUILA (s19)
Pixel attribution	TESScut s59: both deficits centred on the target pixel, centroid within 0.35 px; core depths A 2.58%, B 8.94%; a brighter star about 95″ away does not dip; brightest Gaia neighbour within 60″ is G 18.0
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	The surface-brightness ratio J ≈ 3.4–4.3 needs a 9000–9900 K companion with the TIC T_{eff} (5728 K), too hot for 1.5 $M_\odot$, but 7400–8100 K if the giant is near 5000 K; the TIC T_{eff} is from blended light. A spectrum is needed; the binary interpretation is unaffected
Next eclipses	B: BTJD 4363.773, 4400.890; A: BTJD 4345.215, 4382.332 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3715-870-1 is a giant of about 9 $R_\odot$ with a hot main-sequence companion of 1.4–1.7 $M_\odot$ on a circular 37.1-d orbit. Both eclipses are flat-bottomed and about 31 h long. At B the giant hides the companion completely, so the 8.7% depth is the companion's share of the light. At A the companion crosses the giant's disk. A central eclipse would last 38–42 h; the observed 31 h implies an impact parameter of about 0.74 and predicts the 14-h flat part seen. One geometry fits both kinds of eclipse. A

companion carrying 9% of the light makes Gaia's single-lined classification expected. The only tension is the giant's temperature, which the TIC took from blended light; a spectrum would show the hot companion's lines in the blue.

A39. TYC 3396-1373-1 (TIC 22620802)

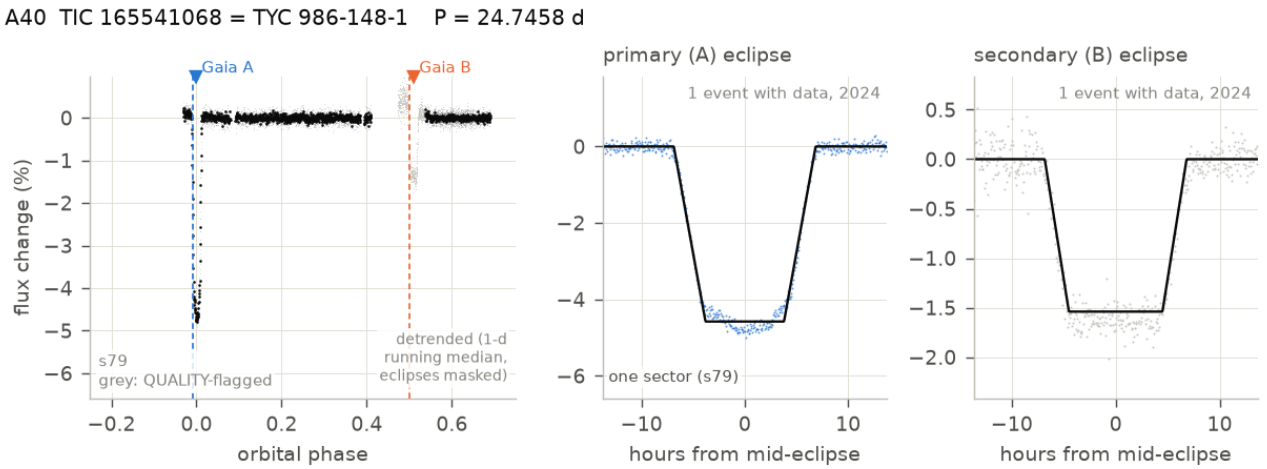


A39. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 35.692330 d, Gaia's predicted conjunctions dashed. Middle: the 2.6%, 8.7-h grazing eclipses at A. Right: the 1.25%, 8.7-h grazing eclipses at B.

Identifiers	TIC 22620802; Gaia DR3 977538409156198272; TYC 3396-1373-1
Position (J2000)	07:11:50.2 +46:52:45.4
Brightness	*G* 10.83; Tmag 10.21
Stars	Subgiant (T_1 = 4908 K) + K dwarf, M_2 = 0.64–0.76 $M_\odot$ (for M_1 = 1.1–1.5), T_2 ≈ 4100 K; mass function 0.0862 $M_\odot$; Gaia SB1 (K_1 28.66 km/s)
Period	35.692330 ± 0.000062 d (Gaia 35.6858 ± 0.0072 d; +0.90σ)
Epoch of primary	T_0(A) = BTJD 2602.95779 ± 0.0007 = BJD_{TDB} 2459602.95779; T_0(B) = BTJD 2582.9649 ± 0.0011
Primary eclipse (A)	2.56–2.58% deep, 8.74–8.76 h, V-shaped (T_{23} only about 2 h), partial: grazing transit (b ≈ 1.07 R_1); three events (s47, s60; s20 complete in TARS, 21 cycles earlier)
Secondary eclipse (B)	1.23–1.26% deep, 8.5–9.0 h, V-shaped, partial; two events (s47, s60)
Orbit	B–A phase 0.43985 ± 0.00004 (Gaia 0.4485 ± 0.0104): e cos ω = −0.0946 against Gaia −0.0810 ± 0.0164 (−0.8σ); equal durations, e sin ω ≈ +0.001 (Gaia +0.010 ± 0.014). O–C within ±1.4 min at every event
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s60); TARS (s20, s47, s60); times within 5 min
Pixel attribution	TESScut s60: deficits fit the out-of-eclipse PSF (A core 2.56 ± 0.06%, B 1.23 ± 0.10%), centroid within 0.15 px; brightest Gaia neighbour within 60″ is *G* 18.5
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	None for the binary; a limb-darkened model would constrain b and k
Next eclipses	A: BTJD 4351.882, 4387.574; B: BTJD 4331.889, 4367.581 (±0.003)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3396-1373-1 is a subgiant grazed by a K dwarf of about 0.7 $M_\odot$ on a nearly circular 35.7-d orbit. A central eclipse would last about 16 h. The observed 8.7 h requires an impact parameter of about 1.07 primary radii, so both eclipses are partial and V-shaped. The depth ratio of the two eclipses, 0.48, approximates the surface-brightness ratio and gives the companion about 4100 K. All measurements are consistent with one geometry: neither eclipse is flat, B lasts 8.5–9 h, and the sector-20 eclipse is complete in TARS and fits the ephemeris to within a minute.

A40. TYC 986-148-1 (TIC 165541068)



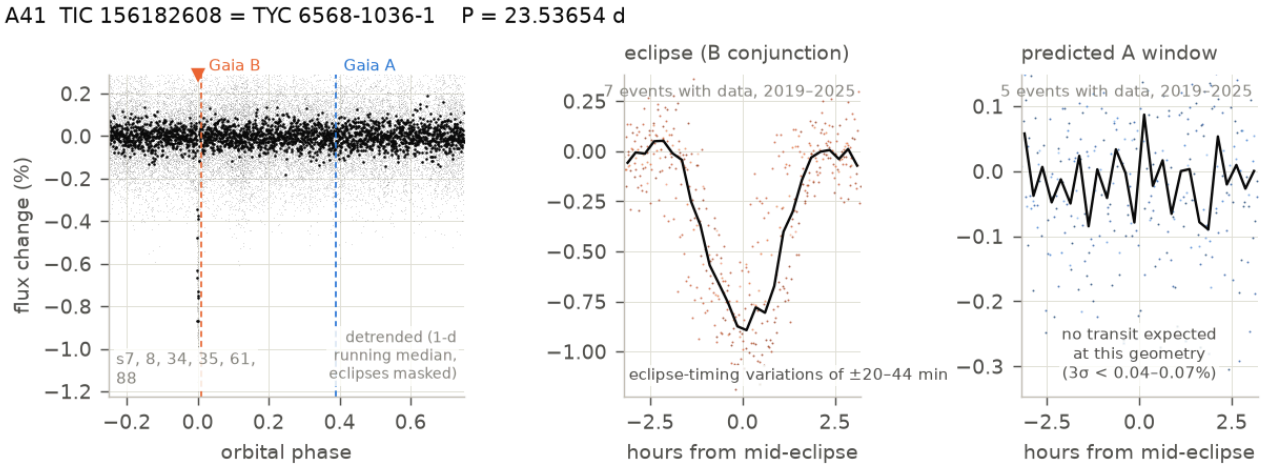
A40. Left: QLP SAP flux of sector 79, the only sector, detrended with a 1-d running median with the eclipses masked, folded on Gaia's P = 24.7458 d, Gaia's predicted conjunctions dashed. Middle: the 4.6%, 13.7-h transit at A on clean data. Right: the 1.5%, 13.7-h flat occultation at B, entirely in flagged cadences (shown in a lighter tone).

Identifiers	TIC 165541068; Gaia DR3 4540690625639510528; TYC 986-148-1
Position (J2000)	17:19:11.3 +12:29:51.2
Brightness	G 10.80; Tmag 10.16
Stars	Subgiant + dwarf, M_2 = 0.61–0.73 $M_\odot$ (for M_1 = 1.1–1.5), $R_2 \approx$ 0.64–0.71 $R_\odot$, 1.54% of the light; mass function 0.0772 $M_\odot$; Gaia SB1 (K_1 31.11 km/s)
Period	Gaia 24.7458 ± 0.0025 d (one event of each kind; no photometric period)
Epoch of primary	A: BTJD 3463.5943 ± 0.0010 = BJD_{TDB} 2460463.5943
Primary eclipse (A)	4.58% deep, T_{14} 13.73 h, T_{23} 7.6 h: transit (b ≈ 0.72, $k \approx$ 0.19–0.22); one event, sector 79, clean data
Secondary eclipse (B)	1.54% deep (TESS-SPOC 1.51%), T_{14} 13.65 h, T_{23} 9.0 h, flat: total occultation; one event, BTJD 3476.1984 ± 0.0031, all 389 cadences within ±0.45 d flagged (bits 2048/4096)
Orbit	A–B gap 12.604 d (P/2 = 12.373): e cos ω = +0.0147 ± 0.0002 against Gaia's +0.0176 ± 0.0057 (−0.5σ); durations give e sin ω ≈ −0.003 (Gaia +0.004 ± 0.007). Gaia's conjunctions within 0.11σ of both events. $i \approx 87°$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (the B event; A in QLP and TARS); TARS (A only; it has a gap over B)
Pixel attribution	TESScut s79: deficit follows the target PSF for both (A core 4.51 ± 0.16%, B 1.62 ± 0.23%), centroid within 0.2 px. None of 23 other QLP stars on the same CCD dips more than 1% in the B window. Gaia neighbours within 60″ are $G \geq$ 17.76 (B would need more than 900%)
Catalogue status	Not listed in: VSX (120″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	**One sector.** The period rests on Gaia; a second epoch of A (4.6%, 13.7 h, readily observable from the ground) would fix it. Mild tension: the companion's implied temperature (3700–3860 K) is below the 4100–4300 K expected, which is not evidence against the eclipses
Next eclipses	A: BTJD 4329.696, 4354.442; B: BTJD 4342.300, 4367.046 (±0.09–0.10, from Gaia's period error)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Probable. One transit and one occultation in a single sector, the occultation wholly in flagged cadences, and the period is Gaia's, not photometric (Appendix A marks it so): the eclipses are on the target and self-consistent, but the protocol's period-from-the-eclipses check is missing until a second epoch. TESS-SPOC confirms the flagged event.

Note: one sector. Both eclipses were seen once, in sector 79, and the B eclipse lies wholly in cadences QLP flagged for scattered light. The eclipses are on the target; the photometric period awaits a second epoch, and until then the status is probable.

TYC 986-148-1 is a subgiant with a dwarf companion of about 0.67 $M_\odot$ on a nearly circular 24.7-d orbit. The flagged B event was tested in detail. It has a sharp ingress, a flat bottom and a sharp egress, while the scattered-light background is a smooth bell that peaks before the event and is back at baseline by egress. An equal bell at 3461.5 produces no dip. The centroids are smooth. The same depth, time and duration are obtained from TESS-SPOC, with its own aperture and background. The pixel deficit follows the target's PSF, and no dip is seen in any other star on the CCD. The occultation's total duration equals the transit's to 0.1 h, as it must for a near-circular orbit, and its phase gives an e cos ω that matches Gaia's independent value. The B eclipse is therefore confirmed despite the quality flags.

A41. TYC 6568-1036-1 (TIC 156182608)



A41. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.53654 d, Gaia's predicted conjunctions dashed. Middle: the seven 0.7–1.0%, 3-h grazing occultations at B, 2018–2025, in time order (the 2025 events are deeper and longer). Right: the A window, flat as the geometry predicts.

Identifiers	TIC 156182608; Gaia DR3 5692749838260791168; TYC 6568-1036-1
Position (J2000)	08:18:03.3 −28:34:59.7
Brightness	G 10.88; Tmag 10.52
Stars	F star + K dwarf, m_2 = 0.64 $M_\odot$ at i = 90°; mass function 0.067 $M_\odot$; a = 43.3 $R_\odot$; Gaia SB1, RUWE 2.04
Period	23.53654 ± 0.00017 d, linear fit (Gaia 23.539603 ± 0.002965 d; 1.0σ)
Epoch of eclipse	T_0(B) = BTJD 1501.940 ± 0.011 = BJD_{TDB} 2458501.940
Eclipse (B)	0.69–0.99% deep (QLP), T_{14} 2.9–3.5 h, T_{23} about 0.8–1.1 h: grazing occultation of the K dwarf by the F star, near periastron; seven events, 2018–2025 (s7, s8, s34, s35, s61, s88 ×2)
Other eclipse (A)	None expected (impact distance at A 2.04 times larger, 3.9 $R_\odot$ against R_1+R_2 = 2.18) and none seen: 100% coverage of the A window, 3σ limits 386–668 ppm
Orbit	At Gaia's epoch the eclipses sit −0.21 ± 0.11 d (−1.9σ) from B and −34σ from A. Separation 0.635a at B, 1.29a at A. Impact d/(R_1+R_2) = 0.88 (s61), 0.83 (s88)
Pipelines agreeing	QLP SAP; TESS-SPOC (s7); eleanor (s7); SPOC 2-min (s34, s35); times agree to about 2 min, so the timing variations are reproduced in three pipelines
Pixel attribution	Difference-image centroids 0.7–4.5″ from the target (s34, s61, s88); the G 13.41 neighbour at 17.8″ would need 7.7% and is excluded by the centroids; the G 12.22 star at 43.7″ would need 2.6% and is about 2 px away; depth falls as the aperture grows
Catalogue status	

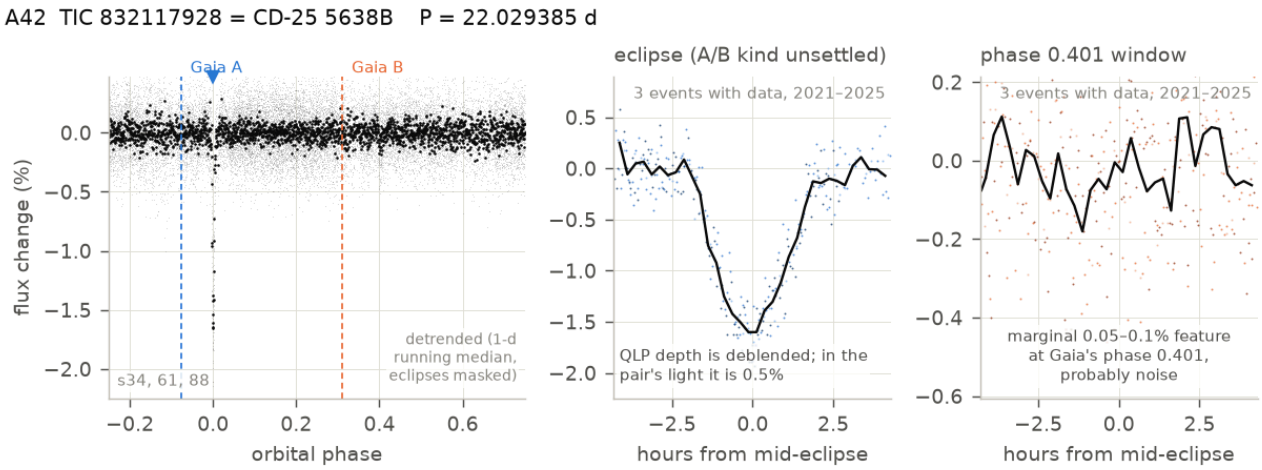
	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP (no TOI/CTOI); Kostov table 2 only. SIMBAD's reference list was not checked. A prior partial detection: an unvetted SPOC s1–s36 TCE at *P* 47.069 d (= 2P)
Open issues	**Eclipse-timing variations:** O–C +21, +20, −6, −11, −44, +17, +11 min, χ^2 = 783 for 6 dof. **Depth change:** the absolute lost flux grew from 76–77 to 106 e/s (1.39×) and T_{14} from 2.9–3.0 to 3.48 h in s88. More eclipse times, RVs for a long-period trend, Gaia DR4 epoch astrometry
Next eclipses	B: BTJD 4349.861, 4373.398, 4396.935 (±0.05 d including the timing variations)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: confirmed, with eclipse-timing variations. The eclipse times vary by tens of minutes in a coherent way, and the 2025 eclipses are deeper and longer than earlier ones. The most likely interpretation, perturbation of the orbit by a third body, is not yet supported by a measured third orbit.

TYC 6568-1036-1 is an F star whose K-dwarf companion is partly hidden behind it once per 23.54-d orbit, near periastron. The central duration would be 6.2 h; the observed 3.0–3.5 h is a graze. At the other conjunction the stars miss by twice the distance, so no transit is expected, and none is seen.

The timing residuals are not scatter, since three other pipelines reproduce them to about 2 min. A light-travel-time orbit is excluded as their cause: it would need a semi-amplitude of at least 3.8 AU with an outer period under about 6 yr, and hence a tertiary of at least 2 $M_\odot$ that would dominate the light. A dynamical perturbation with an outer period of a few years fits, and Gaia's RUWE of 2.04 supports an outer orbit. In fixed apertures the 2025 eclipses lose 1.39 times more flux and last longer, as a grazing eclipse with a shrinking impact parameter does. We interpret this as an inclination change driven by the same third body. Predicted depths for a companion supplying 3.4% of the light are about half those observed: the change is in the observed direction and within a factor of 2.

A42. CD-25 5638B (TIC 832117928)



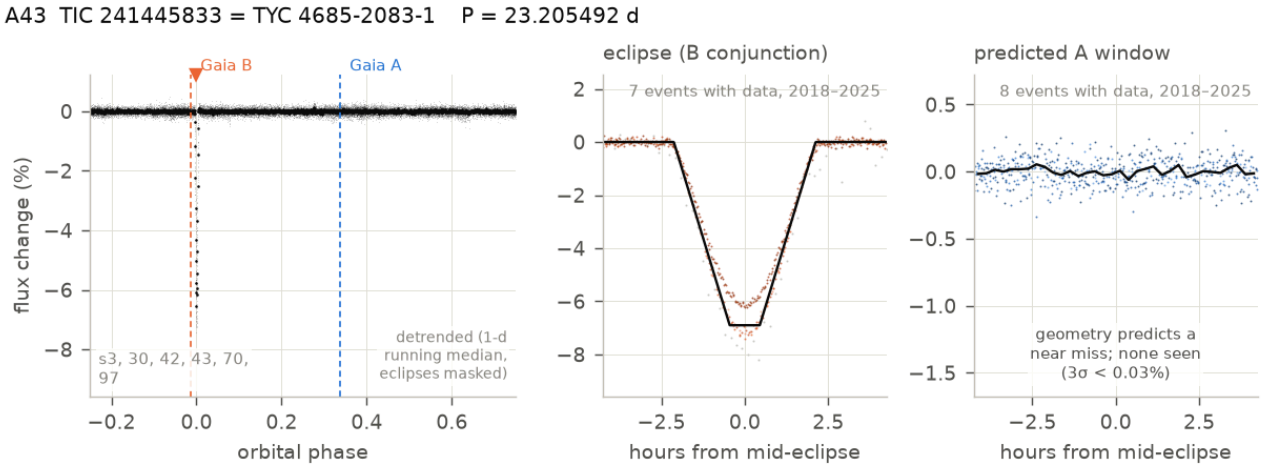
A42. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 22.029385 d, Gaia's predicted conjunctions dashed. Middle: the V-shaped eclipses in QLP's deblended flux (0.5% of the 1.3″ pair's combined light), 2019–2025; the sector-7 event, first found in eleanor-lite, is included where QLP has data. Right: the data near phase 0.40, where a marginal feature is probably noise.

Identifiers	TIC 832117928; Gaia DR3 5694223905397676416; CD-25 5638B (component B of a 1.3″ visual pair)
Position (J2000)	08:09:03.3 −26:17:15.5
Brightness	*G* 10.82; Tmag 10.50

Stars	If on B (Gaia FLAME R 2.24 $R_\odot$, M 1.50 $M_\odot$): companion $m_2 \geq 0.41\ M_\odot$; mass function 0.0194 $M_\odot$. Gaia SB1 on component B
Period	22.029385 ± 0.000018 d (Gaia 22.004 ± 0.060 d; 0.42σ)
Epoch of eclipse	T_0 = BTJD 1508.34974 ± 0.00098 = BJD_{TDB} 2458508.34974
Eclipse	0.56% (eleanor) and 0.40–0.55% (TESScut apertures) of the pair's combined light; QLP's 1.5–1.9% is its deblended value. V-shaped, T_{14} 3.8–4.8 h. Four events: s7 (new, eleanor-lite, predicted to 0.002 d), s34, s61, s88; O–C ≤ 1.4 min
Other eclipse	A marginal feature at phase 0.401 (515–1014 ppm per sector, 3.55 robust σ) at Gaia's predicted separation; the predicted M-dwarf occultation is ≤ 0.045% blended and Gaia's geometry predicts a miss, so it is probably noise
Orbit	Gaia's orbit too poorly constrained to discriminate (*e* 0.28 ± 0.13, ω ±22°, t_p ±1.4 d); at Gaia's epoch the eclipses sit +1.0σ from A and −1.5σ from B. If on B, central duration at A 9.9 h, so the observed ~4 h is a graze
Pipelines agreeing	QLP SAP; eleanor-lite (s7); TESScut apertures
Pixel attribution	The two stars of the pair cannot be separated in TESS pixels; the s61 difference image is centred on the pair (4.7″). QLP centroid shift +0.25 ± 0.21 mpx against +0.26 expected if B is eclipsed and −0.17 if A: favours B at about 2σ. Every other star within 90″ would need ≥ 22%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP; Kostov table 2 only. SIMBAD's reference list was not checked
Open issues	Which component eclipses: the period match favours B strongly, the centroid weakly (2σ); Gaia's RVS spectra of a 1.3″ pair may be blended. Speckle or AO-resolved photometry during an eclipse, or resolved RVs, would resolve it
Next eclipses	BTJD 4328.111, 4350.140, 4372.170 (±0.003 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed in the pair's combined light, with the host component probably B (the period match is strong, the centroid shift favours B at only about 2σ); no single eclipse is measured in two independent pipelines (s7 in eleanor-lite only, the rest in QLP and custom TESScut photometry).

CD-25 5638 is a 1.3″ visual pair. The eclipses are confirmed in the pair's combined light, at a period that matches, to 0.42σ, the radial-velocity orbit Gaia measured on component B. Three lines of evidence bear on which star eclipses. The period match is strong, because a chance eclipsing binary on A within ±0.06 d of that period is very improbable. The centroid shift is weak, favouring B at about 2σ. Gaia's *e* and ω are not decisive. We therefore interpret the system as probably a companion of at least 0.41 $M_\odot$ grazing B. The predicted transit depth (0.72% of B's light) is within a factor of 2 of the observed (about 1.5% of B).

A43. TYC 4685-2083-1 (TIC 241445833)

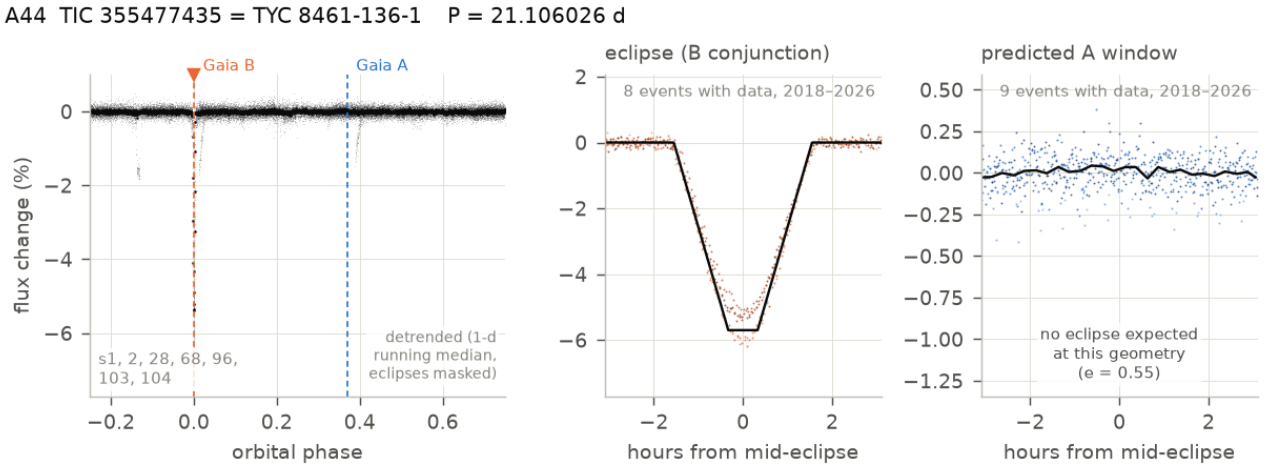


A43. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 23.205492 d, Gaia's predicted conjunctions dashed. Middle: the seven 6.9%, 4.25-h grazing eclipses at B, 2018–2025. Right: the A window, where the near-twin geometry just misses eclipsing and the data are flat (< 300 ppm).

Identifiers	TIC 241445833; Gaia DR3 2508607251322270464; TYC 4685-2083-1
Position (J2000)	01:39:18.9 −01:39:56.3
Brightness	G 10.81; Tmag 10.38
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 0.935, $M_2 \sin^3 i$ = 0.993 $M_\odot$; $a \sin i$ = 42.6 $R_\odot$; radii about 0.95 $R_\odot$ each fit both conjunctions
Period	23.205492 ± 0.000007 d (Gaia 23.19576 ± 0.00395 d; 2.47σ short)
Epoch of eclipse	T_0(B) = BTJD 1389.70430 ± 0.00052 = BJD_{TDB} 2458389.70430
Eclipse (B, periastron side)	6.8–7.0% deep (SPOC 2-min s30, s43 6.82%; TESS-SPOC s70 6.90%), T_{14} 4.25 h, T_{23} 0.9 h, grazing (d/S = 0.74–0.85); seven events, 2018–2025, including s3 and a flagged s42 event; O−C ≤ 1.5 min
Other eclipse (A)	None: 100% coverage, per-epoch −350 to +250 ppm, 3σ limit 280–345 ppm. The geometry predicts no A eclipse, but only narrowly
Orbit	Projected to Gaia's epoch, the eclipse sits +0.283 ± 0.161 d (+1.8σ) from Gaia's B conjunction; the TESS-era 3σ offset is mostly Gaia's period error. Gaia e 0.2711, ω 330.17°. r_A/r_B = 1.31
Pipelines agreeing	QLP SAP; SPOC 2-min (s30, s43); TESS-SPOC (s3, s70); eleanor (s3)
Pixel attribution	Difference-image centroid on the target (0.2–0.4 px); SPOC centroid shifts under 1 mpx at 7% depth; the only stars within 60″ are $G \approx 21$
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP. A prior partial detection: an unvetted SPOC s1–s46 TCE at P 348.08 d (= 15P)
Open issues	A mild t_p/ω tension with Gaia, which is not significant. The A conjunction is a sensitive radius test, where a shallow graze is just possible
Next eclipses	B: BTJD 4336.802, 4360.007, 4383.213 (±0.001 d); A windows +8.13 ± 0.19 d later (4344.93, ...)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4685-2083-1 is a pair of near-twin stars of about 1 $M_\odot$ in a 23.2-d orbit that graze each other near periastron. The central duration there would be 7.3 h; the observed 4.25 h is a graze. At the other conjunction the stars are 1.31 times farther apart. Twins of 1.03 $R_\odot$ would make a 4.2% eclipse with A missing. Twins of 0.9 $R_\odot$ would make a 7.0% eclipse with a 0.16% graze at A, which the data exclude. At about 0.95 $R_\odot$ both observations fit. QLP's 6.0% depth in s97 is a normalisation effect.

A44. TYC 8461-136-1 (TIC 355477435)

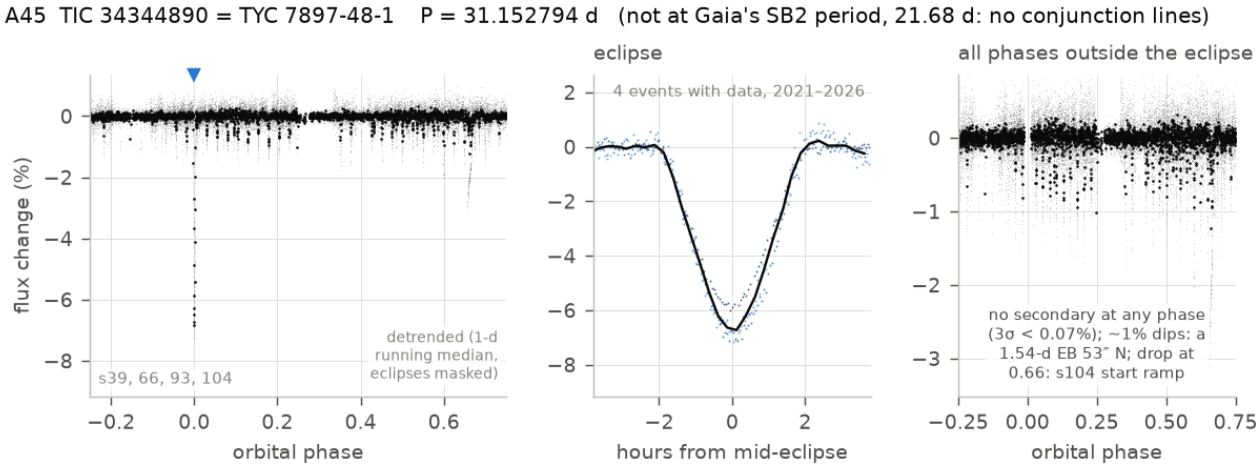


A44. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.106026 d, Gaia's predicted conjunctions dashed. Middle: the eight 5.7%, 3.1-h grazing eclipses at B, 2018–2026 (the last in the January 2026 sectors). Right: the A window; at e = 0.55 no eclipse is possible there.

Identifiers	TIC 355477435; Gaia DR3 6501783310150885376; TYC 8461-136-1
Position (J2000)	23:26:45.3 −50:44:19.7
Brightness	G 10.92; Tmag 10.50
Stars	Gaia SB2 near-twins: $M_1 \sin^3 i$ = 1.138, $M_2 \sin^3 i$ = 1.034 $M_\odot$; $a \sin i$ = 41.6 $R_\odot$; radii about 1.2 and 1.0 $R_\odot$ give the observed depth
Period	21.106026 ± 0.000006 d (Gaia 21.100720 ± 0.003288 d; 1.6σ)
Epoch of eclipse	T_0(B) = BTJD 1325.49469 ± 0.00051 = BJD_{TDB} 2458325.49469
Eclipse (B, periastron side)	5.5–5.9%, constant (QLP's 5.0–5.2% in s103/s104 is a pipeline artefact), T_{14} 3.1 h, T_{23} 0.66 h, grazing (d/S ≈ 0.75); eight events, 2018–2025, including s1 ×2 and s68; O−C ≤ 2.4 min
Other eclipse (A)	Impossible: impact distance at A is 5.1 $R_\odot$ against R_1+R_2 = 2.2 $R_\odot$ (r_A/r_B = 3.13). All epochs flat to ±200 ppm apart from two single-epoch artefacts
Orbit	e = 0.546 (Gaia). At Gaia's epoch the offset from B is −0.016 ± 0.047 d (−0.3σ); the +0.4 to +0.9 d TESS-era offsets are entirely Gaia's period error
Pipelines agreeing	QLP SAP; eleanor (s1); TESScut apertures (lost flux 587 and 579 e/s in s28 and s103)
Pixel attribution	Difference image on the target; the only neighbour within 90″ is G 19.4 at 13.6″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, CCDM, SB9, Gaia DR3 variability tables, ExoFOP
Open issues	None
Next eclipses	B: BTJD 4343.656, 4364.762, 4385.868 (±0.001 d)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 8461-136-1 is a near-twin pair of about 1.1 and 1.0 $M_\odot$ in a highly eccentric (e = 0.55) 21.1-d orbit. The stars graze each other at the periastron-side conjunction; at the other they are more than three times farther apart, so no A eclipse can occur. Eight eclipses over seven years keep a constant depth, and once Gaia's period error is removed the eclipses fall where Gaia's orbit puts conjunction B, to 0.3σ.

A45. TYC 7897-48-1 (TIC 34344890)



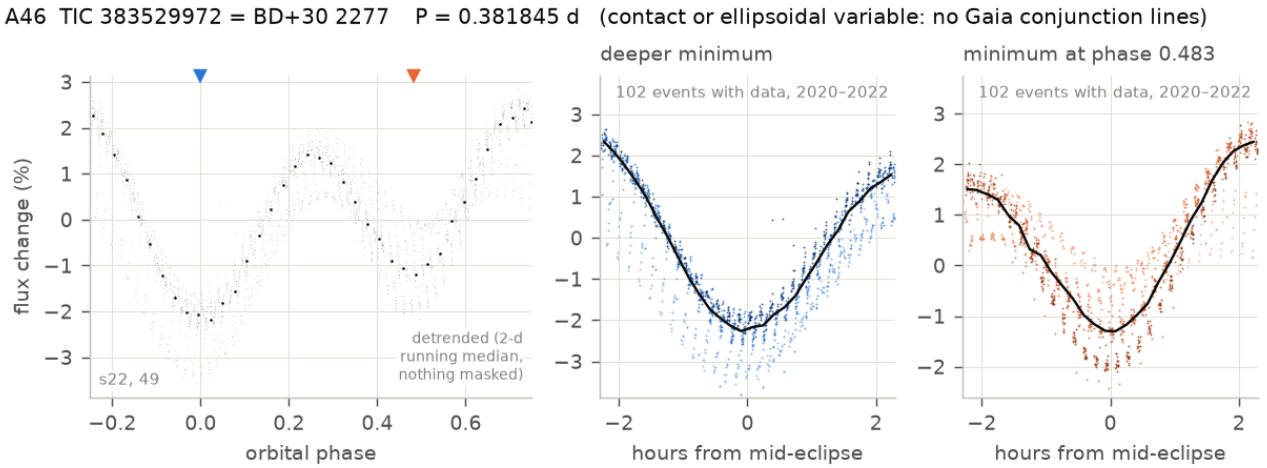
A45. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 31.152794 d. Middle: the four 6–7%, 3.8-h grazing eclipses (sectors 39, 66, 93 and 104, the last in January 2026). Right: all phases outside the eclipse, including phase 0.5; no secondary is seen (< 0.07%).

Identifiers	TIC 34344890; Gaia DR3 5956391835910597888; TYC 7897-48-1
Position (J2000)	17:49:36.3 −43:12:27.7
Brightness	G 10.93; Tmag 10.50
Stars	Gaia SB2 of near-twin stars seen almost edge-on (K_1 56.7, K_2 59.7 km/s, e 0.58, M $\sin^3 i$ 0.98 and 0.93 $M_\odot$) at P = 21.684 d; the eclipsing pair's relation to it is unresolved
Period	31.152794 ± 0.000072 d, the only period the coverage allows (P/2 ruled out). Gaia's SB2 period is 21.683770 ± 0.002520 d
Epoch of eclipse	T_0 = BTJD 2382.7596 ± 0.0037 = BJD_{TDB} 2459382.7596
Eclipse	5.70–6.80% deep, T_{14} 3.7–3.9 h, flat bottom about 0.8 h, close to V (likely grazing); four events, one per sector (s39, s66, s93, s104), 2021–2025. O−C −3.6 to +4.6 min, χ^2 131 for 2 dof (errors underestimated, or real timing variations)
Secondary eclipse	None secure: phase coverage 99%, 3σ limit about 0.07%; one unconfirmed 0.5% dip at phase 0.096
Orbit	The eclipses do not fall at the conjunctions of Gaia's 21.684-d SB2 orbit. 31.15/21.68 = 1.44 cannot be the inner and outer orbits of a stable hierarchy, and the beat period (71.3 d) is not a known alias
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s39)
Pixel attribution	Difference-image centroid within 0.1 px (~2″) of the target in s66 and s93; the G 13.19, 12.66 and 11.73 stars at 57–65″ are excluded; Gaia source 5956391801551111552 at 3.3″, with no G magnitude, cannot be excluded
Catalogue status	Not listed in: VSX, ZTF (Chen et al. 2020), ATLAS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, WDS, CCDM, Hipparcos, SB9, ExoFOP, general web search. The VSX "E" star 53″ north is a separate star (its true period is 1.5432 d, not VSX's 12.3202 d)
Open issues	**Orbit unresolved.** Either Gaia's period is wrong and the 31.15-d pair is the SB2 itself, or the system is a 2+2 quadruple, or the eclipses come from a blend within about 2″. RVs phased at 31.15 d against 21.68 d would distinguish between these
Next eclipses	BTJD 4345.386 ± 0.006
Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)
Status	Confirmed as an eclipsing system on, or within about 2″ of, TYC 7897-48-1, found as by-catch. Its relation to the Gaia SB2 orbit is open (Gaia's period wrong, a 2+2 quadruple, or a blend within 2″), so Gaia's orbit is not counted as contradicted; the Gaia source 3.3″ away without a G magnitude cannot be excluded.

Note: a confirmed eclipser on, or within about 2″ of, the target, with an unresolved orbit. The eclipses are real and on or near TYC 7897-48-1, but they do not follow the Gaia orbit that brought the star into the sample, and the two periods cannot belong to a stable triple; Gaia's orbit is not counted as contradicted.

TYC 7897-48-1 shows one grazing eclipse of 6–7% per TESS sector, four in all, on a single period of 31.15 d. A 3.894-d period is one-eighth of this value. The 0.77-d dips also present are a separate eclipsing binary 53″ north, whose alternating depths give a true period of 1.5432 d. A 3.8-h eclipse fits an $e \approx 0.58$ pair of about 1 $M_\odot$ at either period, so three interpretations remain: Gaia's period is wrong and the eclipsing pair is the Gaia SB2 itself; a 2+2 quadruple; or a blend within about 2″. A Gaia source 3.3″ away without a magnitude cannot be excluded by the pixels.

A46. BD+30 2277 = WDS ES 436 (TIC 383529972)



A46. QLP SAP flux of the 1.87″ pair, detrended with a 2-d running median (nothing masked), folded on P = 0.381845 d (sectors 22 and 49), showing a contact-binary or ellipsoidal light curve with unequal minima (4.6% and 3.4%) and unequal maxima (O'Connell effect). The Gaia 46.3-d orbit of component A is not the period of this variability.

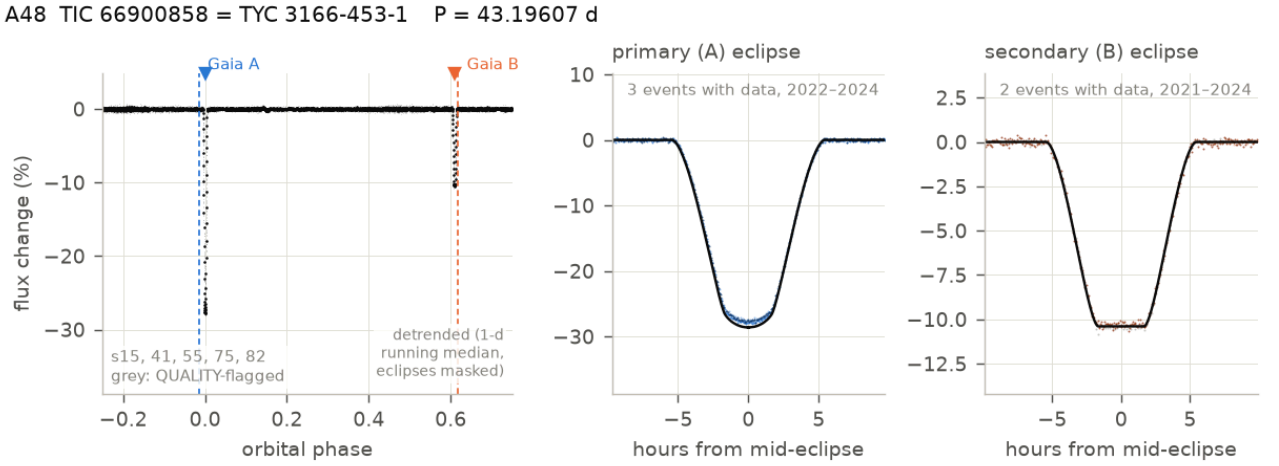
Identifiers	TIC 383529972; Gaia DR3 4011717574379114240 (component A); component B = Gaia DR3 4011717574379114112; BD+30 2277; WDS ES 436
Position (J2000)	12:28:27.5 +29:52:42.3
Brightness	G 10.63 (A), 10.73 (B); Tmag 10.12
Stars	A 1.87″ physical pair (parallaxes 4.02 and 4.15 mas). Component A is a Gaia SB1 at P 46.340 d (K_1 = 3.0 km/s, RUWE 4.6). The variable is probably component B
Period	0.381845 ± ~0.000002 d (Fourier fit 0.3818455; 99 minimum timings 0.3818442 ± 0.0000005); not an alias of the 46.3-d orbit
Epoch of deeper minimum	BTJD 2645.0111 ± 0.0007 = BJD_{TDB} 2459645.0111
Minima	4.6% and 3.4% jointly (4.33/1.97% in s22, 4.56/4.00% in s49), second minimum at phase 0.483; peak-to-peak 4.4–4.7%; O'Connell effect −0.7 to −0.9%; shape changes between sectors. In the variable star's own light, about 9.6% if B, 8.8% if A
Type	A spotted contact binary (EW) or an ellipsoidal variable
Pipelines agreeing	QLP SAP (s22, s49)
Pixel attribution	On the pair: difference image within 0.1 px, excluding the G 10.47 star 76″ away. Which component: the fitted centroid lies 0.03–0.05 px closer to B in all six tests, but systematics are about 0.1 px, so the test is inconclusive. Gaia RVs: A's 17 RVS transits have a robust RV amplitude of only 7.6 km/s, so A is unlikely to be a contact binary; B has no Gaia RV
Catalogue status	Not listed in: VSX, ZTF (Chen et al. 2020), ATLAS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, CCDM, Hipparcos, SB9, ExoFOP, general web search; WDS has the pair only
Open issues	**Host component unresolved.** Speckle or adaptive-optics photometry separating A and B would resolve it
Next minima	Deeper minima: BTJD 4325.511, 4325.893, 4326.275 (+ n × 0.381845)

Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)
Status	By-catch variable, not counted among the eclipsing systems of this paper. A contact-binary or ellipsoidal light curve on an unresolved 1.87″ pair, in one pipeline, with the host component undetermined: a new variable, but not shown to be an eclipsing system.

Note: a new variable on the BD+30 2277 pair, with the host component unresolved. It is a short-period (9.2-h) contact or ellipsoidal variable, not a long-period eclipser, found incidentally rather than at a predicted conjunction, and it is not counted among the 104 eclipsing systems.

BD+30 2277 is a 1.87″ visual pair of two stars of nearly equal brightness at the same distance; component A carries a Gaia single-lined orbit of 46.3 d. TESS shows a 0.3818-d light curve with unequal minima and maxima whose shape changes between 2020 and 2022, typical of a spotted contact binary or an ellipsoidal variable. The pixels cannot tell which star varies. Gaia's radial velocities of A scatter by only 7.6 km/s, too little for a contact binary; B is therefore the more likely host. If B is the variable, the system is a bound 2+2 quadruple, consisting of the Gaia SB1 binary A and the contact binary B.

A48. TYC 3166-453-1 (TIC 66900858)



A48. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 43.19607 d, Gaia's predicted conjunctions dashed; Gaia's A falls about 0.9 d early (Gaia's eccentricity exceeds the photometric value). Middle: the three 27%, 9.7-h annular transits at A. Right: the two 10.4%, 9.9-h flat total occultations at B.

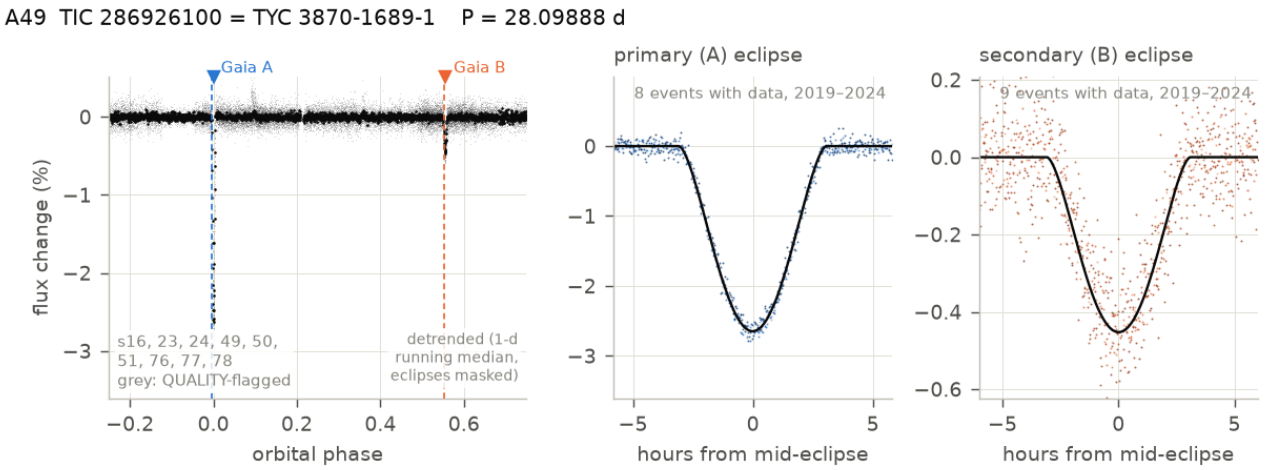
Identifiers	TIC 66900858; Gaia DR3 1871082303289948544; TYC 3166-453-1
Position (J2000)	20:46:48.0 +37:33:55.9
Brightness	G 10.98; Tmag 10.53
Stars	Slightly evolved primary, $M_1 \approx 1.3\ M_\odot$ (Gaia binary_masses 1.06; the parallax luminosity favours 1.3), R_1 = 1.37–1.45 $R_\odot$ + K dwarf, M_2 = 0.73–0.82 $M_\odot$, R_2 = 0.73–0.77 $R_\odot$, about 4600 K, 10.4% of the light; mass function 0.122 $M_\odot$; Gaia SB1
Period	43.19607 ± 0.00003 d (Gaia 43.19275 ± 0.01065 d; +0.3σ)
Epoch of primary	T_0(A) = BTJD 2804.3848 ± 0.0004 = BJD_{TDB} 2459804.3848; T_0(B) = BTJD 2441.9431 ± 0.0008
Primary eclipse (A)	27.0–27.4% deep, 0.403 d (9.7 h), flat part 0.126 d: annular transit of the K dwarf across the primary; three events (2804.38, 3365.93, 3538.72); O–C under 0.1 min
Secondary eclipse (B)	10.3–10.5% deep, 0.414 d (9.9 h), flat part 0.159 d: total occultation of the K dwarf; two events (s41, every QLP cadence flagged but clean in TESS-SPOC; s55). B at phase 0.60938 after A (Gaia 0.6334 ± 0.0085, −2.8σ)
Orbit	Photometric e cos ω = 0.1727, e sin ω = 0.002: e = 0.173, ω = 0.7°; Gaia e cos ω 0.211 ± 0.014, e sin ω −0.005 (Gaia e 0.2111, ω 358.72°). Model: k = 0.528, $i \approx 90°$, χ^2 460/460
Pipelines agreeing	QLP SAP; TESS-SPOC (s41, s55); TARS (all sectors)

Pixel attribution	TESScut difference images: flux lost peaks on the target pixel, centroid 0.03–0.21 px; the QLP centroid shifts at A and B (0.075 and 0.02 px, ratio 3.7) match the depth ratio (3.5), so both eclipses come from the same source; the *G* 12.67 neighbour at 39″ (would need about 130%) is excluded
Catalogue status	Not listed in: VSX, GCVS, NSV, ASAS-3, ASAS-SN, ATLAS, ZTF, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Justesen & Albrecht 2021, Fetherolf et al. 2023, Gaia DR3 variability tables, SB9, ExoFOP, SIMBAD (SB*, one reference), general web search
Open issues	**Gaia eccentricity revised:** Gaia's *e* exceeds the photometric value by about 0.04 (2.8σ). RVs to fix K_1 against the photometric *e* and ω; spectroscopic T_{eff} and log *g*. The Harvard plates (Section 5.6) show the eclipse at this ephemeris (z = 9.4 on 15 plates) but more strongly at P = 43.19588 d, 6σ below the TESS period, a 3.8-h shift of the eclipse centre over 830 cycles: an underestimated period error or apsidal motion
Next eclipses	B: BTJD 4342.570 (2026-10-29 01:37 UTC), 4385.766; A: BTJD 4359.443 (2026-11-14 22:36 UTC), 4402.639
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: the eclipses contradict part of Gaia's orbit. The eclipse spacing gives *e* = 0.173 where Gaia's SB1 has *e* = 0.211 ± 0.014 (2.8σ), which is why Gaia's predicted A times run about 0.9 d early. The timing measurement is model-independent, and light-travel time across the orbit (about a minute) cannot explain a one-day shift.

TYC 3166-453-1 is a slightly evolved star of about 1.3 $M_\odot$ with a K-dwarf companion of about 0.75 $M_\odot$, seen edge-on. At A the dwarf crosses the primary's disk entirely (an annular transit, 27% deep); at B it is hidden completely, and the flat 10.4% depth is its share of the light. Five eclipses fit a single linear ephemeris to within 0.1 min; one of them, in sector 41, is wholly in QLP-flagged cadences but present with quality zero in TESS-SPOC. A companion supplying 10% of the light is consistent with Gaia's single-lined solution.

A49. TYC 3870-1689-1 (TIC 286926100)



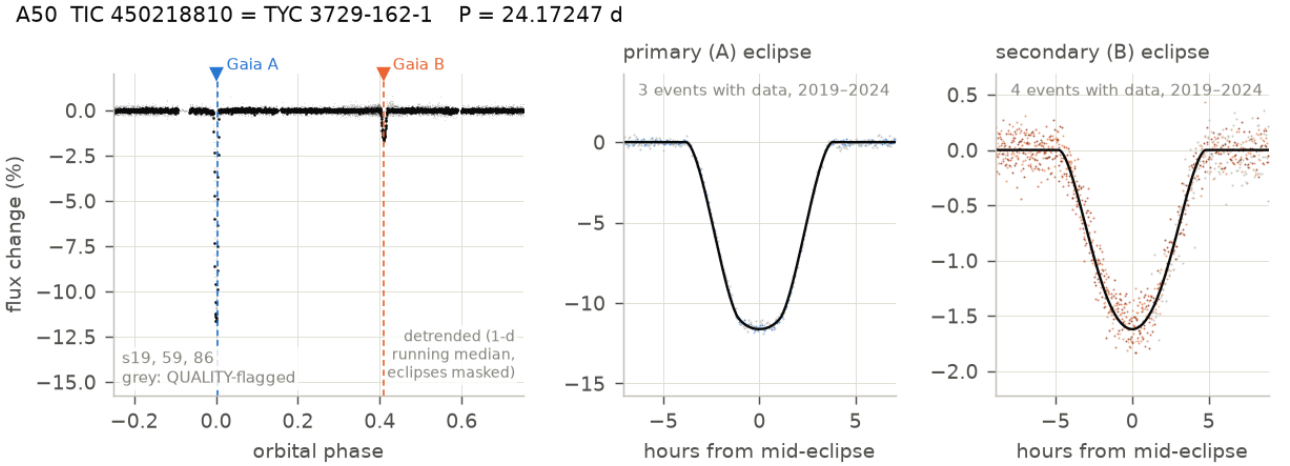
A49. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 28.09888 d, Gaia's predicted conjunctions dashed. Middle: the eight 2.6%, 5.8-h grazing eclipses at A. Right: the nine 0.45%, 6-h grazing eclipses at B.

Identifiers	TIC 286926100; Gaia DR3 1404550929673449216; TYC 3870-1689-1
Position (J2000)	15:49:15.4 +52:32:40.6
Brightness	*G* 10.40; Tmag 10.01
Stars	Primary R_1 = 1.64–1.75 $R_\odot$ (TIC 1.80), M_1 = 1.0–1.25 $M_\odot$ assumed + M dwarf, m_2 = 0.46–0.53 $M_\odot$, R_2 = 0.46–0.49 $R_\odot$, T_2 ≈ 3670 K; mass function 0.0465 $M_\odot$; Gaia SB1; APOGEE entries in SIMBAD

Period	28.09888 ± 0.00001 d (P_A 28.09887 ± 0.00001, P_B 28.09898 ± 0.00004; Gaia 28.1076 ± 0.0067 d, −1.3σ)
Epoch of primary	T_A = BTJD 2645.0156 ± 0.0003 = BJD_{TDB} 2459645.0156; T_B = BTJD 2688.6996 ± 0.0008
Primary eclipse (A)	2.53–2.69% deep, T_{14} 5.8 h, ingress 2.3 h, V-like (flat about 1.2 h): grazing transit of the M dwarf; eight events 2019–2024 (one, s023, wholly in flagged cadences); O−C ≤ 1 min
Secondary eclipse (B)	0.41–0.48% deep, T_{14} mostly 5.8–6.0 h: grazing occultation; nine events; O−C ≤ 5 min. B at phase 0.55468 ± 0.00003 after A (Gaia 0.5567 ± 0.0083)
Orbit	e cos ω = 0.0860 against Gaia's 0.0892 ± 0.013; e sin ω ≈ −0.005 (Gaia −0.029 ± 0.0135, +1.7σ); i = 87.8°, b/(1+k) ≈ 0.80 at both conjunctions
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s50, s76); TARS (s50, s76); within 0–3 min
Pixel attribution	TESScut s50 and s76 difference images (A) peak on the target pixel at 64σ and 79σ, centroid 0.7″ from the out-of-eclipse centroid; the only neighbour (G 15.32 at 53″) would need 247%. Background excursions change sign from event to event while the depth stays constant
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (Gaia multiplicity and APOGEE entries only), general web search
Open issues	P_A and P_B differ by 2.7σ (possibly slow apsidal motion; not significant); the radius-ratio degeneracy of a grazing fit (multi-colour photometry would break it); APOGEE RVs not yet analysed
Next eclipses	A: BTJD 4330.949, 4359.047; B: BTJD 4346.534, 4374.633 (formal 1–2 min; allow about 10 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3870-1689-1 is a star of about 1.7 $R_\odot$ with an M-dwarf companion of about 0.5 $M_\odot$, grazing at both conjunctions of a mildly eccentric 28.1-d orbit. Seventeen eclipses over five years give a period to 0.00001 d, and the eclipse spacing reproduces Gaia's e cos ω. The companion's fitted radius and temperature lie where main-sequence stars of its mass do; the alternative fit with a large companion would need 1.2–1.3 $R_\odot$ for 0.5 $M_\odot$, which is unphysical.

A50. TYC 3729-162-1 (TIC 450218810)



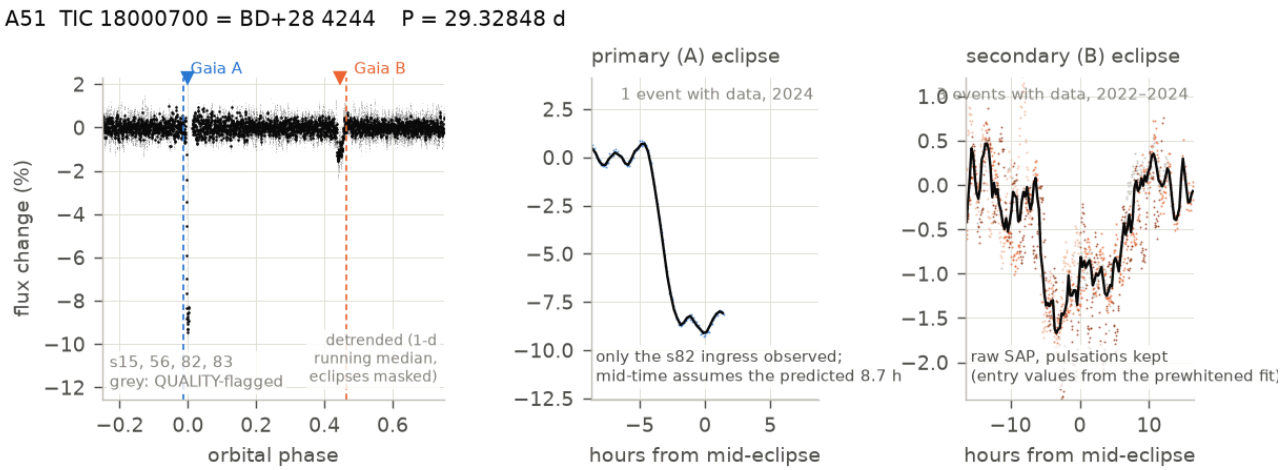
A50. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.17247 d, Gaia's predicted conjunctions dashed. Middle: the A windows with data: three 11.5%, 7.1-h flat-bottomed transits and a fourth window covering only part of an event. Right: the three 1.6%, 8.9-h occultations at B.

Identifiers	TIC 450218810; Gaia DR3 473363571357621760; TYC 3729-162-1
Position (J2000)	03:55:22.1 +59:49:27.7
Brightness	G 10.95; Tmag 10.40
Stars	Primary R_1 = 1.69–1.88 $R_\odot$ (TIC 1.77), M_1 = 0.95–1.2 $M_\odot$ assumed + K/M dwarf, m_2 = 0.59–0.68 $M_\odot$, R_2 = 0.56–0.63 $R_\odot$, T_2 ≈ 3850–4015 K, L_2/L_1 ≈ 0.02–0.025; mass function 0.0879 $M_\odot$; Gaia SB1

Period	24.17247 ± 0.00002 d (Gaia 24.176370 ± 0.007485 d; −0.5σ)
Epoch of primary	T_A = BTJD 2926.6627 ± 0.0004 = BJD_{TDB} 2459926.6627; T_B = BTJD 3274.9793 ± 0.0011
Primary eclipse (A)	11.3–11.9% deep, T_{14} 7.0–7.2 h, flat bottom 2.1 h, ingress 2.5 h: transit of the companion (b_A = 0.57); three events (s19 partly flagged, s59, s86 all flagged); O–C ≤ 0.4 min
Secondary eclipse (B)	1.46–1.72% deep, T_{14} about 8.9 h: occultation of the companion (b_B = 0.83–0.90); three events (s19, s59, s86). B at phase 0.40965 ± 0.00004 after A (Gaia 0.4063 ± 0.0111)
Orbit	e cos ω = −0.1399 against Gaia −0.1434 ± 0.017; e sin ω from the shapes 0.184 (e = 0.231) against Gaia 0.238 ± 0.013; i ≈ 88.2°. B lasts longer because the separation is larger there
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s19, s59); TARS (s19, s59, s86)
Pixel attribution	TESScut s059 A difference image peaks on the target at 224σ, centroid 0.9″ from the out-of-eclipse centroid; B in s019 and s059 also on target; G 15.4 neighbours at 20″ and 56″ would need 680% and 506%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (Gaia multiplicity only), general web search; Kostov table 2 only
Open issues	A mild tension (2.5–3σ) between the shape-based e sin ω and Gaia's (forcing Gaia's e, ω costs $\Delta\chi^2$ ≈ 11), with optimistic errors and partial or flagged eclipses that can bias it; one clean A and B pair in the same sector would test Gaia's e of 0.278
Next eclipses	A: BTJD 4328.666, 4352.839; B: BTJD 4338.568, 4362.741 (a few minutes)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3729-162-1 is a star of about 1.8 $R_\odot$ with a K/M-dwarf companion of about 0.6 $M_\odot$ on an eccentric 24.17-d orbit. At A the companion crosses the primary's disk (11.5%, flat-bottomed); at B it is hidden, and the 1.6% depth is its share of the light. The eclipse timing reproduces Gaia's e cos ω within 0.3σ. Box-fit estimates of the A eclipse (9.7%, 4 h) underestimate the model-fit depth and duration given above.

A51. BD+28 4244 (TIC 18000700)



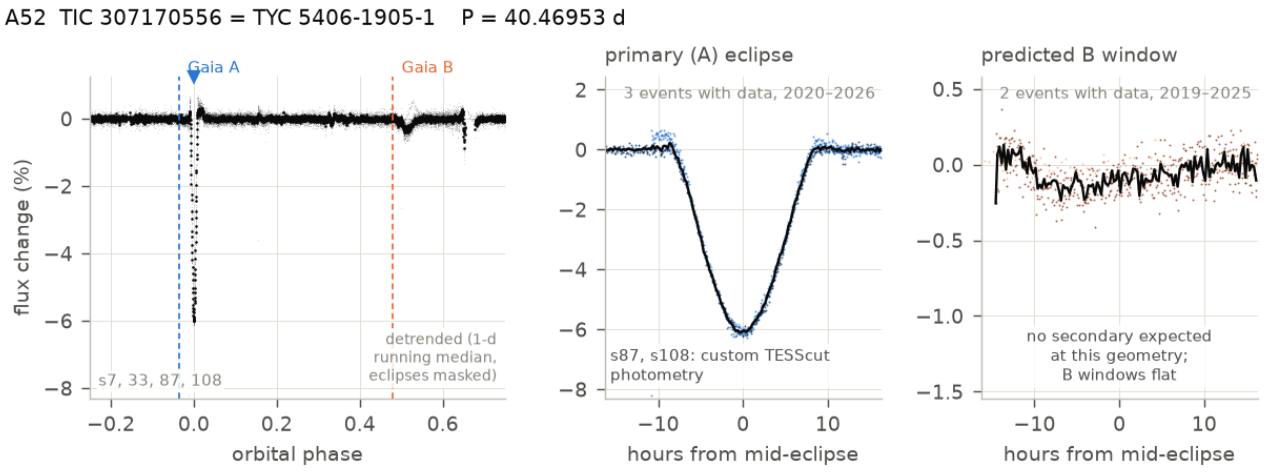
A51. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.32848 d, Gaia's predicted conjunctions dashed; the star pulsates throughout. Middle: the ≥ 8.6% transit ingress at A at the end of sector 82. Right: the four B windows with data (the entry's fit uses all four; one is partly flagged), the 1.1–1.3%, 16.4-h occultations shown in raw SAP flux with the pulsations present (depth and duration from the prewhitened fit; see text).

Identifiers	TIC 18000700; Gaia DR3 1896494009856522240; BD+28 4244
Position (J2000)	21:57:03.2 +29:18:20.1
Brightness	G 10.46; Tmag 10.19
Stars	F0 primary of 2.2 $R_\odot$ (M_1 ≈ 1.6 $M_\odot$ assumed), a hybrid δ Sct/γ Dor pulsator (12.08, 12.37, 4.08, 3.16, 2.70 c/d at ≤ 2.5 ppt) + K dwarf of about 0.7 $R_\odot$, $m_2(i = 90°)$ ≈ 0.73 $M_\odot$, 4500–4800 K; mass function ≈ 0.071 $M_\odot$; a ≈ 53 $R_\odot$; Gaia SB1
Period	P_B = 29.32848 ± 0.00033 d (Gaia 29.3189 ± 0.0156 d; +0.6σ)

Epoch of eclipse	T_0(B) = BTJD 3571.1856 ± 0.0044 = BJD_{TDB} 2460571.1856; A first contact BTJD 3557.9714 ± 0.0012
Eclipse at B	1.12–1.34% deep after prewhitening, T_{14} 15.6–16.5 h, T_{23} about 10 h, flat: total occultation of the K dwarf (its light share), near apastron; four events (s15 all flagged, 1.8%; s56 partly flagged; s82; s83); all O–C under 0.02 d
Eclipse at A	Only an ingress: first contact BTJD 3557.9714, ingress 1.86 h, depth ≥ 8.6% before the data end at the close of s82; transit of the dwarf across the F star (R_2/R_1 ≈ 0.3), predicted full duration 8.7 h near periastron. In all other sectors the A conjunction fell in gaps at sector boundaries
Orbit	A about 16.3 d after B (Gaia predicts 15.26 d); implied $\|e \cos \omega\|$ ≈ 0.09 against Gaia −0.03 ± 0.09; predicted central durations 17.2 h (B) and 8.7 h (A). Every B within 0.3 d of Gaia's prediction
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and TARS (all four measured events, including the A ingress: 8.3% and 9.1%)
Pixel attribution	TESScut: every event on the target (e.g. the A ingress: target loses 9.69%, χ^2/pix 0.80 against 94 for the next source); all neighbours G ≥ 14.4 (one would need ≥ 75%). No common-mode signal at any event (166–297 comparison stars)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP (no TOI/CTOI), SIMBAD (SB*, Gaia DR3 orbit reference only)
Open issues	A full A eclipse (predicted 8.7 h) has not yet been observed; it would give e sin ω. RVs for the K_1 phase
Next eclipses	B: BTJD 4333.73, 4363.05 (±0.01); A: BTJD 4350.02, 4379.35 (±0.06; assumes the predicted duration)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+28 4244 is a pulsating F0 star in an eclipsing binary with a K-dwarf companion. Its δ Sct and γ Dor pulsations, at up to 2.5 ppt, distorted the eclipse profiles. Prewhitening twelve pulsation frequencies reduced the scatter from about 3.9 to 1.5 ppt and left flat-bottomed 16-h occultations. The 9% drop at the end of sector 82 is the start of the transit at the periastron-side conjunction, with a 1.9-h ingress that fits the predicted 8.7-h transit. Every event is on the target in the pixels, reproduced by two other pipelines, and absent from the common mode.

A52. TYC 5406-1905-1 = ASAS J071301-1406.7 (TIC 307170556)



A52. Left: SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 40.46953 d, Gaia's predicted conjunctions dashed (computed with Gaia's 40.543-d period, which the eclipses contradict). Middle: the three 6%, 16-h grazing eclipses at A (sectors 33, 87 under scattered light, and 108). Right: the B windows, where no eclipse is expected at e ≈ 0.31 and none is seen.

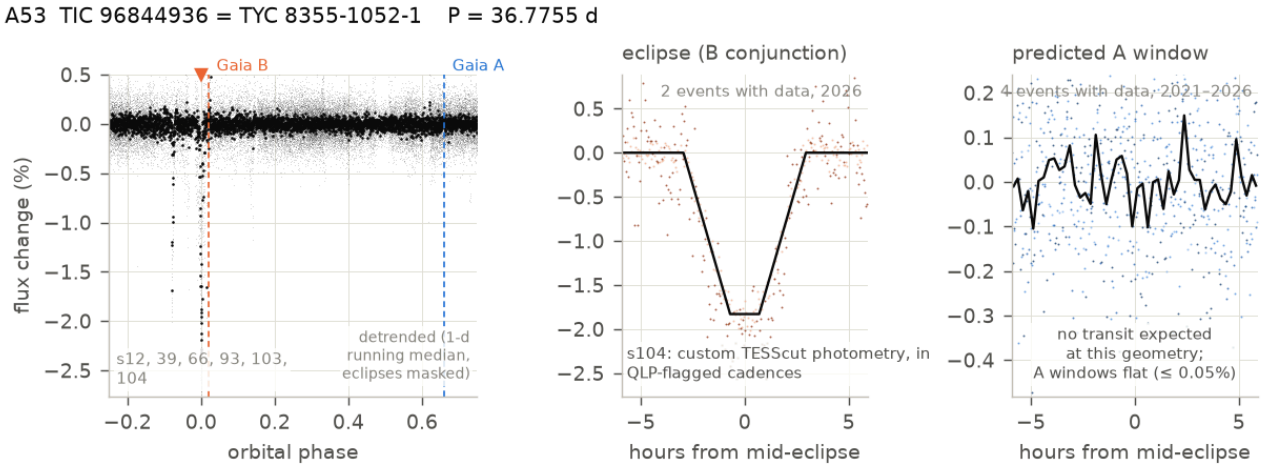
Identifiers	TIC 307170556; Gaia DR3 3032498188045355136; TYC 5406-1905-1; ASAS J071301-1406.7
Position (J2000)	07:13:01.5 −14:06:42.9

Brightness	*G* 10.49; Tmag 9.82
Stars	Gaia SB2 equal-mass pair: K_1 41.0, K_2 40.3 km/s, M $\sin^3 i$ = 0.961 and 0.977 $M_\odot$, $a \sin i$ = 61.8 $R_\odot$. Gaia FLAME (as a single star) R 5.6 $R_\odot$, L 23.5 $L_\odot$: probably two ~1 $M_\odot$ stars on the lower red-giant branch
Period	40.46953 ± 0.00005 d (formal; allowing for red noise the uncertainty is about three times larger). Gaia 40.543005 ± 0.014906 d: −0.0735 d, **−4.9σ**
Epoch of eclipse	T_0 = BTJD 2206.5875 ± 0.0019 = BJD_{TDB} 2459206.5875
Eclipse (A, periastron)	5.92–6.16% deep, T_{14} 16.10–16.36 h, T_{23} 3.4–3.5 h, ingress about 6.4 h: grazing, V/U-shaped; three events: s33, s87 (under a scattered-light surge in flagged cadences, recovered with local-background photometry), s108 (within about 6 min of the time predicted from s33 and s87); O−C after refitting all three +1.0, −3.3, +2.3 min
Other eclipse (B)	None expected (impact parameter at B 1.9× larger) and none seen: the s7 and s87 B windows are flat
Orbit	Gaia *e* 0.3085, ω 85.93°. At Gaia's epoch the eclipse and Gaia's A conjunction agree (+1.7σ); the disagreement is entirely in the period. Gaia goodness-of-fit 11.5
Pipelines agreeing	QLP SAP (s33); custom TESScut aperture photometry (this work) for all three. Gaia epoch photometry: the faintest of 51 G transits falls 3.0 h from mid-eclipse on the TESS ephemeris, 2.84 d out of eclipse on Gaia's period (chance about 0.6%); a single point, +0.12 mag against a scatter of 0.025 mag, about twice what the eclipse can produce at that phase, so supporting rather than independent evidence
Pixel attribution	Difference images of all three events: uniform 5.2–5.3% fractional depth in every core pixel, about 0 off target; no Gaia neighbour within 60″ brighter than *G* 14.5 (would need 245%)
Catalogue status	**VSX type revision MISC → EA.** VSX: ASAS J071301-1406.7, MISC, *P* 79.3 d, at 1″; ASAS-SN "ROT:", *P* 441.8 d; Gaia flags it VARIABLE with classifier RS (score 0.35). Nothing in Gaia's eclipsing-binary table, SB9, WDS or ExoFOP
Open issues	**Gaia period corrected** by 4.9σ; an RV refit at 40.4695 d would test Gaia's solution. The B window in s33 and s108 was not usable
Next eclipses	A: BTJD 4351.473, 4391.942, 4432.412 (±0.01); a B eclipse, if any, near 4372.28 ± 0.93
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: (1) a VSX type revision: the star is in VSX as ASAS J071301-1406.7, type MISC, *P* 79.3 d, and the eclipses are what is new; (2) the eclipses contradict Gaia's period, which exceeds the eclipse period by 4.9σ, while agreeing with Gaia's conjunction phase at Gaia's own epoch.

TYC 5406-1905-1 is a pair of equal-mass stars, likely low-red-giant-branch giants of about 1 $M_\odot$ each, on an eccentric 40.47-d orbit. They graze each other at periastron, where the 16.3-h chord (about 9 $R_\odot$) is consistent with the separation and speed. At the far conjunction they miss. The period rests on three eclipses. One of them lies under a scattered-light surge where QLP's background subtraction fails (12 of 247 stars in that sector show spurious dips there). The eclipse in sector 108 falls exactly at cycle 52 of the TESS period, about 3.8 d from where Gaia's period would put it. One Gaia epoch-photometry point is consistent with the TESS period and not with Gaia's, which supports but does not independently confirm it.

A53. TYC 8355-1052-1 (TIC 96844936)



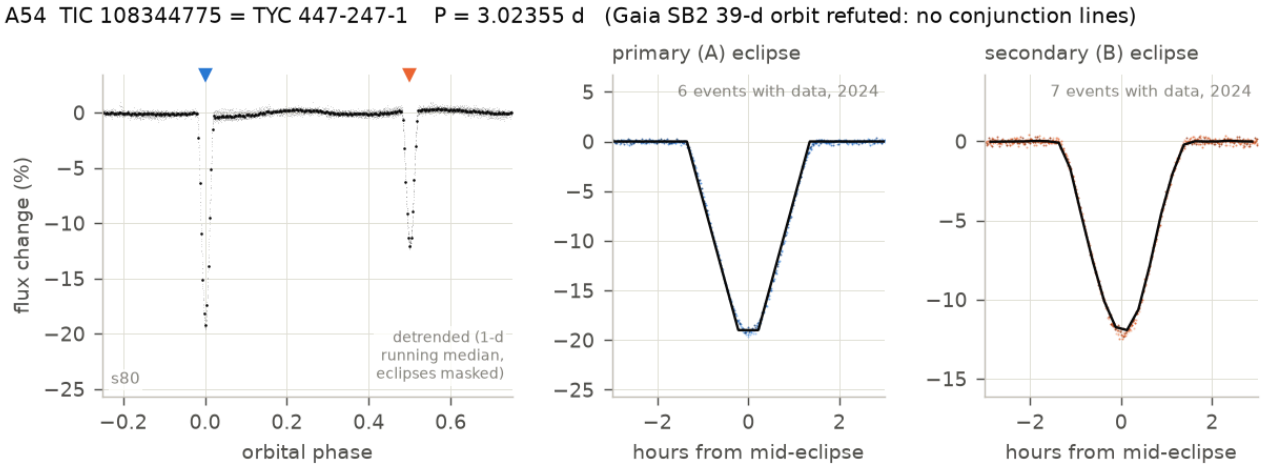
A53. Left: SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 36.7755 d, Gaia's predicted conjunctions dashed. Middle: the two 1.8–2.4%, 5.9-h occultations at B (sector 103 clean; sector 104 in QLP-flagged cadences, shown in a lighter tone). Right: the A windows, flat as the geometry predicts.

Identifiers	TIC 96844936; Gaia DR3 5921931286046605952; TYC 8355-1052-1
Position (J2000)	17:40:45.0 −52:25:24.0
Brightness	*G* 10.94; Tmag 10.46
Stars	Primary of 3.0 $R_\odot$, 1.62 $M_\odot$ (Gaia FLAME) + companion of about 0.95 $M_\odot$, about 6.5% of the TESS light; mass function 0.130 $M_\odot$; Gaia SB1; SIMBAD: GALAH and open-cluster papers
Period	36.7755 ± 0.006 d, the interval between the two events (4192.621 – 4155.8455; the rounded value 36.772 does not reproduce the next-eclipse times given below) (Gaia 36.7818 ± 0.0092 d; −0.7σ)
Epoch of eclipse	BTJD 4155.8455 ± 0.0007 (QLP) = BJD_{TDB} 2461155.8455; our TESScut photometry 4155.8488 ± 0.0008
Eclipse (B)	1.83% (QLP) / 2.44% (our TESScut photometry) deep, T_{14} 5.88 h, T_{23} 1.40 h: partial occultation of the companion (35–39% of its disc covered, b ≈ 0.8, i ≈ 84°). Event 1: s103, clean. Event 2: BTJD 4192.621 ± 0.006 in s104, in QLP-flagged scattered-light cadences, 1.8–2.5% across 12 aperture and background choices in our TESScut photometry
Other eclipse (A)	None expected (projected separation 2.87× larger) and none seen: four A windows (s39, s66, s93, s104) flat to ≤ 0.05 ± 0.03%
Orbit	Gaia e 0.5207, ω 248.07°; central duration at B would be 9.9 h against 5.9 h observed
Pipelines agreeing	QLP SAP (s103); custom TESScut photometry (this work) for both events
Pixel attribution	s103 difference image on target (1.85–2.25% in each core pixel, 0.00% off); s104 difference image noisier (pointing jitter) but strongest in the core. No common mode (228 stars, median −17 ppm). Neighbours: a 3.7″ source with no Gaia photometry, and stars G ≥ 16.1 (would need more than 100%)
Catalogue status	Not listed in: VSX, ASAS-SN, Gaia DR3 eclipsing-binary table, SB9, WDS, ExoFOP (no TOI/CTOI)
Open issues	**One event in flagged data.** Fainter stars in the s104 cutout scatter by ±1–4% in that stretch. One further clean B event would secure the confirmation
Next eclipses	B: BTJD 4339.723, 4376.498 (±0.01–0.03)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable (see below)
Status	Probable. The second event (sector 104) lies in QLP-flagged scattered-light cadences and was recovered only in custom TESScut photometry, where the depth is reduction-dependent (1.8–2.5% across 12 aperture and background choices; an independent re-reduction made for the referee review found 2.0–3.3%, with four of ten reductions degenerate); the period is a single interval, 36.7755 d, with the cycle count fixed by Gaia (two cycles, 18.39 d, is untested); a 3.7″ source without Gaia photometry shares the pixel; a flagged 2.2–2.6% drop at the end of sector 66, inside a B window, is inconclusive. One further clean event would confirm it.

Note: probable, not confirmed. The first occultation (sector 103) is clean and on the target. The second (sector 104) sits in cadences QLP flagged for scattered light and was recovered only in custom TESScut photometry (this work), where its depth depends on the reduction. The period is a single interval, with the cycle count taken from Gaia. The confirmation therefore depends partly on flagged data, and one further clean event is needed.

TYC 8355-1052-1 is a star of about 3 $R_\odot$ and 1.6 $M_\odot$ with a companion of about 0.95 $M_\odot$ on an eccentric 36.8-d orbit. The eclipses fall where Gaia's primary is in front, so they are occultations of the companion. At an impact parameter of about 0.8 the primary covers 35–39% of the companion's disc. For a companion carrying about 6.5% of the light, that reproduces the 2.3–2.5% depth. At the other conjunction the stars are 2.87 times farther apart in projection, and the four covered windows are flat. The second event improves the fit by $\Delta\chi^2 \approx 1150$ over no eclipse, and the brightest other star in the cutout is flat at that time. The 36.7755-d interval agrees with Gaia's period (36.7818 ± 0.0092 d) to 0.7σ.

A54. TYC 447-247-1 (TIC 108344775)



A54. Left: QLP SAP flux of sector 80, detrended with a 1-d running median with the eclipses masked, folded on P = 3.02355 d; the out-of-eclipse wave is a near-synchronous spot modulation. Middle: the 18.8%, 2.7-h primary eclipses (five measured; a sixth window holds partial data). Right: the seven 11.7% secondary eclipses at phase 0.5. Gaia's 39-d orbit is not shown (see text).

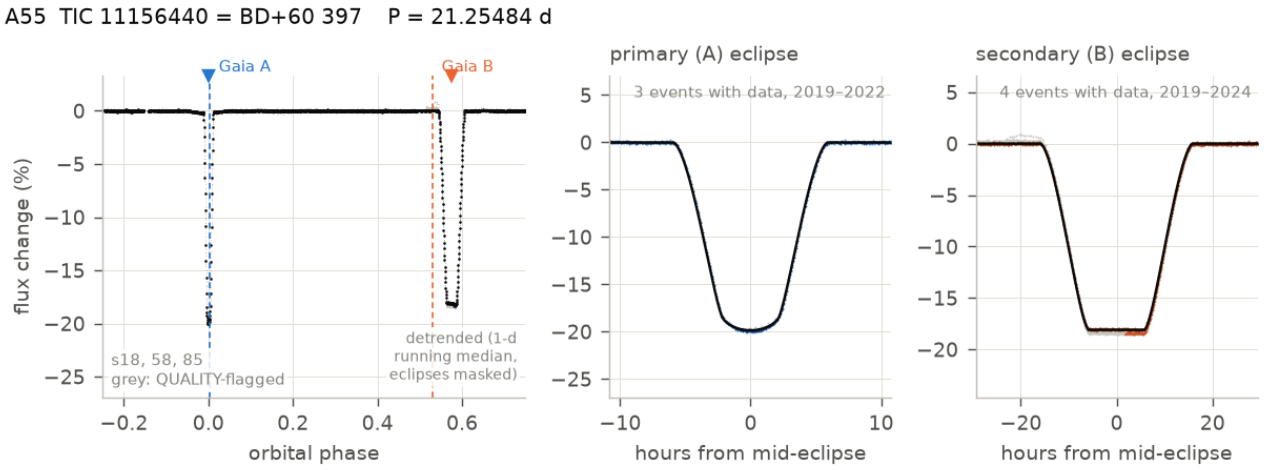
Identifiers	TIC 108344775; Gaia DR3 4279599735525131776; TYC 447-247-1
Position (J2000)	18:39:53.6 +01:43:01.4
Brightness	G 10.95; Tmag 10.29
Stars	K-dwarf pair, about 4800 K (AMBRE: T_{eff} 4837, log g 4.89, Vbroad 13.74 ± 0.29 km/s), radii about 0.8 and 0.7 $R_\odot$; 111 pc, absolute G 5.7. Gaia RUWE 10.58, non_single_star = 3, plus an Acceleration7 astrometric solution (significance 90.7)
Period	3.02348 ± 0.00001 d, from the ASAS-SN V photometry of 2015–2018 (3.023466; bootstrap 16–84%: 3.023454–3.023478) and the Harvard plates (3.023482 ± 0.000003; Section 5.6), with the epoch from sector 80. Sector 80 alone gives 3.02351–3.02358 depending on the reduction and eclipse kind. Gaia's SB2 solution gives 38.956902 ± 0.013672 d, which we show to be spurious
Epoch of primary	T_0 = BTJD 3482.0379 ± 0.0001 = BJD_{TDB} 2460482.0379
Primary eclipse	18.80% (our photometry) / 19.00% (QLP) deep, T_{14} 2.70 h, T_{23} 0.44 h; five events in s80
Secondary eclipse	11.65% / 11.83% deep, T_{14} 2.59 h; seven events in s80; at phase 0.5 within −0.3 ± 0.14 min
Orbit	Circular: $e \cos\omega \approx -1\times10^{-4}$, duration ratio gives $e \sin\omega \approx -0.02$. Gaia's 39-d SB2 ($K_1$+$K_2$ = 190 km/s) would need M $\sin^3 i$ = 14.0 and 13.5 $M_\odot$; the same amplitudes at 3.0235 d give 2.15 $M_\odot$ total against about 1.5 expected
	QLP SAP; TESS-SPOC SAP (18.2% and 11.4%, same times); our TESScut photometry

Pipelines agreeing	Difference images: uniform 17.2–18.0% (primary) and 10.0–11.0% (secondary) in every core pixel, about 0 off target; nearest bright neighbour (*G* 15.9 at 13″) would need more than 100%
Pixel attribution	
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022 (covers sectors 1–26 only), IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 eclipsing-binary table, SIMBAD (not as an EB), ExoFOP, general web search
Open issues	**Gaia orbit refuted.** The 39-d period is not a simple sampling alias of 3.02 d (tested with the epochs of three neighbouring Gaia sources; the RVS epochs themselves were not obtainable). A probable triple: an outer companion with a period much longer than Gaia's 34-month baseline. A few RVs and Gaia DR4 epoch RVs would check the 3.02-d SB2. Only one TESS sector
Next eclipses	Primary: $T_0 + n \times 3.02348$, e.g. BTJD 4328.612 (2026-10-15 02:41 UTC; n = 280, ±0.003)
Discovery	Found incidentally in the conjunction-window search, not at a predicted conjunction; confirmed by independent re-analysis (Section 2.5)

Note: the eclipsing binary is confirmed, and the Gaia orbit that brought it into the sample is refuted. The 3.02-d period differs from Gaia's, so the system was found incidentally rather than at the Gaia period.

TYC 447-247-1 is a detached pair of K dwarfs on a circular 3.02-d orbit, observed in a single TESS sector with twelve eclipses. Three independent facts tie the stars to the short period. The first is the eclipse signal itself. The second is a spot wave of about 2.9 d, from near-synchronous rotation. The third is a spectroscopic line broadening of 13.7 km/s, which equals the rotation speed of a 0.8 $R_\odot$ star at 3.02 d; at 39 d it would be about 1 km/s. Gaia's double-lined solution at 39 d would make these K dwarfs 14 $M_\odot$ stars. At 3.02 d the same velocity amplitudes agree with expectation (about 13% high). The Gaia SB2 solution is therefore spurious: its period is misidentified, although its velocity scale is approximately correct. Gaia's large RUWE and acceleration solution point to a wide third star, though not at a 39-d period.

A55. BD+60 397 (TIC 11156440)



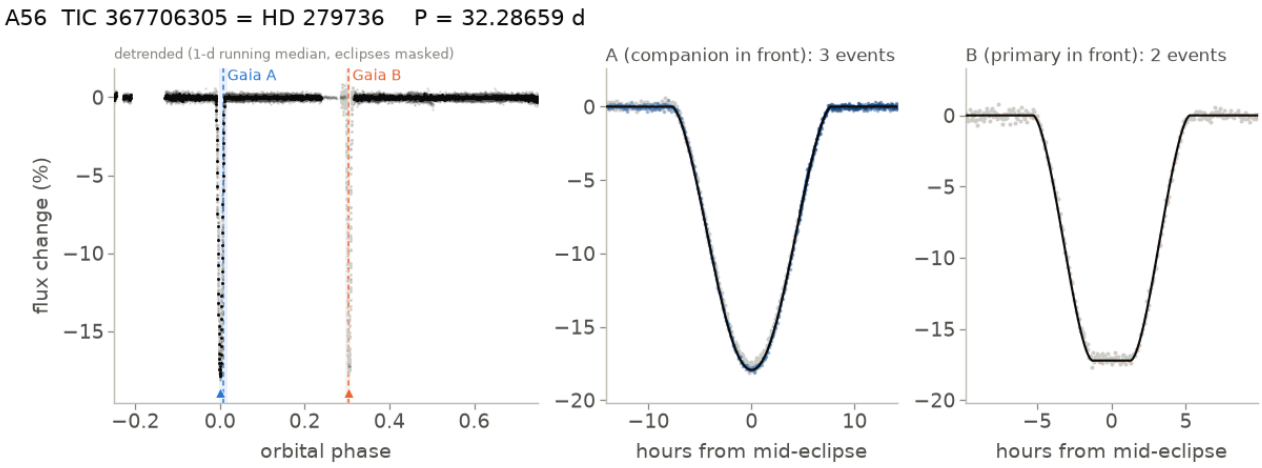
A55. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.25484 d, Gaia's predicted conjunctions dashed. Middle: the 19.5%, 11-h central transits at A. Right: the 18.1%, 29-h flat total occultations at B; the 2.7:1 duration ratio comes from the eccentric orbit.

Identifiers	TIC 11156440; Gaia DR3 508026396981317504; BD+60 397
Position (J2000)	01:58:12.9 +60:43:12.8
Brightness	*G* 10.00; Tmag 9.53
Stars	A-type subgiant, $M_1 \approx 2.6\ M_\odot$, R_1 = 4.1–4.5 $R_\odot$ (4.32 from the split light at 545 pc) + early-F/A companion, M_2 = 1.5–1.8 $M_\odot$, R_2 = 1.9–2.1 $R_\odot$, at nearly the same temperature (L_2/L_1 = 0.221); mass function 0.256 $M_\odot$; SIMBAD A7; Gaia SB1, RUWE 2.93
Period	21.25484 ± 0.00001 d (Gaia 21.257446 ± 0.002133 d; –1.2σ)
Epoch of primary	A: BTJD 1801.3737 ± 0.0005 = BJD_{TDB} 2458801.3737

Primary eclipse (A)	19.23–19.60% deep, T_{14} 10.78–10.84 h, T_{23} 3.95 h, rounded bottom: central transit of the companion (k = 0.466); four events (s18, s58 ×2; s85 flagged, timing only)
Secondary eclipse (B)	18.03–18.13% deep, T_{14} 29.37–29.42 h, T_{23} 12.73–12.75 h, flat: central total occultation of the companion (its light share); four events (s18 start, new; 1813.60; 2897.59; 3620.26). O−C ≤ 2.9 min
Orbit	Duration ratio 2.7 = (1+e sin ω)/(1−e sin ω) with e sin ω = 0.458, against Gaia +0.457 ± 0.039 (0.0σ). B−A phase 0.57510 against 0.527 ± 0.017: e cos ω = +0.105 against Gaia +0.037 ± 0.023 (**+2.9σ**); ω 77° against Gaia 85.3 ± 2.9°. i ≈ 90°, both impact parameters about 0
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s58 A ×2, s58 B, s85 B; t_0 within 0.5 min)
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target's Gaia position; no neighbour within 60″ brighter than G 14.8 (would need more than 100%). Common mode clean
Catalogue status	Not listed in: VSX, ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS, KELT (target list only), WISE variables, Justesen & Albrecht 2021, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, CCDM, Hipparcos, ExoFOP, general web search. The 25″ G 16.0 neighbour's Kostov table 2 flag is almost certainly due to contamination of its aperture by the target's eclipses
Open issues	e cos ω is 2.9σ from Gaia; a bias in the Gaia SB1 solution is plausible because the companion's lines, at nearly the same temperature, blend. A few spectra for K_2 and ω
Next eclipses	A: BTJD 4330.701 (2026-10-17 04:43 UTC), 4351.956; B: BTJD 4342.925 (2026-10-29 10:05 UTC), 4364.179 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+60 397 is an A-type subgiant of about 4.3 $R_\odot$ with an early-F or A-type companion of about 2 $R_\odot$ at nearly the same temperature, seen edge-on in an eccentric 21.25-d orbit. Both eclipses are central. At A the companion crosses the primary's disc (19.5%). At B it is hidden behind the primary for 12.7 h of a 29-h eclipse (18.1%, its share of the light). The occultation lasts 2.7 times as long as the transit because the stars are 0.53a apart at A and 1.44a at B, and this duration ratio reproduces Gaia's e sin ω. The only significant disagreement with Gaia is in e cos ω, and it does not call the eclipses themselves into question.

A56. HD 279736 (TIC 367706305)



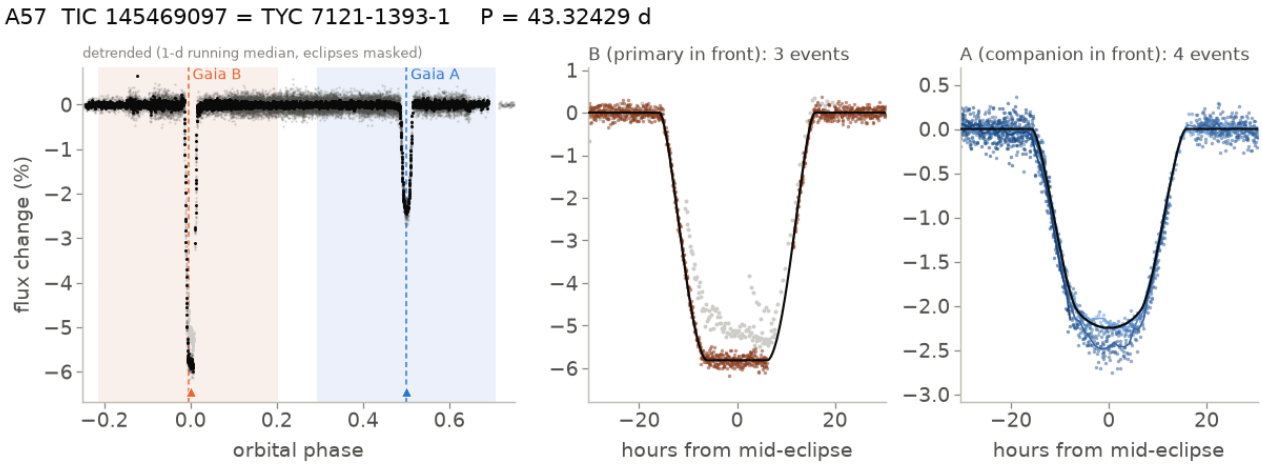
A56. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 32.28659 d, Gaia's predicted conjunctions dashed. Middle: the 17.5%, 14-h partial transits at A. Right: the 17.2%, 9.8-h total occultations at B.

Identifiers	TIC 367706305; Gaia DR3 1774751310738398 72; HD 279736
Position (J2000)	04:21:26.9 +36:36:25.7
Brightness	G 10.86; Tmag 10.35
Stars	

	G0 star (SIMBAD; APOGEE T_{eff} 6322 K, log *g* 4.40, [Fe/H] −0.59), M_1 = 1.0–1.3 $M_\odot$, R_1 = 2.05–2.20 $R_\odot$ + K companion, M_2 = 0.8–0.9 $M_\odot$, R_2 ≈ 1.0 $R_\odot$, L_2/L_1 = 0.21; mass function 0.160 $M_\odot$; Gaia SB1
Period	32.28659 ± 0.00001 d (Gaia 32.290060 ± 0.013184 d; −0.3σ)
Epoch of primary	A: BTJD 2914.8973 ± 0.0005 = BJD_{TDB} 2459914.8973
Primary eclipse (A)	17.53% deep, T_{14} 14.17 h, T_{23} 2.85 h, V-like: partial transit (b/R_1 = 0.61 > 1−*k*); three events (s19 flagged, s59, s86 flagged)
Secondary eclipse (B)	17.24% deep, T_{14} 9.82 h, T_{23} 3.26 h, flat: total occultation (b/R_1 = 0.40); two events (s19; s59 flagged). O−C ≤ 0.5 min
Orbit	B−A phase 0.30391 (Gaia 0.2954 ± 0.0073): *e* cos ω = −0.306 against Gaia −0.320 ± 0.012 (+1.2σ); *e* sin ω = −0.217 (fit) / −0.183 (durations) against −0.206 ± 0.012; Gaia *e* 0.3811. *i* = 88.74°, *k* = 0.474
Pipelines agreeing	QLP SAP; TESS-SPOC (s19 A/B, s59 A); TARS (s59 A/B)
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target; no neighbour within 60″ brighter than *G* 14.8. Common mode clean
Catalogue status	Not listed in: VSX (the ROT entry at 40″ is the *G* 14.8 neighbour Gaia DR3 177475062354364544), ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS (a row at 28″ is a different star), KELT, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, Gaia DR3 variability tables, SB9, WDS, Hipparcos, ExoFOP, general web search
Open issues	APOGEE's single visit (RV 77.42 km/s) is about 10 km/s short of the predicted ~88 km/s; this is not decisive given the uncertain γ and line blending
Next eclipses	A: BTJD 4335.507 (2026-10-22 00:02 UTC), 4367.794; B: BTJD 4345.319 (2026-10-31 19:31 UTC), 4377.606 (±1–2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

HD 279736 is a metal-poor G0 star of about 2.1 $R_\odot$ with a companion of about 0.85 $M_\odot$ and 1 $R_\odot$ on an eccentric 32.3-d orbit. The two eclipses are nearly equal in depth but different in kind: at A the companion crosses only part of the primary's disc (a partial, V-like transit), and at B the primary hides the companion completely (a flat total occultation). The eclipses give *e*, ω and *P* all within 1.2σ of Gaia's independent orbit.

A57. TYC 7121-1393-1 (TIC 145469097)



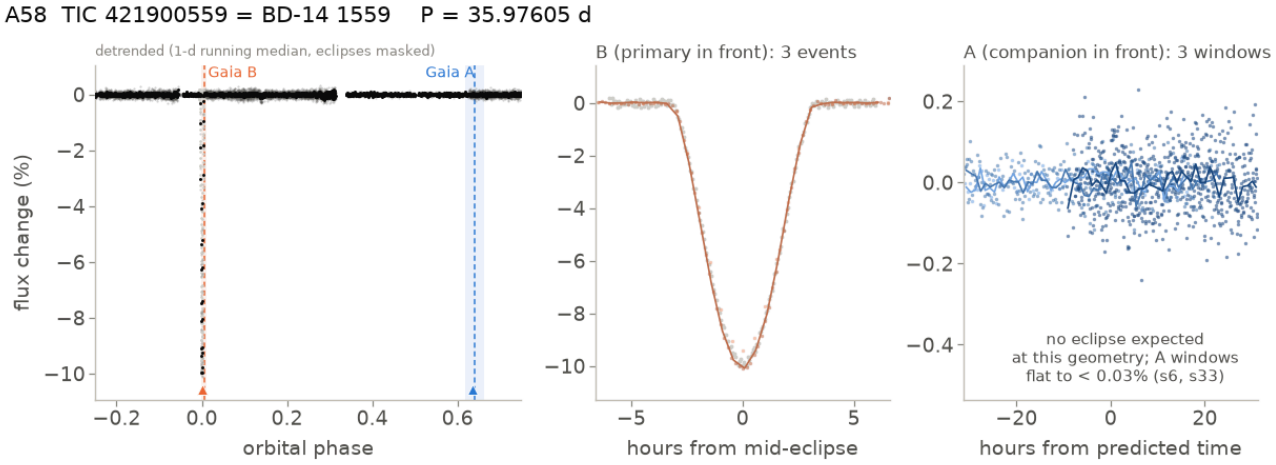
A57. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 43.32429 d, Gaia's predicted conjunctions dashed; the orbit is circular. Middle: the 5.8%, 30-h flat total occultation at B in sector 88, with flagged partials from sectors 8 and 35. Right: the 2.2%, 30-h annular transits of the F dwarf across the giant at A.

Identifiers	TIC 145469097; Gaia DR3 5548179074417798400; TYC 7121-1393-1
Position (J2000)	08:11:16.4 −31:37:04.1
Brightness	*G* 10.62; Tmag 10.02
Stars	

	K giant, M_1 = 1.4–2.6 $M_\odot$, R_1 = 7.7–9.1 $R_\odot$ (TIC 8.0) + hotter F dwarf (about 6800 K), M_2 = 1.0–1.5 $M_\odot$, R_2 = 1.2–1.4 $R_\odot$, L_2/L_1 = 0.062; mass function 0.186 $M_\odot$; Gaia SB1
Period	43.32429 ± 0.00007 d (Gaia 43.336415 ± 0.007917 d; −1.5σ)
Epoch of eclipse	B: BTJD 3696.526 ± 0.001 = BJD_{TDB} 2460696.526; A: BTJD 1508.648 ± 0.002
Deeper eclipse (B)	5.78% deep, T_{14} 30.4 h, T_{23} 15.4 h, flat: total occultation of the F dwarf; one full event (s88, egress across a data gap) plus partials in flagged cadences (s8 egress, about 4.4%; s35 ingress, about 4.7%), all at phase 0.5000
Shallower eclipse (A)	2.15–2.40% deep (s61's 2.40% includes a known common-mode systematic), T_{14} 30.1–30.8 h: annular transit across the giant (b/R_1 = 0.76 < 1–k); three events (s7, s34, s61) plus an ingress at the end of s88, predicted before inspection of the data (A = 3718.187) and matched to 0.07% in every bin. O–C ≤ 1.3 min
Orbit	Circular: B–A phase 0.50004 ± 0.00009 (Gaia 0.4966 ± 0.0039), e cos ω = 0.000 (Gaia −0.006 ± 0.006), equal 30-h durations (e sin ω +0.011/0.000 against +0.002 ± 0.007). i = 85.2°, k = 0.155
Pipelines agreeing	QLP SAP; eleanor-lite, T16, TGLC aperture (s7 A); TESS-SPOC (s61 A). No SPOC-family light curve for s88, so the full B has no independent check
Pixel attribution	TESScut difference images: flux-loss centroid 0.1–0.3 px from the target; no neighbour within 60″ brighter than G 14.8. Same-camera stars flat during the s88 ingress (median −0.04%)
Catalogue status	Not listed in: VSX, ASAS-SN, GCVS, ASAS-3, ZTF, ATLAS, KELT, WISE variables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, Hipparcos, ExoFOP, general web search
Open issues	The one full B has no second-pipeline check; ground photometry of the next B would cover the egress, which fell in the sector-88 data gap
Next eclipses	B (deeper): BTJD 4346.391 (2026-11-01 21:22 UTC), 4389.715; A: BTJD 4368.051, 4411.375 (±7–9 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 7121-1393-1 is a K giant of about 8 $R_\odot$ with a hotter F-dwarf companion of about 1.3 $R_\odot$ on a circular 43.3-d orbit. Both eclipses last 30 h. At A the dwarf crosses the giant's face entirely, an annular transit 2.2% deep. At B the giant hides the dwarf completely. This occultation is deeper than the transit, at 5.8%, because the dwarf is about 2.5 times brighter per unit area in the TESS band. The ingress of an A eclipse at the end of sector 88 was predicted before those data were examined, and it is present. Two further partial B eclipses lie in flagged cadences at phase 0.5000.

A58. BD-14 1559 (TIC 421900559)

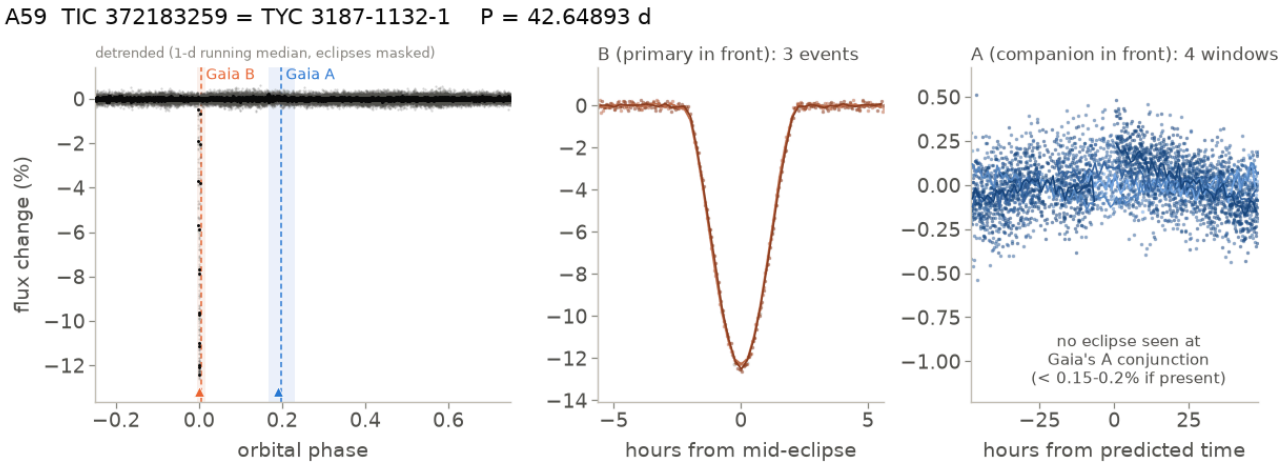


A58. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 35.97605 d, Gaia's predicted conjunctions dashed. Middle: the three 9.8%, 6.0-h partial occultations at Gaia's B conjunction (sector 87 in QLP-flagged cadences). Right: the covered A windows (three with usable data of the four the entry lists), flat as the geometry predicts.

Identifiers	TIC 421900559; Gaia DR3 2947549198126297984; BD-14 1559
Position (J2000)	06:43:55.3 –15:07:36.5
Brightness	*G* 10.03; Tmag 9.56
Stars	G star (GSP-Phot 5830 K; Gaia FLAME R 2.13 $R_\odot$, M 1.25 $M_\odot$); the eclipse fit gives R_1 = 1.56 $R_\odot$, R_2 = 0.83 $R_\odot$, companion 12.1% of the light; companion $m_2(i = 90°) \approx 0.76\ M_\odot$, a K dwarf; mass function 0.108 $M_\odot$; Gaia SB1, K_1 34.1 km/s, RUWE 1.21
Period	35.97605 ± 0.00002 d (Gaia 35.9893 ± 0.0083 d; –1.6σ)
Epoch of eclipse	T_0 = BTJD 2203.38951 = BJD_{TDB} 2459203.38951
Eclipse (B, near periastron)	9.2–10.1% deep, T_{14} 5.99–6.21 h, T_{12} 2.38–2.39 h: partial occultation of the companion, the primary in front (Gaia's B conjunction) (i = 88.1°); three events (s006, s033, s087 with all but 7 of 260 cadences flagged); O–C ≤ 0.5 min. Gaia's B is ν = 27°, about 1 d after periastron
Other eclipse (A)	None possible (sky separation 2.5–2.8 $R_\odot > R_1 + R_2$) and none seen: s006 ≤ 290 ppm, s033 ≤ 190 ppm, s087 about 800 ppm at a scattered-light edge, s108 flat
Orbit	Gaia e 0.438 ± 0.058, ω 243.1 ± 2.6°. With Gaia's period the eclipses sit at B at –1.7σ; carrying Gaia's periastron forward with the eclipse period puts them –0.18 d (–0.77σ) from B and 16σ from A. One eclipse kind only, so e cos ω and e sin ω cannot be derived from timing
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s006, s033); TARS (s006; s087 at 10.09%)
Pixel attribution	TESScut s87 difference image: centroid 0.3″ from the target, 3×3 aperture loses 9.5%; brightest neighbour within 60″ is *G* 17.15 at 34.5″ (would need about 70×)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia NSS reference only), general web search
Open issues	Radius conflict: the fitted R_1 runs 15–25% below Gaia's (about 1.8 $R_\odot$ after the companion's light); with R_1 fixed at 2.13, χ^2 rises from 774 to 1063. An SB2 spectrum would fix which star is in front and R_1
Next eclipses	B: BTJD 4361.952, 4397.928 (±0.001); A (no eclipse expected) 4384.8 ± 0.9
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD-14 1559 is a G star with a K-dwarf companion of about 0.76 $M_\odot$ on an eccentric 36-d orbit. Three identically shaped 9.8%, 6-h eclipses over seven years fall at the conjunction just after periastron. At the far conjunction the stars cannot overlap, and the four covered windows there are flat. The apparent –1.8σ offset from Gaia's prediction is Gaia's period error accumulated over about 50 cycles. Carried forward with the eclipse period, Gaia's periastron puts the eclipses within 0.8σ of B. The sector-87 eclipse sits almost entirely in QLP-flagged cadences and is recovered in TARS and in the pixels. A model with the companion in front would need it to carry 30% of the light, which is implausible.

A59. TYC 3187-1132-1 (TIC 372183259)



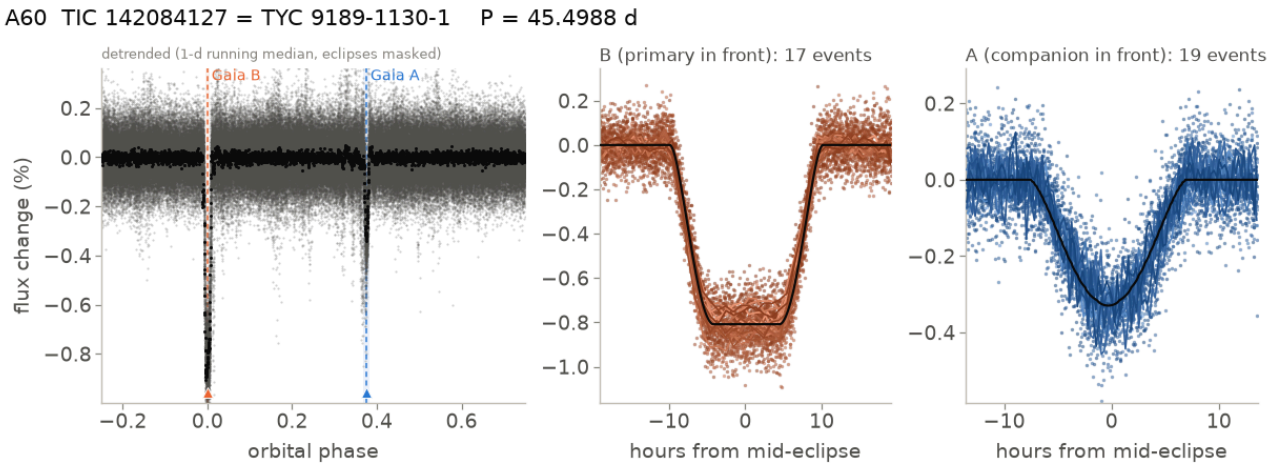
A59. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 42.64893 d, Gaia's predicted conjunctions dashed. Middle: the three 12.3%, 4.05-h partial occultations at Gaia's B conjunction near periastron (sector 16 in QLP-flagged cadences). Right: the A windows, flat to about 0.2%.

Identifiers	TIC 372183259; Gaia DR3 1966468304243902848; TYC 3187-1132-1
Position (J2000)	21:38:30.7 +40:58:26.7
Brightness	G 10.78; Tmag 10.39
Stars	Primary of TIC 6213 K, R 1.65 $R_\odot$, M 1.20 $M_\odot$ at 307 pc (Gaia about 1.64 $R_\odot$) + companion $m_2(i = 90°) \approx 0.69\ M_\odot$; the eclipse fit gives $R_1 = 1.14\ R_\odot$, $R_2 = 0.73\ R_\odot$, companion 12.8% of the light; mass function 0.0914 $M_\odot$; Gaia SB1, K_1 37.1 km/s
Period	42.64893 ± 0.00006 d (Gaia 42.5594 ± 0.0391 d; +2.3σ)
Epoch of eclipse	T_0 = BTJD 3537.57169 = BJD_{TDB} 2460537.57169
Eclipse (B, near periastron)	11.9–12.7% deep, T_{14} 4.03–4.09 h, T_{12} 1.64–1.69 h (b ≈ 0.7): partial occultation of the companion, the primary in front (Gaia's B conjunction) (i = 88.6°); three events (s016, new, every in-eclipse QLP cadence flagged; s082; s083); O−C ≤ 1.1 min. Gaia's B is ν = 319°, 0.75 d before periastron
Other eclipse (A)	Not seen: four A windows (s076, s016, s082, s015) limit any A eclipse to ≲ 0.15–0.2%, which rules out the smallest-R_1 solutions
Orbit	Gaia e 0.672 ± 0.033, ω 310.7 ± 5.4°. With Gaia's period the eclipse lands at A (−0.6σ); with the eclipse period it sits at **B**, −0.35 d (−0.83σ), and A is excluded at 6.4σ. The durations agree: at A (about 50 km/s) a 12% eclipse would last 14–18 h; at B (about 154 km/s) about 6 h central. One eclipse kind only
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP and TARS (s082, s083); TESS-SPOC SAP leaves the s016 eclipse unflagged
Pixel attribution	TESScut difference images put all three on the target (centroid offsets 1.3″, 3.1″, 1.0″; 3×3 losses 9.4–9.9%); brightest neighbour within 60″ is G 16.09 at 19.6″ (would need about 16×)
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia NSS reference only), general web search
Open issues	Unresolved radius conflict: the eclipse shape and duration require $R_1 + R_2 \approx 1.9\ R_\odot$ against about 2.2–2.3 expected (fitted R_1 1.1–1.3 against 1.5–1.65; fixing R_1 at 1.53–1.65 raises χ^2 by 460–1080). Possible causes: Gaia's e or ω off, third light, a non-main-sequence companion, the simple disc model. Spectroscopy and an independent RV orbit would resolve it
Next eclipses	B: BTJD 4347.901, 4390.550 (±0.001); A (≲ 0.2% if present) 4356.0 ± 1.3
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3187-1132-1 is a star of about 1.6 $R_\odot$ with a companion of about 0.7 $M_\odot$ on a highly eccentric ($e \approx 0.67$) 42.6-d orbit. Of its three eclipses, the one in sector 16 lies entirely in QLP-flagged cadences. The period is 42.64893 d (a single-cycle interval gives 42.647). Gaia's period is 2.3σ short. Over 74

cycles that moves the predicted conjunction by about 6 d and would put the eclipses at A. With the eclipse period they fall at B, just before periastron, where the stars move fast enough to make a 4-h eclipse. The eclipses are not in doubt, but the stellar radii they imply are smaller than expected, and this discrepancy remains unresolved.

A60. TYC 9189-1130-1 (TIC 142084127)



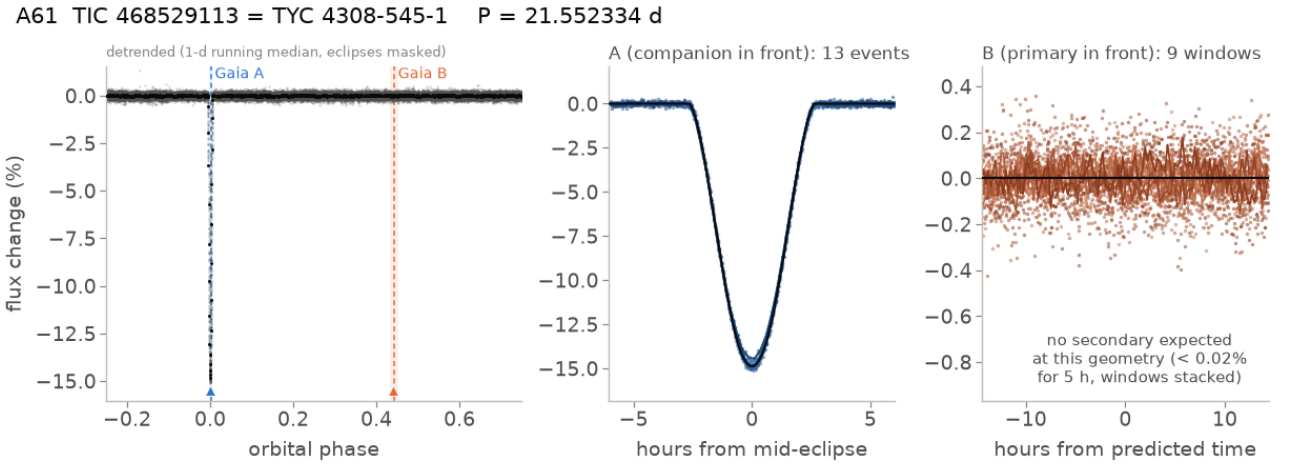
A60. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 45.4988 d, Gaia's predicted conjunctions dashed. Middle: the seventeen 0.8%, 19-h flat total occultations at B. Right: the nineteen 0.3%, 13.6-h V-shaped grazing transits of the K dwarf across the giant at A.

Identifiers	TIC 142084127; Gaia DR3 5261799966596528640; TYC 9189-1130-1
Position (J2000)	06:32:46.2 −74:36:28.9 (southern TESS continuous viewing zone)
Brightness	*G* 10.67; Tmag 10.02
Stars	Oscillating red giant, about 4800 K, R_1 = 5.7–6.3 $R_\odot$ (Gaia L 15.2 $L_\odot$ implies about 5.7) + K dwarf, m_2 = 0.7–0.9 $M_\odot$, R_2 = 0.67–0.78 $R_\odot$, T_2 ≈ 4200 K, 0.81% of the light; mass function 0.0938 $M_\odot$; Gaia SB1. Seismology: Beck et al. 2024 ν_{max} 67, Δν 4.7; Mackereth et al. 2021 pipelines disagree (BHM ν_{max} 125.8, R 7.5; A2Z ν_{max} 7.8)
Period	B: 45.498805 ± 0.000043 d; A: 45.498766 ± 0.000086 d (they agree; no apsidal motion detected). Gaia 45.50477 ± 0.0073 d (−0.8σ)
Epoch of eclipse	B: BTJD 2553.26450 ± 0.00083 = BJD_{TDB} 2459553.26450; A: BTJD 2615.83808 ± 0.00145
Deeper eclipse (B)	0.72–0.86% deep, T_{14} 19.1 h, T_{23} 10.7 h, flat: total occultation of the K dwarf; seventeen events; O−C rms about 5 min
Shallower eclipse (A)	0.28–0.35% deep, T_{14} 13.6 h, T_{23} 3.5 h, V-shaped: partial (grazing) transit of the K dwarf across the giant; nineteen events
Orbit	A–B separation 17.0748 d against Gaia 17.0746 ± 0.22 (0.00σ); e cos ω +0.1957 ± 0.0001 against Gaia +0.1958 ± 0.008. Physical fit to all 36 events: e = 0.2261, ω = 329.7° (Gaia 0.2308 ± 0.008, 328.07 ± 1.74°), e sin ω = −0.114 (Gaia −0.122 ± 0.007), i = 85.30°. The duration ratio cannot give e sin ω here because A is grazing and B total
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s066, both kinds); TGLC PSF flux (s004, same times, diluted)
Pixel attribution	TESS-SPOC s066 TPF difference images: B centroid 1.8″ from the target (S/N 60), A 7.3″ (0.35 px, S/N 8); nearest star brighter than *G* 15 is 117″ away; brightest neighbour within 60″ (*G* 16.75 at 57″) would need 215%
Catalogue status	No eclipse catalogued: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, general web search. Beck et al. 2024 lists the SB1 and its seismology with the inclination blank and no eclipse
Open issues	An RV of the K dwarf, or a seismic mass, would give the red giant a well-measured mass
Next eclipses	B: BTJD 4327.718 (2026-10-14 05:13 UTC), 4373.217 (±3 min); A: BTJD 4344.791, 4390.290 (±5 min)
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 9189-1130-1 is an oscillating red giant of about 6 $R_\odot$ with a K-dwarf companion on a 45.5-d orbit. It lies in the southern continuous viewing zone, so 36 eclipses across 2018–2025 could be measured. The giant hides the dwarf completely at B (0.8%, its share of the light) and the dwarf grazes the giant's limb at A (0.3%). A physical model fitted to all 36 events reproduces Gaia's eccentricity and periastron angle. The star is already in the seismology literature as a binary giant, where pipelines disagree on its oscillation frequencies and so on its radius. For an assumed mass of 1.2–2.0 $M_\odot$, the eclipses fix R_1 at 5.7–6.3 $R_\odot$. That radius matches the Gaia luminosity and argues against the 7.5 $R_\odot$ solution.

A61. TYC 4308-545-1 (TIC 468529113)



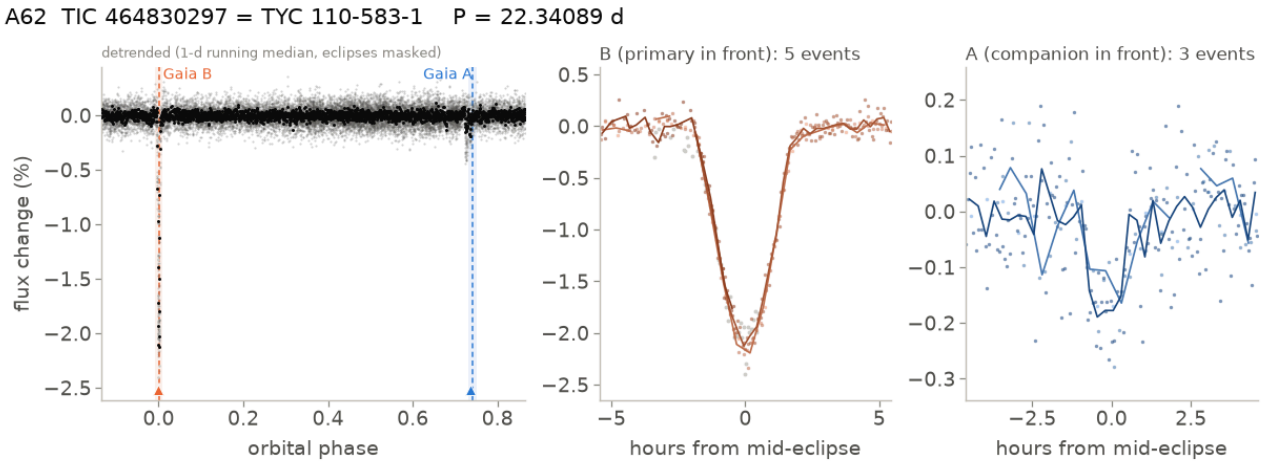
A61. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 21.552334 d, Gaia's predicted conjunctions dashed. Middle: the thirteen 14.6%, 4.9-h transits at A. Right: the covered B windows (nine drawn; the entry's limits use ten), flat as the geometry predicts.

Identifiers	TIC 468529113; Gaia DR3 538862715972209920; TYC 4308-545-1
Position (J2000)	00:58:32.4 +74:58:03.2
Brightness	G 10.40; Tmag 9.98
Stars	F star, $R_1 \approx 1.9\ R_\odot$ ($M_1 = 1.4\ M_\odot$ assumed) + K dwarf, m_2 0.84 $M_\odot$, $R_2 \approx 0.79\ R_\odot$ (k 0.41); a 42.6 $R_\odot$; mass function 0.1177 $M_\odot$; Gaia SB1
Period	21.552334 ± 0.000007 d (Gaia 21.55579 ± 0.0036 d; −0.95σ)
Epoch of primary	T_0 = BTJD 2481.89522 ± 0.00022 = BJD_{TDB} 2459481.89522
Eclipse (A)	14.2–15.0% deep, T_{14} 4.9 h, T_{23} 1.0 h: transit of the K dwarf; thirteen events s018–s086 (s079 partial); χ^2/dof 1.1/11, all O−C under 1 min
Other eclipse (B)	None possible: sky separation at B is 1.43 (R_1+R_2) (1.37–1.47 in every refit); a B eclipse would need i > 87.5°. Ten covered windows give stacked 3σ limits of 0.03% (3 h), 0.02% (5 h), 0.013% (8 h); a total occultation, had it happened, would be about 7%
Orbit	Fit with Gaia's e = 0.497, ω = 99.4° fixed: i = 86.5°; r_A = 0.506a, r_B = 1.477a. Observed minus Gaia −0.25 to −0.55 d, a constant −1.0σ, all from the period
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (s059, depth 14.4% after crowding correction); TARS (same times)
Pixel attribution	Difference-image centroid 2.8–3.5″ from the target and 49″ from the only neighbour bright enough to produce the dip (G 11.59 at 45.3″, which would need a 45% dip), at S/N 200–300
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, VizieR all-catalogue cone, SIMBAD ([SB*] only), general web search
Open issues	No photometric question remains open; spectroscopic detection of the companion
Next eclipses	A: BTJD 4335.396 (2026-10-21 21:25 UTC), 4356.948 (2026-11-12 10:44 UTC), 4378.501 (2026-12-04 00:00 UTC) (±1 min). No B expected
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4308-545-1 is an F star of about 1.9 $R_\odot$ with a K-dwarf companion of about 0.84 $M_\odot$ on an eccentric ($e \approx 0.50$) 21.55-d orbit. All thirteen transits from 2019 to 2025 are measured, with every O–C under a minute. The other conjunction falls near apastron, where the stars pass 1.43 times their summed radii apart on the sky, so no occultation is expected. Ten covered windows show none, to a limit of 0.02%; a total occultation of a K dwarf would have been about 7% deep.

A62. TYC 110-583-1 (TIC 464830297)



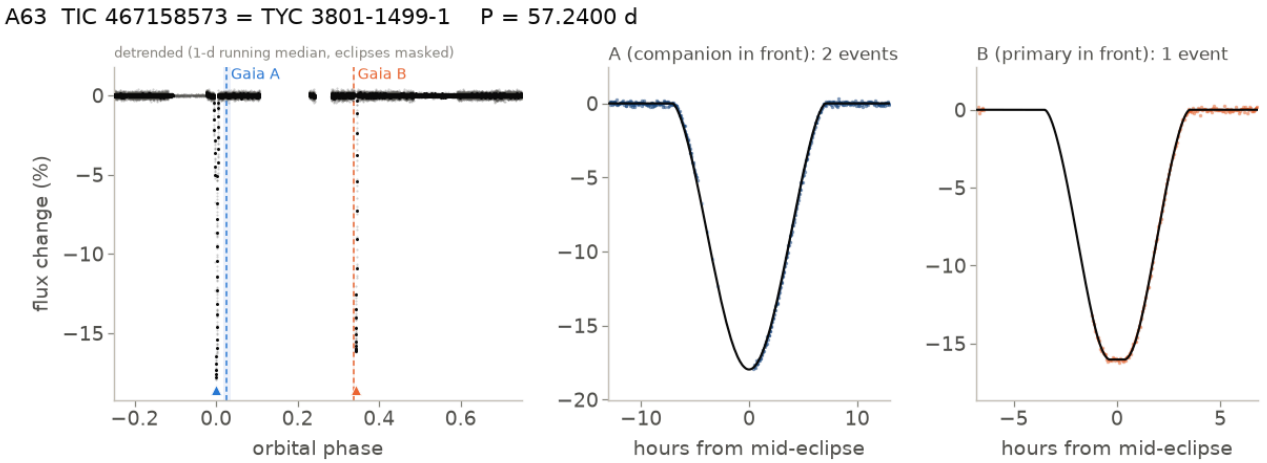
A62. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 22.34089 d, Gaia's predicted conjunctions dashed. Middle: the five 2.1%, 3.7-h grazing occultations at B. Right: the 0.15%, 2-h graze at A in the three covered windows.

Identifiers	TIC 464830297; Gaia DR3 3240092992751580416; TYC 110-583-1
Position (J2000)	05:00:47.7 +05:55:39.4
Brightness	G 10.76; Tmag 10.35
Stars	Primary (TIC radius) + companion of minimum mass 0.78 $M_\odot$; the depths need $k \approx 0.45$, $R_2 \approx 0.61\ R_\odot$, about 4650 K; mass function 0.1255 $M_\odot$; Gaia SB1
Period	22.34089 ± 0.00001 d (Gaia 22.329766 ± 0.005097 d; +2.2σ)
Epoch of eclipse	B: T_0 = BTJD 1440.0266 ± 0.0004 = BJD_{TDB} 2458440.0266
Deeper eclipse (B)	1.99–2.19% deep, near-V, 3.7 h: grazing occultation of the companion (primary in front at B); four full events (1440.03; 1462.37; 2177.28; 4009.23) and a partial near 4031.57 in flagged data, on the ephemeris; O–C under 1 min
Shallower eclipse (A)	About 0.15% deep, about 2 h: grazing transit ($b_A = 1.27\ b_B$); detected in all three covered A windows at the same offset: 1456.452 (1450 ± 216 ppm), 2193.701 (1606 ± 204), 4025.654 (1582 ± 184), where the largest dip elsewhere in those sectors is 839–1123 ppm
Orbit	A–B = 16.425 d gives e cos ω = −0.376 against Gaia −0.386 ± 0.015 (+0.7σ); Gaia e 0.4042, ω 197.15°. Propagated back 47 cycles, the B ephemeris lands −0.07σ from Gaia's B conjunction at Gaia's epoch
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS; eleanor-lite; TGLC aperture and PSF (s5 eclipses at 1.9–2.35%)
Pixel attribution	Difference-image centroids within 0.03–0.05 px of the out-of-eclipse centroid; the G 16.80 star at 10.5″ would need 434%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	A mild tension: a main-sequence 0.75 $R_\odot$ companion would eliminate the A eclipse, so the companion is slightly small or the primary radius is in error (FFI smearing not modelled)
Next eclipses	B: BTJD 4344.342, 4366.683 (±0.001); A: BTJD 4338.426, 4360.767 (±0.005)
Discovery	

	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 110-583-1 has a companion of at least 0.78 $M_\odot$ on an eccentric 22.34-d orbit, and the two stars only graze at both conjunctions. The 2% eclipses fall where Gaia's primary is in front (occultations of the companion). A 0.15% graze appears at the other conjunction in all three covered windows. Each is at the same offset, and each is larger than any other dip in its sector. Together with the main eclipses, the graze gives e cos ω within 0.7σ of Gaia. A further full occultation is present in 2018.

A63. TYC 3801-1499-1 (TIC 467158573)



A63. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 57.2400 d, Gaia's predicted conjunctions dashed. Middle: the 18%, 13-h V-shaped transit at A in sector 60, with the ingress at the end of sector 20. Right: the 16%, 6.4-h flat occultation at B in sector 74.

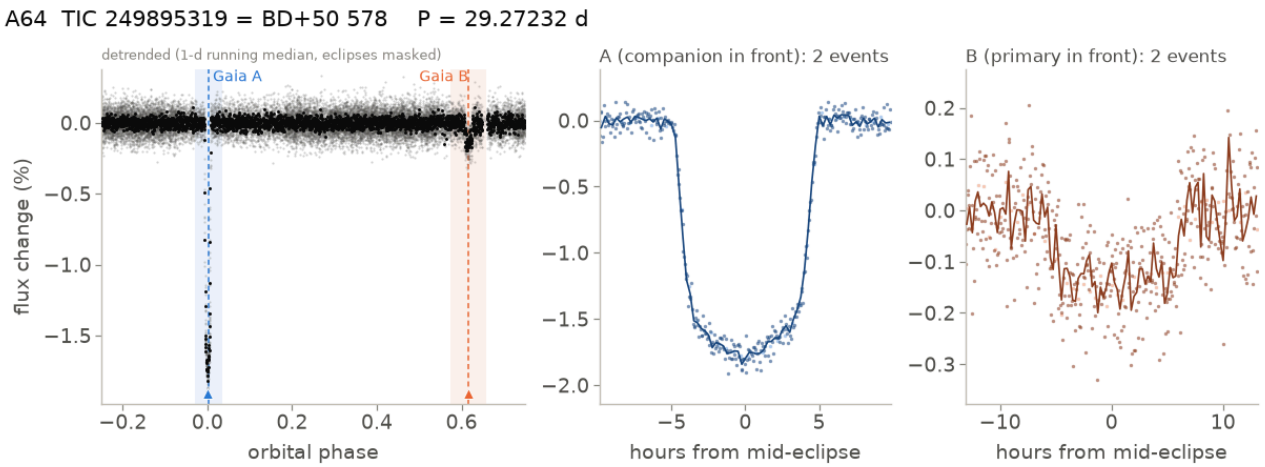
Identifiers	TIC 467158573; Gaia DR3 1037239725881771008; TYC 3801-1499-1
Position (J2000)	08:47:33.6 +56:45:12.2
Brightness	G 10.95; Tmag 10.50
Stars	Primary $R_1 \approx 1.22\ R_\odot$ + K2–3 dwarf of about 0.75 $M_\odot$ (mass-function minimum 0.69), $R_2 \approx 0.73\ R_\odot$, about 4900 K, 19% of the light; Gaia SB1
Period	57.2400 ± 0.0003 d over 19 cycles (Gaia 57.249210 ± 0.016133 d; −0.57σ)
Epoch of primary	A ingress fitted with the s60 template: BTJD 1868.8865 ± 0.005 = BJD_{TDB} 2458868.8865; s60 A centre about 2956.45, the midpoint of ingress and egress (the minimum falls in a data gap)
Primary eclipse (A)	≥ 17.4% observed (18.0% modelled), V-shaped, $T_{14} \approx$ 12.7–13 h: transit; s60 (ingress 2956.19, egress ends 2956.72) and an ingress at the end of s20 (QUALITY 0, identical in TESS-SPOC SAP and PDCSAP)
Secondary eclipse (B)	16.0% deep, flat bottom, $T_{14} \approx$ 6.4 h assuming symmetry: occultation of the companion; one event (s74, BTJD 3319.528, ingress in a gap)
Orbit	One geometry fits both kinds: e 0.457, ω 240.0°, i = 89.40°, k = 0.60; Gaia e 0.497 ± 0.012 (−3.3σ), ω 236.6 ± 1.3° (+2.6σ); e cos ω −0.229 against −0.274 (about 3.8σ); e sin ω −0.396 against −0.415 (about 1.5σ). Fixing e, ω at Gaia's values puts B 1.8 d early. Gaia goodness-of-fit 2.84
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (16.9–18.4% in s60; 15.3–16.0% in s74)
Pixel attribution	Difference-image centroid offsets 0.04 px (s60), 0.05 px (s20), 0.17 px (s74, post-gap settling); no neighbour could produce 16%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	

	Gaia orbit tension: e cos ω about 3.8σ from Gaia's, whose errors may be underestimated. Three eclipses in all; no other conjunction in covered data (s20, s47, s60, s74). **Not seen on the Harvard plates** (Section 5.6): the nine plates inside the A window, 1893–1989, give 0.066 ± 0.036 mag against the 0.21 mag the TESS depth predicts, 4σ short, with injected eclipses recovered 10 times in 10 and no signal within ±40 period errors; the eclipse was shallower or absent before TESS, or the period is off by more than 0.012 d
Next eclipses	A: BTJD 4330.207, 4387.447 (±0.008); B: BTJD 4349.848, 4407.088 (±0.007)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)
Status	Confirmed, with caveats. No eclipse is completely observed (the s60 A minimum falls in a gap, s20 is an ingress only, the s74 B ingress is in a gap); one B event; the cycle count of 19 between the s20 ingress and the s60 transit is fixed by Gaia's period (18 or 20 cycles would give 60.42 or 54.38 d); three pipelines agree on what is seen; e cos ω is 3.8σ from Gaia's.

Note: confirmed with caveats, in tension with Gaia's orbit. One eclipse geometry fits both the transit and the occultation, with e = 0.457 against Gaia's 0.497 ± 0.012. Forcing Gaia's e and ω misplaces the occultation by 1.8 d. No eclipse is completely observed, and the cycle count between sectors 20 and 60 is fixed by Gaia's period.

TYC 3801-1499-1 is a star of about 1.2 $R_\odot$ with a K-dwarf companion of about 0.75 $M_\odot$ on an eccentric 57.24-d orbit, the longest period in this set. Three eclipses are seen: a deep V-shaped transit in sector 60, the ingress of an earlier one at the end of sector 20, and a 16% flat occultation in sector 74. The occultation is nearly as deep as the transit, so the companion carries about 19% of the light, as a K2–3 dwarf would beside this primary. The period rests on 19 cycles between the sector-20 ingress and the sector-60 transit.

A64. BD+50 578 (TIC 249895319)



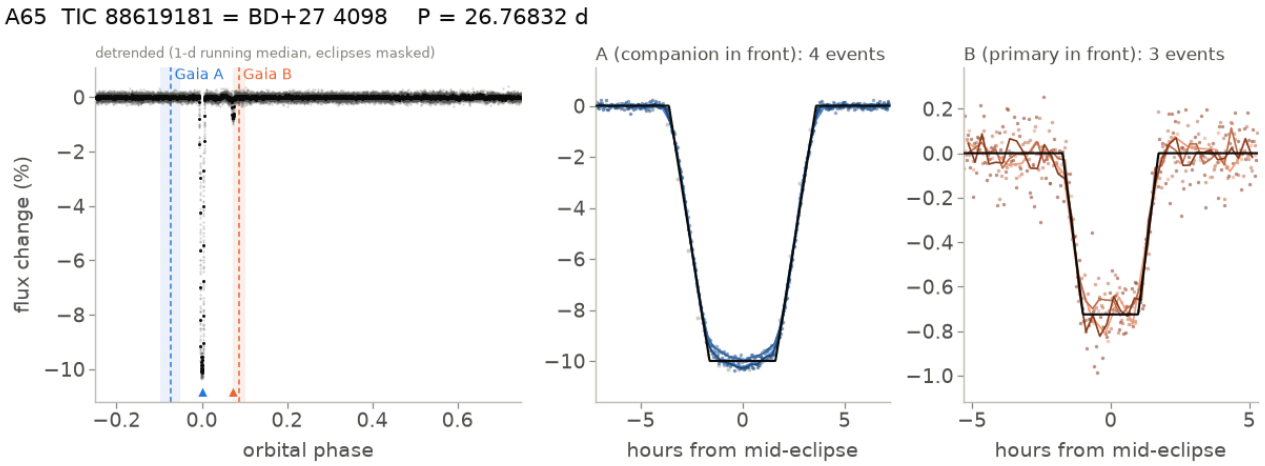
A64. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.27232 d, Gaia's predicted conjunctions dashed. Middle: the two 1.67%, 9.9-h flat-bottomed transits of the M dwarf at A. Right: the 0.15%, 13-h occultation of the M dwarf at B, stacked from both windows.

Identifiers	TIC 249895319; Gaia DR3 451309635844189056; BD+50 578
Position (J2000)	02:33:53.5 +51:36:39.1
Brightness	G 10.37; Tmag 9.77
Stars	Giant of 2.34 $R_\odot$ + M dwarf of about 0.26 $M_\odot$, R_2 = 0.29 $R_\odot$ (k = 0.123), about 3000 K from the occultation depth; mass function 0.0081 $M_\odot$; Gaia SB1
Period	29.27232 ± 0.00005 d over 38 cycles (the 37- and 39-cycle aliases, 30.06 and 28.52 d, excluded by Gaia); Gaia 29.285986 ± 0.019640 d (−0.70σ)
Epoch of primary	A: BTJD 2906.2943 ± 0.0005 = BJD_{TDB} 2459906.2943; also 1793.9463 ± 0.0007

Primary eclipse (A)	1.665–1.680% deep, T_{14} 9.89–9.90 h, flat 6.89–6.90 h: transit of the M dwarf (b ≈ 0.4); two events, identical in shape (s18, s58)
Secondary eclipse (B)	Stacked 1476 ± 70 ppm, T_{14} 13.1 ± 0.6 h (13.5 predicted), flat 9.8 h: occultation of the M dwarf; same offset in both B windows (1811.947: 1408 ± 75 ppm; 2895.063: 1429 ± 68, TESS-SPOC 1542 ± 65). Relative to the sector rms, each dip is about 3σ, together about 4σ
Orbit	B–A = 18.027 d gives e cos ω = +0.1805 (Gaia +0.177 ± 0.039); duration ratio 1.32 gives e sin ω ≈ +0.14 (Gaia +0.160). −0.11σ from Gaia's conjunction at Gaia's epoch
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (1.75–2.04%)
Pixel attribution	Centroids within 0.01 px (s58) and 0.13 px (s18); the G 15.03 star at 53″ would need 127%
Catalogue status	Not listed in: VSX, ASAS-SN, WDS, Hipparcos, SB9, Latham et al. 2002, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	Two transits and two occultation windows; the s85 transit was not covered and its B fell in flagged scattered light
Next eclipses	A: BTJD 4340.638, 4369.910 (±0.005); B: BTJD 4329.393, 4358.665 (±0.03)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

BD+50 578 is a giant of about 2.3 $R_\odot$ with an M-dwarf companion of about 0.26 $M_\odot$ on a 29.27-d orbit. Two transits of identical shape, four years apart, fix the period at 38 cycles. A 0.15% occultation of the M dwarf appears at the same phase in both windows that cover it, and also in TESS-SPOC. The occultation's timing and duration reproduce Gaia's e cos ω and e sin ω. The offset of about 1 d between the observed transits and the times predicted from Gaia's elements is the accumulated error of Gaia's period.

A65. BD+27 4098 (TIC 88619181)



A65. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 26.76832 d, Gaia's predicted conjunctions dashed (Gaia's e and ω put B 4.3 d after A; it falls 1.95 d after). Middle: the four 10%, 7.2-h flat transits of the M dwarf at A. Right: the three 0.72%, 3.4-h flat occultations at B.

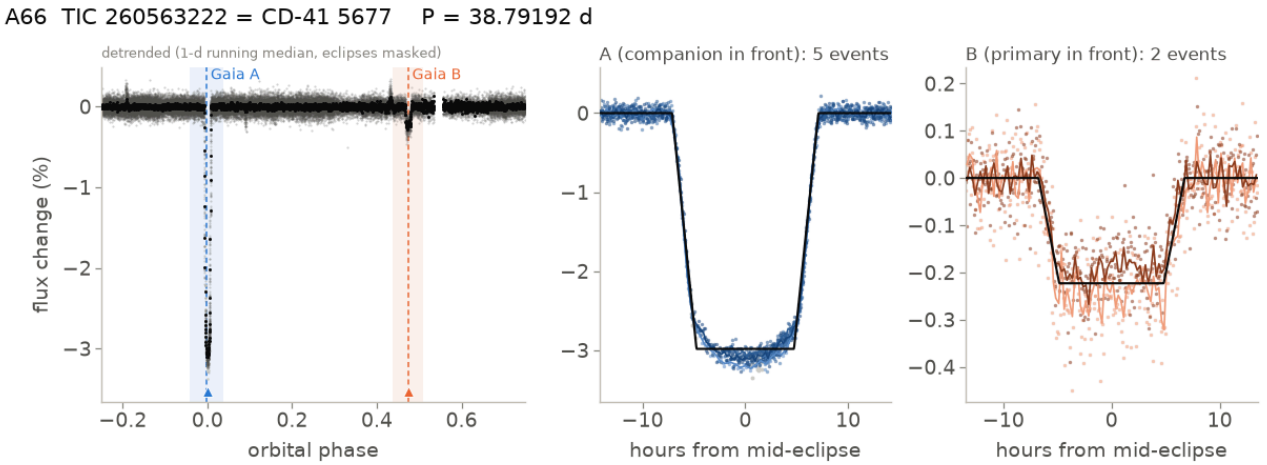
Identifiers	TIC 88619181; Gaia DR3 1848755826318803968; BD+27 4098
Position (J2000)	21:33:50.8 +27:43:45.3
Brightness	G 10.45; Tmag 10.02
Stars	F0 star (SIMBAD; Gaia GSP-Spec 6333 K, log g 3.79), R_1 ≈ 1.37 $R_\odot$ (TIC 1.54) + M dwarf, m_2 ≈ 0.29–0.33 $M_\odot$, R_2 ≈ 0.41 $R_\odot$ (k = 0.302), about 3300–3400 K, L_2/L_1 = 0.0070; mass function 0.0107 $M_\odot$; Gaia SB1, RUWE 7.8
Period	26.76832 ± 0.00001 d (s55–s82; $P_A = P_B$); with the s15 eclipse 26.76842 ± 0.000005 d. Gaia 26.76904 ± 0.023 d (−0.03σ)
Epoch of primary	A: BTJD 2819.5596 ± 0.0002 = BJD_{TDB} 2459819.5596
Primary eclipse (A)	9.71–10.18% deep, T_{14} 7.21–7.23 h, T_{23} 3.2 h, flat (b ≈ 0.4): transit of the M dwarf across the F star; four events (1722.0515, in s15 flagged cadences; 2819.56; 2846.33; 3542.30)

Secondary eclipse (B)	0.70–0.73% deep, T_{14} 3.41–3.46 h, T_{23} 2.0 h, flat: total occultation of the M dwarf; three events; B–A constant at 1.9527–1.9534 d (phase 0.07296)
Orbit	Joint fit: e cos ω = −0.691, e sin ω = −0.383, ***e* = 0.790, ω = 209°**, i = 88.7° (the joint fit was stable only for e sin ω between −0.40 and −0.35; no formal errors on e and ω); Gaia e = 0.688 ± 0.017 (+6σ), ω = 223.8 ± 4.3° (−3.4σ). Gaia's e, ω predict phase 0.161 ± 0.017 (4.30 d), −5.1σ from observed, and with them fixed the model cannot place B (χ^2 6948 against 2008). Periastron 2821.12 against Gaia 2821.44 ± 0.40 (−0.8σ): the disagreement is confined to e and ω
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s56, s82: 9.7–9.8%, 0.69–0.71%)
Pixel attribution	TESScut difference images on target for both kinds (A 1.7–3.2″ off at S/N 400–600, within the ~3″ floor; B 5.1″ at S/N 74, 20–60″ from every Gaia neighbour); the only feasible neighbour (G 12.33 at 58″, would need 57%) is 59″ from the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity and JSDC only), general web search
Open issues	**Gaia orbit shape corrected.** The four A times do not fit a line (χ^2 156 for 2 dof; O–C −2.4, +1.8, +1.6, −2.0 min): possible eclipse-timing variations, consistent with a third body (RUWE 7.8). R_2 is 10–15% above the main sequence for 0.3 $M_\odot$. RVs measuring e directly (predicted about 0.79); high-resolution imaging or Gaia DR4 astrometry for the third body
Next eclipses	A: BTJD 4345.354, 4372.122 (±0.001; ETVs of ±2–10 min possible); B: BTJD 4347.307, 4374.075
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

Note: confirmed at Gaia's period, with Gaia's orbit shape contradicted. The 1.95-d spacing between transit and occultation, together with the two durations, needs e = 0.790 and ω = 209°, against Gaia's 0.688 ± 0.017 and 224 ± 4°. With Gaia's values the occultation cannot be placed at all. The transit times also deviate from a linear ephemeris by about ±2 min, a hint of a third body. The e cos ω disagreement comes from the eclipse timing and is secure; the 6σ figure for e rests on a fit without formal errors.

BD+27 4098 is an F0 star with an M dwarf of about 0.3 $M_\odot$ on a very eccentric 26.77-d orbit. The transit and the occultation are only 1.95 d apart because periastron falls between them, 1.6 d after the transit and 0.4 d before the occultation. The durations test the fitted geometry. With Gaia's e and ω a central transit would last 12–12.5 h. The observed flat transit lasts 7.2 h ($b \approx 0.4$), which matches the fitted orbit. The fitted orbit, however, predicts a 3.9–4.1-h occultation, against the 3.4–3.5 h observed. Both eclipses are flat-bottomed, and the occultation depth gives the M dwarf's share of the light directly; that share is in line with the main sequence. The fourth transit, in 2019, lies in QLP-flagged cadences.

A66. CD-41 5677 (TIC 260563222)

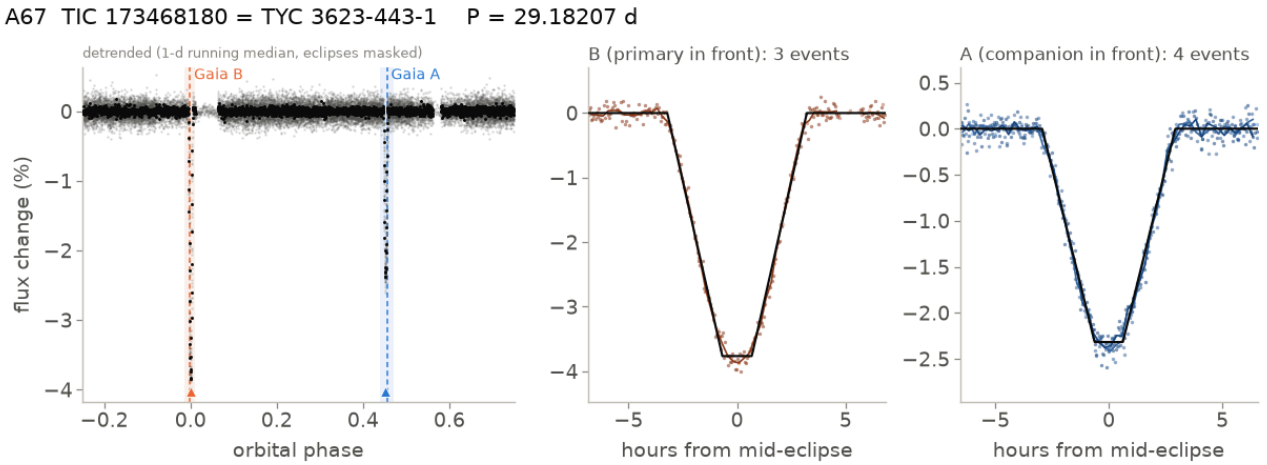


A66. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 38.79192 d, Gaia's predicted conjunctions dashed. Middle: the 3.0%, 14-h flat transits of the M dwarf at A. Right: the two 0.22%, 13.5-h flat occultations at B.

Identifiers	TIC 260563222; Gaia DR3 5416799976193456256; CD-41 5677
Position (J2000)	10:12:10.2 −42:30:11.4
Brightness	G 10.06; Tmag 9.65
Stars	Primary $R_1 \approx 2.3\ R_\odot$ (TIC 2.66) + M dwarf, $m_2 \approx 0.37$–$0.43\ M_\odot$, $R_2 \approx 0.39\ R_\odot$ (k = 0.165, main sequence), L_2/L_1 = 0.0022; mass function 0.0235 $M_\odot$; Gaia SB1; SIMBAD: GALAH DR2/DR3
Period	38.79192 ± 0.00002 d (Gaia 38.795818 ± 0.007172 d; −0.54σ)
Epoch of primary	A: BTJD 3027.5910 ± 0.0006 = BJD_{TDB} 2460027.5910
Primary eclipse (A)	2.95–3.08% deep, T_{14} 14.2 h, T_{23} 9.5 h: transit of the M dwarf; four full events (1553.50, 2290.55, 3027.59, 3764.64) and a partial (3725.874); s63 event 39 of 172 cadences flagged, TESS-SPOC time within 0.7 min; O−C within ±1 min
Secondary eclipse (B)	0.204–0.242% deep, T_{14} 13.4–13.6 h, T_{23} 9.5–9.8 h, flat: total occultation of the M dwarf; two events (3744.2316, 4054.5667)
Orbit	Phase separation 0.47395 (Gaia 0.4765 ± 0.0067, −0.4σ); $e \cos\omega = -0.0409$, $e \sin\omega = -0.044$ (±0.003 statistical, about ±0.01 with red noise): e = 0.060, ω = 227° against Gaia 0.061 ± 0.011, 232 ± 10°. i = 89.7°
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP; TARS (B in s89: 0.32–0.36%, against 0.20–0.24% in QLP and TESS-SPOC, so TARS disagrees on the depth)
Pixel attribution	TESScut difference images on target for both kinds (B 2.6″ and 4.3″; neighbours beyond 52″); the only feasible neighbour (G 13.68 at 59″, would need 84%) is 57–62″ from the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	The occultation gives L_2/L_1 = 0.22% directly, implying $T_2 \approx 3300$ K against about 3800 K from the main-sequence relation: a mild tension (a factor of 2 in L_2). One RV epoch, or space-quality photometry of the next B; whether the T_2 tension holds depends on which B depth is right
Next eclipses	B: BTJD 4364.902 (2026-11-20 09:40 UTC), 4403.694; A: BTJD 4346.517, 4385.309 (±0.002)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

CD-41 5677 is a star of about 2.3 $R_\odot$ with an M dwarf of about 0.4 $M_\odot$ on a nearly circular 38.8-d orbit. Both eclipses are flat-bottomed and long: a 3% transit of the M dwarf and a 0.22% total occultation, whose depth is the dwarf's share of the light. The eclipse timing and shapes give e = 0.060 and ω = 227°, matching Gaia's independent orbit within 1σ.

A67. TYC 3623-443-1 (TIC 173468180)

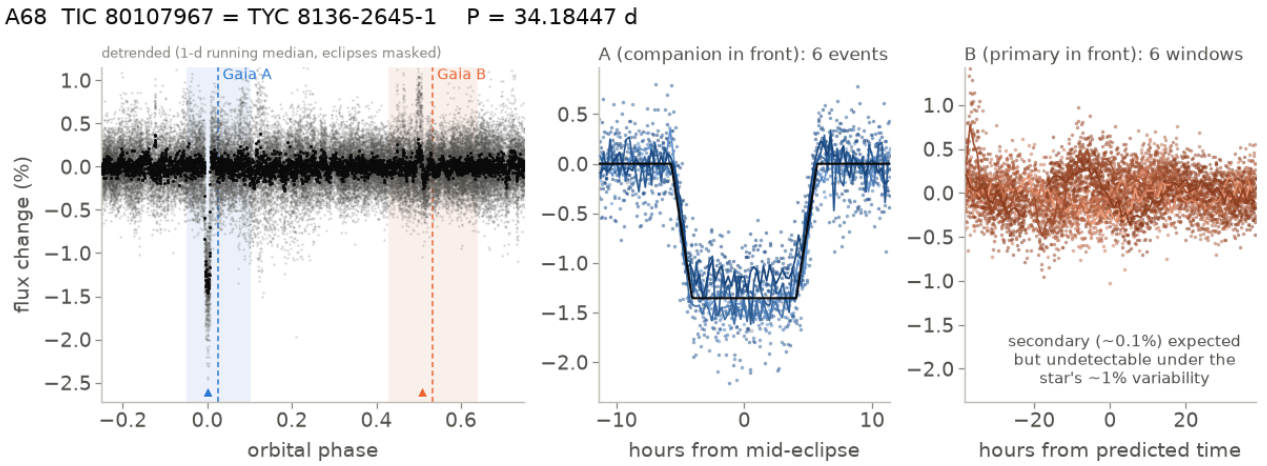


A67. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 29.18207 d, Gaia's predicted conjunctions dashed. Middle: the three 3.8%, 6.5-h near-total occultations at B, deeper than the transits. Right: the four 2.3%, 5.9-h grazing transits at A.

Identifiers	TIC 173468180; Gaia DR3 1941146757609930240; TYC 3623-443-1
Position (J2000)	23:12:27.6 +45:52:28.9
Brightness	*G* 10.82; Tmag 10.30
Stars	Primary $R_1 \approx 1.6\ R_\odot$ (FLAME 1.73) + K dwarf, $m_2 \approx 0.67$–$0.76\ M_\odot$, $R \approx 0.75\ R_\odot$, 4700–5000 K, $L_2/L_1 \approx 0.07$–0.10; mass function 0.115 $M_\odot$; Gaia SB1
Period	29.18207 ± 0.00001 d (Gaia 29.179837 ± 0.002830 d; +0.8σ)
Epoch of primary	A: BTJD 2878.8658 ± 0.0005 = BJD_{TDB} 2459878.8658
Shallower eclipse (A)	2.25–2.36% deep, T_{14} 5.8–5.9 h, T_{23} 1.2–1.4 h, nearly V-shaped: grazing transit of the primary's darkened limb ($b \approx 1.2\ R_1$); four events (1740.76, 1769.95, 2878.87, 3608.42); O−C ≤ 0.4 min
Deeper eclipse (B)	3.75–3.77% deep, T_{14} 6.4–6.5 h, T_{23} 1.3–1.4 h: near-total occultation of the companion ($b \approx 0.95\ R_1$), removing roughly half its light; three events (1756.76, 1785.94, 3595.23)
Orbit	Phase separation 0.54816 (Gaia 0.5433 ± 0.0049, +1.0σ); $e \cos \omega$ = 0.0745–0.0756 against Gaia 0.0676 ± 0.0077 (+1σ); $e \sin \omega$ poorly constrained (−0.20 to −0.05), Gaia's −0.119 in the middle. The duration-ratio estimate (+0.05) is wrong because the impact parameters differ. $i \approx 88°$; stars 1.11a apart at A, 0.88a at B
Pipelines agreeing	QLP SAP; TESS-SPOC (B in s16, s17, s84: 3.66–3.83%)
Pixel attribution	TESScut difference images: A and B 0.1–2.3″ from the target; every neighbour would need more than 180%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	A spectrum for the K-dwarf lines ($K_2 \approx 51$ km/s for $M_1 = 1.05$); the system should be detectable as SB2
Next eclipses	A: BTJD 4337.969, 4367.151 (±0.001); B: BTJD 4353.966, 4383.148
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3623-443-1 is a star of about 1.6 $R_\odot$ with a K dwarf of about 0.7 $M_\odot$ on a mildly eccentric 29.18-d orbit. The occultation is deeper than the transit. Gaia's orbit puts the stars farther apart at A (1.11a) than at B (0.88a), so at A the companion only clips the primary's limb-darkened edge, giving a 2.3% eclipse. At B the primary covers about half of the companion, which carries 7–9% of the light, giving a 3.8% eclipse. Both chords are near grazing, so both eclipses last about 6 h against 11–13 h for central ones.

A68. TYC 8136-2645-1 (TIC 80107967)



A68. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 34.18447 d, Gaia's predicted conjunctions dashed; the star varies at about 1%. Middle: the six 1.3%, 11.4-h flat transits at A. Right: the B windows, where the expected 0.1% occultation is below the star's variability.

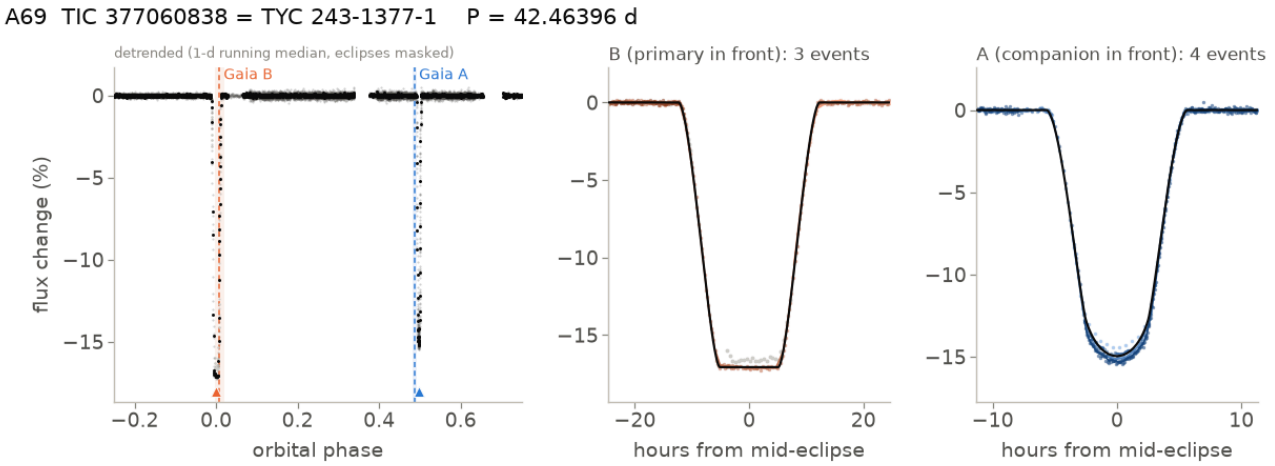
Identifiers	TIC 80107967; Gaia DR3 5520398813627036288; TYC 8136-2645-1
Position (J2000)	08:04:48.2 −45:32:43.4
Brightness	*G* 11.00; Tmag 10.58
Stars	Primary $R_1 \approx 2.6\ R_\odot$ (TIC 2.45) + M dwarf, $m_2 \approx 0.25$–$0.28\ M_\odot$, $R_2 \approx 0.27$–$0.29\ R_\odot$ (k = 0.111); mass function 0.0066 $M_\odot$; Gaia SB1. A *G* 14.46 star (TIC 818814154) lies 4.4″ away
Period	34.18447 ± 0.00008 d (Gaia 34.164065 ± 0.036796 d; +0.55σ)
Epoch of primary	T_0 = BTJD 1521.3074 = BJD_{TDB} 2458521.3074
Eclipse (A)	1.20–1.49% deep, T_{14} 11.1–11.7 h, T_{23} 8.0–8.5 h, flat: transit of the M dwarf near periastron (predicted central duration 11.2–11.7 h); six events 2018–2025; χ^2_{red} 3.2, O−C up to +8.5 min (s35, partly flagged; red noise from the star's variability); phases ≤ 0.27σ from Gaia's A
Other eclipse (B)	Not detected, and not expected to be: geometry allows a 15–17-h total occultation, but $L_2/L_1 \approx$ 0.10–0.15% sits under the star's red noise (spurious box-fit dips up to 0.25–0.67%)
Orbit	No B measured, so e cos ω is unmeasurable; a depth/duration fit with Gaia's e, ω fixed is as good as free (χ^2 2421.1 against 2421.0). $i \approx 89°$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s8, s34, s35: 1.20%, 1.39–1.43%); TARS (single-event times within ±5–8 min of QLP)
Pixel attribution	**Not decisive.** The 4.4″ neighbour would need a 33% eclipse, which is physically possible. TGLC's aperture light curve of the neighbour shows exactly the 32.8% contamination expected if the target dims 1.3%; TGLC PSF photometry splits the dip (target 0.9–1.3%, neighbour 13–18%); difference-image centroids favour the target (0.5″ against 4.6″ in s34; 1.5″ against 3.0″ in s62), within the ~3″ systematic floor. The *G* 14.78 star at 35″ is excluded by the centroid
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*), general web search
Open issues	Host attribution rests on the period and phase match (a chance match of a background eclipsing binary in both period and phase, not quantified) and self-consistent physics, not pixels. AO or speckle imaging of the 4.4″ neighbour plus ground photometry with a ~1.5″ aperture during a transit would resolve the attribution: 1.3% on the target, about 33% on the neighbour
Next eclipses	A: BTJD 4358.619, 4392.803 (±0.007); B (about 0.1%, undetectable from TESS): 4341.79, 4375.98 (±1.46)
Discovery	Conjunction-window search (Section 2.5); checked by independent re-analysis with separately written code (Section 2.5); status probable (see below)
Status	

Probable. The host is not pixel-proven: the *G* 14.46 star 4.4″ away could host a 33% eclipse, TGLC PSF photometry splits the dip between the two stars, and the difference-image centroids favour the target only within the ~3″ systematic floor. The attribution rests on the match of period and phase to Gaia's orbit and on self-consistent physics; the chance of a background eclipsing binary matching both has not been quantified. Resolved imaging, or photometry through a ~1.5″ aperture during a transit, would resolve it.

Note: probable, not confirmed. The system eclipses at Gaia's period, but the host is not pixel-proven. A *G* 14.5 star 4.4″ from the target could host a 33% eclipse, and TESS pixels (21″ across) cannot separate the two. The target is adopted as the host on physical consistency and on the period and phase match, not on pixel data.

TYC 8136-2645-1 is a star of about 2.6 $R_\odot$ with an M dwarf of about 0.26 $M_\odot$ on a 34.18-d orbit. Six flat transits from 2018 to 2025 fall within 0.27σ of Gaia's predicted conjunction, with a period 0.55σ from Gaia's. A background eclipsing binary would have to match both the period and the conjunction phase by chance. That is unlikely, but the probability has not been quantified. The 11.4-h duration, the flat bottom, the 1.3% depth and the companion mass from the mass function all fit an M dwarf crossing this star near periastron. Its occultation, about 0.1%, is lost under the star's own 1% variability, as expected.

A69. TYC 243-1377-1 (TIC 377060838)



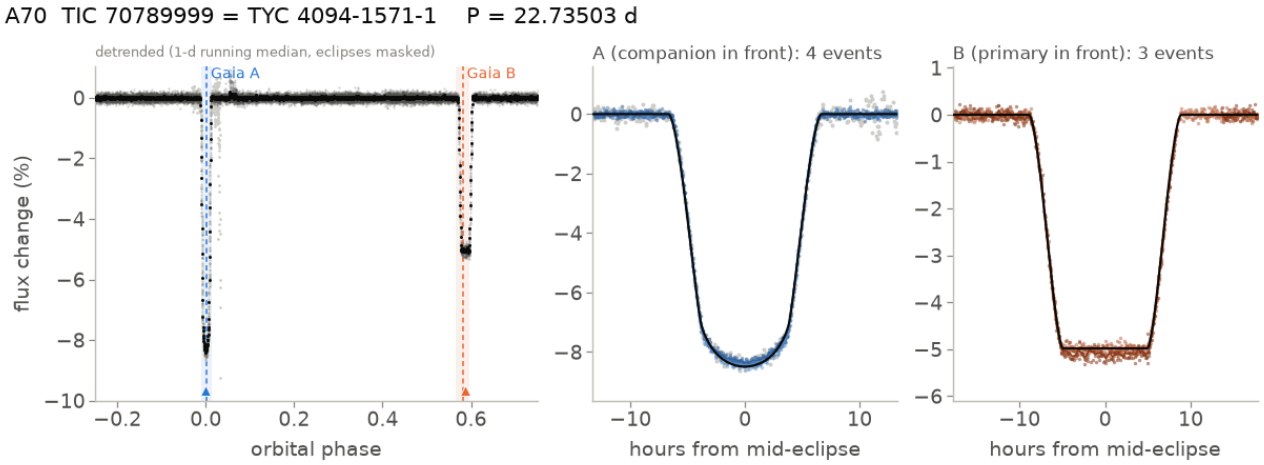
A69. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 42.46396 d, Gaia's predicted conjunctions dashed. Middle: the 17%, 24.6-h flat total occultations at B, deeper and longer than the transits. Right: the four 14.6%, 11.3-h full transits at A.

Identifiers	TIC 377060838; Gaia DR3 3853435449561968000; TYC 243-1377-1
Position (J2000)	09:51:56.8 +06:29:19.4
Brightness	*G* 10.63; Tmag 10.16
Stars	G subgiant, R_1 ≈ 2.6–2.7 $R_\odot$, M_1 ≈ 1.4–1.6 $M_\odot$, T_1 ≈ 5400 K + smaller, hotter dwarf, m_2 0.94–1.0 $M_\odot$, R_2 = 0.92–1.04 $R_\odot$, about 6000 K, surface-brightness ratio S_2/S_1 = 1.40; total about 7 $L_\odot$ (Gaia FLAME 7.2 $L_\odot$, 1.42 $M_\odot$); mass function 0.150 $M_\odot$; Gaia SB1
Period	42.46396 ± 0.00002 d (Gaia 42.46599 ± 0.01286 d; −0.2σ); B alone 42.46396 ± 0.00007
Epoch of primary	$T_{0,A}$ = BTJD 1538.63354 = BJD_{TDB} 2458538.63354; $T_{0,B}$ = BTJD 1517.53332
Primary eclipse (A)	14.1–15.0% deep, T_{14} 11.3 h, nearly flat bottom 5.0 h, U-shaped from limb darkening: full transit of the companion (b_A ≈ 0.0, *k* = 0.384); four events (s008, s035, s046, s072); every O−C under 0.2 min
Secondary eclipse (B)	17.0% deep, T_{14} 24.6 h, flat 10.3 h: total occultation of the companion near apastron (b_B = 0.24); two full events (s045; s008 in flagged cadences) plus an ingress in flagged cadences at the end of s046
Orbit	

	B–A phase 0.50310 ± 0.00002 (Gaia 0.5205 ± 0.0090, –1.9σ); *e* cos ω +0.0045 (Gaia +0.0295); *e* sin ω +0.378 (Gaia +0.403); *e* 0.378 against 0.404 ± 0.0145 (1.8σ); ω 89.3° against 85.8 ± 1.8°. The fit tied to Gaia's duration ratio fits B's ingress and egress noticeably worse
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (A s35); TESS-SPOC (A s35, s72); eleanor-lite (A s8); TARS (A ×4, B s45); every refit within 0–3 min
Pixel attribution	SPOC TPF and TESScut scene fits: lost flux on the target in all four eclipses tested (13.8–16.3%); the only bright neighbour (*G* 11.37 at 45″, would need 30–34%) gets +2.0, −0.8, +1.4, −0.1%; $\Delta\chi^2$ 3.5×10^5 to 2×10^6 in favour of the target
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, WDS, CCDM, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, Gaia DR3 multiplicity only), general web search
Open issues	A mild tension: a 1.0 $M_\odot$ star at 6000 K is slightly hot for its mass (possibly low metallicity), easing if the primary is cooler (Gaia 5937 K against TIC 5446 K). No formal error on *e* sin ω (the free-ρ fit failed). Spectra for an SB2 detection and the companion's temperature
Next eclipses	A: BTJD 4341.255, 4383.719; B: BTJD 4362.619, 4405.083 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 243-1377-1 is a G subgiant of about 2.6 $R_\odot$ with a dwarf companion of about 1 $M_\odot$ and 1 $R_\odot$ on an eccentric 42.46-d orbit. The occultation is deeper than the transit, even though the transit is full. The companion is smaller and hotter, with a surface about 1.4 times brighter than the primary's. Hiding it entirely therefore removes more light (17%) than covering part of the primary with it (14.6%). The occultation also lasts more than twice as long because it falls near apastron, where the stars move 2.2 times slower. One geometry fits all five eclipses, with white residuals.

A70. TYC 4094-1571-1 (TIC 70789999)



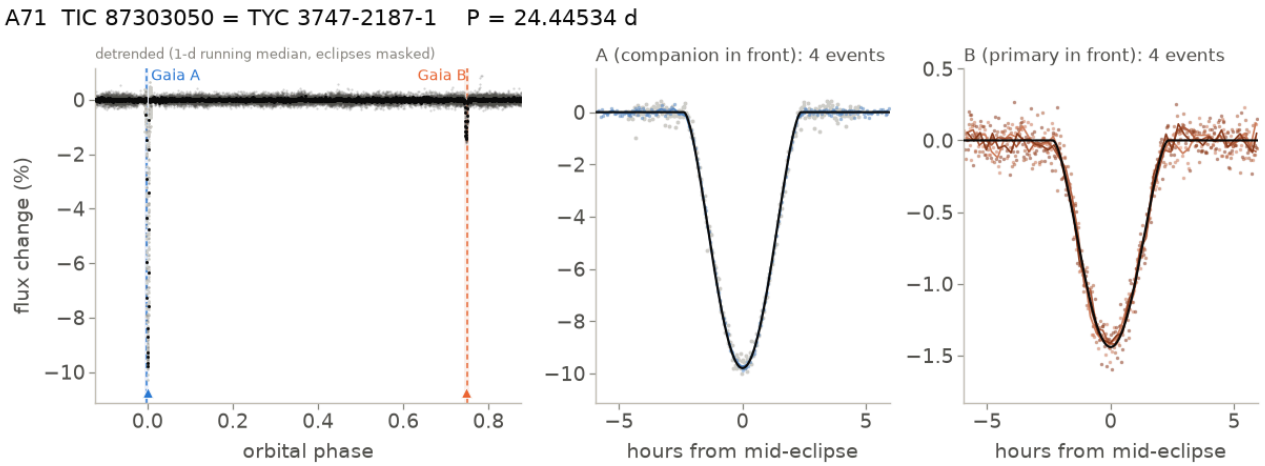
A70. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on *P* = 22.73503 d, Gaia's predicted conjunctions dashed. Middle: the 8.2%, 13.3-h flat transits at A. Right: the three 5.1%, 17.8-h flat total occultations at B.

Identifiers	TIC 70789999; Gaia DR3 2822584818250071O4; TYC 4094-1571-1
Position (J2000)	05:53:58.9 +60:19:36.1
Brightness	*G* 10.17; Tmag 9.76
Stars	Subgiant, R_1 = 3.0–3.4 $R_\odot$ (TIC 3.60; Gaia 3.21–3.57, FLAME 1.83 $M_\odot$), T_1 6300–6600 K + G dwarf, m_2 0.9–1.0 $M_\odot$, R_2 = 0.83–0.94 $R_\odot$, T_2 = 5600–5900 K, L_2 = 5% of the light; mass function 0.120 $M_\odot$; Gaia SB1; SIMBAD: 2019 TESS asteroseismic target list
Period	22.73503 ± 0.00001 d (Gaia 22.73174 ± 0.00324 d; +1.0σ)
Epoch of primary	$T_{0,A}$ = BTJD 1823.53837 = BJD_{TDB} 2458823.53837; $T_{0,B}$ = BTJD 1836.88172
Primary eclipse (A)	

	8.09–8.35% deep, T_{14} 13.3 h, 7.6 h flat, U-shaped: full transit ($b_A \approx 0$, $k = 0.275$); four events (s019, s059, s060; s073 in flagged cadences, clean); plus a further A eclipse at 2937.555 on a scattered-light feature, unusable for timing; O–C ≤ 0.1 min for the events in the fit; the flagged s073 event is −3 min
Secondary eclipse (B)	5.00–5.13% deep, T_{14} 17.8 h, 9.9 h flat: total occultation ($b_B = 0.19$); three events (s019, s059, s073)
Orbit	B–A phase 0.58691 ± 0.00003 (Gaia 0.5794 ± 0.0099, +0.8σ); e cos ω +0.135 (Gaia +0.124); e sin ω +0.148 (+0.140); e = 0.201 against 0.187 ± 0.020; ω = 47.5° against 48.4 ± 3.4°. The Gaia-tied fit is as good as the free one
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (A s19, s59, s60; B s19, s59, s73; 0–1 min); TARS (all six within 1 min, B s73 within 6 min in noisy data)
Pixel attribution	TESScut s59 scene fit, A and B: lost flux on the target (8.2%, 5.1%); $\Delta\chi^2 \approx 3\times10^5$ for target over neighbour; centroid within 0.08 px; neighbours within 60″ (G 15.6–15.7) would need more than 100%
Catalogue status	Not a catalogued EB: nothing in VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, WDS, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP or a general web search. Flagged by Kostov et al. 2025 table 2 (unvetted) and by a TESS-SPOC multi-sector TCE at P = 11.3675 d (half the true period), depth 8.4%, never followed up
Open issues	Spectroscopy for SB2 and the companion's temperature
Next eclipses	B: BTJD 4337.735, 4360.470; A: BTJD 4347.126, 4369.862 (±1 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 4094-1571-1 is a subgiant of about 3.2 $R_\odot$ with a G dwarf of about 1 $M_\odot$ on a mildly eccentric 22.74-d orbit. Both eclipses are central and flat-bottomed: a full transit of the dwarf (8.2%) and its total occultation (5.1%, the dwarf's share of the light). The eclipses reproduce Gaia's e and ω within 1σ. A TESS-SPOC multi-sector search had already detected the transit as a threshold-crossing event, at half the true period. It was not followed up, and we do not treat it as a prior catalogue entry.

A71. TYC 3747-2187-1 (TIC 87303050)



A71. Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.44534 d, Gaia's predicted conjunctions dashed. Middle: the four 9.7%, 4.5-h V-shaped grazing eclipses at A. Right: the four 1.4% grazing eclipses at B.

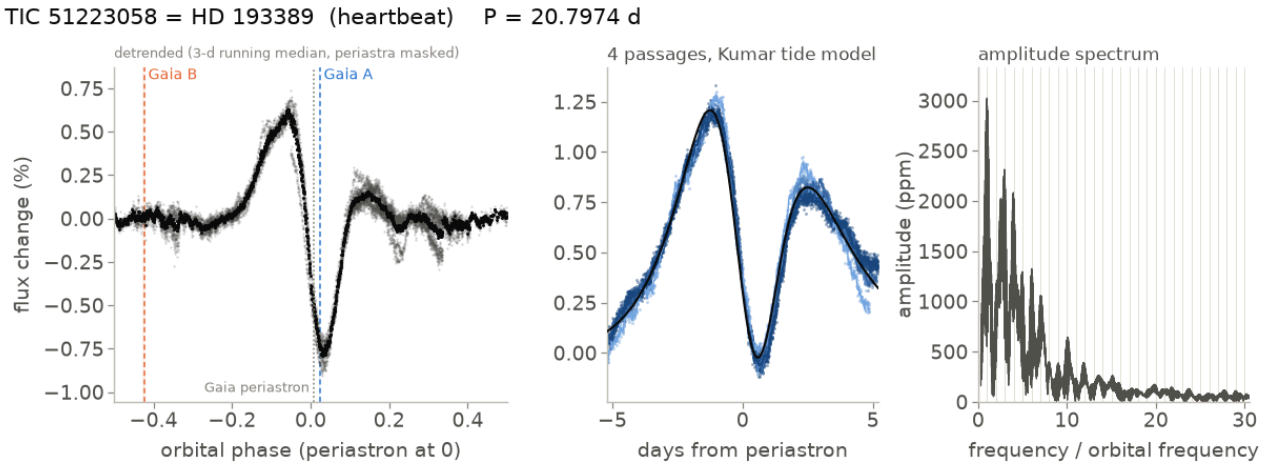
Identifiers	TIC 87303050; Gaia DR3 2809691025802535 68; TYC 3747-2187-1
Position (J2000)	05:15:50.8 +58:13:29.8
Brightness	G 10.62; Tmag 10.20
Stars	G0 star (LAMOST; RV −25.2 km/s), R_1 from the crossing speed 1.0–1.06 $R_\odot$, depending on b (TIC 1.18; Gaia 1.24–1.25) + K dwarf, m_2 = 0.62–0.68 $M_\odot$; a = 42–44 $R_\odot$; mass function 0.089 $M_\odot$; Gaia SB1
Period	24.44534 ± 0.00001 d (Gaia 24.44347 ± 0.00331 d; +0.6σ)

Epoch of primary	$T_{0,A}$ = BTJD 1833.11980 = BJD_{TDB} 2458833.11980; $T_{0,B}$ = BTJD 1826.93956
Primary eclipse (A)	9.6–9.8% deep, V-shaped, 4.3–4.7 h: grazing transit of the companion; four events (s019, s059; s073 and s086 in flagged cadences, clean)
Secondary eclipse (B)	1.36–1.47% deep, V-shaped, grazing; four events (s019, s059, s073, s086); O−C ≤ 2 min
Orbit	B−A phase 0.74718 ± 0.00003 (Gaia 0.7514 ± 0.0044, −1.0σ); e cos ω = +0.399 (Gaia +0.406); e sin ω unconstrained by the durations; e = 0.399–0.41 against Gaia 0.409 ± 0.0075; i ≈ 87.6–87.8°
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s19: A 9.5%, B 1.33–1.38%); TESS-SPOC (A s19, s59; B s19, s59, s73); TARS (all six within 0–3 min)
Pixel attribution	SPOC TPF s19 and TESScut s59: lost flux on the target (A 8.7%, 9.1%; B 1.2%, 1.4%); centroid shift ≤ 0.03 px (A), ≤ 0.1 px (B); nearest bright neighbour (G 16.07 at 41″) would need more than 100%
Catalogue status	Not a catalogued EB: nothing in VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, WDS, Hipparcos, SB9, Gaia DR3 variability tables, ExoFOP or a general web search; Kostov et al. 2025 table 2 only (unvetted)
Open issues	Both chords graze, so k, b and third light are not separable: the free fit (k = 0.83, b = 1.37) and the Gaia-tied fit (k = 0.66) are statistically identical; the free-fit companion (R_2 ≈ 0.85 $R_\odot$, about 3700 K) is implausible for 0.65 $M_\odot$, and the expected k ≈ 0.55 presumably lies within the degeneracy (the k-fixed fits failed). Multi-band photometry or an SB2 detection would break it
Next eclipses	A: BTJD 4350.990 (2026-11-06 11:38 UTC), 4375.435; B: BTJD 4344.809, 4369.255 (±2 min)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 3747-2187-1 is a G0 star with a K dwarf of about 0.65 $M_\odot$ on an eccentric (e ≈ 0.40) 24.45-d orbit. The two stars graze at both conjunctions: both eclipses are V-shaped and last 4.5 h, against 7–8 h for central ones. Eight eclipses over six years, two of them recovered from QLP-flagged cadences, fit a linear ephemeris to within 2 min. Their spacing gives Gaia's e cos ω. The 1.4:9.7 depth ratio fits a K dwarf behind a G0 star; the grazing geometry leaves the radius ratio open.

S3.1.3 Heartbeat stars

HD 193389 (TIC 51223058)



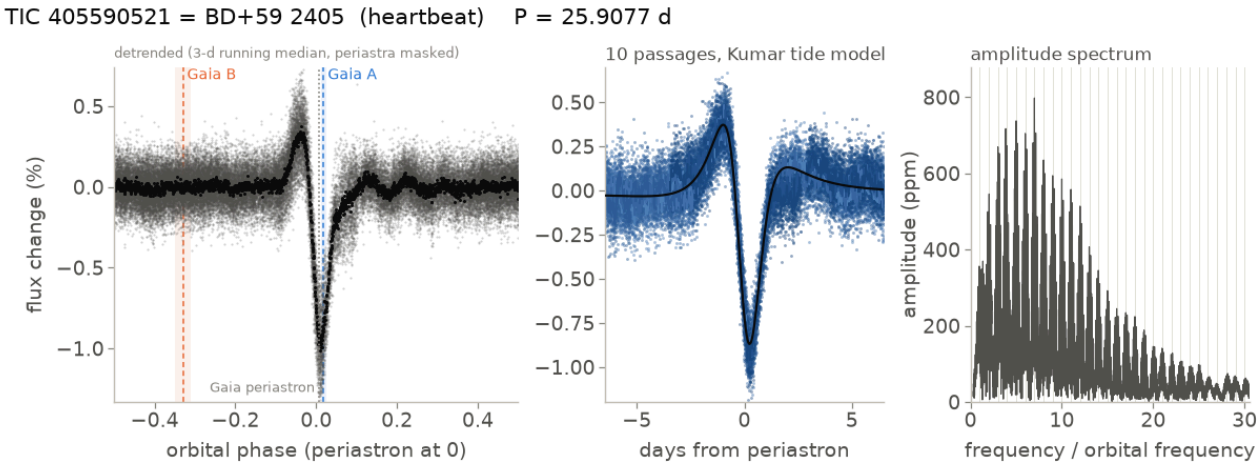
HD 193389. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 20.797 d, periastron and Gaia's conjunctions marked. Middle: the 1.2% peak-to-trough heartbeat at each covered periastron (four passages overlaid) with the Kumar et al. (1995) model (i ≈ 35°). Right: the amplitude spectrum of the light curve, against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 51223058; Gaia DR3 1808373856612878464; HD 193389; WDS HDS 2905
Position (J2000)	20:19:31.5 +15:32:42.2
Brightness	G 7.45; Tmag 7.06

Stars	F5.5 II-III bright giant (SIMBAD), L ≈ 210 $L_\odot$, R_1 ≈ 11 $R_\odot$ (8–16 from the Hipparcos parallax range; TIC 13.3), M_1 ≈ 3.5 $M_\odot$ assumed + companion m_2 ≈ 2.3 $M_\odot$ at i = 35° (about 10% of the light); mass function 0.0655 $M_\odot$; Gaia SB1, K_1 33.4 km/s, RUWE 2.06. A 0.3″ visual companion (HDS 2905, ΔHp ≈ 3.7) probably explains the RUWE
Period	20.7974 ± 0.0015 d from the troughs (Gaia 20.79663 ± 0.00058 d; 0.5σ)
Epoch	Trough BTJD 2778.057 = BJD_{TDB} 2459778.057 (troughs 2778.057, 2798.834, 2819.718, 3526.775; O−C ≤ 0.05 d); the trough falls about 0.6 d after periastron
Periastron signature	A 4–5-d rise to a peak 1.05–1.23 d before periastron, a steep fall, a trough 0.54–0.63 d after periastron, then a lower secondary maximum and slow decline; peak to trough 12,160–12,410 ppm in all three fully covered passages (s54, s55, s81)
Tidal model *e*, ω vs Gaia	Orbit free: e = 0.371, ω = 77.6°, i = 35.5° against Gaia e 0.3555 ± 0.0044, ω 77.26 ± 0.86°; from the light curve alone, ω agrees with Gaia's to 0.3° and e to 0.016 (3.2σ by Gaia's quoted error). Periastron shifted +0.28 d (about 3σ), attributed to the simplified model, with apsidal motion also possible. Sharp χ^2 minimum at i ≈ 35°; fitted S = 5.4×10^{-3} against 4.3×10^{-3} predicted for an equilibrium tide at that inclination
Eclipse	None: the heartbeat model beats a trapezoid eclipse by ΔBIC ≈ 67,400; an eclipse at A would need $i \gtrsim 70°$; any eclipse on top of the heartbeat is ≲ 0.05% against the 1.2% depth of the candidate eclipse; B windows flat
Oscillations	Not searched; a search for tidally excited oscillations at orbital harmonics remains to be made
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s81: 11,136 and 11,514 ppm against QLP's 11,621)
Pixel attribution	TESScut difference images (peak minus trough): the 5×5 core loses 10,300–11,100 ppm, centred on the target; the brightest Gaia neighbour (G 12.54 at 16.6″) would need 132%. HDS 2905 B (0.3″) would need about 37% and cannot be separated in pixels; the signal is locked to HD 193389's own RV periastron and reproduces that orbit
Catalogue status	Not listed as a heartbeat or eclipsing binary in: VSX (only an unrelated G 18.2 variable at 24″), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, SB9, Gaia DR3 eclipsing-binary table (Gaia's variability classification: ACV|CP|MCP|ROAM|ROAP|SXARI), ExoFOP, SIMBAD's 22 references, general web search
Open issues	A heartbeat model fitted jointly with the Gaia RVs; spectra for the about 2.3 $M_\odot$ companion. Only three TESS sectors
Next periastra	BTJD 4337.24, 4358.04, 4378.83 (A conjunction 4337.57 ± 0.12, 4358.36, 4379.16)
Discovery	Conjunction-window search (Section 2.5), as a candidate eclipse; reclassified as a heartbeat by independent re-analysis with separately written code (Section 2.5)

HD 193389 is an F-type bright giant of about 11 $R_\odot$ on a 20.8-d orbit with e = 0.36. The 1.2% dip after periastron is not an eclipse at the A conjunction. The same asymmetric peak-and-trough pattern recurs at every periastron, with the trough 0.2–0.3 d after the conjunction rather than on it. A tidal model fitted to the light curve alone reproduces Gaia's orbit. At periastron the giant's radius is about 0.3 of the separation, so a heartbeat of about this size is expected. The eclipse interpretation is refuted, and the heartbeat has not been reported before.

BD+59 2405 (TIC 405590521)



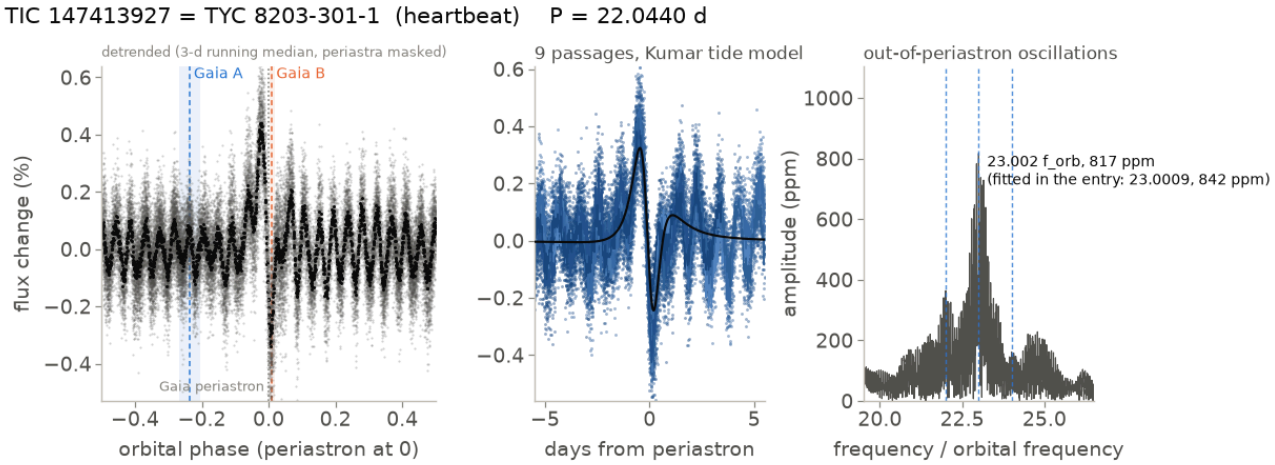
BD+59 2405. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 25.9077 d, periastron and Gaia's conjunctions marked. Middle: the +0.3–0.4% brightening and 1.2% dip at each periastron (ten passages overlaid; seven were fitted) with the Kumar et al. (1995) model ($i \approx 60$–$65°$). Right: the amplitude spectrum of the light curve, against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 405590521; Gaia DR3 2203762361132796800; BD+59 2405
Position (J2000)	21:37:16.0 +60:29:23.1
Brightness	*G* 10.40; Tmag 9.89
Stars	A2 star (SIMBAD), member of Trumpler 37, $R_1 \approx 7\ R_\odot$ (TIC 6.96), $M_1 \approx 2.5\ M_\odot$ assumed + companion $m_2 \approx 1.5$–$1.85\ M_\odot$; mass function 0.2225 $M_\odot$; Gaia SB1
Period	25.9077 ± 0.0004 d from the tidal fit (Gaia 25.8987 ± 0.0077 d; +1.2σ)
Epoch	Periastron t_p = BTJD 3562.21 ± 0.07 = BJD_{TDB} 2460562.21 (Gaia 3561.28 ± ~0.95, +1σ)
Periastron signature	A +0.3–0.4% brightening and about 1.2% dip in every orbit, in seven fitted periastron windows (s17, 56, 57, 76, 77, 83, 84); the model's flux minimum falls at the A conjunction (2837.14, 3562.56, 3588.45); per-window timing shifts under 0.03 d
Tidal model *e*, ω vs Gaia	e = 0.5846 ± 0.0033 against Gaia 0.587 ± 0.025; ω = 69.4 ± 4.7° against 70.0 ± 2.1°; i = 60.5 ± 5.4° (65 ± 7° without s17); S = 465–515 ppm; χ^2 8620 for 3046 points. With Gaia's orbit fixed the fit is poor (i = 37°, χ^2 155k). Amplitude check: $(m_2/M_1)(R_1/d_{peri})^3 \approx 1.6\%$ for $R_1 \approx 7\ R_\odot$
Eclipse	None: residuals in a 0.3-d box at each A conjunction are at the red-noise level, so any A eclipse is shallower than about 0.1%; A needs $i \gtrsim 69°$ and B $i > 84°$, and the fitted 60–65° predicts none
Oscillations	None reported; residual rms 460–880 ppm against about 350 white. An RV test for tidally excited oscillations remains to be made
Pipelines agreeing	QLP SAP; TESS-SPOC s83 (SAP dip 1.03% at 3562.48, brightening +0.21%; PDCSAP 0.88% / +0.27%)
Pixel attribution	Not tested separately; the signal is locked to the star's own RV periastron and reproduces that orbit's e and ω
Catalogue status	No prior heartbeat claim: VSX, ASAS-SN, Solanki et al. 2025 (no source within 30″), Callahan et al. 2025 (112 heartbeat systems from Gaia SBs + TESS; not included); the Kirk et al. 2016, Shporer et al. 2016 and Wrona et al. 2022 tables cover Kepler and OGLE fields. No TOI/CTOI; a general web search found nothing
Open issues	A joint RV and light-curve fit; an RV test for tidally excited oscillations
Next periastra	Heartbeat minima BTJD 4339.79, 4365.70, 4391.60 (±0.08)
Discovery	Conjunction-window search (Section 2.5), flagged as a heartbeat candidate; confirmed by independent re-analysis with separately written code (Section 2.5)

BD+59 2405 is an A2 star of about 7 $R_\odot$ in the young cluster Trumpler 37, with a companion of about 1.6 $M_\odot$ on a 25.9-d orbit with e = 0.58. Its light rises by about 0.3% and falls by about 1.2% at every periastron. A tidal model fitted to the light curve alone returns Gaia's eccentricity and periastron angle

to within 0.3σ at an inclination of 60–65°, where no eclipse is possible. The residuals at the conjunction limit any eclipse to about 0.1%. The star is not in the heartbeat catalogues we checked (Callahan et al. 2025; Solanki et al. 2025).

TYC 8203-301-1 (TIC 147413927)



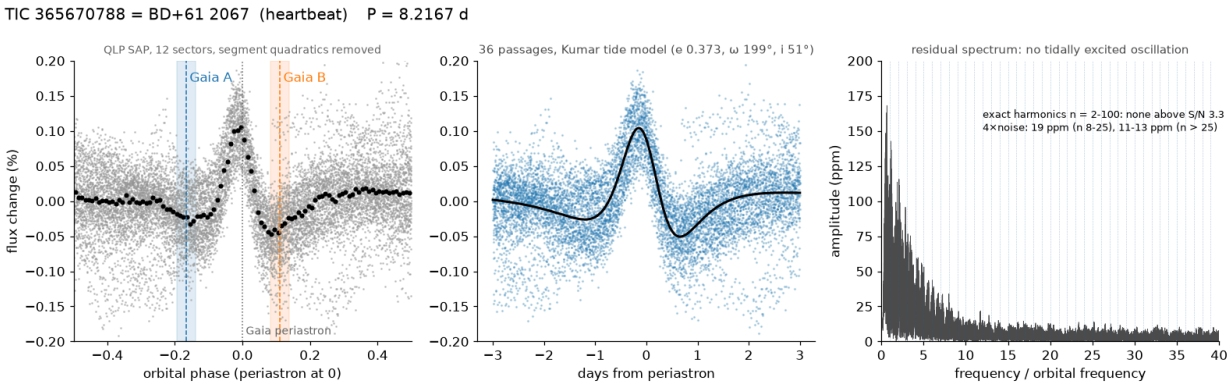
TYC 8203-301-1. Left: QLP SAP flux, detrended with a 3-d running median with the periastra masked, folded on P = 22.0440 d, periastron and Gaia's conjunctions marked. Middle: the heartbeat at each covered periastron (nine passages overlaid) with the Kumar et al. (1995) model ($i \approx 45°$); the oscillation is present in the plotted data. Right: the amplitude spectrum of the out-of-periastron data, with 22, 23 and 24 times the orbital frequency marked; the peak at 23 times the orbital frequency is 817 ppm in this panel, against the fitted 842 ppm given in the entry.

Identifiers	TIC 147413927; Gaia DR3 5362949228883726 08; TYC 8203-301-1
Position (J2000)	10:52:52.3 –47:42:04.8
Brightness	*G* 10.44; Tmag 10.13
Stars	Late-main-sequence star in the γ Dor instability strip: Gaia FLAME R 3.12 $R_\odot$, M 2.07 $M_\odot$, L 28 $L_\odot$; GSP-Phot 7530 K, log *g* 3.67 (TIC 7255 K, 3.19 $R_\odot$, 1.63 $M_\odot$) + companion m_2 = 0.94–1.08 $M_\odot$ at i = 44.6°; a = 45–48.5 $R_\odot$; mass function 0.0440 $M_\odot$; Gaia SB1, K_1 36.6 km/s
Period	22.0440 ± 0.0005 d from the tidal fit (Gaia 22.0393 ± 0.0080 d; 0.6σ); the three oscillation harmonics imply 22.0432, 22.0438, 22.0435 d
Epoch	Periastron t_p (N = 0) = BTJD 399.972 ± 0.078 = BJD_{TDB} 2458399.972 (Gaia 399.948 ± 0.136; 0.15σ)
Periastron signature	Peak 0.47 d before periastron, trough 0.07 d after; peak to trough 7.0–7.8 ppt in QLP; five passages fully covered (s10, s63, s90 twice, s100)
Tidal model *e*, ω vs Gaia	e = 0.670 ± 0.007 against Gaia 0.680 ± 0.029 (0.3σ); ω = 244.5 ± 4.5° against 241.1 ± 4.2° (0.55σ); i = 44.6 ± 4.6°; S = 2.05×10^{-4} against 1.3–1.9×10^{-4} predicted for an equilibrium tide; reduced χ^2 2.9, jackknife errors over the five passages
Oscillations	Main peak 1.0434063 ± 0.0000136 c/d (842 ppm), coherent over 2550 d, at **23.0009 ± 0.0006** times the orbital frequency (+1.4σ from 23); weaker peaks at 22.0002 ± 0.0006 (about 170 ppm) and 24.0006 ± 0.0008 (about 130 ppm): tidally excited oscillations at n = 22 and 24, or orbital sidebands of n = 23; in either case they are locked to the orbit. Other peaks not at harmonics: 0.15–0.19 c/d (γ Dor-like) and 0.644 c/d
Eclipse	None possible: at i = 44.6° an eclipse needs i > 75° (near-periastron conjunction) or > 86° (far); residual boxes at both conjunctions are at the noise level; no eclipse ≳ 0.05%
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s10, s63: peak to trough 6.9–7.7 ppt)
Pixel attribution	TESScut s63 difference image (peak minus trough) is PSF-shaped and on the target, 6.4 ppt in the 3×3 aperture (centroid 0.4 px from the reference, read as jitter on a sub-percent signal); neighbours within 60″ (*G* 15.76–16.85) would need about 100–270% variation
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, WDS, Hipparcos, SB9, CCDM, Gaia DR3 variability tables, ExoFOP, SIMBAD (SB*, one reference), general web search; not in Callahan et al. 2025, the APOGEE heartbeat search (Callahan et al. 2026), Li et al. 2024a,b or Solanki et al. 2025, whose searches cut at $P \lesssim 12$ d

Open issues	The s99/s100 2-min data; a PHOEBE fit with the Gaia RVs; spectroscopy
Next periastra	BTJD 4345.85 (peak 4345.38, trough 4345.92), 4367.90, 4389.94 (±0.12)
Discovery	Conjunction-window search (Section 2.5); confirmed by independent re-analysis with separately written code (Section 2.5)

TYC 8203-301-1 is a star of about 7300–7500 K and 3.1 $R_\odot$ on a 22.04-d orbit with e = 0.68, seen at about 45°. The heartbeat reproduces Gaia's orbit to within 0.6σ. The light curve also carries a coherent 842-ppm oscillation at 23.0009 times the orbital frequency, with weaker peaks at exactly 22 and 24 times it. The three harmonics each imply the orbital period independently, and all three agree with the heartbeat's. An oscillation held at an exact harmonic of the orbit over 2550 d is the signature of a tidally excited oscillation, and that is how we read it. The published TESS heartbeat searches stop at periods of about 12–13 d, which is why a 22-d system would not be in them.

BD+61 2067 (TIC 365670788)



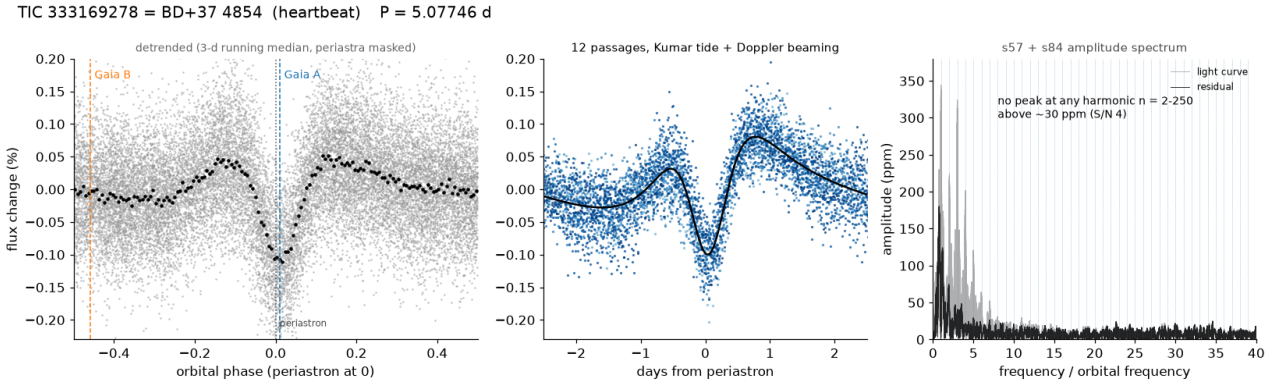
BD+61 2067. Left: QLP SAP flux, 12 sectors, with a quadratic removed per segment, folded on Gaia's P = 8.2167 d, Gaia's periastron and conjunctions marked. Middle: 36 periastron passages overlaid with the Kumar et al. (1995) model (e 0.373, ω 199°, i 51°). Right: the residual amplitude spectrum against frequency in units of the orbital frequency, with the orbital harmonics marked.

Identifiers	TIC 365670788; Gaia DR3 2195886868574308608; BD+61 2067; TYC 4251-1269-1; SAO 19082
Position (Gaia DR3, J2016)	20:53:08.12 +62:10:15.7; 226 pc
Brightness	G 9.01; Tmag 8.65
Stars	F8 (SIMBAD), T_{eff} 6170–6400 K, R_1 2.55–2.65 $R_\odot$, M_1 1.3–1.6 $M_\odot$ + M dwarf, $m_2 \approx$ 0.23–0.26 $M_\odot$ at i = 51°; mass function 0.00244 $M_\odot$; Gaia SB1. A co-moving G 15.38 star 25.3″ away (about 5,700 au) makes it a probable hierarchical triple
Period	8.216284 ± 0.000094 d from twelve per-sector periastron times (Gaia 8.216686 ± 0.000589 d; −0.7σ)
Epoch	Periastron BTJD 2953.1308 ± 0.0083 = BJD_{TDB} 2459953.1308
Periastron signature	A +1.0-ppt hump peaking 0.15 d before periastron and a −0.55-ppt dip 0.66 d after; 1.45–2.14 ppt peak to trough in each of 12 sectors (2019–2024), 36 passages
Tidal model *e*, ω vs Gaia	From the light curve alone, P fixed: e = 0.373 ± 0.017 against Gaia 0.364 ± 0.063 (0.1σ); ω = 198.7 ± 6.5° against 192.3 ± 4.0° (0.9σ, compared mod 180°); i = 51 ± 6°; periastron +0.04 ± 0.05 d from Gaia's. S = 2.65×10^{-4} against 3.1–3.7 × 10^{-4} expected for an equilibrium tide. Errors from a bootstrap over sectors
Eclipse	None: an eclipse needs i > 80°; any eclipse is shallower than 130–150 ppm (3σ) at either conjunction
Oscillations	No tidally excited oscillation: no exact harmonic n = 2–100 above S/N 3.5; limits 19 ppm (n = 8–25) and 11–13 ppm (n > 25)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (12 sectors: e 0.357, ω 198.0°); SPOC 2-min (s57, s58); own TESScut photometry
Pixel attribution	

	The per-pixel heartbeat image is star-shaped and centred on the target to ≤ 0.5″ in s57, s76 and s84. The brightest neighbour within 60″ (the co-moving *G* 15.38 star) would need 58% variability; GJ 809 at 109″ shows none
Catalogue status	Not in VSX, ASAS-SN, SB9, the Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Kirk et al. 2016, Shporer et al. 2016 or Wrona et al. 2022; SIMBAD SB*, 13 references, none on variability
Open issues	ω compared mod 180° only (a beaming term does not choose the branch here); a joint RV and light-curve fit; RVs of the wide companion
Next periastra	BTJD 4333.467, 4341.683, 4349.899 (±0.018); no eclipse expected
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

BD+61 2067 is an F8 star of about 2.6 $R_\odot$ with an M-dwarf companion on an 8.2-d orbit with e = 0.37. At every periastron its light rises by 0.1% and falls by half that, in all twelve TESS sectors from 2019 to 2024. A tidal model fitted to the light curve alone returns Gaia's eccentricity and periastron angle within 1σ, at an inclination where no eclipse is possible. It is the clearest of the four.

BD+37 4854 (TIC 333169278)



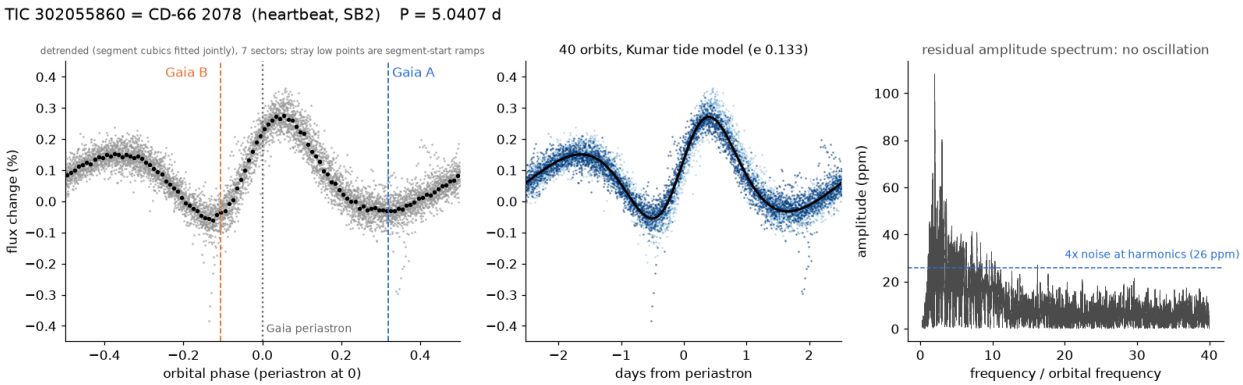
BD+37 4854. Left: the detrended QLP light curve folded on Gaia's P = 5.0776 d, periastron and Gaia's conjunctions marked. Middle: 12 periastron passages with the tide-plus-beaming model. Right: the residual amplitude spectrum (s57 and s84) in units of the orbital frequency.

Identifiers	TIC 333169278; Gaia DR3 1919980437220007296; BD+37 4854; TYC 3231-1540-1; SAO 73302
Position (J2000)	23:30:10.38 +38:24:56.0; 253 pc
Brightness	*G* 9.68; Tmag 9.38
Stars	F2 (SIMBAD), T_{eff} 6670–7280 K, R_1 1.8–1.95 $R_\odot$, M_1 1.5–1.6 $M_\odot$ + M dwarf, $m_2 \approx$ 0.48–0.51 $M_\odot$ at i = 51.5°; mass function 0.0140 $M_\odot$; Gaia SB1, K_1 31.5 km s^{-1}. RUWE 4.2 is far more than this orbit can cause, so a third body is possible
Period	5.077463 ± 0.000021 d from Gaia's periastron and three TESS sectors (Gaia 5.077644 ± 0.000140 d; −1.3σ)
Epoch	Periastron BTJD 2870.807 ± 0.008 = BJD_{TDB} 2459870.807
Periastron signature	An “M”: a hump 0.53 d before periastron, a dip 0.04 d after it on Gaia's A conjunction, and a higher hump 0.77 d after; peak to trough 1540–1720 ppm in QLP s16, s57, s84 and 1570–1780 ppm in TESS-SPOC; present at every covered periastron
Tidal model *e*, ω vs Gaia	Tide plus Doppler beaming, from the light curve alone: e = 0.304 ± 0.009 against Gaia 0.315 ± 0.010 (−0.8σ); ω = 79.5 ± 2.7° against 81.7 ± 2.2° (−0.6σ, not mod 180°); i = 51.5 ± 6.4°; S = 3.9 ± 0.8 × 10^{-4} (3.9–5.8 expected); beaming 3.4 ± 0.5 × 10^{-4} (2.9 expected); χ^2/dof 2.1. The pure tide returns the mirror angle 100 ± 4°; adding beaming lowers χ^2 by 1807 and an injection reproduces the mirror
Eclipse	None: an eclipse needs i > 77°; any eclipse is ≤ 160 ppm at A and ≤ 100 ppm at B (3σ)
Oscillations	No tidally excited oscillation (limit about 30 ppm at harmonics n = 2–250); no δ Sct or γ Dor pulsation

Pipelines agreeing	QLP SAP; TESS-SPOC SAP (*e* 0.307, ω 79.7°) and PDCSAP; own TESScut photometry
Pixel attribution	TESScut hump-minus-dip images: the 3×3 core changes by 1380–1470 ppm in s16, s57, s84 against 1395 expected, PSF-shaped, centroid ≤ 0.16 px; the brightest neighbour (*G* 13.93, 31.5″) would need 7% and its pixels do not vary
Catalogue status	Not in VSX, ASAS-SN, SB9, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Wrona et al. 2022, Kirk et al. 2016, Shporer et al. 2016 or the Gaia DR3 variability tables; SIMBAD SB*, 4 references
Open issues	A joint RV and light-curve fit; the cause of RUWE 4.2; inclination loosely set; four sectors, one dominated by scattered light
Next periastra	BTJD 4328.041, 4333.118, 4338.196 (±0.008)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

BD+37 4854 is an F2 star with an M-dwarf companion on a 5.08-d orbit with e = 0.31. Its periastron pulse is an M-shape of 0.17%. Near ω = 90° a pure tidal model cannot tell the true periastron angle from its mirror image and picks the mirror. Doppler beaming of the F star, whose size the Gaia velocity fixes, is antisymmetric about periastron and settles it: with that term the light curve returns Gaia's ω to 0.6σ and its e to 0.8σ.

CD-66 2078 (TIC 302055860)



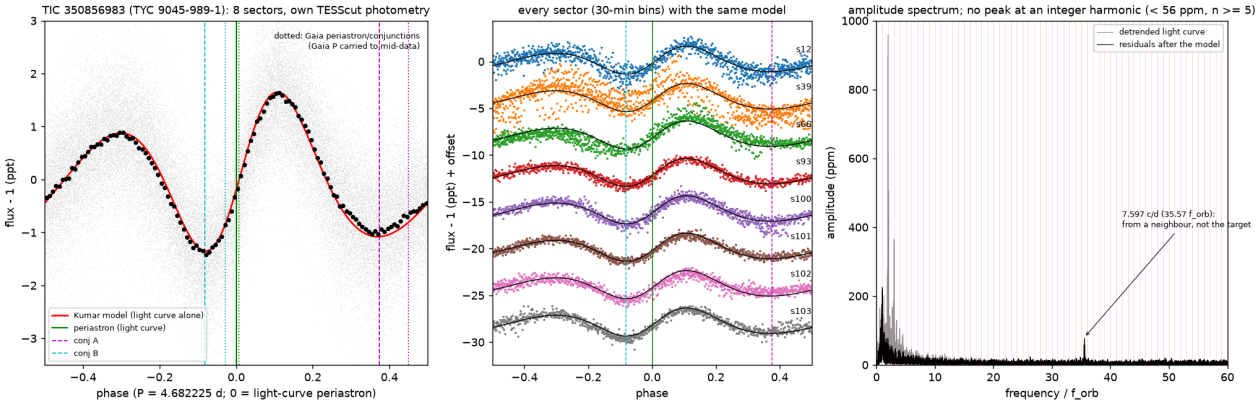
CD-66 2078. Left: QLP SAP flux with baselines removed, folded on Gaia's P = 5.0407 d, Gaia's periastron and conjunctions marked. Middle: 40 orbits against time from periastron, with the model. Right: the residual amplitude spectrum in units of the orbital frequency, with the 26-ppm limit.

Identifiers	TIC 302055860; Gaia DR3 5813326715771172992; CD-66 2078; CPD-66 3086; TYC 9064-655-1
Position (Gaia DR3, J2016)	17:20:15.59 −66:45:26.2; 364 pc
Brightness	*G* 9.90; Tmag 9.55
Stars	Two near-twin F stars (Gaia two-star fit 6338 and 6291 K), each about 1.9 $R_\odot$ and 1.4–1.5 $M_\odot$; Gaia **SB2**, K_1 80.2, K_2 73.5 km s^{-1}, M $\sin^3 i$ 0.88 and 0.96 $M_\odot$, so i ≈ 56–63°
Period	5.040480 ± 0.000030 d (TESS; Gaia 5.040691 ± 0.000063 d, −3.0σ, or −1.3σ with Gaia's error scaled by its χ^2/dof)
Epoch	Periastron BTJD 3412.468 ± 0.021 = BJD_{TDB} 2460412.468
Periastron signature	A hump 0.21 d after periastron and dips at both conjunctions, the deeper at B; 3.05–3.76 ppt peak to trough in each of 7 sectors
Tidal model *e*, ω vs Gaia	e = 0.134 ± 0.005 (stat) ± 0.004 (sys) against Gaia 0.152 ± 0.003 (−2.6σ; −1.8σ with Gaia's error scaled by its χ^2/dof of 6.2); ω = 137.8 ± 2.9° (mod 180°) against 140.7 ± 1.8° (0.9σ). The pure tide wants i = 40° against 56–63° from the masses; adding irradiation of each twin by the other fits at i = 59° with the expected tidal amplitude and e 0.130
Eclipse	None: an eclipse needs i > 75°; any eclipse is shallower than about 75 ppm
Oscillations	No tidally excited oscillation above 26 ppm (harmonics n = 9–60); nothing above S/N 4 to 80 c/d

Pipelines agreeing	QLP SAP (*e* 0.134); SPOC 2-min s12–13 (0.142 SAP, 0.132 PDCSAP); TESS-SPOC (0.139); own TESScut regression
Pixel attribution	TESScut template regression in s13, s39, s101: PSF-shaped, centred 0.9–1.7″ from the target in the 200-s and 10-min sectors; the brightest neighbour within 60″ (*G* 16.11) would need 98%
Catalogue status	Not in VSX, ASAS-SN, SB9, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, the Gaia DR3 variability tables, Solanki et al. 2025, Callahan et al. 2025, Li et al. 2024a,b, Wrona et al. 2022, Kirk et al. 2016 or Shporer et al. 2016; SIMBAD SB*, 1 reference
Open issues	The *e* tension with Gaia, whose SB2 solution has χ^2/dof 6.2 and a period 3σ off; independent RVs or Gaia DR4 epoch velocities would settle it
Next periastra	BTJD 4329.835, 4334.876, 4339.916 (±0.022)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; confirmed by independent re-analysis with separately written code

CD-66 2078 is a pair of near-twin F stars on a 5.04-d orbit with e = 0.15, which Gaia resolved as a double-lined binary. Its 0.3% periastron pulse is in all seven TESS sectors and on the target. The light curve returns Gaia's periastron angle within 1σ and an eccentricity 0.018 lower, a gap typical of heartbeat fits against Gaia orbits (Callahan et al. 2025) and within twice Gaia's error once that is scaled to its own fit quality.

TYC 9045-989-1 (TIC 350856983)



TYC 9045-989-1. Left: own TESScut photometry folded on the photometric period 4.6822 d with the model, light-curve and Gaia periastron and conjunctions marked. Middle: all eight sectors with the same model. Right: the amplitude spectrum in units of the orbital frequency.

Identifiers	TIC 350856983; Gaia DR3 5828538665312585856; TYC 9045-989-1; 2MASS J16281728-6345236
Position (Gaia DR3, J2016)	16:28:17.26 −63:45:24.1; 395 pc
Brightness	*G* 9.72; Tmag 9.37
Stars	Two near-twin F stars (q = 0.95), T_{eff} ≈ 6400–6600 K, each about 2.3–2.5 $R_\odot$; Gaia **SB2**, M $\sin^3 i$ 0.82 and 0.78 $M_\odot$
Period	4.682234 ± 0.000021 d (photometric, eight sectors; Gaia 4.682539 ± 0.000076 d, 4σ long)
Epoch	Light minima at conjunction A BTJD 3800.117 and B 3802.654 (±0.003); periastron 3798.36 ± 0.10
Periastron signature	A dip of 1.4 ppt at conjunction B, a hump of 1.65 ppt 0.52 d after periastron and a second dip at A; 2.8–3.6 ppt peak to trough in every sector; the second orbital harmonic (ellipsoidal) is 1.0–1.2 ppt
Tidal model *e*, ω vs Gaia	e = 0.109 ± 0.005 against Gaia 0.119 ± 0.006 (−1.3σ); ω = 127 ± 8° (mod 180°) against 106.6 ± 3.3°: **+20°, 2.4σ**, the same in three independent data sets and with the period free; not the mirror branch, beaming, reflection or apsidal motion. Carried back with the photometric period, the B conjunction lands on Gaia's to 0.3 h
Eclipse	None: an eclipse needs i > 68° against the fitted 40–44°; limit about 200 ppm
Oscillations	

	None at the orbital harmonics (limit 56 ppm for n ≥ 5); a 7.597 c/d signal belongs to a neighbour, since it grows with the aperture
Pipelines agreeing	QLP SAP, TESS-SPOC SAP and PDCSAP, TARS, eleanor-lite, own TESScut photometry
Pixel attribution	TESScut difference images in five sectors have the target's PSF shape; the amplitude does not depend on aperture size; the brightest neighbour within 60″ (*G* 15.28) would need 49%
Catalogue status	In no variable-star, eclipsing-binary or heartbeat catalogue checked (as for H6). Callahan et al. (2025) inspected every Gaia DR3 SB2 and did not list it
Open issues	ω 20° from Gaia's; at e = 0.12 the signal is largely ellipsoidal; the inclination from the amplitude implies masses heavy for the luminosity; an RV orbit would settle all three
Next periastra	BTJD 4327.45, 4332.14, 4336.82 (±0.10); conjunction B minima 4331.747, 4336.429 (±0.01)
Discovery	Gaia-orbit fold at P < 20 d, G 9–10 (the continuation of Section 2.4), set aside as a non-eclipse feature and found locked to Gaia's periastron; kept as a candidate by independent re-analysis with separately written code
Status	Candidate. The pulse is on the target, in every sector and locked to Gaia's conjunctions, and e agrees, but ω does not

TYC 9045-989-1 is a pair of near-twin F stars on a 4.68-d orbit with e = 0.12. Its light curve is mostly the ellipsoidal variation of the distorted stars, with a periastron asymmetry of a few tenths of a part per thousand on top. The asymmetry is locked to Gaia's orbit and returns its eccentricity, but places periastron 20° from where Gaia does. Gaia's solution has a poor fit and a period 4σ off, so the discrepancy may be Gaia's; until an RV orbit settles it, the star is a candidate.

S3.2 Eclipses on edge-on Gaia astrometric orbits (A19, A72–A74)

A19 TIC 219766411 = Gaia DR3 1440245818754478976

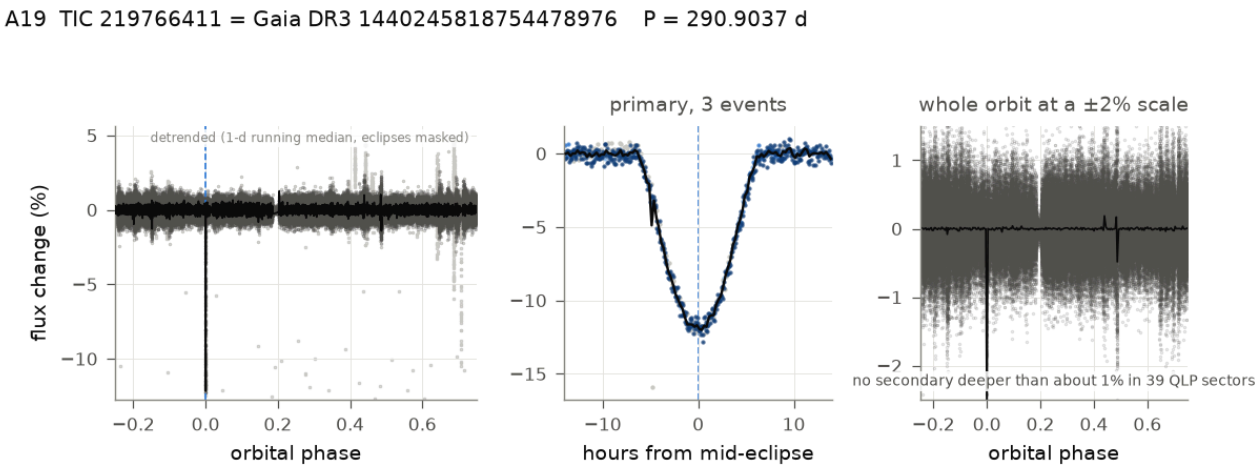


Figure A19. TIC 219766411: the three ~125-ppt, ~7-h eclipses at BTJD 1936.36 (sector 23), 2809.07 (sector 55, in flagged cadences) and 3390.88 (sector 76), on P = 290.9037 d. Left: the flux, detrended with a 1-d running median with the eclipses masked, folded on P. Middle: the primary eclipse (3 events) against hours from mid-eclipse. Right: the whole orbit at a ±2% scale; no secondary deeper than about 1% in 39 QLP sectors.

Identifiers	TIC 219766411; Gaia DR3 1440245818754478976
Brightness	Tmag 12.39 (G 12.8)
Stars	TIC: T_{eff} 6190 K, 1.26 $R_\odot$; a faint companion (secondary eclipse < ~1%)
Period	290.9037 ± 0.0006 d (formal, three events) (Gaia astrometric: 292.7769 ± 2.7548 d; agreement 0.7σ)
Epoch	BTJD 1936.3573 ± 0.0028 = BJD_{TDB} 2458936.3573
Primary eclipse	124, 127 and 125 ppt (Gaussian fits, raw QLP SAP), FWHM ~7 h; O–C −21 ± 9, +2 ± 2, −1 ± 1 min (sector 23 is 30-min cadence)
Secondary eclipse	None deeper than about 1% in hourly bins over all 39 QLP sectors, flagged cadences included

Orbit	Gaia: i = 91.190° (derived from Thiele–Innes), e = 0.418, ω = 15° (mod 180°). The eclipses fall 26 d (68%: 16–36 d) before the nearer predicted conjunction; 0.6% of Monte Carlo draws reach zero, about 2.5σ
Pipelines agreeing	QLP SAP (three events); TESS-SPOC recovers the sector-23 event at 85 ppt (no TESS-SPOC products for the other two sectors)
Pixel attribution	Sector-23 difference image: deficit 0.19 TESS pixel from the target; no other Gaia source within 30″
Catalogue status	Not listed in: VSX, Prša et al. (2022), IJspeert et al. (2024), Kostov tables 3 and 4, or the eccentric-EB catalogue of Tang et al. (2025; J/AJ/170/356), with no row within 30″; no Gaia variability or EB row; SIMBAD "*". Kostov table 2 only
Open issues	The 2.5σ conjunction offset, recorded as a caveat on Gaia's error model rather than as confirmation. Spectroscopy (an SB1 at Gaia's period with a consistent K) would be decisive
Next eclipse	BTJD 4554.491 ± 0.005 (2027 May 28), computed here from the three-event ephemeris (formal error; the three-event fit has $\chi^2 \approx 7.4$ for one degree of freedom, so the true error is about 2.7 times larger)
Method	Edge-on astrometric-orbit search (Section 2.6); third event and Gaia-covariance check by independent re-analysis (Section 2.5)
Status	Probable. The period agrees with Gaia's and the events repeat, but the conjunction is 2.5σ off, the sector-55 event exists only in flagged cadences and the sector-76 event only in QLP, the one independent depth (TESS-SPOC, sector 23) is 30% shallower than QLP's, the neighbour check reaches 30″ only, and the independent re-analysis itself names spectroscopy as the decisive test. The same protocol passed TIC 316045876, since withdrawn (Section 3.8)

The third event was in sector 55, where every in-eclipse cadence carries QLP quality flag 4096. A search restricted to QUALITY = 0 shows only two events. Gaia's period agrees to 0.7σ and the eclipses repeat three times. We therefore interpret the 26-d offset as a measurement of Gaia's error model for a G = 12.8 star over 34 months rather than as evidence of a different orbit. The timing should not be cited as confirmation.

A72 TIC 279636729 = Gaia DR3 4720373709202359808

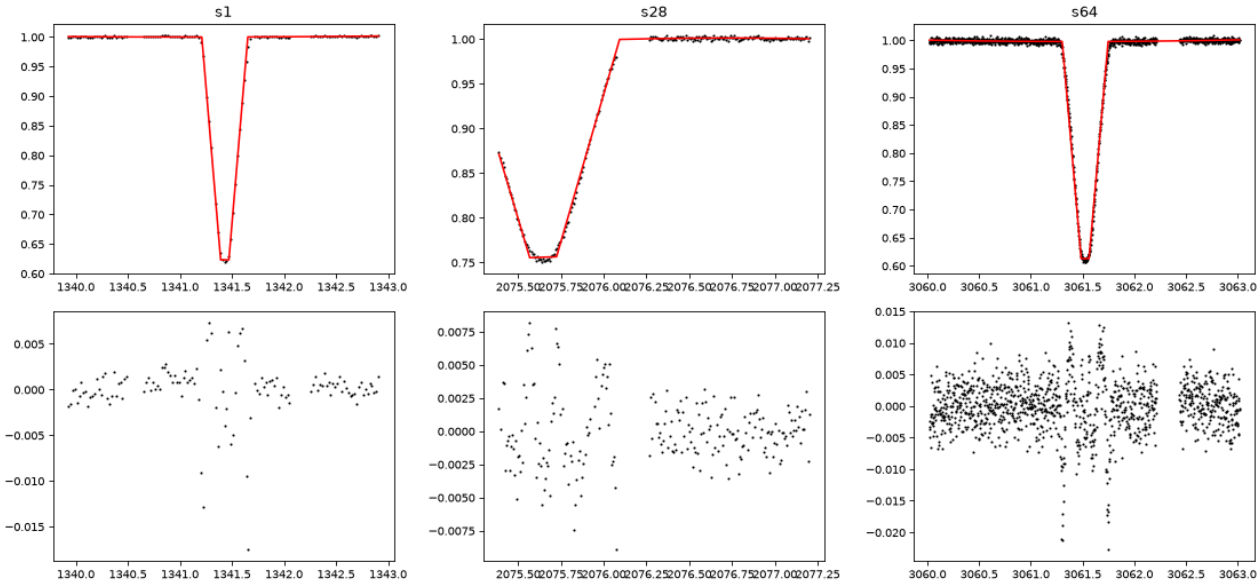


Figure A72. TIC 279636729, raw QLP SAP flux with trapezoid fits (red; top) and residuals (bottom) for sectors 1, 28 and 64 (left to right): the 38% primaries in sectors 1 and 64 and the 24%, 21-h secondary in sector 28, whose data begin part-way through ingress, just after a mid-sector gap. The third primary, in sector 96 under scattered-light flags, is not shown.

Identifiers	TIC 279636729; Gaia DR3 4720373709202359808
Position (J2000)	48.475926, −64.367416 deg (TIC)

Brightness	Tmag 12.01; G 12.43
Stars	F-type (TIC: T_{eff} 6081–6213 K, 1.3–1.4 $R_\odot$, ~1.2 $M_\odot$) at 560 pc with a companion of comparable size and temperature supplying 20–25% of the light (from the depths and the Gaia photocentre orbit, a_0 = 0.351 mas); total mass ~2.2 $M_\odot$, a ≈ 0.91 AU
Period	215.0116 ± 0.0002 d from three primaries over 12 cycles (χ^2 0.01); Gaia astrometric `Orbital` solution 217.22 ± 2.04 d, agreement at −1.1σ. P/2 and P/4 are excluded by flat data at their predicted times; P/3 is excluded only by Gaia's period
Epochs	Primary T_0 = BTJD 1341.4321 ± 0.0009 = BJD_{TDB} 2458341.4321; primaries at 1341.4321, 3061.5252 and 3921.565–3921.576 (O−C 0.0, +0.1, −0.9 min); secondary at 2075.6456 ± 0.0012, phase 0.4148
Primary eclipse	37.7–38.8%, total duration 10.6 h, ingress 4.4 h, flat bottom 1.9 h
Secondary eclipse	24.2%, 21.4 h, at phase 0.4148: e cos ω between −0.14 and −0.06; the duration ratio of 2.0 gives e sin ω ≈ 0.34, so e ≈ 0.36, ω ≈ 111°, against Gaia's e = 0.40 ± 0.11 and $\lvert e \cos \omega \rvert$ at Gaia's 84th percentile: agreement at about 1σ
Predicted event	Sectors 1 and 64 alone predict the sector-96 primary at 3921.5717; every QLP cadence there is flagged on the tail of a scattered-light episode, and the raw flux holds the eclipse bottom and a clean egress at 3921.565–3921.576 with the sector-64 shape. The seven other scattered-light episodes of that sector, aligned on their end, show the flux already recovered where this one sits at 0.6; custom cutout photometry (this work) with a separately estimated background reproduces the eclipse
Pixel attribution	Difference images from TESScut cutouts for all four events: deficit centroid within 0.02–0.2 px of the Gaia position each time, while the out-of-eclipse light centroid is pulled 0.2–0.6 px toward the G 13.09 neighbour at 41″, whose own pixel loses 0.4–4.8% (it would need 71%). The only other star within 60″ (G 17.07) would need 2768%
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability and eclipsing-binary tables, SB9, WDS, ExoFOP, SIMBAD, general web search; ADS full text (TIC and Gaia identifiers) none. Kostov table 2 only
Gaia conjunctions	The primaries fall 2–29 d before Gaia's predicted conjunction, the spread growing with cycle because Gaia's period is 2.2 d longer than the eclipse period; consistent within the Monte Carlo ranges. The test is weak
Next eclipses	Primary BTJD 4351.595 ± 0.002 (2026-11-07), then 4566.607, 4781.618; secondary 4440.773 ± 0.003 (rests on one event)
Method	Short-period search of the edge-on astrometric sample (Section 2.6); confirmed by independent re-analysis (Section 2.5)

Both stars are luminous: a 38% primary needs the eclipsed star to supply at least 39% of the light, and one impact parameter of about 0.7 fits both the 10.6-h primary and the 21-h secondary once the eccentric geometry is allowed for. The star should be double-lined, and with the astrometric orbit a spectroscopic one would give the masses directly. The next secondary, in late February 2027, would fix e cos ω. In the period-window search of Section 2.6 the system scored z = 2.6, below the detection threshold.

A73 TYC 4590-364-1 = TIC 315721826

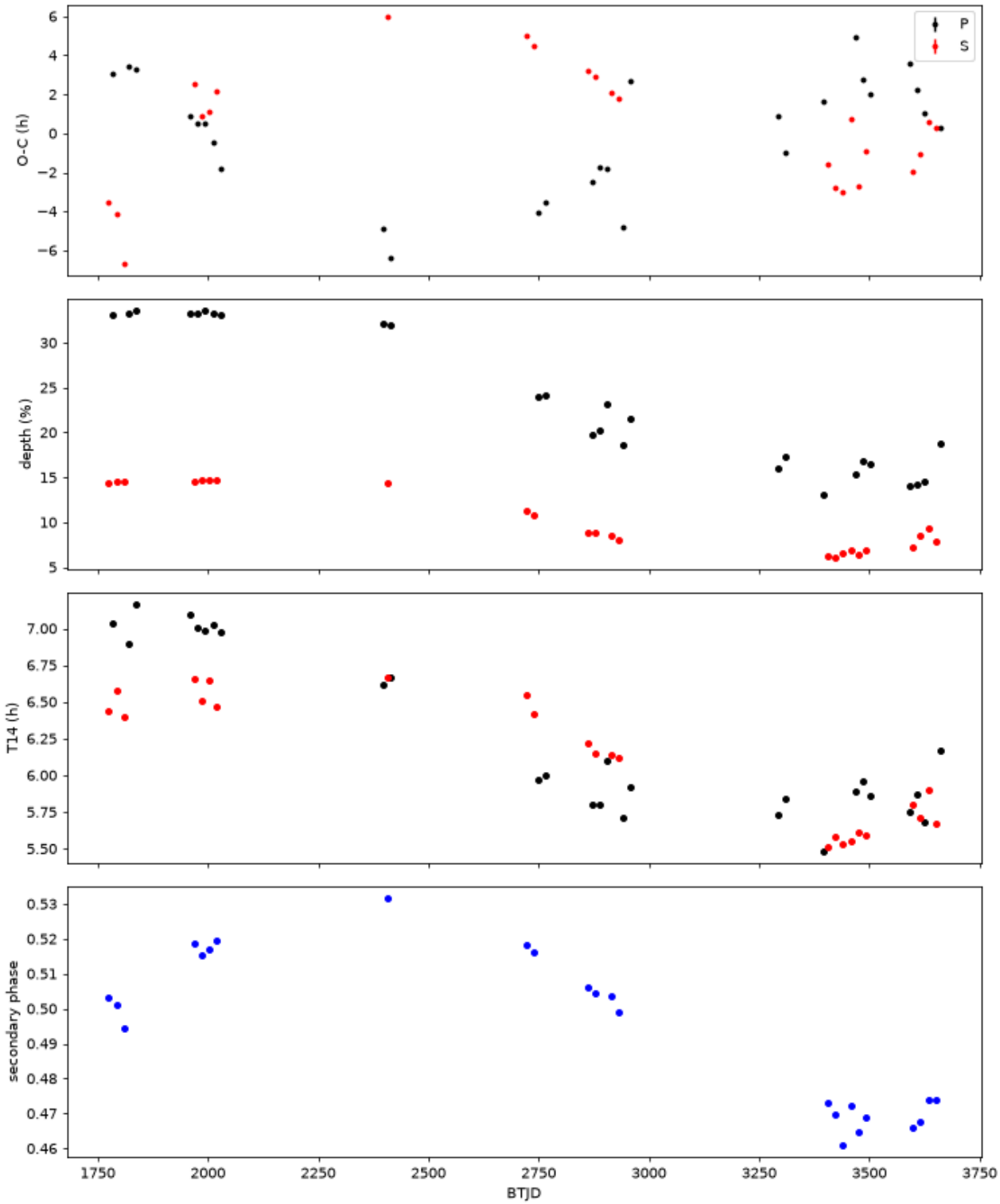


Figure A73. TYC 4590-364-1, from fits to 27 primaries (black) and 24 secondaries (red) in 21 QLP sectors, 2019–2024, from top to bottom: timing residuals against a linear 17.54-d ephemeris, eclipse depths, total durations and the phase of the secondary (blue). The eclipses fade from 33% to 13% and narrow, the timings vary by hours, and the secondary's phase rises then falls.

Identifiers	TIC 315721826; TYC 4590-364-1; Gaia DR3 (RUWE 1.93)
Position (J2000)	310.554255, +78.340072 deg (TIC)
Brightness	Tmag 10.81; G 11.28; V ≈ 11.6
Stars	

	TIC: T_{eff} 5932 K, 1.46 $R_\odot$, at 282 pc (parallax 3.55 mas), BP−RP 0.90; inner pair of about 2 $M_\odot$ in total, $a_{in} \approx 0.167$ AU. Gaia astrometric `Orbital` solution: $P_{out} = 124.72 \pm 0.24$ d, $e_{out} = 0.41 \pm 0.08$, $a_0 = 0.719$ mas, implying a third body of about 0.9 $M_\odot$ if dark
Period	17.538 d (mean of the primaries; the secondaries give 17.531 d because $e \cos \omega$ is changing); single-cycle intervals range from 17.386 to 17.851 d. Not Gaia's period: the eclipsing pair is the inner binary of a triple whose outer orbit Gaia measured
Epochs	27 primaries and 24 secondaries fitted in sectors 17–86, median timing error 0.8 min; against a linear ephemeris the primaries' O−C spans −6.4 to +5.0 h (rms 2.9 h), the secondaries' −6.7 to +6.0 h. The largest single jump, 17.851 d between the clean eclipses at BTJD 2940.800 and 2958.651, is real
Primary eclipse	Fading: 33.1–33.6% (2019–20), 32.0% (2021), 24.0% (2022), 18.6–20.2% (late 2022), 12.7–17% (sectors 73–80, December 2023 to July 2024), 14.1–14.9% (sectors 84–86, late 2024); total duration 7.0 h falling to 5.5–5.9 h
Secondary eclipse	14.4–14.8% falling to 6–9%; its phase 0.500 (2019), 0.515–0.520 (2020), 0.517 (2022), 0.504–0.509 (late 2022), 0.465–0.494 (sectors 77–80), 0.466–0.474 (sectors 84–86): $e \cos \omega$ rises then falls by about 0.05
Attribution of the fading	Custom cutout photometry (this work) at an early, a middle and a late epoch gives primary/secondary depths of 31.7/14.1% (sector 19), 18.8/8.6% (58) and 12.3/6.3% (79), the same within 0.5% for 1×1 to 7×7-pixel apertures; the star's flux against a fixed reference star stays constant (ratios 1.856, 1.954, 1.843), so no added light dilutes it, and the eclipses narrow, which dilution would not do
Pixel attribution	Difference images at sectors 19, 58 and 79: deficit centroid within 0.6″ of the light centroid, depths at the target pixel 31.5%, 18.3%, 12.2%. The nearest Gaia source (G 17.95 at 8.5″) would need an eclipse 150 times deeper than its whole light; nothing brighter than G 17.8 within 60″
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability and eclipsing-binary tables, SB9, WDS, ExoFOP, general web search; SIMBAD has no references; ADS full text (TIC and Tycho identifiers) none. Kostov table 2 only
Third body	Required: timing variations of hours that change cycle to cycle, $e \cos \omega$ changing by 0.03–0.05 in a year, and the depth and width falling together with no change in total flux (an inclination or conjunction-geometry change); relativity and tides are orders of magnitude too slow. Gaia's 124.7-d companion fits the scale (expected dynamical timing amplitude ~4 h against the observed rms of 3 h; light-travel time only ~2 min; nodal precession on ~8 yr/cos i_{mut}) but is in tension with stability: $a_{out}/a_{in} \approx 4.2$ against the stability limit of Mardling & Aarseth (2001), 6–7 at e_{out} 0.41. The perturbations do not localise to Gaia's periastron. A third body is required; its identification with Gaia's companion is not established
Next eclipses	Primaries BTJD 4344.09, 4361.62, 4379.16; secondaries 4334.60, 4352.13 (±0.3 d because of the timing variations)
Method	Short-period search of the edge-on astrometric sample (Section 2.6); confirmed by independent re-analysis (Section 2.5)

A detached 17.5-d pair whose eclipses are disappearing within the TESS era is rare in itself. Two things would resolve its nature. The first is a photodynamical fit of the times, depths and widths, with Gaia's outer orbit first as a prior and then left free; the fit must reproduce the jump at BTJD 2958 and remain stable. The second is spectroscopy of the systemic velocity over about 125 d. Before TESS, ASAS-SN V photometry (527 epochs, 2014–2018, one or two points per eclipse) catches the primary eclipse in every season, 11–20 h after the TESS-era linear ephemeris predicts and at least 33% deep in June 2018, so the fade began after mid-2018. Whether the eclipses were deeper or shallower in 2014–2016, and so whether the fade is periodic, those data cannot say. Harvard plates (DASCH DR7, 1894–1990; 1,939 after cleaning) show no eclipse signal at any period; eclipses of the 2019 depth injected into the plates are recovered 60–93% of the time and those of the 2023 depth never, so the plates disfavour a century at the 2019 depth and say nothing about the shallower state. The late-2024 depths may show the start of a rebound. The period-window search of Section 2.6 detected the system at an alias period of 125.27 d.

A74 TIC 427345302 = Gaia DR3 2226943914530555776

A74 TIC 427345302 = Gaia DR3 2226943914530555776 P = 173.8755 d

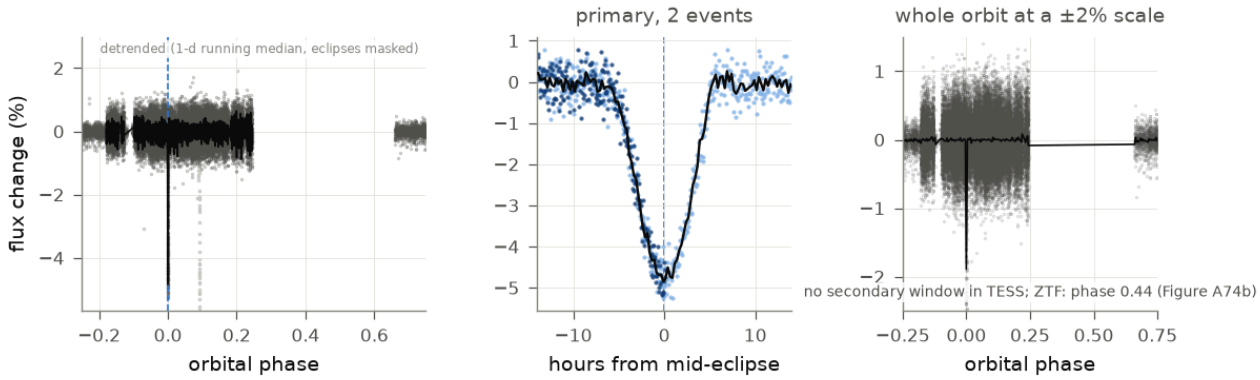


Figure A74. TIC 427345302: the two 4.6%, 10.8-h grazing eclipses at BTJD 2884.986 (sector 58) and 3406.613 (sector 77), three cycles of 173.8755 d apart. Left: the QLP SAP flux of nine sectors, detrended with a 1-d running median with the eclipses masked, folded on *P*. Middle: the two events against hours from mid-eclipse. Right: the whole orbit; no TESS sector covers the secondary window.

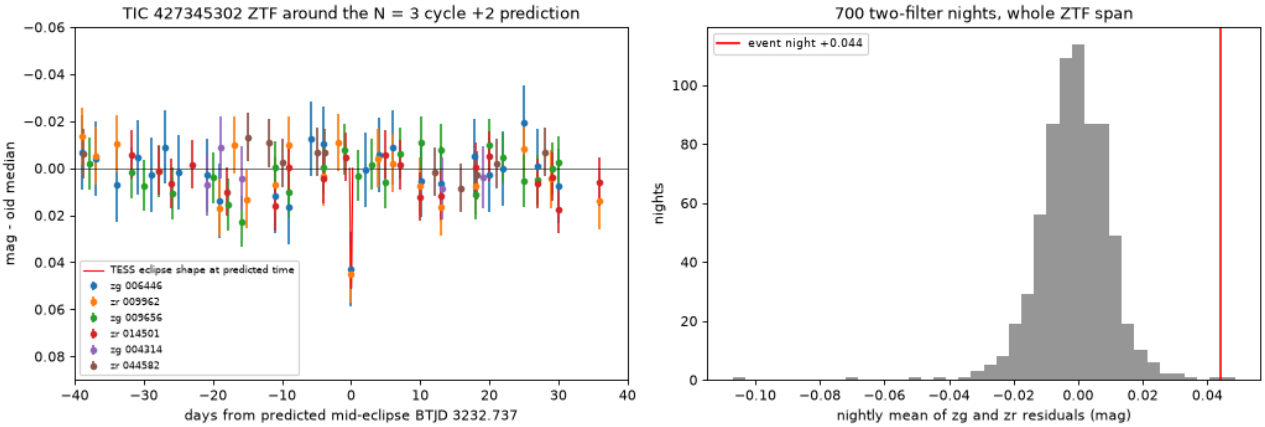


Figure A74b. Left: ZTF g and r photometry of TIC 427345302 within 40 d of the predicted mid-time of the eclipse between the two TESS events, BTJD 3232.737 (2023 October 15), each point a ZTF epoch relative to its object's median, with the TESS eclipse shape at the predicted time. Right: the distribution of the nightly mean of the g and r residuals over the 700 ZTF nights with both filters; the event night is the red line.

Identifiers	TIC 427345302; Gaia DR3 2226943914530555776 (RUWE 1.86); no other catalogue name
Position (J2000)	344.194757, +71.598251 deg (TIC)
Brightness	Tmag 12.10; G 12.55
Stars	TIC: T_{eff} 6168 K, 1.40 $R_\odot$, 1.18 $M_\odot$, at 540 pc (parallax 1.85 mas). Gaia astrometric `Orbital` solution: P = 173.19 ± 0.87 d, e = 0.062 ± 0.078, a_0 = 0.401 mas (0.217 AU), significance 25
Period	173.8755 ± 0.0005 d: the interval between the two TESS eclipses divided by three. TESS alone allowed 1, 2, 3, 5 or 6 cycles between them (the other divisors put an eclipse in flat observed data); ZTF excludes 5 and 6 (flat on predicted nights) and shows the eclipse at the 3-cycle prediction, while 1 and 2 cycles predict nothing on that night. Gaia 173.19 ± 0.87 d (+0.8σ)
Epochs	BTJD 2884.9860 ± 0.0013 = BJD_{TDB} 2459884.9860 (sector 58); 3406.6126 ± 0.0006 (sector 77)
Eclipse	4.62% (sector 58) and 4.6–5.3% (sector 77) deep, T_{14} 10.8 h, ingress about 4.2 h (T_{12}/T_{14} ≈ 0.4), V-shaped: grazing. Both events fully covered; no flagged cadences at the first; the second ends 0.3 d before a 10-d gap. No other dip deeper than 3% in nine sectors
Third eclipse (ZTF)	

	On the night BTJD 3232.68–3232.72 (2023 October 15), with mid-eclipse predicted at 3232.737 ± 0.002, ZTF reads the star 0.043 ± 0.016 mag faint in g and 0.045 ± 0.012 mag faint in r (0.051 and 0.048 expected from the TESS shape); the nights 21 h before and 23 h after are at baseline. Among the 700 ZTF nights with both filters, one other is as faint, so the chance that the one two-filter night inside a predicted window is this faint by accident is 0.1–0.3%. ZTF resolves the star from every neighbour (nearest G 15.4 at 40″). Comparison stars from the same exposures were not checked (the IRSA service failed during the check); that is the item that keeps the status at probable
Secondary (ZTF)	The only other faint ZTF nights, BTJD 1743.9 (five epochs, 0.04–0.06 mag, two fields and both filters) and 2091.9 (four epochs, 0.03–0.07 mag), are 2.00 cycles apart at phase 0.438, inside Gaia's predicted secondary window (phase 0.38–0.50 at 95%); the nights around them are at baseline. A secondary as deep as the primary, about 5% in g. Not a tested prediction: it was found in the same archival check
Orbit	The secondary phase gives $e \cos \omega \approx -0.10$ ($e \geq 0.10$; Gaia 0.062 ± 0.078). The TESS eclipses fall at Gaia's second conjunction (offsets +13 and +15 d against an uncertainty of −4 to +31 d)
Pixel attribution	TESS difference images of both events: 4.7% and 4.5% at the target pixel, deficit centroid within 0.15 px of the target; the neighbours that could host it (G 14.22 at 59″, G 15.41 at 40″) lose 0.1–0.7% where they would need 21% and 64%. The ZTF dimming is of the resolved target
Catalogue status	Not listed in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP, general web search; SIMBAD has no entry; ADS full text (TIC and Gaia identifiers) none. Kostov table 2 only. A TESS-SPOC data-validation report for sector 52 carries a threshold-crossing event at 1.143 d, 0.3% deep, unrelated to the eclipses (Table A1)
Open issues	The companion. For a dark companion the photocentre orbit gives $m_2 \approx 0.5\ M_\odot$, but a secondary as deep as the primary needs a companion of comparable surface brightness, so the photocentre understates the orbit and the companion is more massive; a near-equal pair grazing each other at both conjunctions would give the similar depths seen. Not modelled here
Next eclipses	Primaries BTJD 4449.87 (2027-02-13), 4623.74 (2027-08-06); secondaries 4352.11 (2026-11-07), 4525.98 (2027-04-30); ±0.05 d. No TESS sector covers the star before 2027; at Dec +71.6° the November 2026 secondary is observable from northern sites
Method	Short-period search of the edge-on astrometric sample (Section 2.6; the inspection of the deepest single events); checked by independent re-analysis (Section 2.5); third eclipse from ZTF archival photometry

Note: probable. The third eclipse rests on one ZTF night whose comparison stars have not been checked, and the secondary identification comes from the same archival check rather than a tested prediction. One clean eclipse of either kind from the ground would make the system confirmed.

An F-type star with a 173-d Gaia astrometric orbit seen nearly edge-on shows two 4.6% grazing eclipses in TESS, three cycles apart. ZTF, read at the predicted time of the cycle between them, shows the star faint by the same amount for one night. The two TESS events alone allowed five spacings: one, two, three, five or six cycles between them. The ZTF record is flat on the nights that five or six cycles predict, which removes those two. The ZTF eclipse falls on the night that three cycles predict; one or two cycles predict nothing on that night. The same ZTF data hold two nights of a secondary as deep as the primary, at one phase, 0.438. A secondary that deep means the companion is a star of comparable surface brightness, not the faint dwarf that a dark-companion reading of the photocentre would give. The system is 12th magnitude with 5% eclipses lasting 11 h, in reach of a small telescope. The next secondary falls on 2026-11-07 and the next primary on 2027-02-13.

S3.3 Eclipsing systems from the Gaia-orbit period folds (A5–A12,

A5 HD 106770 = TIC 22388285

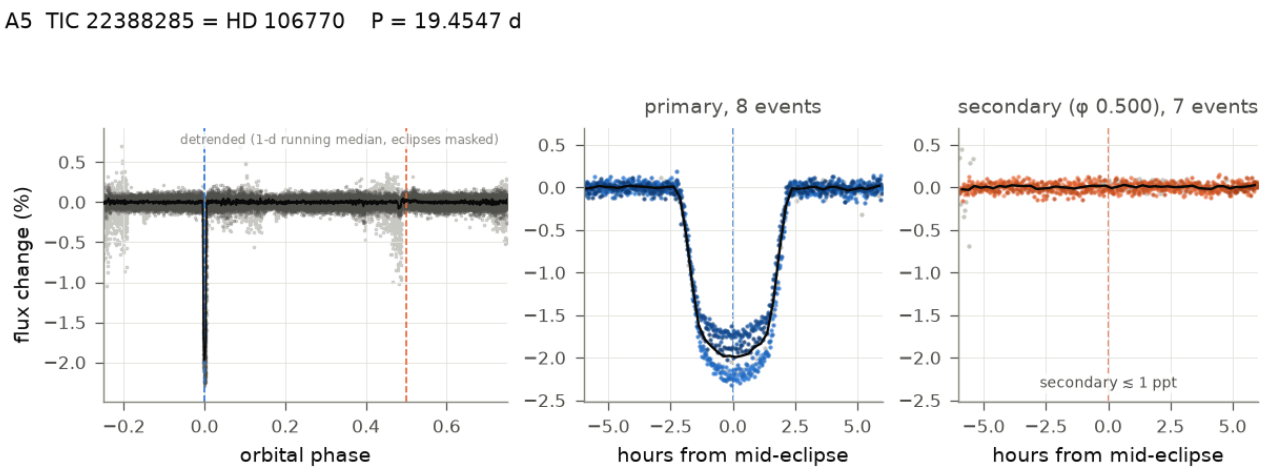


Figure A5. HD 106770, QLP SAP flux folded on the search period P = 19.4547 d (left panel detrended with a 1-d running median, eclipses masked): a ~20-ppt eclipse (eight events in the fold; the search's box gave 19 ppt and 2.9 h; the raw-flux timing fit to the seven fully covered events gives 20.3 ppt and 4.4 h); any secondary is ≲ 1 ppt.

Identifiers	TIC 22388285; HD 106770; TYC 8234-1406-1; Gaia DR3 6130370305216737408
Position (J2000)	184.19851, −47.74244 deg (TIC)
Brightness	Tmag 8.98; G 9.43
Stars	G5V (TIC: T_{eff} 5651 K, 1.14 $R_{\odot}$) at 97 pc, with a spectroscopic companion
Period	19.454722 ± 0.000010 d, timing fit of seven events (search fold 19.4547 d; Gaia SB1C: 19.4564 ± 0.0031 d)
Epoch	BTJD 3494.45294 ± 0.00039 = BJD_{TDB} 2460494.45294 (timing fit; the search's box epoch was 1587.8864)
Primary eclipse	20.3 ppt, 4.4 h in raw QLP SAP flux (timing fit; the search's box gave 19 ppt, 2.9 h)
Secondary eclipse	≲ 1 ppt
Orbit	Gaia's solution is circular (SB1C); no conjunction-phase test is possible
Catalogue status	In no variable catalogue. SIMBAD "**", known SB, zero references. Not in any catalogue searched (Section 2.7); Algol was recovered as a positive control
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. The period agreement with Gaia is by construction of the fold; the orbit is circular, so no conjunction-phase test is possible; one pipeline; no pixel test. A difference image or a second pipeline would confirm it

A G dwarf at 97 pc whose spectroscopic orbit Gaia published, and whose 19-ppt eclipses are uncatalogued. The contamination ratio is 0.003, so the aperture holds little light from anything else, and the period agrees with Gaia's to within its quoted error. Status: probable (see the Status row).

A6 BD−13 1046 = TIC 348773952

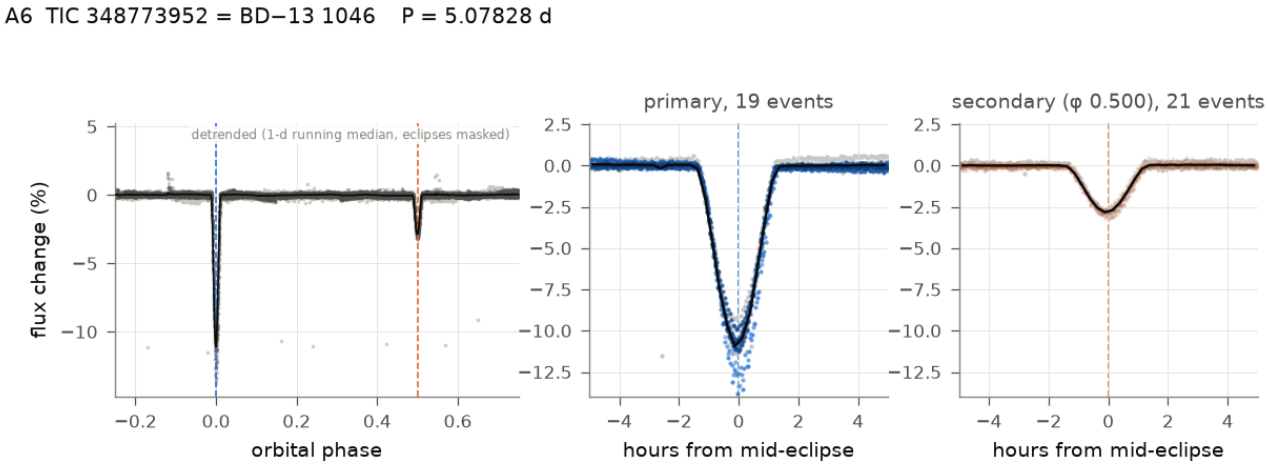


Figure A6. BD−13 1046, QLP SAP flux folded on the search period P = 5.07828 d (left panel detrended with a 1-d running median, eclipses masked): a ~110-ppt primary and a ~28-ppt secondary at phase 0.5.

Identifiers	TIC 348773952; BD−13 1046; TYC 5329-1207-1; Gaia DR3 2987862104761137280
Position (J2000)	75.15913, −13.71899 deg (TIC)
Brightness	Tmag 9.60; G 10.18
Stars	K dwarf (TIC: T_{eff} 4912 K, 0.74 $R_{\odot}$) at 63 pc, with a spectroscopic companion
Period	5.0782783 ± 0.0000010 d, timing fit of 13 primaries (search fold 5.07828 d; Gaia SB1: 5.078108 d)
Epochs	Primary BTJD 3609.56921 ± 0.00016 = BJD_{TDB} 2460609.56921; secondary BTJD 3657.81301 ± 0.00016 (timing fits; the search's box epoch was 1441.1441)
Primary eclipse	110 ppt, 2.55 h in raw QLP SAP flux (timing fit; the search's box gave 114 ppt in a 1.9-h box)
Secondary eclipse	28 ppt, 2.55 h, at phase 0.5
Orbit	Eclipse falls on Gaia's predicted conjunction: Δφ = +0.006
Catalogue status	No eclipsing classification anywhere; RAVE lists solar-like activity and Schmitt et al. (2026) a period only. Known SB. Not in any catalogue searched (Section 2.7)
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. The eclipse falls on Gaia's predicted conjunction, but the period agreement is by construction of the fold, and there is one pipeline and no pixel test

The deepest of A5–A7, a K dwarf 63 pc away eclipsing by about 11%, with a secondary eclipse at phase 0.5. It had been catalogued only for activity and a period, not as an eclipsing binary.

A7 UCAC4 130-138342 = TIC 271321097

A7 TIC 271321097 = UCAC4 130-138342 P = 13.9107 d

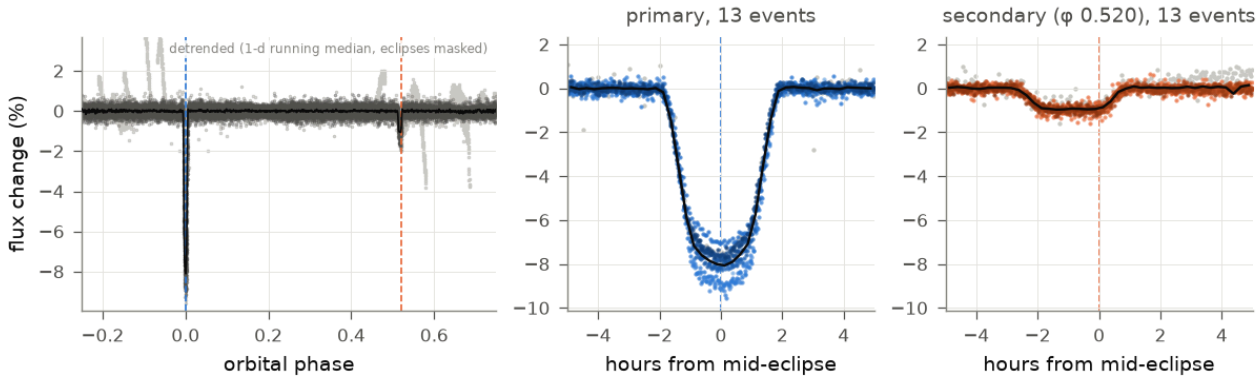


Figure A7. UCAC4 130-138342, QLP SAP flux folded on the search period P = 13.9107 d (left panel detrended with a 1-d running median, eclipses masked): a primary of ~70–80 ppt (in a contaminated aperture; the search's box gave 68 ppt, the raw-flux fit 80 ppt) and a ~10-ppt secondary at phase 0.52.

Identifiers	TIC 271321097; UCAC4 130-138342; Gaia DR3 5826591602034536320
Position (J2000)	233.04765, −64.14559 deg (TIC)
Brightness	Tmag 10.93; G 11.69
Stars	K dwarf (TIC: T_{eff} 4223 K, 0.71 $R_{\odot}$) at 80 pc, with a spectroscopic companion
Period	13.9106635 ± 0.0000071 d, timing fit of 12 primaries (search fold 13.9107 d; Gaia SB1: 13.915147 d)
Epochs	Primary BTJD 3685.73790 ± 0.00034 = BJD_{TDB} 2460685.73790; secondary BTJD 3887.68375 ± 0.0007 (timing fits; the search's box epoch was 1626.9584)
Primary eclipse	80 ppt, 3.8 h in raw QLP SAP flux (timing fit; the search's box gave 68 ppt; TIC contamination ratio 0.41, so diluted)
Secondary eclipse	10 ppt, 3.4 h, at phase 0.52
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = −0.017
Catalogue status	Only a Gaia DR2 large-amplitude-variable flag, untyped. The VSX eclipsing stars 107″ away have other periods. Not in any catalogue searched (Section 2.7). **Prior report (ADS full text):** Bryant et al. (2023), vetting giant-planet candidates around low-mass stars, identified TIC 271321097 as an eclipsing binary from the extended-mission 10-min light curve (the primary-mission 30-min data had shown only the secondary eclipses, the primaries falling in gaps) and excluded it; no period, epoch or catalogue entry was published. This is the only one of the 104 with a prior mention of its eclipses
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. 41% of the aperture light is foreign; the eclipse falls on Gaia's conjunction, but the period agreement is by construction of the fold, and there is one pipeline and no pixel test. Bryant et al. (2023) saw the eclipses in TESS-SPOC data, which supports their reality but not the attribution

The aperture is crowded (41% of its light is attributed to neighbours), so the depth is a lower limit on the eclipse of the star itself. The attribution rests on the period agreeing with Gaia's orbit and the eclipse falling on Gaia's conjunction; no pixel-level attribution test is reported.

A8 HD 131614 = TIC 402800288

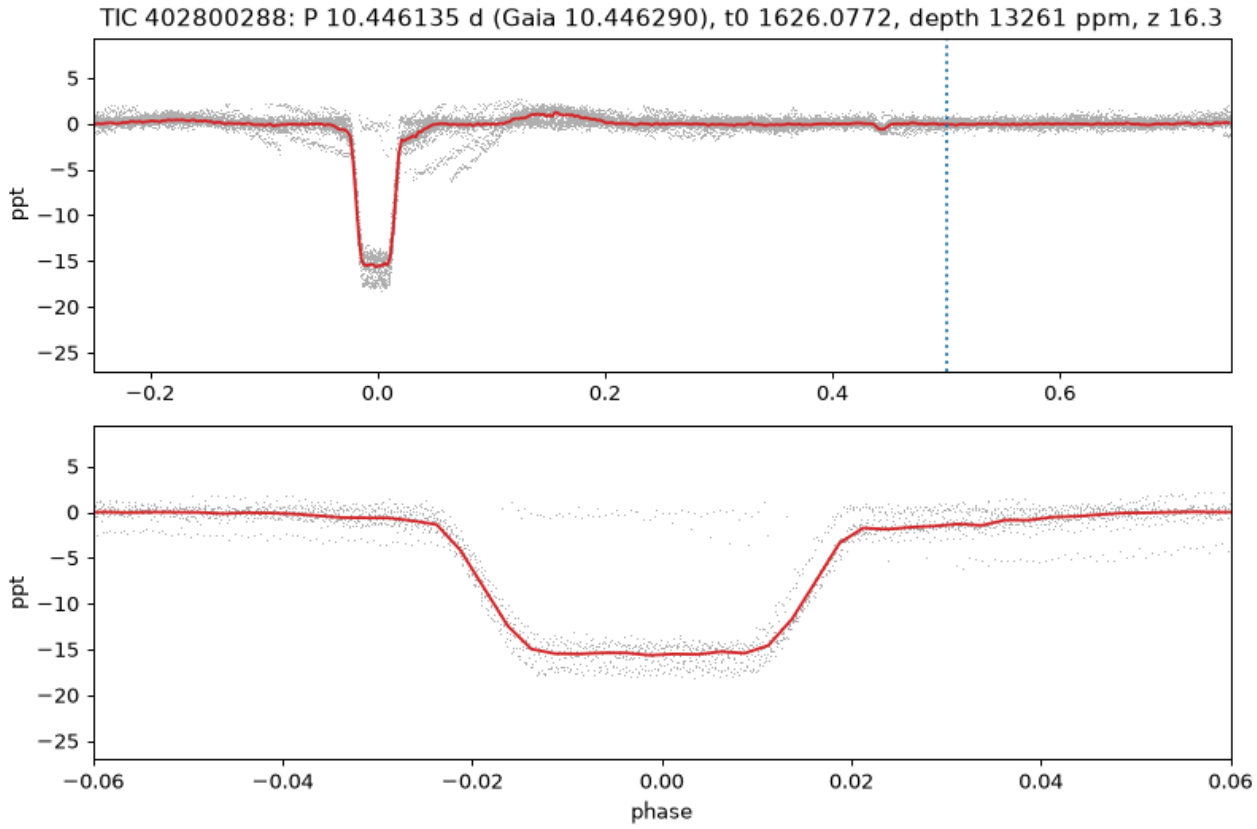


Figure A8. HD 131614, the discovery fold on P = 10.4461 d in the search's detrended flux, in which the eclipse is suppressed toward zero in the sectors after 56 (the points near 0 ppt inside the eclipse); in raw SAP flux with per-event baselines it is 18.2 ppt and about 11 h (Section 3.3), stable across all ten sectors.

Identifiers	TIC 402800288; HD 131614; TYC 9266-1012-1; Gaia DR3 5793878347940289152
Position (J2000)	224.85938, −72.53302 deg (TIC)
Brightness	Tmag 8.47; G 8.91
Stars	Evolved primary (TIC: T_{eff} 5972 K, 5.8 $R_{\odot}$, giant) at 405 pc, with a spectroscopic companion
Period	10.4462098 ± 0.0000098 d, timing fit of 21 events (search fold 10.4461 d; Gaia: 10.446290 d)
Epoch	BTJD 3391.45705 ± 0.00071 = BJD_{TDB} 2460391.45705 (timing fit; the search's box epoch, 1626.0772, was 0.12 d off the fitted line)
Primary eclipse	18.2 ppt against flux 0.75–1.75 durations away (15.1–15.7 ppt with a local quadratic baseline: the flux curves into the eclipse), about 11 h (trapezoid fits in QLP, TESS-SPOC and TARS give 10.9–11.2 h; the 13.2 h first quoted from per-event boxes is not reproduced), stable across all 10 sectors (13.3 ppt in the search's fold)
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = +0.004 (0.9σ, Monte Carlo through Gaia's errors)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s65, s66: 15.1 ppt, 10.9 h, mid-times within 0.1–0.3 min of QLP's); TARS; SPOC's own multi-sector search detected the eclipse (MES 91) with a centroid 1.4 ± 2.8″ from the catalogue position. The QLP detrended flux reduces the eclipse depth from 14.7 to 0.8 ppt in some sectors
Pixel attribution	TESScut difference images of two eclipses each in s65 and s38: the deficit has the star's own image shape (residual 3–4%), its centroid 0.2″ and 0.7″ from the light centroid, and the same depth in 1×1, 3×3 and 5×5 pixel apertures. No Gaia source within 60″ is brighter than G 14.62; the brightest would need a 293% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not in any catalogue searched (Section 2.7): VSX, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025), tables 3 and 4; confirmed by independent cone searches. Known SB
Method	Gaia-orbit period fold (Section 2.4); checked by independent re-analysis (Section 2.5)
Status	

	Confirmed: 21 events in ten sectors at Gaia's period on the eccentric orbit's conjunction phase (0.9σ), the deficit on the target in difference images, and TESS-SPOC agreeing with QLP (both added after the pre-submission review)

A8 is the clearest case of the detrending artefact of Section 2.8. In QLP detrended flux the eclipse depth collapses in some sectors; in raw SAP flux with a local baseline around each event it is ~18 ppt in all ten sectors. The same raw-flux re-measurement rejects the three shallowest candidates of the fold (Section 3.8).

A9 HD 7601 = TIC 399009076

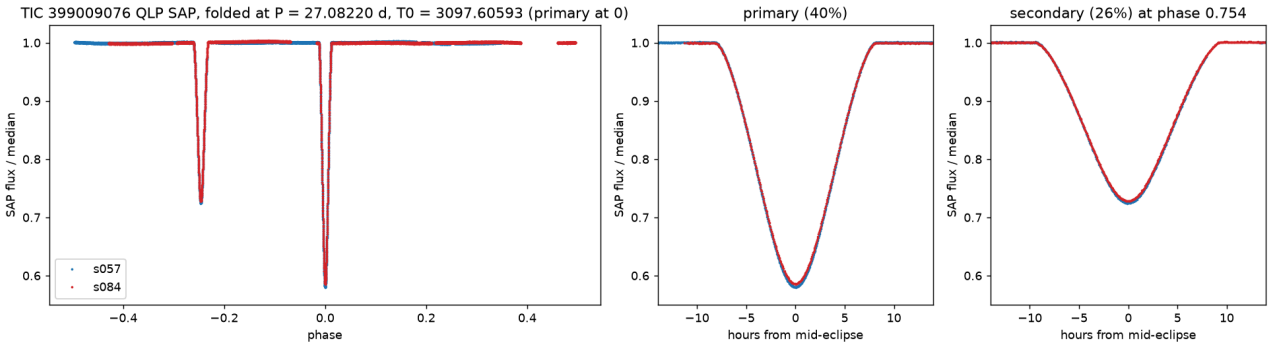


Figure A9. HD 7601, raw QLP SAP flux of sectors 57 and 84 folded on P = 27.0822 d: a 40%, 14-h V-shaped primary and a 26%, 16-h secondary at phase 0.754. The 13.4166-d period of the search's two-sector fold is an alias.

Identifiers	TIC 399009076; HD 7601; TYC 1747-1864-1; Gaia DR3 293150213945293824
Position (J2000)	19.09573, +23.78509 deg (TIC)
Brightness	Tmag 8.56; G 8.97
Stars	F-type (TIC: T_{eff} 6152 K, 3.5 $R_\odot$) at 274 pc; Gaia DR3 lists a double-lined (SB2) orbit
Period	27.082201 ± 0.000018 d from the five eclipses in sectors 57 and 84 (timing fit); the search's 13.4166 d (Gaia's 13.4234 d) does not phase the eclipses: the 13.42-d prediction at BTJD 2867.45 falls on 48 good cadences at flux 1.000
Epochs	T_0(primary) = BTJD 3097.60593 ± 0.00023 = BJD_{TDB} 2460097.60593; T_0(secondary) = BTJD 3253.43045 ± 0.00028, phase 0.754
Primary eclipse	39.8% (trapezoid; ~41% at minimum), 14.3 h, V-shaped (partial); three events (2853.866, 2880.948, 3585.086)
Secondary eclipse	26.1% (~27% at minimum), 16.5 h, at phase 0.754; two events (2874.280, 3605.499). From the phase and durations $e \approx 0.41$, $\omega \approx 10°$
Long baseline	ASAS-3 V (2003–2009, ~200 cycles earlier) folded on the TESS ephemeris: its three faintest points (0.62, 0.51, 0.34 mag; 0.62 mag = 43% = the TESS primary depth) fall at phases −0.001 to −0.005 and seven more at 0.748–0.758; at either 13.42-d period the same points scatter at random
Gaia orbit	Not a confirming match. Gaia's SB2 solution (P 13.4234 ± 0.0012 d, e 0.073, K_1 37.8, K_2 33.0 km/s, 14 transits, F2 = 12.8) fits the eclipses at no period: $2P_{Gaia}$ = 26.847 d differs from 27.082 by 100σ, the eclipse $e \cos \omega$ of 0.41 contradicts Gaia's 0.035, and $(M_1+M_2)\sin^3 i$ = 0.49 $M_\odot$ is far too small for an eclipsing G = 9.0 pair at 274 pc. Gaia's period is erroneous; the exact alias is not identified (DR3 publishes no epoch velocities for this star)
Pixel attribution	Difference images of all five events (sectors 57 and 84): the lost flux is a copy of the target's PSF with its centroid 0.09–0.44 px from the out-of-eclipse centroid, pointing away from the G 11.55 neighbour 6 px west; nothing within 36″ brighter than G 19.8. Only an unresolved companion inside the PSF cannot be excluded
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches; ADS full text). Known SB (Gaia)
Next eclipses	Primary: BTJD 4343.387 (2026-10-29 21:16 UT), 4370.469; secondary: 4336.718, 4363.800
Method	Gaia-orbit period fold (Section 2.4), retained under the supplementary criterion for deep signals (Section 2.4); period from the eclipse-timing fit (Appendix A)

A9 has the deepest eclipses found by the fold, but the fold assigned it an incorrect period. Its eclipses are so deep that they inflate the star's own random-period null. Its z score (3.6) therefore fell below the cut, and it was retained under the supplementary criterion for deep, high-S/N signals (Section 2.4). The fold at Gaia's period produced one smeared 20% dip. The conjunction-phase test, which compares the events with Gaia's conjunction *at Gaia's period*, passed spuriously, because the fold merged two different eclipses. An eclipse-timing fit (Appendix A) gives the 27.08-d ephemeris, which the raw sectors, ASAS-3 photometry two decades earlier and the pixels confirm. A9 is kept as a confirmed eclipsing binary at 27.08 d, with Gaia's orbit recorded as a discrepancy rather than a confirmation.

A10 HD 99010 = TIC 300395201

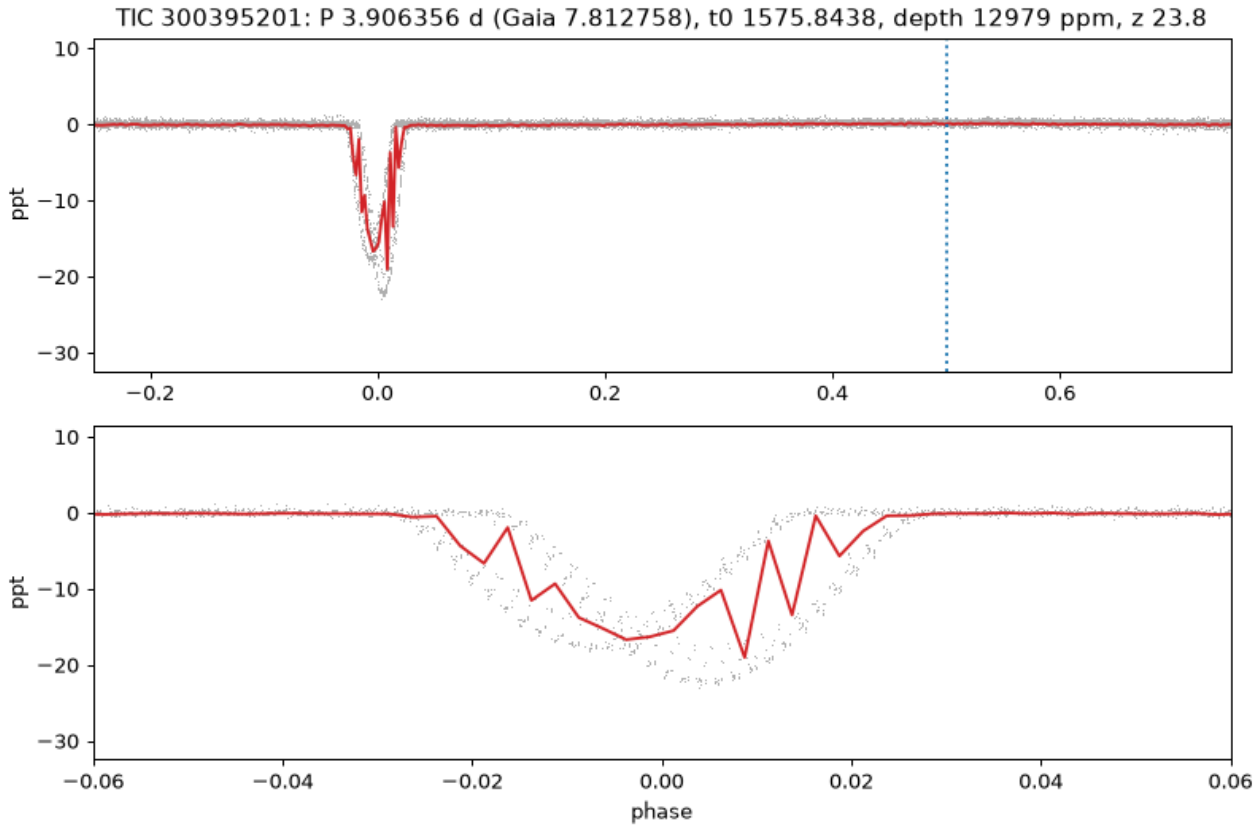


Figure A10. HD 99010, the discovery fold at P/2 = 3.906356 d, half the 7.8128-d orbital period, which stacks the two eclipses into one dip; the raw-flux timing fit gives 22.0 and 17.3 ppt for the two eclipses.

Identifiers	TIC 300395201; HD 99010; TYC 8617-227-1; Gaia DR3 5348342423752784384
Position (J2000)	170.78242, −53.10540 deg (TIC)
Brightness	Tmag 8.45; G 8.81
Stars	F-type primary (TIC: T_{eff} 6372 K, 2.2 $R_{\odot}$) at 175 pc; Gaia double-lined circular orbit (SB2C)
Period	7.8127421 ± 0.0000013 d, timing fit of 17 primaries and 20 secondaries (search fold 7.8128 d; Gaia: 7.812758 d)
Epochs	Primary BTJD 3384.50567 ± 0.0001 = BJD_{TDB} 2460384.50567; secondary BTJD 3427.43069 ± 0.00016 (timing fits; the search's box epoch was 1579.7618)
Eclipses	22.0 and 17.3 ppt, 4.0 and 3.8 h, alternating, in raw SAP flux (timing fit; the 18.3 and 9.3 ppt of the independent re-analysis came from detrended flux)
Orbit	Circular SB2C: no conjunction-phase test is possible
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches). Known SB
Method	Gaia-orbit period fold (Section 2.4)
Status	Probable. With a circular orbit no conjunction-phase test is possible, and the period agreement, the only attribution, is guaranteed by construction of the fold; one pipeline, no pixel test

Two eclipses per orbit at Gaia's double-lined period. With a circular orbit the attribution rests on the period agreement alone, which is why it is listed as probable.

A11 BD+40 3254 = TIC 159892657

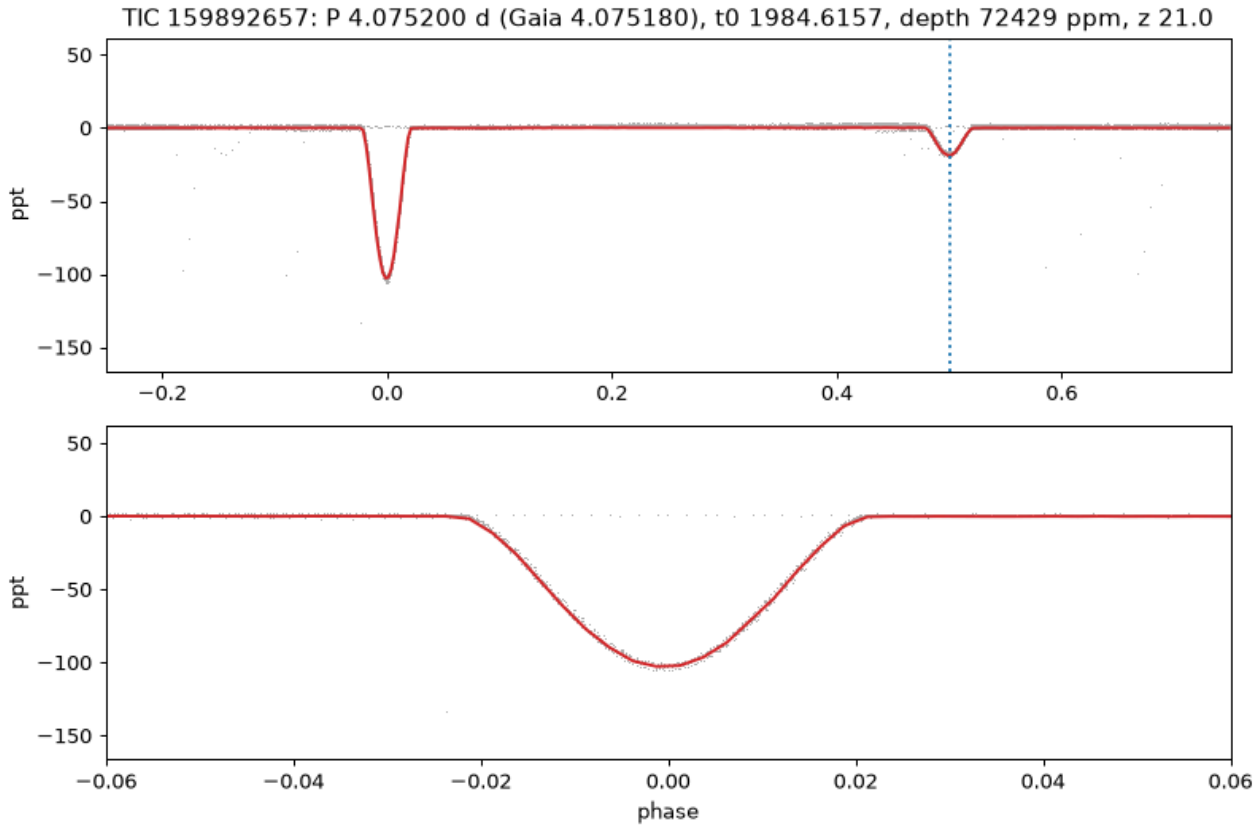


Figure A11. BD+40 3254, the discovery fold on P = 4.0752 d: a ~9% primary (the panel's own box depth reads 7.2%) and a 2% secondary at phase 0.498 (detrended search flux; the raw-flux fit gives 101 and 19 ppt).

Identifiers	TIC 159892657; BD+40 3254; HIP 87971 (TIC cross-match); TYC 3093-922-1; Gaia DR3 4611458011500418176
Position (J2000)	269.54374, +40.11201 deg (TIC)
Brightness	Tmag 8.42; G 8.79
Stars	F-type primary (TIC: T_{eff} 6192 K, 1.7 $R_{\odot}$) at 129 pc, with a spectroscopic companion
Period	4.0751904 ± 0.0000009 d, timing fit of 46 primaries and 40 secondaries (search fold 4.0752 d; Gaia: 4.075180 d)
Epochs	Primary BTJD 3133.82003 ± 0.00011 = BJD_{TDB} 2460133.82003; secondary BTJD 3184.76020 ± 0.00014 (timing fits; the search's box epoch was 1984.6157)
Primary eclipse	101 ppt, 4.0 h in raw SAP flux (timing fit); ~9% in 7 of 8 sectors in the independent re-analysis (sector 26's core diluted by the detrend; its minimum still 3.7%); 7.2% in the search's fold
Secondary eclipse	19 ppt, 3.8 h, at phase 0.498
Orbit	Eclipse on Gaia's predicted conjunction: Δφ = +0.003
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP and TESS-SPOC (s74, s79, s80: primary 97.8–98.3 ppt, secondary 17.7–17.9 ppt, 3.94 h, secondary at phase 0.4998, mid-times within 0.1 min of QLP's in s74 and s79); SPOC's own search detected both eclipses (MES 4630). Against the SPOC products the 2024 primaries fall 0.9 ± 0.15 min earlier than this entry's ephemeris and the period differs by 3×10^{-6} d (3.3σ): this star's QLP sector 80 light curve runs 2.7 min late against SPOC 2-min and TESS-SPOC on the same seven events (one star, one sector: not a statement about the pipeline), and it entered the timing fit, so the errors quoted here are understated by about a minute at 2024 epochs
Pixel attribution	

	TESScut difference images of seven primaries in s74 and six in s40: the deficit has the star's own image shape (residual 2%), its centroid 0.1″ from the light centroid and within 0.2 px of the catalogue position. All 12 Gaia sources within 60″ are G ≥ 17.3 and would need eclipses of 25,000% or more. A companion within about 1″ is not excluded
Catalogue status	Listed only in Kostov et al. (2025), table 2, as an unvetted machine-learning candidate. Known SB
Method	Gaia-orbit period fold (Section 2.4)
Status	Confirmed: 86 eclipses in eight sectors at Gaia's period on the conjunction phase, the deficit on the target in difference images, and SPOC 2-min and TESS-SPOC agreeing with QLP (both added after the pre-submission review)

A 9% eclipse on an eighth-magnitude star, listed only as an unvetted machine-learning candidate (Kostov et al. 2025). Both the conjunction phase and the near-0.5 secondary agree with Gaia's nearly circular orbit.

A12 HD 217492 = TIC 436758044

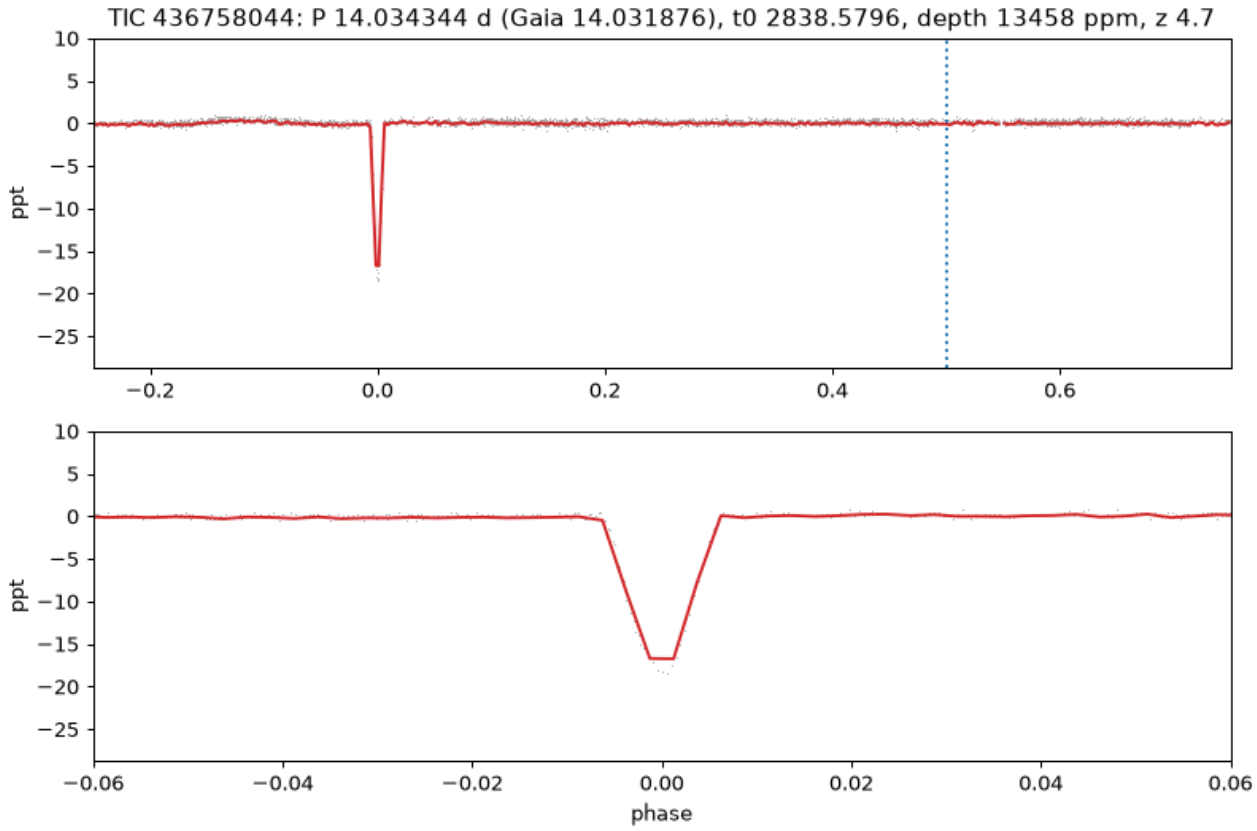


Figure A12. HD 217492, the discovery fold on P = 14.0343 d (detrended QLP search flux; the panel title gives z 4.7): a narrow eclipse whose box depth is 13.5 ppt and whose deepest bins reach about 17 ppt; the raw-flux fit gives 18.2 ppt, and TESS-SPOC confirms the events.

Identifiers	TIC 436758044; HD 217492; HIP 113630 (TIC cross-match); TYC 2242-1176-1; Gaia DR3 2843881366341547520
Position (J2000)	345.19954, +26.50218 deg (TIC)
Brightness	Tmag 8.33; G 8.66
Stars	F-type primary (TIC: T_{eff} 6526 K, 2.5 $R_{\odot}$) at 193 pc; Gaia double-lined orbit (SB2)
Period	14.0343984 ± 0.0000110 d, timing fit of the two QLP events with the cycle count (51) from the search period (search fold 14.0343 d; Gaia: 14.031876 d; 4σ apart)
Epoch	BTJD 3203.47040 ± 0.00028 = BJD_{TDB} 2460203.47040 (timing fit; the search's box epoch was 2838.5796). TESS-SPOC events at BTJD 2852.61, 3568.36 and 3582.40
Primary eclipse	

	18.2 ppt, 4.0 h in raw QLP SAP flux (trapezoid fits to the two events); 13.5 ppt in the search's detrended fold (minimum bin 14.1 ppt at phase −0.003); 8.2, 8.9 and 9.1 ppt in a ±0.10-d core in TESS-SPOC, ~16 ppt undiluted (fold minimum 15.98 ppt)
Orbit	Conjunction offset +0.043 ± 0.008: a **5.5σ tension** with Gaia's SB2 elements
Pipelines agreeing	QLP, TESS-SPOC
Pixel attribution	In the SPOC target-pixel files all three events dip the summed flux 13.3–14.2 ppt while the flux-weighted centroid moves ≤ 0.0015 px: the eclipsing source is within ~0.1 px (~2″) of HD 217492
Catalogue status	Not in any catalogue searched (Section 2.7; independent cone searches). Known SB
Open issues	The period and conjunction tensions are with Gaia's SB2 elements, or indicate an unresolved third body. The Gaia epoch RVs that would allow re-deriving the orbit are not public: both Gaia epoch-RV tables hold no rows for this source, so resolution requires the Gaia DR4 epoch RVs
Method	Gaia-orbit period fold (Section 2.4), retained at $z = 4.7$ under the supplementary criterion for deep signals (Section 2.4); confirmed by independent re-analysis with the orbit caveat, after an epoch-RV check (Section 2.5)

The eclipse is real in two pipelines and falls on the target to within ~2″, so it is attributed to HD 217492. Either Gaia's double-lined elements for this star are biased, or a third star in the system perturbs the orbit or the line profiles. The 5.5σ conjunction offset appears only when the uncertainties of all of Gaia's elements are propagated into the conjunction phase (Section 2.2).

A75 TYC 3255-1824-1 = TIC 202282515

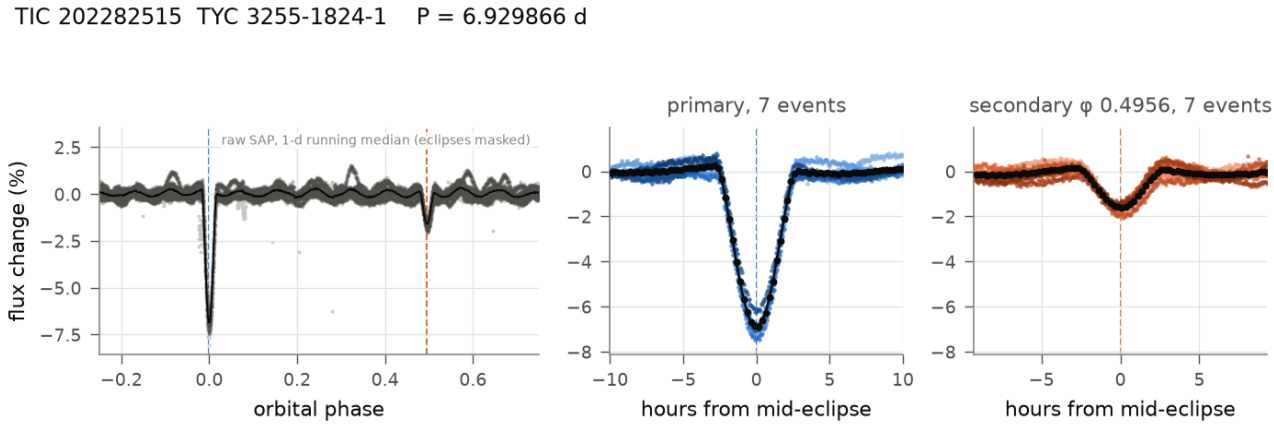


Figure A75. TYC 3255-1824-1, raw QLP SAP flux of sectors 57 and 84, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on $P = 6.929866$ d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.4956 (7 events of each kind plotted; the 6 fully covered primaries are those in the timing fit): a 68-ppt, 5.0-h V-shaped primary and a 16-ppt, 4.6-h secondary.

Identifiers	TIC 202282515; TYC 3255-1824-1; Gaia DR3 394192545103488768
Position (J2000)	5.13060, +49.24466 deg (TIC)
Brightness	Tmag 9.66; G 9.95
Stars	TIC: T_{eff} 6840 K, 2.12 $R_\odot$ (Gaia GSP-Phot 6978 K) at 325 pc; Gaia single-lined orbit (SB1, e = 0.012 ± 0.006, K_1 46.3 km/s). The mass function, 0.071 $M_\odot$, gives $M_2 \geq 0.70$ $M_\odot$ for $M_1 = 1.48$ $M_\odot$; the depth ratio, 0.23, indicates a K-dwarf companion of roughly 4500–5000 K
Period	6.929866 ± 0.000015 d, timing fit of 6 primaries; 7 secondaries fitted at that period (Gaia: 6.929767 ± 0.000185 d; +0.5σ)
Epochs	Primary BTJD 3111.42326 ± 0.00065 = BJD_{TDB} 2460111.42326; secondary BTJD 3114.85767 ± 0.00293 (phase 0.49560 ± 0.00042)
Primary eclipse	67.6 ppt, 5.00 h (ingress 1.95 h), V-shaped, in raw QLP SAP flux (trapezoid fit to the fold of 6 events). The sector-84 events are 8% shallower than those of sector 57 (medians 64.4 and 70.4 ppt) in every pipeline and in TESScut photometry, a sector-dependent dilution; alternate events within sector 57 agree to within the event scatter (about 3 ppt)

Secondary eclipse	15.5 ppt, 4.6 h, at phase 0.4956 (7 events). The secondary mid-times scatter by about ±10 min (O−C rms 9.3 min against formal errors of 1–2 min), event by event identically in QLP, TESS-SPOC and TARS
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at +0.0049 ± 0.0028 (+1.7σ), secondary on conjunction B at −0.0054 ± 0.0023 (−2.3σ); carried to the TESS epochs with Gaia's period, +0.0103 ± 0.0103 (+1.0σ) and −0.0002 ± 0.0117 (0.0σ). The mean of the two offsets is on Gaia's conjunctions to 0.0003 in phase. $e \cos \omega = -0.0069 \pm 0.0007$ from the eclipse spacing, against Gaia's +0.0087 (+0.0071/−0.0065): 2.3σ apart, which is the opposite-signed split of the two offsets
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s57, s84: 68.6 and 16.7 ppt against QLP's 69.1 and 16.7 ppt; matched mid-times +0.03 ± 0.13 min over 6 primaries and +0.15 ± 0.49 min over 7 secondaries); SPOC 2-min SAP and PDCSAP (s57: −0.01 ± 0.14 min over 4 primaries); TARS. Depths agree within 1.5% across pipelines. SPOC's s57 search detected the primary as a TCE at 6.92985 ± 0.00014 d (MES 159), 0.09 ± 2.50″ from the out-of-transit centroid, and the secondary as a second TCE (MES 39); the TESS-SPOC s84 search detected the primary (MES 95). None was flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 4 primaries in s57 and 2 in s84: deficit centroid 0.1″ from the light centroid in both, a scaled copy of the star's image (residual 4%), depth 61 ppt (s57) and 55 ppt (s84) alike in 1×1, 3×3 and 5×5 pixel apertures (the eclipse-core median of a V-shaped event). The brightest Gaia source within 60″, G 15.62 at 57″, would need a 1250% eclipse
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	The 2.3σ difference between the photometric and Gaia values of $e \cos \omega$, and the ±10-min scatter of the secondary mid-times, which the predicted secondary times below do not include
Next eclipses	Primary: BTJD 4331.080 (2026-10-17 13:47 UTC), 4338.010, 4344.939; secondary: BTJD 4327.584 (2026-10-14 01:54 UTC), 4334.514 (allow ±12 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A detached F-type star with a K-dwarf companion on a nearly circular 6.93-d orbit, seen at a moderately high impact parameter (b ≈ 0.75 of the sum of the radii), so that the 68-ppt primary is partial and V-shaped. Six primaries and seven secondaries in two sectors fall on Gaia's two conjunctions, the deficit is on the target in difference images, and TESS-SPOC and SPOC 2-min reproduce QLP's depths and mid-times to 0.1 min. The photometric $e \cos \omega$ differs from Gaia's by 2.3σ, but both give $e < 0.02$, and the difference appears only as an opposite-signed split of the two conjunction offsets. The secondary times scatter by ±10 min in every pipeline, so the scatter lies in the star or in the local baseline of the shallow eclipse, not in QLP; further sectors would show whether it is periodic.

A76 TYC 3256-1533-1 = TIC 202710940

TYC 3256-1533-1 = TIC 202710940 P = 3.306372 d

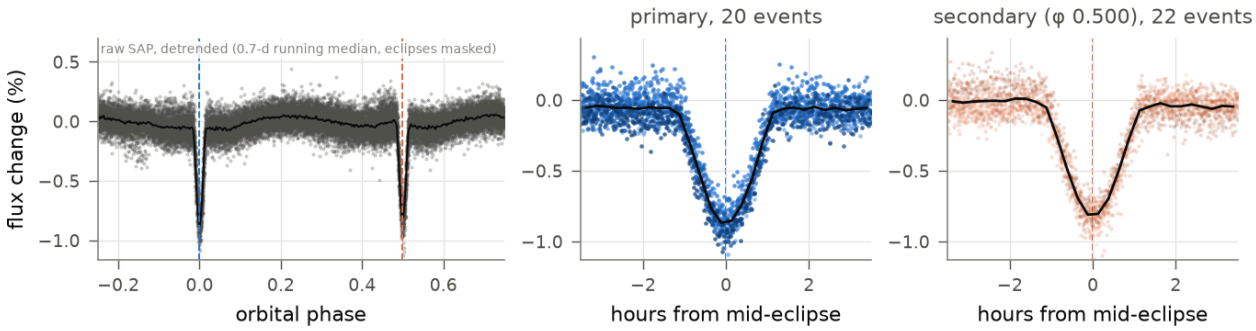


Figure A76. TYC 3256-1533-1, raw QLP SAP flux of sectors 57, 84 and 85, sector medians removed and detrended by a 0.7-d running median with the eclipses masked, folded on P = 3.3063718 d (left), with zooms on the primary and on the secondary at phase 0.500, each event normalised to its own baseline (20 primaries and 22 secondaries plotted; the 18 and 20 fully covered events are those in the timing fit): two nearly equal grazing eclipses, 8.1 and 7.8 ppt deep and both 2.25 h long. The out-of-eclipse flux carries 2–3 ppt of spot modulation.

Identifiers	TIC 202710940; TYC 3256-1533-1; Gaia DR3 391596224492510592
Position (J2000)	7.91761, +49.89143 deg (TIC)
Brightness	Tmag 9.48; G 9.92
Stars	TIC: T_{eff} 5923 K, 2.50 $R_\odot$ (the combined light of both stars) at 277 pc; Gaia double-lined orbit (SB2, e = 0.010 ± 0.004, K_1 95.5, K_2 96.8 km/s): a near-twin pair (q = 0.99) with M $\sin^3 i$ = 1.23 and 1.21 $M_\odot$, two slightly evolved G0-type stars of 1.6–1.8 $R_\odot$ each; at the inclination implied by the grazing eclipses, $i \approx 75$–77°, the masses are about 1.3–1.4 $M_\odot$. Spot modulation of 2–3 ppt semi-amplitude with a period of 3.2–3.7 d per sector, consistent with synchronous rotation
Period	3.3063718 ± 0.0000018 d, timing fit of 18 primaries; 20 secondaries give 3.3063711 ± 0.0000018 d (Gaia: 3.306368 ± 0.000030 d; +0.13σ). Twin eclipses: the period is twice the fold period, 1.653192 d. The two eclipses have the same shape and duration and differ in depth by only 4%, so the photometry alone does not separate P from P/2; Gaia's double-lined orbit at 3.306 d does
Epochs	Primary BTJD 3299.60466 ± 0.00020 = BJD_{TDB} 2460299.60466; secondary BTJD 3294.64483 ± 0.00020 (phase 0.49992)
Primary eclipse	8.11 ± 0.33 ppt, 2.25 h, V-shaped (grazing), in raw QLP SAP flux (trapezoid fits to 18 events)
Secondary eclipse	7.82 ± 0.44 ppt, 2.25 h, at phase 0.49992 (20 events). The depth difference, 0.29 ± 0.13 ppt (2.2σ), is 0.5–2.5% in the other pipelines, so which eclipse is called primary is a 2σ choice
Orbit	At Gaia's epoch with the TESS period (880 cycles back): primary on conjunction B at +0.0030 ± 0.0013 (2.3σ), secondary on conjunction A at +0.0023 ± 0.0013 (1.7σ); carried to the TESS epochs with Gaia's period, +0.0041 ± 0.0080 (0.5σ) and +0.0034 ± 0.0077 (0.4σ). Both eclipses follow Gaia's conjunctions by a common 0.0027 in phase (13 min). The eclipse spacing, 0.4999, matches Gaia's predicted 0.4995 ± 0.0022 (e cos ω = −0.0001), so the common shift is not an e cos ω effect. With q = 0.99 the two stars are interchangeable
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s57, 7 events of each kind: median mid-time differences 0.03–0.21 min, depths within 5%); TESS-SPOC SAP and PDCSAP (s57, s84, s85, 17 primaries and 19 secondaries: within 0.21 min, depths within 4.3%); TARS. SPOC's s57 search detected the eclipses as a TCE at the half period, 1.653144 ± 0.000031 d (MES 123.5, 7986 ppm, 2.36 h, impact parameter 0.98), 0.52 ± 2.50″ from the out-of-transit centroid; with both eclipses folded together it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images in s57 and s84, 14 eclipses of both kinds stacked in each: deficit centroid 1.0″ and 0.9″ from the light centroid, within 0.10 px of the catalogue position, a scaled copy of the target's image (residual 8–9%), depth 6.6–7.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 32 Gaia sources within 60″, only G 14.12 at 23.5″ (would need a 39% eclipse) and G 14.33 at 45.4″ (47%) are bright enough; either would pull the deficit centroid by more than 20″, and both are excluded. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 table 3, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; north of ASAS-3 coverage. Kostov et al. 2025 table 2 lists only the neighbour TIC 202710934 (Tmag 13.4, 24″); no period is compared there, so this may, but need not, be this star's eclipses in the neighbour's aperture. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	The common 13-min (2σ) delay of both eclipses relative to Gaia's conjunctions. A light-travel-time effect from a third body cannot be tested against Gaia's single epoch
Next eclipses	Primary: BTJD 4327.886 (2026-10-14 09:09 UTC), 4331.193, 4334.499; secondary: BTJD 4329.539 (2026-10-16 00:49 UTC), 4332.846 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A near-twin pair of slightly evolved G stars (q = 0.99) on a circular 3.31-d orbit, seen at an impact parameter of about 0.93–0.95 of the sum of the radii ($i \approx 75$–$77°$ for radii of 1.6–1.8 $R_\odot$): the eclipses are grazing, which is why two equal stars give only 8-ppt dips. Because the two eclipses are equal to within 4% in every pipeline, the photometry alone cannot distinguish the period from its half; the double-lined Gaia orbit at 3.306 d, which a 1.65-d binary could not produce, fixes it, and the TESS period agrees with Gaia's to 0.13σ. Both eclipses fall on Gaia's conjunctions with a common 13-min delay at the 2σ level, which does not affect the attribution. The spot modulation near the orbital period is that expected of a tidally synchronised, active pair.

A77 BD+44 219 = TIC 196475343

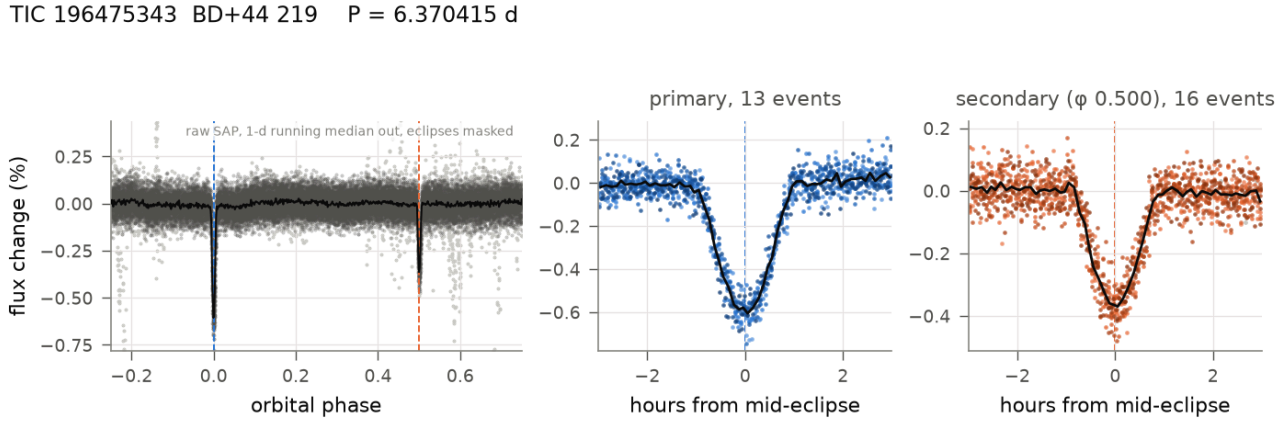


Figure A77. BD+44 219, raw QLP SAP flux of sectors 17, 57, 84 and 85, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on P = 6.370415 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.500, each event normalised to its own baseline (13 primaries and 16 secondaries plotted, including sector 17; the 11 and 12 fully covered events are those in the timing fit): a 6.1-ppt, 1.9-h primary and a 3.7-ppt, 1.7-h secondary, both grazing.

Identifiers	TIC 196475343; BD+44 219; TYC 3263-811-1; Gaia DR3 377002711596378752
Position (J2000)	15.62474, +45.18042 deg (TIC)
Brightness	Tmag 9.12; G 9.51
Stars	G0 (SIMBAD; TIC: T_{eff} 6196 K, 1.79 $R_\odot$) at 185 pc; Gaia double-lined orbit (SB2, e = 0.011 ± 0.005, K_1 75.4, K_2 75.0 km/s): twins with q = 1.004 and M sin³i = 1.120 and 1.125 $M_\odot$, two G0 dwarfs of about 1.15 $M_\odot$ and 1.2 $R_\odot$
Period	6.370415 ± 0.000005 d, timing fit of 11 primaries; 12 secondaries give 6.370416 ± 0.000003 d (Gaia: 6.370535 ± 0.000148 d; −0.8σ). The fold ran at half the orbital period, 3.1851971 d; the alternate eclipses differ in depth (6.07 ± 0.07 against 3.74 ± 0.10 ppt, errors of the means: 20σ) and in duration
Epochs	Primary BTJD 3407.33044 ± 0.00027 = BJD_{TDB} 2460407.33044; secondary BTJD 3321.33200 ± 0.00016 (phase 0.50034 ± 0.00005)
Primary eclipse	

	6.07 ppt (event-to-event scatter 0.22 ppt), 1.9–2.0 h, in raw QLP SAP flux (trapezoid fits to 11 events in s57, s84 and s85; the 30-min cadence of s17 does not resolve the eclipse well enough for the timing fit)
Secondary eclipse	3.74 ppt (event-to-event scatter 0.34 ppt), 1.7 h, at phase 0.50034 (12 events)
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at −0.0009 ± 0.0019 (0.5σ), secondary on conjunction B at +0.0013 ± 0.0022 (0.6σ). Eclipse spacing 0.50034 against Gaia's predicted 0.4982 ± 0.0035 (+0.6σ); $e \cos \omega$ = +0.0005 against Gaia's −0.003 ± 0.005. The deeper, longer eclipse falls at conjunction A, the nearer one for Gaia's $e \sin \omega$ = +0.009: for two equal stars in grazing geometry the overlap areas at the two conjunctions are in the ratio 0.607, against an observed depth ratio of 0.616
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s57, s84, s85, 22 matched events: mean mid-time differences 0.0–0.9 min per sector, depths 5.82 and 3.66 ppt against QLP's 5.72 and 3.63 ppt with the same quadratic baseline); TARS (5.98 and 3.25 ppt). There are no SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of 3 primaries in s57 and of 4 primaries and 4 secondaries in s84: deficit centroid 0.4–0.8″ from the light centroid, with the target's image shape (residual 6–10%), depth 4.7, 4.7, 4.8 ppt (s57) and 5.1, 5.2, 5.2 ppt (s84) in 1×1, 3×3 and 5×5 pixel apertures, and 3.2 ppt for the secondaries. The brightest Gaia source within 60″, G 16.89 at 31″, would need a 547% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, the Gaia DR3 eclipsing-binary table, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. The Gaia DR3 variability summary has a row (39 epochs, G range 0.022 mag) with no classification. SIMBAD's three references (two stellar-diameter catalogues and the Gaia DR3 non-single-star catalogue) include none on variability. Known SB (Gaia DR3)
Next eclipses	Primary: BTJD 4331.041 (2026-10-17 12:50 UTC), 4337.411, 4343.781; secondary: BTJD 4327.858 (2026-10-14 08:27 UTC), 4334.228 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

Two equal G0 dwarfs (q = 1.004) in a nearly circular 6.37-d orbit, seen so close to the grazing limit (b ≈ 0.95 of the sum of the radii) that the eclipses last under 2 h and remove only a few parts per thousand. The two eclipses are nevertheless unequal, 6.1 against 3.7 ppt. Gaia's small $e \sin \omega$ of +0.009 makes the stars 1.9% closer at conjunction A than at B, and in a grazing geometry this changes the overlap area enough to reproduce both the depth ratio (0.616 observed, 0.607 predicted) and the duration ratio for a single inclination. The unequal eclipses are thus consistent with the spectroscopic $e \sin \omega \approx 0.009$, a check on the orbit rather than an independent measurement of it. Both eclipses fall on Gaia's conjunctions within 0.6σ. Radial velocities at higher precision and a joint fit would give the inclination and both radii of a grazing twin.

A78 HD 6217 = TIC 268587594

TIC 268587594 HD 6217 P = 9.140474 d

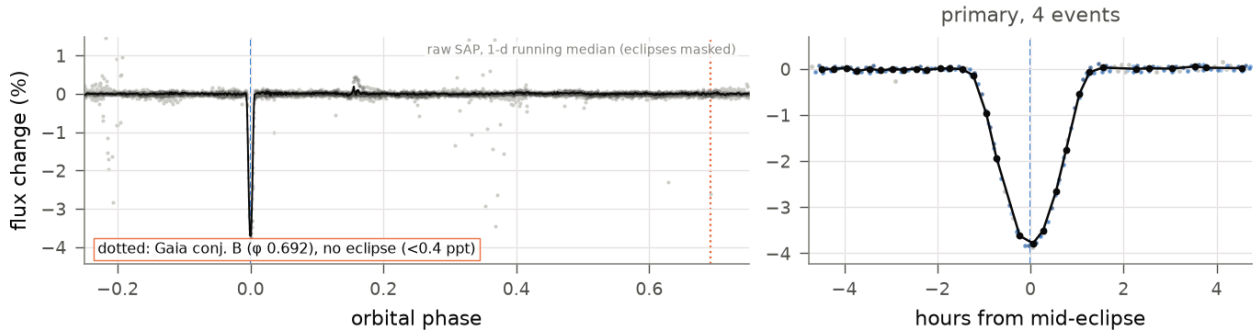


Figure A78. HD 6217, raw QLP SAP flux of sectors 3 and 30, sector medians removed and detrended by a 1-d running median with the eclipse masked, folded on P = 9.140474 d (left; flagged cadences grey), with a zoom on the primary, each event normalised to its own baseline: a 37-ppt, 2.4-h V-shaped eclipse at Gaia's conjunction A. The dotted line marks Gaia's conjunction B at phase 0.692, where no eclipse is seen (< 0.4 ppt).

Identifiers	TIC 268587594; HD 6217; CD-24 458; CPD-24 106; TYC 6422-1324-1; Gaia DR3 2345174743458805760
Position (J2000)	15.73469, −23.88401 deg (TIC)
Brightness	Tmag 8.84; G 9.27
Stars	G2/3V (SIMBAD; TIC: T_{eff} 5887 K, 1.36 $R_\odot$) at 115 pc; Gaia double-lined orbit (SB2, e = 0.4017 ± 0.0045, ω = 42.7 ± 0.7°, K_1 73.6, K_2 64.8 km/s): M $\sin^3 i$ = 0.90 and 1.03 $M_\odot$, two G dwarfs of about 0.9 and 1.0 $M_\odot$
Period	9.140474 ± 0.000004 d, timing fit of 4 primaries in s3 and s30 (with a fifth, in flagged cadences, 9.140477 ± 0.000004 d) (Gaia: 9.140352 ± 0.000190 d; +0.6σ)
Epoch	Primary BTJD 1764.02178 ± 0.00018 = BJD_{TDB} 2458764.02178
Primary eclipse	36.5 ppt, 2.38 h (ingress 0.88 h), V-shaped, in raw QLP SAP flux (trapezoid fit to the fold of 4 events); 37.7 ppt in s30; 37.3 ppt in SPOC 2-min
Secondary eclipse	Not detected (< 0.4 ppt, 3σ) at Gaia's conjunction B, predicted at phase 0.6922 ± 0.0030 and covered in four cycles; no dip above S/N 3.5 at any phase. In the e = 0.40 orbit the projected separation at B (1.152 a) is 1.75 times that at A (0.659 a), so the impact parameter of about 0.84 (in units of the sum of the radii) implied by the primary's duration becomes about 1.47 at B: the stars do not eclipse there
Orbit	At Gaia's epoch with the TESS period: primary on conjunction A at −0.0012 ± 0.0007 (−1.7σ; −0.27 ± 0.16 h); carried to the TESS epochs with Gaia's period, +0.0008 ± 0.0035 (+0.2σ). Conjunction B is 80σ away. With one eclipse per orbit, $e \cos \omega$ is not tested by the eclipse spacing
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s3: 37.3 ppt; mid-times within 0.6 min on two events at QLP's 30-min cadence); TESS-SPOC SAP and PDCSAP (s3, s30: 36.6–36.8 ppt; QLP − TESS-SPOC +0.05 ± 0.08 min on 3 events); TARS; TESScut photometry of the 200-s images of sector 107 (this work): two primaries at O−C +0.74 ± 0.15 and +1.22 ± 0.14 min, within the ephemeris's propagated error (1.6 min) about 2100 d after the last QLP sector, 37.6 ppt. SPOC's s3 search detected the eclipse as a TCE at 9.14050 ± 0.00013 d (MES 395), 0.41 ± 2.50″ from the catalogue position; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 1 primary in s3 (30-min cadence) and 2 in s30: deficit centroid 0.7″ and 0.4″ from the light centroid, a scaled copy of the star's image (residual 3.6–3.7%), depth 31.8, 31.7 and 31.8 ppt in 1×1, 3×3 and 5×5 pixel apertures (s30; the eclipse-core median of a V-shaped event). The only Gaia source within 60″ is G 20.18 at 53″ (would need 84,000%); the brightest within 90″, G 16.10 at 66″, would need about 2000%
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. SIMBAD's eight references include none on variability. Known SB (Gaia DR3)

Next eclipses	Primary: BTJD 4332.495 (2026-10-18 23:44 UTC), 4341.635, 4350.776 (allow ±5 min); no secondary eclipse
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A pair of G dwarfs in an e = 0.40 double-lined orbit that eclipse at only one conjunction. At conjunction A, on the periastron side, the stars are 0.66 a apart in projection and overlap in a grazing eclipse, at an impact parameter of about 0.84 of the sum of the radii from its duration. At conjunction B they are 1.15 a apart and miss each other, and the light curve is flat there to 0.4 ppt. The single eclipse is attributed to the Gaia orbit by its period (+0.6σ) and its phase (1.7σ at Gaia's epoch), and the 2018–2020 baseline gives the period to 4×10^{-6} d. The ephemeris was tested out of sample: two eclipses in sector 107 fall within its propagated error. Because only one eclipse per orbit occurs, the photometry alone constrains neither e nor ω; the radii and inclination require a joint fit with radial velocities.

A79 BD+64 520A = TIC 416277351

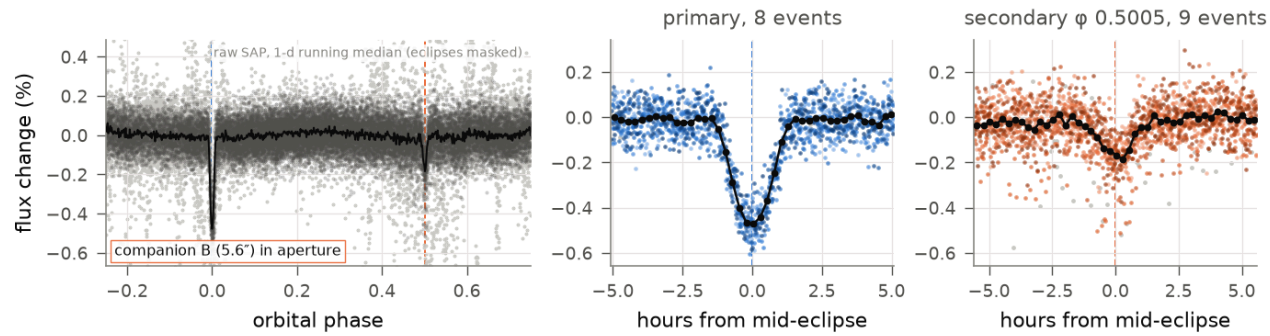


Figure A79. BD+64 520A, raw QLP SAP flux of sectors 59, 73 and 86, sector medians removed and detrended by a 1-d running median with the eclipses masked, folded on P = 6.607651 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.5005, each event normalised to its own baseline: a 4.5-ppt, 2.5-h V-shaped primary and a 1.7-ppt, 2.8-h secondary. The aperture includes the visual companion BD+64 520B, 5.6″ away; QLP's SAP depths here are already corrected for its light.

Identifiers	TIC 416277351; BD+64 520A; HIP 25046; TYC 4088-1339-1; Gaia DR3 479996547047333120; WDS J05218+6454A (STF 659A)
Position (J2000)	80.44749, +64.90443 deg (TIC)
Brightness	Tmag 9.36; G 9.75
Stars	F2 (SIMBAD; TIC: T_{eff} 6506 K, 2.61 $R_\odot$) at 308 pc; Gaia single-lined orbit (SB1, e = 0.009 ± 0.005, K_1 49.4 km/s). The mass function, 0.082 $M_\odot$, gives $M_2 \geq 0.70$ $M_\odot$; the secondary-to-primary depth ratio, 0.35, indicates a companion of roughly 5000 K, a K dwarf of about 0.8 $M_\odot$. The visual companion BD+64 520B (TIC 416277353; G 10.62, 5.6″) has the same parallax and colour and 0.445 of A's G-band flux; Gaia DR3 sees its radial velocity as constant (robust amplitude 6.2 km/s over 21 transits)
Period	6.607651 ± 0.000012 d, timing fit of 8 primaries; 9 secondaries give 6.607705 ± 0.000030 d (1.7σ) (Gaia: 6.607711 ± 0.000099 d; −0.6σ)
Epochs	Primary BTJD 3149.97660 ± 0.00045 = BJD_{TDB} 2460149.97660; secondary BTJD 3153.28375 ± 0.00125 (phase 0.50050 ± 0.00019)
Primary eclipse	4.54 ppt, 2.54 h (ingress 0.93 h), V-shaped (grazing), in raw QLP SAP flux (trapezoid fit to the fold of 8 events). QLP's SAP depth is corrected for B's light (it exceeds SPOC's uncorrected SAP depth by a factor of 1.52 on every matched event) and is the depth on A; in the combined light of A and B the eclipse is about 3.0 ppt (SPOC 2-min SAP 2.95 ppt, crowding 0.69; PDCSAP 4.26 ppt)
Secondary eclipse	1.69 ppt, 2.80 h, at phase 0.50050 (9 events of 1.3–2.8 ppt); 1.48 ppt in SPOC 2-min PDCSAP

Orbit	At Gaia's epoch with the TESS period: primary on conjunction A (A eclipsed) at −0.0024 ± 0.0020 (−1.2σ), secondary on conjunction B at +0.0037 ± 0.0019 (+1.9σ); carried to the TESS epochs with Gaia's period, −0.0061 ± 0.0065 (−1.0σ) and +0.0002 ± 0.0070 (0.0σ). $e \cos \omega$ = +0.0008 ± 0.0003 from the eclipse spacing, against Gaia's −0.0082 (+0.0050/−0.0055): 1.7σ apart, both circular to about 0.01
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s59, s73: matched mid-times −0.20 ± 0.70 min over 6 primaries and +0.07 ± 2.31 min over 6 secondaries); TESS-SPOC SAP and PDCSAP (s59, s73: 4.33 ppt in PDCSAP); TARS. SPOC's multi-sector search (s14–s78) detected the primary as a TCE at 6.607629 ± 0.000018 d (MES 56, 4458 ppm), 1.48 ± 2.51″ from the out-of-transit centroid, and the secondary as a second TCE (MES 15.6); neither was flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 4, 3 and 1 primaries in s59, s73 and s86: deficit centroid 1.7″, 1.8″ and 0.4″ from the light centroid and within 0.11 px of the catalogue position, a scaled copy of the star's image (residual 17–23%, from the faintness of a 3-ppt signal), depth 2.7–2.8 ppt in 3×3 and 5×5 pixel apertures; the 1×1 depths are not stable (3.42, 3.04 and 2.15 ppt). The G 15.60 star at 38″ would need an eclipse of about 95% of its light and would move the deficit centroid by more than a pixel; only B (would need 9.7 ppt) is bright and close enough. Along the A–B axis the deficit-minus-light centroid shift, −0.78 ± 0.57″ over three sectors, lies 1.7σ from the −1.7″ expected for an eclipse on A and more than about 5σ from the +3.9″ expected for one on B (null distribution from six stacks per sector at off-eclipse phases, so the error of the null is itself uncertain by about 30%)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, ExoFOP (no TOI or CTOI), general web search; no ASAS-3 data. WDS lists the visual pair STF 659 AB (5.6″). SIMBAD's 13 references include none on variability. Known SB (Gaia DR3)
Open issues	The pixel separation of A from B rests on a 0.27-px centroid shift; the attribution to A rests equally on the eclipses' agreement with A's Gaia orbit and on B's constant radial velocity
Next eclipses	Primary: BTJD 4332.746 (2026-10-19 05:50 UTC), 4339.354, 4345.961; secondary: BTJD 4329.446 (2026-10-15 22:37 UTC), 4336.053 (allow ±10 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+64 520A, an F2 star and Hipparcos target, is the A component of the visual pair STF 659, with a companion of 0.45 times its brightness 5.6″ away, inside every TESS aperture. The eclipses are shallow and grazing (b ≈ 0.96 of the sum of the radii): 4.5 ppt on A and 1.7 ppt at phase 0.5005, from a K-dwarf companion on a circular 6.61-d orbit. Three independent lines place them on A rather than B. They have the period of A's Gaia orbit and fall on both of its conjunctions; B is radial-velocity constant in Gaia to a few km/s, whereas a companion able to produce a 9.7-ppt eclipse on B would move it by tens of km/s; and the difference-image centroid along the A–B axis falls where an eclipse on A puts it. The entry's depths are those of A alone; in the combined light the primary is about 3.0 ppt. Adaptive-optics or speckle-resolved photometry of one eclipse would separate A and B directly.

A8o TYC 1310-1984-1 = TIC 19702288

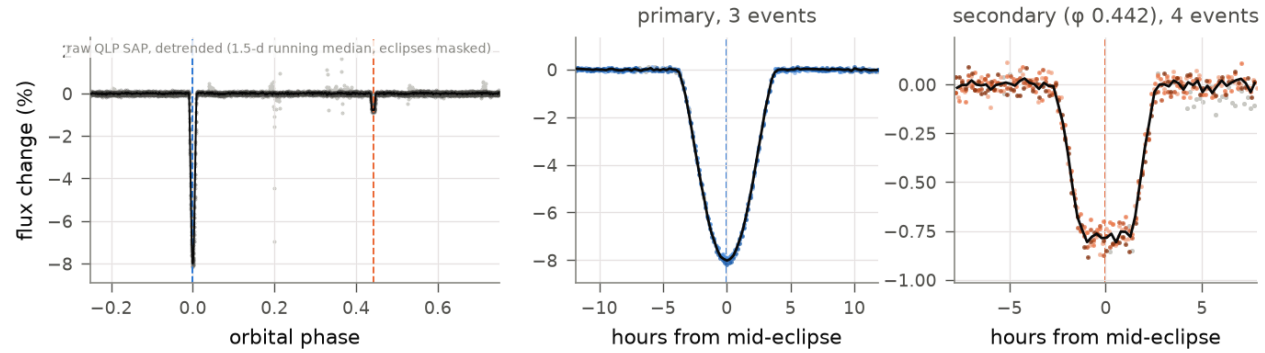


Figure A80. TYC 1310-1984-1, raw QLP SAP flux of sectors 43–45, sector medians removed and detrended by a 1.5-d running median with the eclipses masked, folded on P = 17.02161 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.442, each event normalised to its own baseline: an 80-ppt, 7.4-h primary with a rounded, limb-darkened bottom (transit of the companion) and a 7.7-ppt, 4.9-h flat-bottomed secondary (total occultation).

Identifiers	TIC 19702288; TYC 1310-1984-1; Gaia DR3 3404186685061564288
Position (J2000)	84.34853, +22.45937 deg (TIC)
Brightness	Tmag 9.48; G 9.86
Stars	TIC: T_{eff} 6225 K, 1.62 $R_\odot$ at 200 pc; Gaia single-lined orbit (SB1, e = 0.267 ± 0.011, ω = 245.6 ± 3.2°, K_1 27.4 km/s). The companion is an M dwarf: the mass function, 0.032 $M_\odot$, gives $M_2 \geq$ 0.45 $M_\odot$ for M_1 = 1.20 $M_\odot$; the primary depth gives $R_2/R_1 \approx 0.28$ ($R_2 \approx 0.46$ $R_\odot$); the occultation depth corresponds to a TESS-band surface-brightness ratio of about 0.1 (roughly 3600 K)
Period	17.02161 ± 0.00032 d, timing fit of 3 primaries; 4 secondaries give 17.02189 ± 0.00045 d (Gaia: 17.02361 ± 0.00114 d; –1.7σ). ASAS-3 V photometry of 2004–2009, about 353 cycles earlier, shows the primary above 5σ only for 17.0215 < P < 17.0221 d (peak 7.9σ at 17.02195 d) and nothing at Gaia's period
Epochs	Primary BTJD 2516.54908 ± 0.00042 = BJD_{TDB} 2459516.54908; secondary BTJD 2524.0685 (phase 0.44176 ± 0.00004)
Primary eclipse	79.8 ppt (78.7–80.7 ppt per event), 7.4 h, rounded and limb-darkened ($T_{23}/T_{14} \approx 0.2$), in raw QLP SAP flux (trapezoid fits to 3 events)
Secondary eclipse	7.7 ppt (7.3–8.0 ppt per event), 4.9 h, flat-bottomed ($T_{23}/T_{14} \approx 0.6$), a total occultation, at phase 0.44176 (4 events). It falls before phase 0.5 because $e \cos \omega < 0$, and is shorter than the primary because conjunction B is the nearer one in projection (0.75 a against 1.23 a at A)
Orbit	At Gaia's epoch (the TESS ephemeris carried 125 cycles back, where Gaia's period and the TESS period give the same result): primary on conjunction A at –0.0082 ± 0.0070 (1.2σ), secondary on conjunction B at +0.0057 ± 0.0034 (1.7σ); conjunction B is excluded for the primary (130σ). The eclipse spacing, 0.44176, gives $e \cos \omega = -0.0915$ against Gaia's –0.110 ± 0.013 (1.5σ); the duration ratio, 0.66, compares with about 0.61 predicted for equal impact parameters from Gaia's $e \sin \omega = -0.243$
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s43–s45: 3 primaries at 76.2 and 79.3 ppt, 4 secondaries at 7.4 and 7.7 ppt; mean mid-time differences from QLP +0.19 ± 0.55 and +0.42 ± 0.84 min); TARS (79.8 and 8.1 ppt); aperture photometry on TESScut images (this work; s44, s45: 77.9 ppt); ASAS-3 (the primary at +72 mmag, 7.0σ, at the TESS period). The TESS-SPOC s45 search detected a single primary as a TCE at 17.0347 d (79.8 ppt, 7.79 h, MES 431), 0.45 ± 2.50″ from the out-of-transit centroid; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of the single primaries in s44 and s45: deficit centroid 0.8″ and 0.5″ from the light centroid, a scaled copy of the star's image (residual 9–11%), depth 67.8, 68.4 and 65.9 ppt in 1×1, 3×3 and 5×5 pixel apertures (s44; the median over the rounded bottom). The two s44 secondaries, stacked, are also on the target (7.3 ppt, 2.4″ from the light centroid). No Gaia source within 60″ is bright enough: the brightest, G 13.73 at 43″, would need a 289% eclipse. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), the SuperWASP false-positive catalogue (Schanche et al. 2019), general web search. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Open issues	TESS has observed the star in one season only (2021): in sectors 71 and 72 it falls off the science pixels. The period is therefore known to ±0.00032 d, and the predicted times carry ±49 min (1σ). A joint fit of the TESS and ASAS-3 timings would reduce the period error
Next eclipses	Primary: BTJD 4337.861 (2026-10-24 08:34 UTC), 4354.883; secondary: BTJD 4328.359 (2026-10-14 20:32 UTC), 4345.381 (allow ±148 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

An F-type star (6200 K) with an M-dwarf companion of about 0.45 $M_\odot$ and 0.46 $R_\odot$ on an eccentric (e = 0.27) 17.02-d orbit. The 80-ppt primary is the M dwarf crossing the F star, with a rounded, limb-darkened bottom; the 7.7-ppt secondary is its total occultation, flat-bottomed. Both eclipses fall on Gaia's conjunctions within 1.7σ at Gaia's own epoch, and one geometry with Gaia's e and ω accounts for the secondary's phase (0.442) and its shorter duration at the nearer conjunction; the depth ratio is set by the surface-brightness ratio of the two stars. ASAS-3 recovers the primary more than a decade before TESS and restricts the period to 17.0215–17.0221 d, in agreement with the TESS period, which differs from Gaia's 17.02361 d by 1.7σ. With only three primaries from one TESS season, the next eclipses are uncertain by about 50 min (1σ); a single timed eclipse from the ground would reduce this to minutes, and the total occultation makes the system suitable for a direct measurement of the M dwarf's temperature.

A81 CD−24 5530 = TIC 109061795

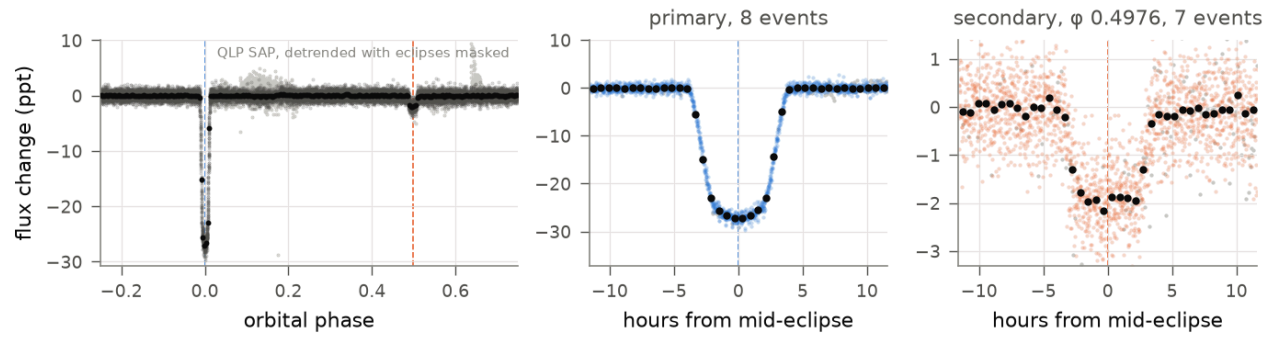


Figure A81. CD-24 5530, raw QLP SAP flux of sectors 7, 34, 61 and 88, sector medians removed and detrended per sector by 1-d bin medians with the eclipses masked, folded on P = 13.2271715 d (left; flagged cadences grey), with zooms on the primary and on the secondary at phase 0.4976 (binned medians in black): a flat-bottomed 25.6-ppt, 7.25-h primary and a 1.9-ppt secondary of the same duration.

Identifiers	TIC 109061795; CD-24 5530; CPD-24 2516; TYC 6542-2411-1; Gaia DR3 5615242068166737280
Position (J2000)	112.91182, −24.52605 deg (TIC)
Brightness	Tmag 9.34; G 9.77
Stars	TIC: T_{eff} 5899 K, 2.32 $R_\odot$ (Gaia GSP-Phot 5925 K) at 247 pc; Gaia single-lined orbit (SB1, e = 0.039 ± 0.011, K_1 23.1 km/s). The companion is an M dwarf: the mass function, 0.0168 $M_\odot$, gives $M_2 \geq 0.32\ M_\odot$ for M_1 = 1.07 $M_\odot$; the flat-bottomed primary gives $R_2/R_1 \approx 0.16$ ($R_2 \approx 0.37\ R_\odot$, about 0.33–0.35 $M_\odot$); the depth ratio, 0.074, gives about 3200 K. $i \approx 86$–88°
Period	13.2271715 ± 0.0000023 d, timing fit of 8 primaries; 7 secondaries give 13.227137 ± 0.000023 d (−1.5σ) (Gaia: 13.22633 ± 0.00115 d; +0.7σ)
Epochs	

	Primary BTJD 3040.42781 ± 0.00013 = BJD_{TDB} 2460040.42781; secondary BTJD 3047.00989 (phase 0.49762 ± 0.00011)
Primary eclipse	25.64 ± 0.14 ppt, 7.25 h, flat-bottomed ($T_{23}/T_{14} \approx 0.6$), in raw QLP SAP flux (trapezoid fit to the fold of 8 events); alternate events 26.03 and 25.76 ppt. The QLP depth per sector falls from 27.8 ppt (s7) to 25.4 ppt (s88) at constant duration and depth ratio, while TESS-SPOC SAP is constant at 24.4–25.0 ppt and PDCSAP at 26.7–27.2 ppt: the change is in the dilution of QLP's aperture by the G 13.94 neighbour, not in the binary
Secondary eclipse	1.89 ± 0.04 ppt, 7.25 h, at phase 0.49762 (7 events of 1.7–2.2 ppt; 1.85–2.19 ppt in every pipeline)
Orbit	At Gaia's epoch with the TESS period (200 cycles back): primary on conjunction A at −0.0014 ± 0.0025 (0.6σ), secondary on conjunction B at −0.0036 ± 0.0030 (1.2σ); carried to the TESS epochs with Gaia's period, +0.0110 ± 0.0165 (0.7σ) and +0.0085 ± 0.0170 (0.5σ). $e \cos \omega$ = −0.0037 ± 0.0002 from the eclipse spacing, against Gaia's −0.0006 ± 0.0071 (0.4σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s7, s34, s61, 6 primaries: matched mid-times −0.21 ± 0.24 min from QLP's, 24.9 ppt in SAP and 26.9 ppt in PDCSAP); TARS (8 primaries, 25.9 ppt); TGLC aperture photometry (s7: 27.9 ppt, −0.51 ± 0.19 min); ASAS-3 V (2001–2009, about 497 cycles earlier): +21.0 ± 5.0 mmag in the eclipse core on the TESS ephemeris against 28.2 mmag expected, consistent with the eclipse but only about 2.5σ against the scatter of trial periods (sd 7.8 mmag). The TESS-SPOC s61 search detected the eclipses as a TCE at P/2, 6.61375 d (primaries and secondaries folded together; 27.0 ppt, MES 95), 0.12 ± 2.5″ from the catalogue position; it was not flagged as an eclipsing binary
Pixel attribution	TESScut difference images of 2 primaries each in s34, s61 and s88: the deficit centroid is 0.6–2.0″ from the catalogue position, while the out-of-eclipse light centroid is pulled up to 0.20 px toward the G 13.94 neighbour 16.9″ (0.8 px) away. The depth is largest on the target's pixel (27.3, 25.1 and 23.8 ppt in 1×1, 3×3 and 5×5 pixel apertures in s34) and falls in larger apertures, which take in the neighbour's constant light. That neighbour would need a 122% eclipse; none of the 75 Gaia sources within 60″ can produce the dip. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), general web search. Listed in Kostov et al. (2025), table 2, as an unvetted machine-learning candidate. SIMBAD's single reference is the Gaia DR3 non-single-star catalogue. Known SB (Gaia DR3)
Next eclipses	Primary: BTJD 4336.691 (2026-10-23 04:33 UTC), 4349.918; secondary: BTJD 4330.046 (2026-10-16 13:05 UTC), 4343.273 (allow ±2 min for primaries and ±6 min for secondaries, whose phase is less well fixed; the secondaries also scatter by 4.2 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A slightly evolved G star (2.3 $R_\odot$) with an M-dwarf companion of about 0.35 $M_\odot$ on a near-circular 13.23-d orbit. The primary is a flat-bottomed transit of the M dwarf, and the 1.9-ppt secondary at phase 0.4976 is its occultation, implying a temperature near 3200 K. Both eclipses fall on Gaia's conjunctions within 1.2σ, and the TESS period agrees with Gaia's to 0.7σ. A G 13.94 star 17″ away shares the TESS pixels: the difference images place the deficit on CD-24 5530 while the light centroid is displaced toward the neighbour, and the slow decrease of the QLP depth from sector to sector, absent from TESS-SPOC, is a change in that neighbour's contribution to QLP's aperture. ASAS-3 photometry two decades earlier is consistent with the eclipse where the TESS ephemeris carried back 497 orbits places it.

A82 CD−83 126 = TIC 317250623

TIC 317250623 CD-83 126 P = 19.60528 d

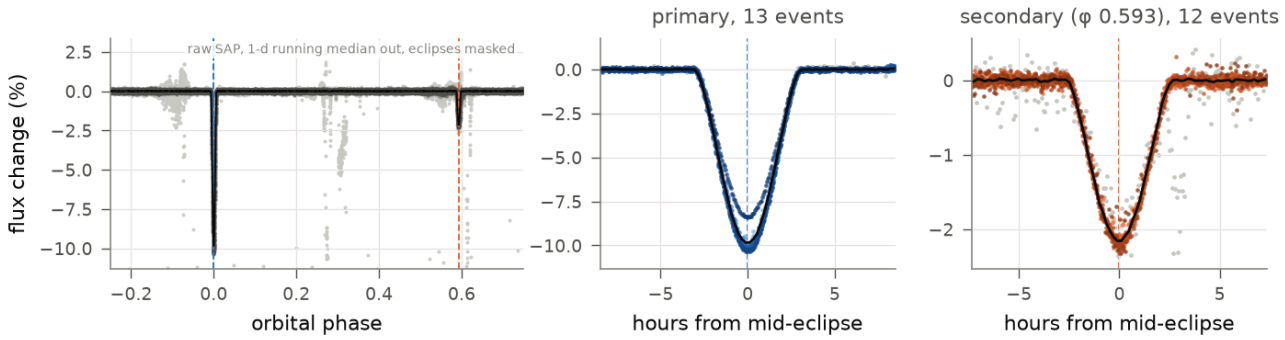


Figure A82. CD-83 126, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 19.605278 d (flagged cadences grey). Left: the 98-ppt primary at phase 0 and the 20.6-ppt secondary at phase 0.593. Middle: the 5.65-h primary, each event normalised to its own out-of-eclipse flux; the shallower track is the sector-94 event in QLP flux, which TARS gives at full depth. Right: the 4.9-h secondary, aligned on its own linear ephemeris. The panels plot 13 primaries and 12 secondaries, including partly covered events; the timing fits use the 10 primaries and 11 secondaries that are fully covered.

Identifiers	TIC 317250623; CD-83 126; CPD-83 220; TYC 9506-410-1; Gaia DR3 5193353340543454848
Position (J2000)	122.99276, −84.23232 deg (TIC)
Brightness	Tmag 8.98; G 9.41
Stars	G0 (SIMBAD; TIC: T_{eff} 5301 K, 2.34 $R_\odot$ for the combined light) at 159 pc; Gaia SB2 orbit (e = 0.1862 ± 0.0053, ω = 36.0 ± 2.5°, K_1 = 49.57 and K_2 = 48.74 km/s): a near-twin pair of about 0.92 $M_\odot$ each at $i \approx 86°$, a = 37.4 $R_\odot$. Gaia RUWE 6.7
Period	19.605278 ± 0.000014 d, timing fit of 10 primaries (Gaia: 19.60653 ± 0.00145 d; −0.9σ). The secondaries alone give 19.604896 ± 0.000013 d (see Open issues)
Epochs	Primary BTJD 3060.10199 ± 0.00051 = BJD_{TDB} 2460060.10199; secondary BTJD 3071.72505 ± 0.00040 (phase 0.5929; 0.5945 in sector 11, 0.5928 in sector 65)
Primary eclipse	98 ppt, 5.65 h in raw QLP SAP flux (trapezoid fits to 10 events); the sector-94 event is 81.9 ppt in QLP and 96.1 ppt in TARS
Secondary eclipse	20.6 ± 1.1 ppt, 4.9 h, at phase 0.593 (11 events). It is a grazing eclipse (b ≈ 0.9) at conjunction B, where the stars are 40.6 $R_\odot$ apart against 32.6 $R_\odot$ at conjunction A; this, not a difference between the stars, makes it much shallower than the primary
Orbit	Primary on Gaia's conjunction A: Δφ = −0.0022 ± 0.0021 (1.1σ); secondary on conjunction B: −0.0054 ± 0.0023 (2.3σ; an upper bound, since it carries the secondary's non-constant period back 125 cycles). Eclipse separation 0.59285 against Gaia's 0.5960 ± 0.0041 (0.8σ): $e \cos \omega$ = +0.146 against Gaia's +0.1505 ± 0.0065. The observed drift carried back to Gaia's epoch gives a separation of 0.5957
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s65–s67: mid-times within 0.0–0.7 min of QLP's, depths within 2–6%); eleanor-lite (s11–s13) and TGLC aperture photometry (s11) within 0.7 min (TGLC PSF 1.7 min); TARS (all sectors, 20 events, −1.3 to +1.5 min). The TESS-SPOC multi-sector search (s56–s69) produced TCEs on both eclipses, at 19.6042 d (MES 1472, 98.9 ppt) and 19.6053 d (MES 302, 21.0 ppt), with centroids 0.53 ± 2.54″ and 0.19 ± 2.54″ from the TIC position; neither is flagged as an eclipsing binary. ASAS-3 (2000–2009) shows the primary at the TESS period (+94 mmag, SNR 9.7), so the primary ephemeris holds from 2000 to 2023 to about $\pm 3 \times 10^{-5}$ d
Pixel attribution	TESScut difference images in s27 (two primaries) and s66 (one primary, one secondary): the peak deficit is in the target's pixel, the deficit centroid 3.4–3.6″ (0.16 px) from the light centroid; the deficit has the star's image shape (residual 19–23%, from uneclipsed diffuse light in the cutouts) and primary depths of 84/83/78 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest of the nine Gaia sources within 60″ is G 18.15 at 42.8″; every neighbour would need an eclipse of more than 30,000%. A companion within about 1″ is not excluded
Catalogue status	

	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, G0); its SIMBAD references (RAVE, APASS, diameter catalogues, Gaia DR3) do not mention variability. TESS-SPOC TCEs at both eclipses, not promoted to TOIs (Table A1)
Open issues	The secondary drifts against the primary. The two linear periods differ by 0.00038 ± 0.00002 d (18σ); measured within single sectors, the secondary-minus-primary interval is 11.65577 d in sector 11 and 11.62176 d in sector 65, a change of −49.0 min in 75 cycles, also seen in TARS and in eleanor-lite and TGLC against TESS-SPOC. The secondary O−C is curved. As apsidal motion this would need dω ≈ 0.020° per cycle (1.47° in 75 cycles at 33.4 min per degree for e = 0.186, ω = 36°; U ≈ 990 yr), more than 100 times the general-relativistic rate, and the tidal term is negligible (R/a ≈ 0.035); a third body perturbing the orbit is the likely cause, consistent with RUWE 6.7 for a near-twin. QLP depths rise from Year 1 to Year 5 (secondary 19.0 to 21.3 ppt), but TARS shows a much smaller rise and none in the primary, so a depth change is not established
Next eclipses	Primary: BTJD 4334.445 (2026-10-20 22:41 UTC), 4354.050 (allow ±5 min); secondary, from its own linear ephemeris: BTJD 4346.043 (2026-11-01 13:02 UTC), 4365.648 (allow ±15 min, given the drift)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-83 126 is an eccentric (e = 0.19) pair of nearly equal G stars of about 0.92 $M_\odot$, seen at i ≈ 86°. The 98-ppt primary occurs at the closer conjunction; at the wider one the chord is nearly grazing, so the secondary is only 21 ppt deep. Both eclipses fall on Gaia's double-lined conjunctions, their separation reproduces Gaia's e cos ω, the deficit is on the target, and SPOC's own search detected both eclipses. Between 2019 and 2023 the secondary moved 49 min earlier relative to the primary, with a curved O−C. As apsidal motion that would be more than a hundred times faster than general relativity allows for this orbit, and together with a Gaia RUWE of 6.7 for a near-twin it points to a third body. Continued timing of both eclipses and the Gaia DR4 epoch radial velocities can test this; because the secondary is near grazing, its depth is also a sensitive monitor of any change in inclination.

A83 BD+13 1919 = TIC 7060775

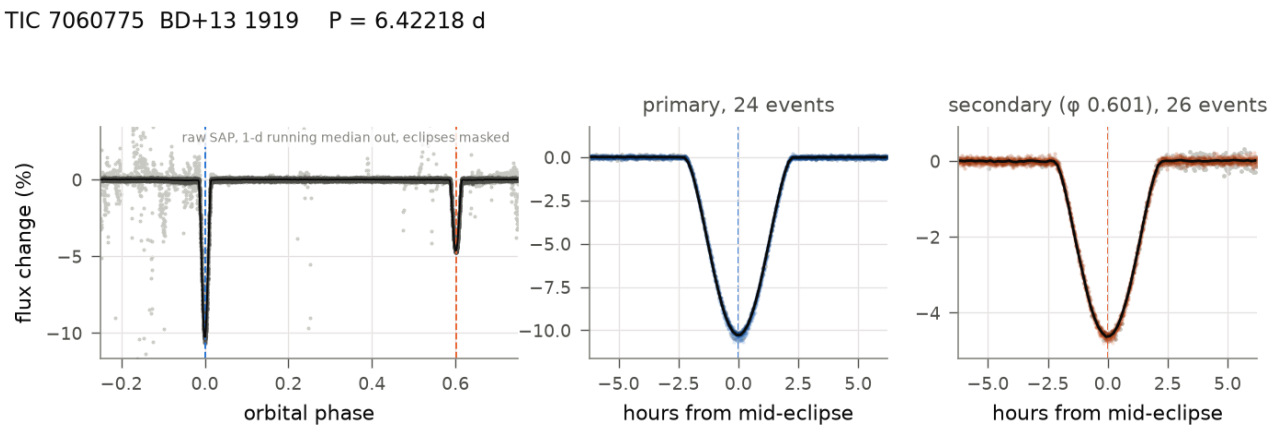


Figure A83. BD+13 1919, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 6.422180 d (flagged cadences grey). Left: the 101-ppt primary at phase 0 and the 45.9-ppt secondary at phase 0.601. Middle and right: the two eclipses, each event normalised to its own out-of-eclipse flux; both last 4.15 h. The panels plot 24 primaries and 26 secondaries, including partly covered events; the timing fits use the 23 and 24 that are fully covered.

Identifiers	TIC 7060775; BD+13 1919; TYC 804-821-1; Gaia DR3 650833814750768768
Position (J2000)	126.92111, +12.76900 deg (TIC)
Brightness	Tmag 9.16; G 9.55
Stars	

	F7 (SIMBAD; TIC: T_{eff} 6133 K, 1.89 $R_\odot$ for the combined light) at 198 pc; Gaia SB2 orbit (e = 0.234 ± 0.019, ω = 50.0 ± 1.9°; K_1/K_2 = 1.19, so Gaia's secondary is the heavier star): about 0.98 and 1.16 $M_\odot$ at i ≈ 85°, a ≈ 18.7 $R_\odot$. The two eclipse durations give $R_1 + R_2 \approx 2.4\ R_\odot$, two late-F dwarfs. RUWE 1.06
Period	6.422180 ± 0.000001 d, timing fit of 23 primaries; 24 secondaries give 6.422175 ± 0.000001 d (Gaia: 6.421481 ± 0.000299 d; +2.3σ)
Epochs	Primary BTJD 2741.55165 ± 0.00012 = BJD_{TDB} 2459741.55165; secondary BTJD 2841.74306 ± 0.00011 (phase 0.60080). O–C rms 0.5 min for each kind
Primary eclipse	101 ± 2 ppt, 4.15 h in raw QLP SAP flux (trapezoid fits to 23 events); SPOC 2-min SAP 96.6 ppt, PDCSAP 98.2 ppt
Secondary eclipse	45.9 ± 0.7 ppt, 4.15 h, at phase 0.60080 (24 events); SPOC 2-min SAP 43.2 ppt, PDCSAP 44.2 ppt. Impact parameters b ≈ 0.57 at conjunction A (separation 15.0 $R_\odot$) and b ≈ 0.82 at conjunction B (21.5 $R_\odot$) reproduce the equal durations and the depth ratio
Orbit	Primary on Gaia's conjunction A: Δφ = +0.0023 ± 0.0030 (0.8σ); secondary on conjunction B: +0.0061 ± 0.0060 (1.0σ). Eclipse separation 0.60080 against Gaia's 0.5970 ± 0.0085 (+0.4σ): e cos ω = +0.158 against Gaia's +0.150 ± 0.013
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s34, s44; 13 events, mid-times within 0.2 min of QLP's); TESS-SPOC SAP (s7, s34, s44, s71; 25 events, within 0.1 min); eleanor-lite (s7, 0.1 min); TARS. QLP depths are 2–6% deeper than the other pipelines'. The SPOC multi-sector search (s1–s46) produced TCEs on both eclipses, at 6.4221771 d (MES 2600, 101.0 ppt) and 6.422174 d (MES 1017, 45.3 ppt), with centroids 0.48 ± 2.54″ and 0.55 ± 2.52″ from the TIC position; neither is flagged as an eclipsing binary. ASAS-3 (2000–2009) is consistent with the TESS ephemeris at SNR ≈ 3
Pixel attribution	TESScut difference images in s45 (four primaries, three secondaries) and s71 (three primaries): the deficit centroid is 0.2–0.4″ from the light centroid (both share a 0.2–0.3 px offset from the catalogue position, which lies in the cutouts' astrometry); the deficit is PSF-shaped (residual 9–23%), with primary depths of 87.1/86.7/86.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest of six Gaia sources within 60″, G 14.60 at 16.0″, would need a 1067% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, F7). SIMBAD links the star to Whitney (1958, AJ 63, 243), a study of the eclipsing binary TW Cnc 34′ away, most likely as a comparison star; that paper itself was not examined. SPOC TCEs at both eclipses, not promoted to TOIs (Table A1)
Open issues	The secondaries give a period shorter than the primaries' by $(5 \pm 1.4) \times 10^{-6}$ d, the size expected from apsidal motion in this orbit: general relativity gives about 4.9×10^{-6} rad per cycle and the tidal term about 1.6×10^{-6} rad per cycle, together about 4.7×10^{-6} d per cycle in the eclipse spacing. At about 3σ, with mid-times quantised at 1 min, this is a hint of apsidal motion only
Next eclipses	Primary: BTJD 4327.830 (2026-10-14 07:56 UTC), 4334.252, 4340.674 (allow ±2 min); secondary: BTJD 4331.689 (2026-10-18 04:32 UTC), 4338.111 (allow ±4 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+13 1919 is an eccentric (e = 0.23) double-lined pair of late-F dwarfs of about 1.0 and 1.2 $M_\odot$, with 10% and 4.6% eclipses of equal duration. The deeper eclipse falls at the closer conjunction, where the chord is nearer the centre; the shallower one, at the wider conjunction, is close to grazing. Both fall on Gaia's conjunctions within 1σ, the eclipse spacing reproduces Gaia's e cos ω, the 47 eclipses fit linear ephemerides with 0.5-min residuals, and SPOC's own search detected both eclipses at very high significance without either becoming a TOI. The TESS period is 2.3σ longer than Gaia's, a difference of the size seen in several other Gaia orbits tested here. The two eclipse kinds give a hint of apsidal motion, to which general relativity and tides both contribute; further seasons of TESS timing can test it.

A84 TYC 3805-1342-1 = TIC 456339196

TYC 3805-1342-1 = TIC 456339196 P = 7.106115 d

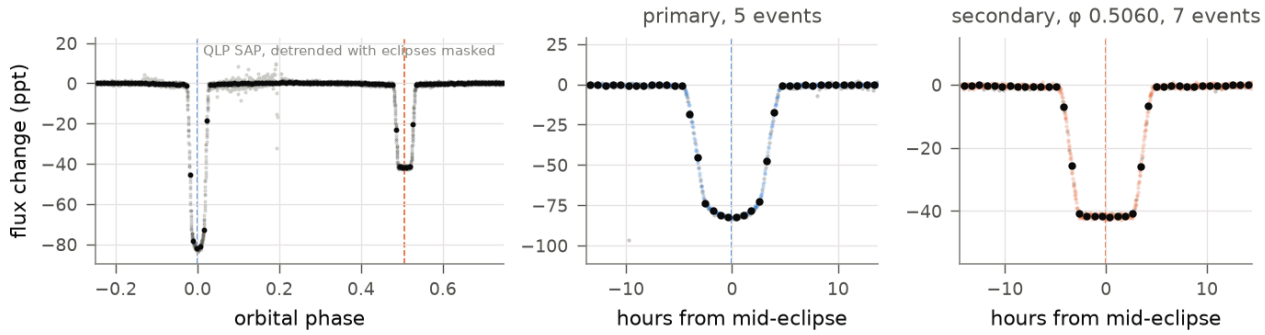


Figure A84. TYC 3805-1342-1, QLP SAP flux detrended with the eclipses masked and folded on P = 7.106115 d (binned points black). Left: the 78.6-ppt primary at phase 0 and the 41.3-ppt secondary at phase 0.5060. Middle: the 8.6-h primary, with the rounded bottom of a transit across a limb-darkened star. Right: the 9.0-h secondary, flat-bottomed: a total occultation of the companion.

Identifiers	TIC 456339196; TYC 3805-1342-1; Gaia DR3 1017462707153422976
Position (J2000)	135.49647, +52.53628 deg (TIC)
Brightness	Tmag 9.57; G 9.92
Stars	Slightly evolved F-type primary (TIC: T_{eff} 6384 K, 2.89 $R_\odot$) at 388 pc; Gaia SB1 orbit (e = 0.008 ± 0.005, K_1 = 52.66 km/s, f(M) = 0.107 $M_\odot$). The eclipses make the companion a G9–K0 dwarf of T_{eff} ≈ 5300 K, giving 4.1% of the TESS light, and of about 0.78 $M_\odot$ (the mass-function minimum is 0.77 $M_\odot$). The radii await a light-curve model: the long flat bottom of the occultation requires a nearly central eclipse and a radius ratio of about 0.25 or less
Period	7.1061147 ± 0.0000043 d, timing fit of 5 primaries; 7 secondaries give 7.106109 ± 0.000002 d (Gaia: 7.105995 ± 0.000157 d; +0.8σ). The fold ran at half the orbital period, 3.5531 d. The alternate eclipses differ in depth (78.6 and 41.3 ppt) and are 0.506, not 0.5, apart in phase
Epochs	Primary BTJD 2530.89350 ± 0.00012 = BJD_{TDB} 2459530.89350; secondary BTJD 2534.48917 (phase 0.50600 ± 0.00005)
Primary eclipse	78.6 ± 0.3 ppt, 8.6 h in raw QLP SAP flux (trapezoid fits to 5 events), rounded bottom: the transit of the companion across the F star
Secondary eclipse	41.3 ± 0.1 ppt, 9.0 h, at phase 0.50600 (7 events), flat-bottomed: the total occultation of the companion. The 0.006 offset from phase 0.5 (0.0426 d, 61 min) is the orbit's small eccentricity
Orbit	With the TESS period carrying the eclipses back 301 cycles to Gaia's epoch: primary on Gaia's conjunction A, Δφ = −0.0022 ± 0.0018 (1.2σ); secondary on conjunction B, +0.0002 ± 0.0018 (0.1σ). $e \cos \omega$ = +0.0094 ± 0.0001 from the secondary's phase, against Gaia's +0.0069 ± 0.0049 (0.5σ). The deeper eclipse is at conjunction A, where the radial-velocity star is behind, as expected when the visible spectrum is the larger, hotter star's
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s21, s47: primary 78.4–78.6 ppt, secondary 40.6–40.7 ppt; matched mid-times −0.07 ± 0.16 and +0.39 ± 0.64 min from QLP's); TARS (81.3 and 41.8 ppt). The TESS-SPOC search in s47 produced one TCE at 3.55304 d, half the orbital period (MES 929, mean depth 42.2 ppt), not flagged as an eclipsing binary, with its centroid 0.10 ± 2.5″ from the TIC position
Pixel attribution	TESScut difference images in s21 (two primaries, four secondaries) and s47 (three primaries, four secondaries): both eclipse kinds have deficit centroids 0.1–0.7″ from the light centroid, as scaled copies of the star's image (residual 1.3–3.1%), with depths flat across apertures (primary 79.1/79.0/79.2 ppt in 1×1, 3×3 and 5×5 pixels in s21). Only two Gaia sources lie within 60″: G 16.76 at 54.8″ and G 17.27 at 58.7″, which would need eclipses of 4306% and 6886%. A companion within about 1″ is not excluded
Catalogue status	

	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, from Gaia DR3 only, no spectral type). No ASAS-3 coverage. TESS-SPOC TCE at half the period, not promoted to a TOI (Table A1)
Next eclipses	Primary: BTJD 4328.741 (2026-10-15 05:46 UTC), 4335.847, 4342.953; secondary: BTJD 4332.336 (2026-10-18 20:03 UTC), 4339.442 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3805-1342-1 is a slightly evolved F star with a G9–K0 dwarf companion in a 7.1-d, nearly circular orbit. The two eclipses are both deep but unequal: a 78.6-ppt transit with a limb-darkened bottom and a 41.3-ppt flat-bottomed occultation whose depth is the companion's 4.1% share of the light, giving a surface-brightness ratio of 0.53 and T_{eff} ≈ 5300 K against 6400 K. The secondary sits at phase 0.5060, so no half-period fold can align the two, and its offset gives $e \cos \omega = 0.0094$, within 0.5σ of Gaia's. Both eclipses lie on Gaia's conjunctions, the deficit is on the target, and TESS-SPOC reproduces the depths to 1%. Radial velocities of the companion's lines, if detectable at 4% of the light, would give both masses.

A85 CD−27 6336 = TIC 38388435

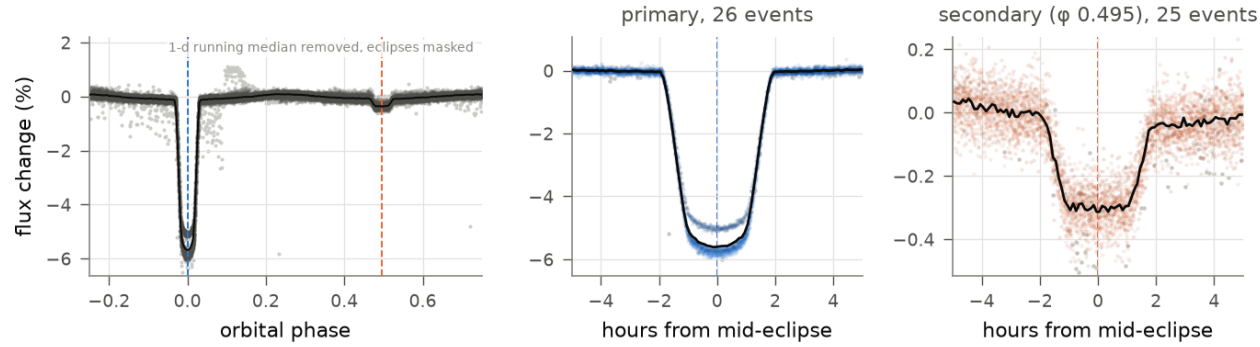


Figure A85. CD-27 6336, QLP SAP flux with sector medians removed, detrended by a 1-d running median with the eclipses masked and folded on P = 2.6164745 d (flagged cadences grey). Left: the 55-ppt primary at phase 0 and the 2.8-ppt secondary at phase 0.495. Middle: the 3.6-h primary, each event normalised to its own out-of-eclipse flux; the shallower track is sector 99, where the QLP flux is diluted (about 50 ppt). Right: the flat-bottomed secondary, 3.7 h. The panels plot 26 events of each kind, including partly covered events; the timing fits use the 25 that are fully covered.

Identifiers	TIC 38388435; CD-27 6336; CPD-27 3674; SAO 177090; TYC 6598-160-1; Gaia DR3 5636144020418779008
Position (J2000)	137.87691, −28.14134 deg (TIC)
Brightness	Tmag 9.14; G 9.46
Stars	F-type primary (TIC: T_{eff} 6772 K, 1.72 $R_\odot$, 1.45 $M_\odot$) at 202 pc; Gaia SB1 orbit (e = 0.025 ± 0.011, ω = 167 ± 21°, K_1 = 40.36 km/s, f(M) = 0.0178 $M_\odot$). The companion is an M dwarf of about 0.39 $M_\odot$ (0.32–0.46 for the range of primary masses), R ≥ 0.40 $R_\odot$, T_{eff} ≈ 3300–3500 K. A G 14.30 Gaia source 2.2″ away contributes about 1.1% of the light. RUWE 0.90
Period	2.6164745 ± 0.0000005 d, timing fit of 25 primaries; 25 secondaries give 2.6164734 ± 0.0000019 d (Gaia: 2.616527 ± 0.000044 d; −1.2σ)
Epochs	Primary BTJD 3518.67725 ± 0.00008 = BJD_{TDB} 2460518.67725; secondary BTJD 3519.97302 (phase 0.49524 ± 0.00008). Primary O−C rms 0.21 min
Primary eclipse	55 ppt, adopted from the difference images (55.1–55.4 ppt in s62 and s99) and from SPOC 2-min and TESS-SPOC (54.5–56.0 ppt); 3.6 h (trapezoid fits to 25 events in raw QLP SAP flux; 3.5–3.7 h by fitting method). QLP trapezoid fits give 56.6, 55.3 and 48.7 ppt in sectors 62, 89 and 99; the sector-99 QLP flux is diluted, since the sector-99 difference image gives 55.4 ppt

Secondary eclipse	2.8 ppt (2.7 ppt in QLP, 2.8–2.9 ppt in SPOC), 3.7 h, flat-bottomed, at phase 0.49524 (25 events): the total occultation of the M dwarf
Orbit	With the TESS period carrying the eclipses to Gaia's epoch: primary on Gaia's conjunction A, $\Delta\varphi = -0.0045 \pm 0.0037$ (1.2σ); secondary on conjunction B, $+0.0054 \pm 0.0034$ (1.6σ); conditioning Gaia's elements on the TESS period gives 1.8σ and 1.9σ. $e \cos \omega = -0.0075 \pm 0.0001$ from the secondary's phase, against Gaia's -0.023 ± 0.010 (1.5σ)
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP and TESS-SPOC SAP and PDCSAP (s62; 10 shared primaries, median mid-time differences 0.00–0.04 min, rms ≤ 0.11 min); TARS (s62, s89; +0.5 min). The SPOC 2-min search in s62 produced a TCE at 2.616486 d (MES 504, 57.2 ppt), its centroid 0.79 ± 2.5″ from the TIC position, and a second TCE on the secondary (3.0 ppt, MES 29); TESS-SPOC gives MES 759 and 37. Neither is flagged as an eclipsing binary. ASAS-3 V photometry (2000s, about 2700 cycles before the TESS epoch) folded on the TESS ephemeris shows the eclipse: +65 ± 4 mmag in 21 points in the eclipse core, against 61 mmag expected, and the TESS period gives the deepest signal among trial periods within ±4 eclipse durations of drift
Pixel attribution	TESScut difference images in s62 (10 events) and s99 (7 events): the deficit centroid is 0.2″ from the light centroid, with the peak deficit in the target's pixel; PSF-fit residuals 2.3% and 3.4%; depths 55.4/55.1/55.3 and 55.9/55.4/55.4 ppt in 1×1, 3×3 and 5×5 pixel apertures. None of the 22 Gaia sources within 60″ can make the eclipse: the G 14.30 star at 2.2″ would need 473%, and G 15.99 at 28.6″ 2225%. A source brighter than G 12.6 within about 1″ is not excluded; Gaia shows none
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary`, SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, no spectral type, three references). SPOC and TESS-SPOC TCEs in s62, not promoted to TOIs (Table A1)
Next eclipses	Primary: BTJD 4329.784 (2026-10-16 06:52 UTC), 4332.401, 4335.017; secondary: BTJD 4328.464 (2026-10-14 23:10 UTC), 4331.080 (allow ±2 min; the secondaries scatter by 2.4 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-27 6336 is an F star eclipsed every 2.62 d by an M dwarf of about 0.4 $M_\odot$. The 55-ppt primary is a transit at impact parameter about 0.5–0.65, and the flat-bottomed 2.8-ppt secondary is the dwarf's total occultation. The orbit is nearly circular: the secondary at phase 0.4952 gives $e \cos \omega = -0.0075$, 1.5σ from Gaia's value, and the two eclipses fall on Gaia's conjunctions within 1.6σ. The deficit is on the target in two sectors, SPOC 2-min and TESS-SPOC timings agree with QLP's to 0.1 min, and ASAS-3 data from the 2000s show the eclipse on the TESS ephemeris. With a Tmag 9.1 primary and a deep transit by a low-mass star, the system is suited to radial-velocity work on the companion's mass and radius.

A86 BD−01 2238 = TIC 170891257

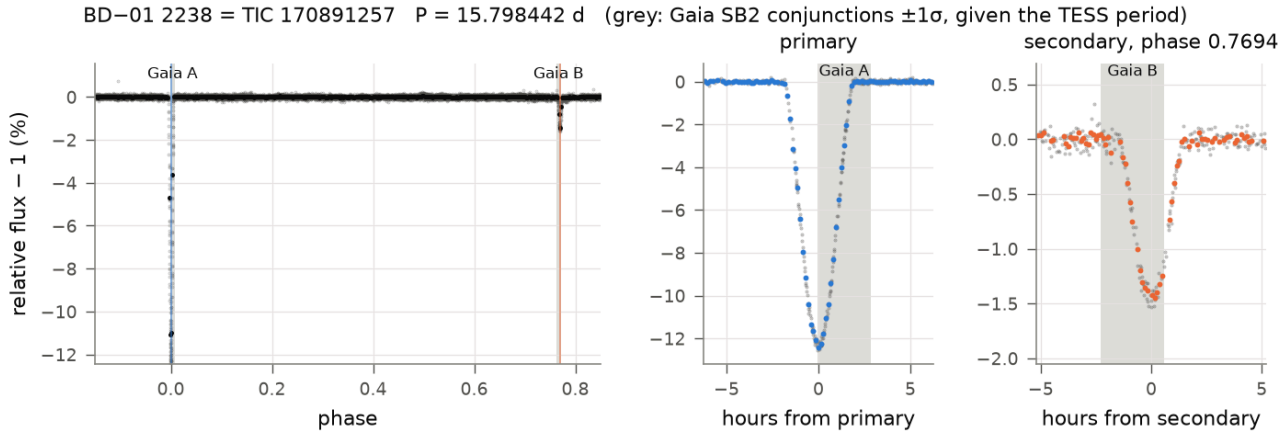


Figure A86. BD-01 2238, QLP SAP flux with sector medians removed, detrended by a running median with the eclipses masked and folded on P = 15.798442 d; individual cadences grey, binned points coloured. Grey bands mark Gaia's predicted conjunctions A and B (±1σ) carried with the TESS period. Left: the 120-ppt primary at phase 0 and the 14-ppt secondary at phase 0.7694. Middle: the V-shaped 3.5-h primary. Right: the 2.8-h grazing secondary.

Identifiers	TIC 170891257; BD-01 2238; TYC 4884-610-1; Gaia DR3 3838406607374037632
Position (J2000)	139.65855, −2.55169 deg (TIC)
Brightness	Tmag 9.41; G 9.87
Stars	F9 (SIMBAD; TIC: T_{eff} 5732 K, 1.28 $R_\odot$ for the combined light) at 135 pc; Gaia SB2 orbit (e = 0.4313 ± 0.0121, ω = 23.89 ± 1.45°, K_1 = 57.8 and K_2 = 57.6 km/s, so q = 1.00): two equal dwarfs of about 0.93 $M_\odot$ at $i \approx 87°$ (the F9 type and the TIC temperature describe their combined light), a sin i = 32.5 $R_\odot$. RUWE 1.10
Period	15.798442 ± 0.000003 d, timing fit of 4 secondaries spanning 2543 d; 4 primaries give 15.798447 ± 0.000013 d, in agreement (Gaia: 15.789859 ± 0.002086 d; +4.1σ)
Epochs	Primary BTJD 2811.62273 ± 0.00024 = BJD_{TDB} 2459811.62273; secondary BTJD 2744.78595 (phase 0.76941)
Primary eclipse	120 ppt (118.8–122.6), 3.5 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 4 events in s35 and s61); TESS-SPOC 115–117 ppt
Secondary eclipse	14 ppt, 2.8 h, at phase 0.76941 (4 events: two in s61, one in s99 in flagged but clean cadences, and one in s08). The stars are nearly equal; the secondary is shallow because it is grazing (b ≈ 0.90) at conjunction B, where the separation is 0.986a against 0.693a at conjunction A (b ≈ 0.63)
Orbit	Primary near conjunction A, secondary near conjunction B. Conditioned on the TESS period, which lies 4.1σ from Gaia's, through Gaia's full covariance: −0.0037 ± 0.0039 (0.9σ) and +0.0022 ± 0.0036 (0.6σ); with the correlations ignored, −0.0214 ± 0.0064 (3.3σ) and +0.0001 ± 0.0082 (0.0σ); with Gaia's own period carried to the TESS epochs (72–153 cycles), +0.062 ± 0.016 (3.8σ) and +0.081 ± 0.020 (4.1σ). The eclipse spacing, 0.7694, against Gaia's predicted 0.748 ± 0.008, differs by 2.7σ; it gives e cos ω = +0.4308 at Gaia's e sin ω, against Gaia's +0.394 ± 0.014 (2.5σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s35, s61: 3 primaries and 2 secondaries, same-event mid-times within 0.3 min of QLP's, depths 115–117 and 13.7–13.8 ppt); SPOC 2-min (the s08 secondary, 13.8 ppt); TARS (117.0 and 14.1 ppt). The SPOC search in s61 produced TCEs on both eclipses: the primary at 15.79788 d (MES 632, 120.4 ppt), its centroid 0.44 ± 2.50″ from the TIC position, and the secondary (MES 71, 13.2 ppt); the s08 2-min search produced a TCE on the s08 secondary. None is flagged as an eclipsing binary. ASAS-3 shows the primary on the TESS ephemeris 387 cycles earlier (5 points at 103 mmag, SNR 11)
Pixel attribution	TESScut difference images of the primary in s61 (one event) and s35 (two) and of the secondary in s61 (two): the deficit centroid is 0.1–0.6″ from the light centroid, with the peak deficit in the target's pixel each time; PSF-fit residuals 1.4–1.5% (primary) and 5.7% (secondary); primary depths 103.1/103.5/103.2 ppt in 1×1, 3×3 and 5×5 pixel apertures in s61. The brightest Gaia source within 90″, G 15.87 at 41.5″, would need eclipses of 3033% (primary) and 367% (secondary); the other eight are G ≥ 17.37. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, the ASAS-SN variable-star catalogue, Gaia DR3 variability tables (target and the four sources within 60″), SB9, WDS (120″), Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, F9; also typed Em*); its references (RAVE, APASS, LAMOST, Gaia DR3) report no eclipses. SPOC TCEs in s08 and s61, not promoted to TOIs (Table A1)
Open issues	Both the period (4.1σ) and the eclipse spacing (2.7σ) are in tension with Gaia's orbit, and the agreement of the conjunctions depends on conditioning Gaia's covariance on the TESS period; a light-curve model fitting e and ω from TESS alone is not yet made. $R_1 + R_2 = 1.58\ R_\odot$ from the trapezoid durations is a little below the ~1.75 $R_\odot$ expected for two 0.93-$M_\odot$ dwarfs
Next eclipses	Primary: BTJD 4328.273 (2026-10-14 18:36 UTC), 4344.072; secondary: BTJD 4340.429 (2026-10-26 22:18 UTC), 4356.227 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD-01 2238 is a pair of equal-mass dwarfs of about 0.93 $M_\odot$ (q = 1.00) in a 15.8-d orbit with e = 0.43, seen at $i \approx 87°$. The orbit alone sets the difference between the 120-ppt and 14-ppt eclipses of two equal stars: the deep one is at the close conjunction and the grazing one at the far conjunction, at phase 0.7694. The TESS period is 0.0086 d (4.1σ) longer than Gaia's. Conditioned on the TESS period through Gaia's full covariance, both eclipse kinds fall on Gaia's conjunctions within 1σ, but with the correlations ignored the primary is 3.3σ off, and the eclipse spacing differs from Gaia's prediction by 2.7σ; the period and the spacing are both in tension with the Gaia orbit. The deficit is on the target for both eclipses, TESS-SPOC and SPOC 2-min agree with QLP, and ASAS-3 shows the primary on the same ephemeris two decades earlier. A light-curve and radial-velocity model would resolve the 2.5σ difference in $e \cos \omega$ and give the radii of two nearly identical stars.

A87 HD 89725 = TIC 463896609

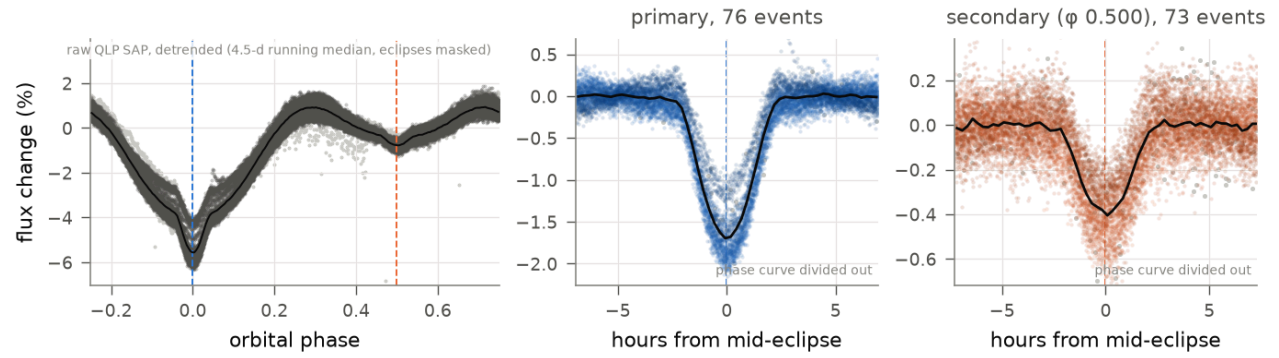


Figure A87. HD 89725, QLP SAP flux with sector medians removed, detrended by a 4.5-d running median with the eclipses masked and folded on P = 2.2366733 d (flagged cadences grey). Left: the full orbit, with the 6% peak-to-peak phase curve and the eclipses at phases 0 and 0.500. Middle and right: the V-shaped 4.3-h primary (16 ppt) and the 3.6-ppt secondary, each with the phase curve divided out. The panels plot 76 primaries and 73 secondaries, including partly covered events; the timing fit uses the 70 fully covered primaries.

Identifiers	TIC 463896609; HD 89725; CD-57 3153; CPD-57 2988; SAO 237919; TYC 8608-545-1; Gaia DR3 5258656806462084096
Position (J2000)	154.91715, −58.09636 deg (TIC)
Brightness	Tmag 8.89; G 9.13
Stars	

	F0/3IV/V (SIMBAD; Gaia GSP-Phot T_{eff} 7159 K; the TIC's 11,639 K is inconsistent with BP−RP 0.55 and the spectral type) at 371 pc, in a crowded Carina field; Gaia SB1 orbit (e = 0.050 ± 0.018, ω = 134 ± 24°, K_1 = 46.75 km/s, f(M) = 0.0236 $M_\odot$: $m_2 \geq 0.50\ M_\odot$ for $M_1 \approx 1.8\ M_\odot$), RUWE 1.68. The out-of-eclipse light curve has a reflection term (cos 2πφ, −18.3 ppt) and an ellipsoidal term (cos 4πφ, −14.6 ppt). The ellipsoidal term is of the size the F star's own tidal distortion gives in a detached F + M pair (about 11–15 ppt); the reflection term is too large for a detached main-sequence companion and suggests, as a hypothesis, a semi-detached, Algol-like system with an irradiated, Roche-lobe-filling companion
Period	2.2366733 ± 0.0000008 d, timing fit of 70 primaries, with errors scaled by 4.0 because the individual times scatter on the steep phase curve (O−C rms 3.1 min, no trend) (Gaia: 2.236642 ± 0.000026 d; +1.2σ)
Epochs	Primary BTJD 3195.87831 ± 0.00025 = BJD_{TDB} 2460195.87831; secondary BTJD 3196.9962 (phase 0.49982 ± 0.00036)
Primary eclipse	16.4 ppt (median of 70 events; sd 1.6), 4.3 h, V-shaped (grazing; SPOC fits give b = 0.99), in raw QLP SAP flux against a Fourier phase-curve model. The QLP depth varies by sector with dilution: 12.8 ppt in s99, 14.6–15.1 ppt in s9–s10, 17–18 ppt in s36–s64. TESS-SPOC SAP 14.5 ppt, PDCSAP 16.8 ppt
Secondary eclipse	3.6 ± 0.1 ppt (65σ) in a box at phase 0.5000 in the folded light curve against the phase-curve model; a trapezoid fit gives 4.0 ppt, 4.6 h, at phase 0.4998: the occultation of the companion
Orbit	With the TESS ephemeris carried 1255 cycles back to Gaia's epoch: secondary on Gaia's conjunction B, Δφ = −0.0016 ± 0.0064 (0.2σ); primary −0.0210 ± 0.0049 from conjunction A (4.3σ; 68 min, inside the eclipse half-width of 0.040 in phase). e cos ω = −0.0003 ± 0.0006 from the eclipse separation, against Gaia's −0.030 ± 0.015 (2.0σ); Gaia places B 0.481 ± 0.009 after A, against 0.4998 observed
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s9–s64, 62 events; 51 matched mid-times differ from QLP's by +0.15 ± 0.74 min); TARS (75 events, 15.7 ppt); TGLC (18.6 ppt aperture, 16.0 ppt PSF) and eleanor-lite (14.2–14.5 ppt) in s9–s10. TESS-SPOC searches produced TCEs in s36, s37, s63, s64 and s56–s69 (2.23660 d, 16.2 ppt, MES 26–43), centroids 0.9–1.3 ± 2.5″ from the TIC position, none flagged as an eclipsing binary. ASAS-3 V photometry folded on the TESS ephemeris (about 3000 cycles back) is consistent with the eclipse: +32 ± 5 mmag in 27 points in the eclipse core, a figure that includes the phase-curve minimum at the primary; the broad minimum does not single out the TESS period among trial periods
Pixel attribution	TESScut difference images in s63 and s90 (11 events each): the deficit centroid is 1.4″ and 2.6″ (0.06 and 0.12 px) from the catalogue position, with the peak deficit in the target's pixel; the light centroid is pulled 3–4″ away by neighbours (152 Gaia sources within 60″). Depths fall with aperture (24.3/23.8/22.5 ppt in 1×1, 3×3 and 5×5 pixels in s63; larger than the light-curve depth because the short windows include phase-curve curvature), as dilution predicts when the target is the source. The PSF-fit residual is 36–37%, from the crowding and the phase curve. Two neighbours are bright enough in principle, G 13.06 at 34″ (would need 63%) and G 13.28 at 57″ (77%), but their pixels lose only 18 and 4–5 ppt, the target's wing, and the deficit centroid excludes them. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (only a neighbour 47″ away, type L), the ASAS-SN variable-star catalogue, Gaia DR3 variability tables (only the same neighbour, as a long-period variable), SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP TOIs and CTOIs, the SuperWASP false-positive catalogue (Schanche et al. 2019), or a general web search. Known SB (SIMBAD SB*, F0/3IV/V; references are photometric catalogues only). TESS-SPOC TCEs, not promoted to TOIs (Table A1)
Open issues	The primary falls 4.3σ before Gaia's conjunction A while the secondary is on conjunction B. Gaia's eccentricity (2.8σ from zero for a 2.24-d binary that is expected to be circular) compresses the A-to-B interval and accounts for part of this; the mean offset of the pair, −0.011 ± 0.004, leaves about 2.8σ unexplained, possibly distortion of the radial velocities by the irradiated companion (RUWE 1.68)
Next eclipses	Primary: BTJD 4327.635 (2026-10-14 03:17 UTC), 4329.872, 4332.108; secondary: BTJD 4328.753 (2026-10-15 06:07 UTC), 4330.990 (allow ±2 min for primaries and ±4 min for secondaries; the primaries scatter by 3.1 min rms from event to event)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 89725 is an F0 subgiant with a 2.24-d companion whose grazing 16-ppt eclipse and 3.6-ppt occultation are superimposed on a 6% peak-to-peak phase curve. The phase curve is the notable feature. Its 15-ppt ellipsoidal term is about what the F star's own distortion gives, but its 18-ppt reflection term, with its minimum at the primary eclipse, is too large for a detached F + M pair; a semi-detached, Algol-like system in which the companion fills its Roche lobe and is strongly irradiated is the hypothesis that would explain it. The eclipses are on the target in two sectors despite the crowded field, and TESS-SPOC reproduces the timings. The secondary falls on Gaia's conjunction B, while the primary is 0.021 in phase early relative to conjunction A, partly because Gaia's small nonzero eccentricity compresses the predicted spacing; the mean offset of the two eclipses still leaves about 2.8σ unexplained. Spectroscopy to detect the companion's lines and a combined light-curve model would establish whether mass transfer is under way.

A88 TYC 3827-170-1 = TIC 302351416

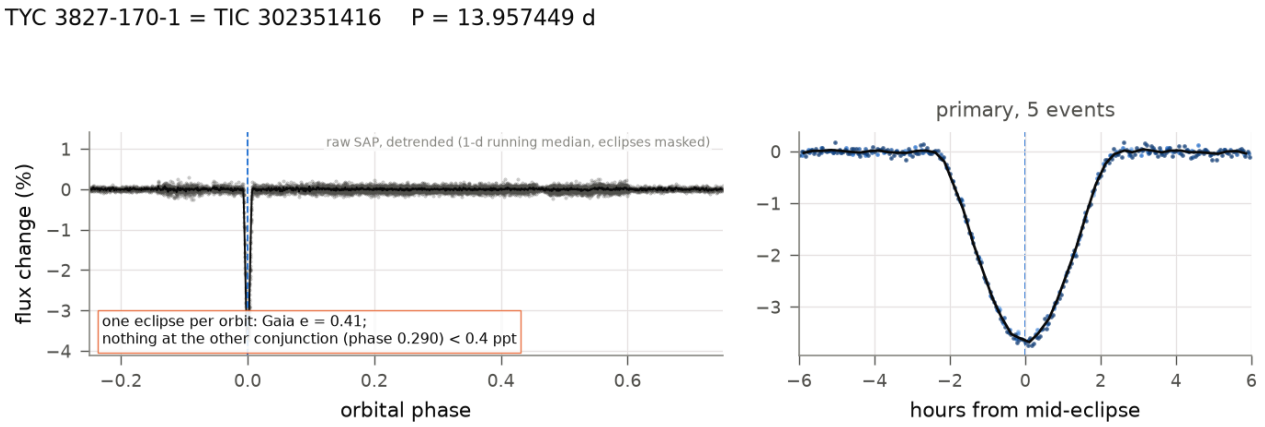


Figure A88. TYC 3827-170-1, QLP SAP flux (unflagged cadences), each sector divided by its median, detrended by a 1-d running median with the eclipse masked and folded on P = 13.9574485 d. Left: the full orbit, with one eclipse per orbit; Gaia's other conjunction, at phase 0.29 (not marked), shows no eclipse (< 0.4 ppt). Right: the five events of the 35.7-ppt, 4.5-h V-shaped eclipse.

Identifiers	TIC 302351416; TYC 3827-170-1; Gaia DR3 856255301585891840
Position (J2000)	168.61584, +56.20059 deg (TIC)
Brightness	Tmag 9.42; G 9.81
Stars	Slightly evolved F9–G0 primary (TIC: T_{eff} 6112 K, 2.03 $R_\odot$) at 240 pc; Gaia SB1 orbit (e = 0.4118 ± 0.0030, ω = 318.60 ± 0.81°, K_1 = 45.57 km/s, f(M) = 0.104 $M_\odot$: $m_2 \geq 0.71\ M_\odot$ for M_1 = 1.15 $M_\odot$, a ≈ 30 $R_\odot$)
Period	13.9574485 ± 0.0000066 d, timing fit of 5 eclipses, errors scaled by 2.7 (Gaia: 13.956758 ± 0.000425 d; +1.6σ). Odd and even eclipses agree (35.82 and 35.50 ppt) and nothing is seen at phase 0.5, so the period is P
Epochs	Primary BTJD 2928.45698 ± 0.00023 = BJD_{TDB} 2459928.45698
Primary eclipse	35.7 ± 0.4 ppt, 4.5 h, V-shaped (flat fraction 0.2), in raw QLP SAP flux (trapezoid fits to 5 events in s21, s48 and s75); TESS-SPOC 34.7 ppt
Secondary eclipse	Not detected (< 0.4 ppt, 3σ, at Gaia's conjunction A, phase 0.29 after the eclipse; nothing elsewhere in phase above 0.32 ± 0.46 ppt). At conjunction A the stars are 1.75 times farther apart (1.141a against 0.653a), so for the impact parameter of the observed eclipse (b ≈ 0.78) the other conjunction has b ≈ 1.4 and no eclipse
Orbit	The eclipse is on Gaia's conjunction B (ν = 270° − ω): Δφ = +0.0000 ± 0.0014 (0.0σ) with the conjunction at Gaia's epoch carried by the TESS period; +0.009 ± 0.006 (1.5σ) with Gaia's own period carried 181 cycles. Conjunction A is excluded (−0.299 ± 0.002). With the standard radial-velocity sign convention the RV star is in front at conjunction B, which would make this eclipse an occultation of the companion. A grazing occultation of a companion that supplies about 5% of the light could hardly remove 36 ppt, which mildly favours the reading as a transit of the radial-velocity star. Single-eclipse system: no *e* cos ω test

Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s21, s48, s75; 5 events: 34.7–34.8 ppt, 4.40 h, mid-times within 0.0–0.6 min of QLP's event by event); TARS (36.0 ppt). No SPOC 2-min data and no SPOC TCE
Pixel attribution	TESScut difference images in s21, s48 and s75 (2, 2 and 1 events): the deficit centroid is 0.2–0.7″ from the light centroid and 0.04–0.15 px from the catalogue position, with the peak deficit in the target's pixel and the target's image shape (residual 2.2–7.8%); depths 29.7–33.7 ppt in 1×1, 3×3 and 5×5 pixel apertures (the core window of a V-shaped eclipse is shallower than the fitted depth). Gaia DR3 has no source within 60″; the nearest within 90″ is G 19.98 at 65.3″, and only a star brighter than G 13.42 could make the eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, the ASAS-SN variable-star catalogue, Gaia DR3 `vari_eclipsing_binary` and `vari_summary` (target and neighbours within 90″), SB9, WDS, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2 and 3, ExoFOP TOIs and CTOIs, or a general web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit; three catalogue references). No ASAS-3 data
Open issues	The five mid-times scatter about the linear ephemeris by more than their errors (O–C −0.8 to +1.5 min, χ^2 22.6 for 3 degrees of freedom); the sector-21 offsets of +1.0 and +1.5 min are reproduced by TESS-SPOC. Which star is eclipsed rests on the radial-velocity sign convention (the depth mildly favours a transit of the radial-velocity star), and the 36-ppt depth of a grazing eclipse constrains the companion only weakly
Next eclipses	Primary: BTJD 4338.159 (2026-10-24 15:50 UTC), 4352.117 (allow ±3 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3827-170-1 is a slightly evolved F9–G0 star with a companion of at least 0.71 $M_\odot$ in a 14.0-d orbit with e = 0.41. Only one eclipse occurs per orbit: a 35.7-ppt, 4.5-h V-shaped grazing eclipse at conjunction B, where the stars are closest, while at conjunction A they are 1.75 times farther apart and miss each other, consistent with the < 0.4-ppt limit there. The eclipse falls on Gaia's conjunction B, at +0.0000 ± 0.0014 in phase, when Gaia's epoch is carried by the TESS period, the deficit is on the target in three sectors in an otherwise empty field, and TESS-SPOC reproduces the depth and timings. The system shows how an eccentric orbit can hide one of its two eclipses entirely; radial velocities that resolve the companion's lines would determine which star is eclipsed and how massive it is.

A89 CPD−85 289 = TIC 388287047

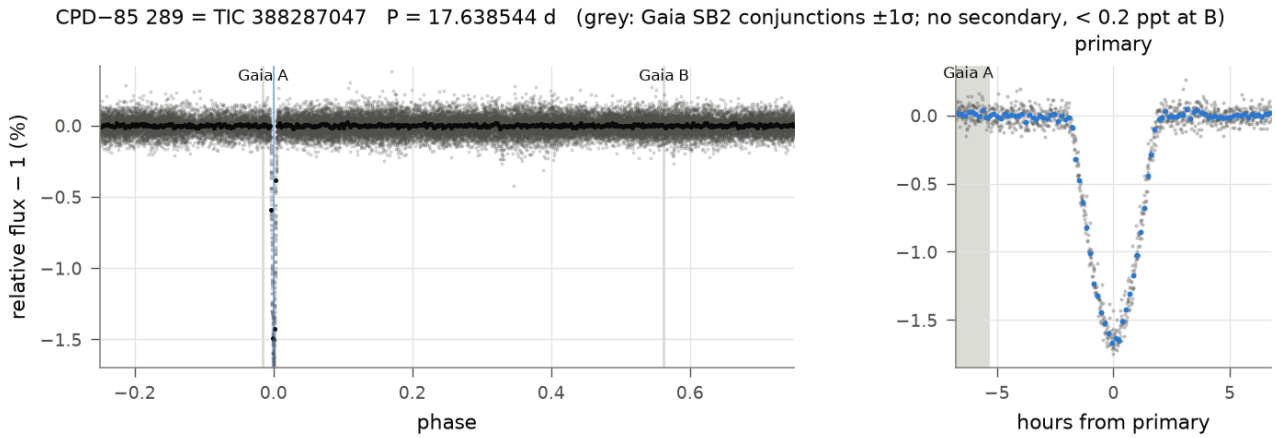


Figure A89. CPD-85 289, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 17.638544 d; grey bands are Gaia's predicted conjunctions A and B (±1σ). Left: the full orbit, with no secondary eclipse (< 0.2 ppt at B). Right: the 16-ppt, 3.9-h V-shaped primary, which falls about 6 h after the predicted conjunction A (grey band at the left edge).

Identifiers	TIC 388287047; CPD-85 289; TYC 9511-2297-1; Gaia DR3 5190617098481961984
Position (J2000)	169.88291, −85.71758 deg (TIC)
Brightness	Tmag 9.32; G 9.73

Stars	G0 (SIMBAD; TIC: T_{eff} 6120 K, 2.24 $R_\odot$, combined light of both stars) at 244 pc; Gaia SB2 orbit (e = 0.2082 ± 0.0052, ω = 54.6 ± 1.9°, goodness of fit 4.5). $M_1 \sin^3 i$ = 1.32 $M_\odot$ and $M_2 \sin^3 i$ = 1.10 $M_\odot$ (q = 0.83), so about 1.33 + 1.11 $M_\odot$ at i ≈ 85°, the companion giving roughly a third of the light; a sin i = 38.3 $R_\odot$
Period	17.638544 ± 0.000019 d, timing fit of 7 primaries in five sectors (Gaia: 17.63675 ± 0.00102 d; +1.8σ). Odd and even depths agree (15.85 ± 0.77 and 15.61 ± 0.51 ppt)
Epochs	Primary BTJD 3032.65988 ± 0.00069 = BJD_{TDB} 2460032.65988
Primary eclipse	16.1 ± 0.2 ppt at 200-s cadence (15.2 ppt at 10 min and 14.0 ppt at 30 min, smeared), 3.8–3.9 h, V-shaped (flat fraction about 0.2), in raw QLP SAP flux (trapezoid fits to 7 events); a grazing eclipse (impact parameter about 0.9)
Secondary eclipse	Not detected: 0.09 ± 0.04 ppt at Gaia's conjunction B (< 0.2 ppt, 3σ) and −0.04 ± 0.04 ppt at phase 0.5. None is expected: at B the stars are 1.152a apart against 0.818a at A, and the inclination that gives the observed grazing eclipse at A puts the impact parameter at B at 1.25
Orbit	Primary 0.0147 ± 0.0022 in phase (6.2 ± 0.9 h) after Gaia's predicted conjunction A, conditioned on the TESS period: 6.7σ with Gaia's full covariance, 3.3σ with Gaia's errors doubled, 2.2σ with the correlations ignored. Closing the gap needs ω = 47.4° at fixed T_p (−3.8σ) or T_p later by 0.25 d at fixed ω (2.7σ). The eclipse cannot be at B: an eclipse there would force one at A (impact parameter 0.67). No secondary, so no e cos ω test
Pipelines agreeing	QLP SAP (7 events, 15.98 ppt, 3.78 h); TESS-SPOC SAP and PDCSAP (5 events, 15.19 and 15.50 ppt, 3.79 h, mid-times within 0.1 min of QLP's on the same events); SPOC 2-min (s38: 14.51 ppt, 3.71 h, mid-time within 0.1 min of QLP's); TARS (7 events). SPOC multi-sector TCE (s56–s69): P 17.63837 ± 0.00023 d, depth 15.87 ppt, 3.97 h, impact parameter 0.97, MES 232, not flagged as an eclipsing binary, centroid 0.13 ± 2.57″ from the TIC position; single-sector TCEs in s66 (MES 183) and s38 (MES 92)
Pixel attribution	TESScut difference images in sectors 66 (2 events), 13 (2) and 38 (1): deficit centroid 0.3″, 0.5″ and 0.1″ from the light centroid, PSF-shaped (residual 6%), the peak deficit on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (14.0, 13.5, 13.3 ppt in s66). The brightest of 27 Gaia sources within 60″ (G 16.01 at 48.7″) would need 522%; the G 13.10 star at 80.5″ would need 37% and loses only 0.2–3.0 ppt at its own pixel. A companion within about 1″ is not excluded (RUWE 1.00)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and all 27 sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB: SIMBAD type SB*, in the APOGEE double-lined list of Kounkel et al. (2021); no eclipses reported. SPOC TCEs as above, not promoted to a TOI
Open issues	The conjunction offset, whose significance rests on Gaia's ω–T_p correlation (0.968) and a poorly fitting SB2 solution. The primary O–C runs about +3 min (2019), −7 min (2021; the s38 event is 6.8 min early in QLP, TESS-SPOC and SPOC 2-min alike) and about 0 (2023–25), rms 3.1 min against fitted errors about four times smaller; a light-travel-time signal from a third body is possible but untested with seven events
Next eclipses	Primary: BTJD 4337.912 (2026-10-24 09:54 UTC), 4355.551 (allow ±7 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CPD-85 289 is a known double-lined binary of two stars of about 1.33 and 1.11 $M_\odot$ in a 17.6-d orbit with e = 0.21, and the eclipses are new. Only conjunction A, where the stars are closest, is eclipsed, in a 16-ppt grazing eclipse; at B the projected separation is too large for any eclipse, which accounts for the missing secondary. The eclipses are on the target in three sectors of difference images and are reproduced by TESS-SPOC, SPOC 2-min and SPOC's own transit search, and the period agrees with Gaia's to 1.8σ. The primary falls 6.2 h after Gaia's predicted conjunction; the formal 6.7σ offset depends on the strong ω–T_p correlation of a poorly fitting SB2 solution, and is most likely an underestimated error in Gaia's ω rather than a misattribution. Gaia DR4 epoch radial velocities would test that directly, and further TESS sectors would show whether the timing residuals are a light-travel-time effect.

A90 CD−73 685 = TIC 454747287

CD−73 685 = TIC 454747287 P = 5.634764 d

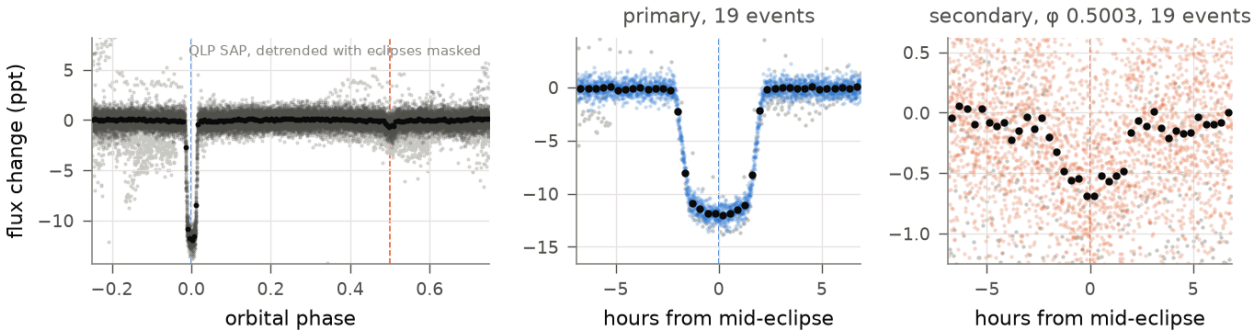


Figure A90. CD-73 685, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 5.634764 d (flagged cadences in light grey). Left: the full orbit. Middle: the 19 flat-bottomed 11.7-ppt, 4.3-h primaries. Right: the 0.49-ppt, 3.9-h secondary at phase 0.5003 (19 events).

Identifiers	TIC 454747287; CD-73 685; CPD-73 881; TYC 9238-667-1; Gaia DR3 5226756297640451840
Position (J2000)	175.63397, −73.86180 deg (TIC)
Brightness	Tmag 9.35; G 9.89
Stars	TIC: T_{eff} 5480 K, 2.10 $R_\odot$ (Gaia GSP-Phot 6270 K), a slightly evolved F/G star, at 183 pc; Gaia SB1 orbit (e = 0.019 ± 0.020, K_1 = 14.13 km/s, f(M) = 0.00165 $M_\odot$). The companion is an M dwarf: m_2 ≈ 0.13–0.14 $M_\odot$, with a radius ratio of at least 0.108. The flat bottom (T_{23}/T_{14} ≈ 0.6) and the 4.3-h duration together imply an impact parameter of about 0.75 and a primary radius of about 1.7 $R_\odot$, below the TIC's 2.1 $R_\odot$, so R_2 ≈ 0.18–0.23 $R_\odot$; and the secondary-to-primary depth ratio (0.042) matches a 2800–3000 K companion
Period	5.6347635 ± 0.0000015 d, timing fit of 19 primaries (Gaia: 5.634718 ± 0.000250 d; +0.18σ). Odd and even depths agree (11.67 and 11.62 ppt)
Epochs	Primary BTJD 2691.10727 ± 0.00013 = BJD_{TDB} 2459691.10727; secondary BTJD 2693.92630 (phase 0.50029 ± 0.00024)
Primary eclipse	11.70 ± 0.06 ppt, 4.30 h, flat-bottomed (T_{23}/T_{14} ≈ 0.6), in raw QLP SAP flux (trapezoid fits to 19 events; 11.0–12.0 ppt per event); the transit of the M dwarf across the primary
Secondary eclipse	0.49 ± 0.03 ppt, 3.9 h, at phase 0.50029 ± 0.00024 (fold fit to 19 events, each about 3σ); the occultation of the M dwarf
Orbit	Primary on Gaia's predicted conjunction A: +0.0041 ± 0.0058 in phase (0.7σ), secondary on conjunction B: +0.0032 ± 0.0057 (0.6σ), both at Gaia's epoch with the TESS period. e cos ω = +0.0005 ± 0.0004 from the eclipse spacing, Gaia +0.0005 ± 0.017 (0.0σ): circular
Pipelines agreeing	QLP SAP (19 events, 11.47 ppt, 4.32 h); TESS-SPOC SAP and PDCSAP (18 events, 11.13 and 11.46 ppt, mid-times −0.07 ± 0.23 min from QLP's on 14 matched events); SPOC 2-min SAP and PDCSAP (s39, 11.08 and 11.24 ppt, mid-times −0.14 ± 0.17 min); TGLC aperture flux (s11, 12.75 ppt). SPOC multi-sector TCE (s56–s69): P 5.63481 d, 11.84 ppt, 4.42 h, MES 217, not flagged as an eclipsing binary, centroid 0.11 ± 2.51″ from the TIC position; a second TCE is the secondary (P 5.6345 d, 531 ppm, MES 9.9, at phase 0.500)
Pixel attribution	TESScut difference images in sectors 39 and 65 (four events each): deficit centroid 0.2″ and 0.3″ from the light centroid, PSF-shaped (residual 7–8%), the peak deficit on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (11.8, 11.6, 11.4 ppt in s39). The brightest of 46 Gaia sources within 60″ (G 14.98 at 36.4″) would need 128%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit). ASAS-3 photometry (15–18 mmag scatter, 12 points in the eclipse core) is too noisy to detect the eclipse. SPOC TCEs as above, not promoted to a TOI
Next eclipses	

	Primary: BTJD 4330.823 (2026-10-17 07:47 UTC), 4336.458, 4342.093; secondary: BTJD 4328.008 (2026-10-14 12:13 UTC), 4333.642 (allow ±2 min for primaries and ±6 min for secondaries; the individual secondaries, each about 3σ, scatter by 30 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

CD-73 685 is a slightly evolved F/G star eclipsed by an M dwarf of about 0.13 $M_\odot$ in a circular 5.63-d orbit. The 11.7-ppt flat-bottomed transit and the 0.49-ppt occultation at phase 0.5003 fall on Gaia's two predicted conjunctions, the period agrees with Gaia's to 0.2σ, and the eclipse spacing gives the same e cos ω as Gaia's orbit. Difference images in two sectors, three further pipelines and SPOC's own transit search all place the signal on the target. The deep eclipse is at the conjunction where the radial-velocity star is behind its companion, as expected for a single-lined system. With radial velocities and a better primary temperature (TIC and Gaia differ by about 800 K), the system would give a mass and radius for a very-low-mass star.

A91 HD 311112 = TIC 311635507

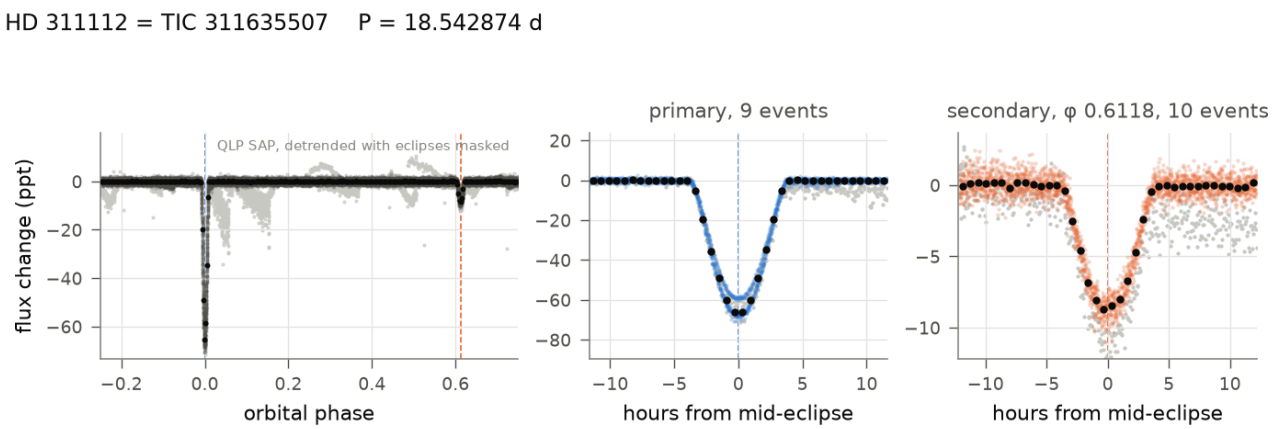


Figure A91. HD 311112, QLP SAP flux with sector medians removed and a running-median detrend with the eclipses masked, folded on the fitted P = 18.542874 d (flagged cadences in light grey). Left: the full orbit, with the secondary at phase 0.6118. Middle: the nine 67-ppt, 7.3-h near-V primaries; the two depth levels are the sectors before and after the depth step of sectors 99–100 (59 ppt). Right: the ten 8.7-ppt, 7.6-h secondaries.

Identifiers	TIC 311635507; HD 311112; CD-65 1201; CPD-65 1787; TYC 8986-1004-1; Gaia DR3 5860649009076840320
Position (J2000)	181.56340, −66.42961 deg (TIC)
Brightness	Tmag 9.38; G 9.84
Stars	F5 (SIMBAD; TIC: T_{eff} 5919 K, 2.33 $R_\odot$, slightly evolved) at 239 pc; Gaia SB1 orbit (e = 0.184 ± 0.014, ω = 6.2 ± 6.2°, K_1 = 29.32 ± 0.36 km/s). f(M) = 0.046 $M_\odot$, so $m_2 \geq 0.48\ M_\odot$ for M_1 = 1.08 $M_\odot$; radius ratio ≥ 0.26 ($R_2 \geq 0.6\ R_\odot$) and the secondary-to-primary depth ratio (0.13) indicate a K/M dwarf of about 3600 K. Crowded field (322 Gaia sources within 60″; TIC contamination ratio 0.10)
Period	18.5428744 ± 0.0000049 d, timing fit of 9 primaries; the 10 secondaries give 18.542873 ± 0.000009 d (Gaia: 18.54250 ± 0.00509 d; +0.07σ)
Epochs	Primary BTJD 3479.18152 ± 0.00017 = BJD_{TDB} 2460479.18152; secondary BTJD 3490.52516 (phase 0.61175 ± 0.00003)
Primary eclipse	66.74 ± 0.19 ppt, 7.27 h, nearly V-shaped ($T_{23}/T_{14} \approx 0.2$), in raw QLP SAP flux (trapezoid fits to 9 events). The QLP depth is 67.3–68.4 ppt in sectors 37–65 and 58.9–59.5 ppt in sectors 99–100, with unchanged duration and the secondary scaled in proportion: a change of dilution or background in this crowded field, not of geometry. A fixed 3×3 TESScut aperture shows a smaller step (59.1 ppt in s65, 55.6 ppt in s99), so the change is not confined to QLP's aperture. Crowding-corrected TESS-SPOC PDCSAP gives 69.5 ppt
Secondary eclipse	8.70 ± 0.06 ppt, 7.63 h, at phase 0.61175 ± 0.00003 (10 events); nothing at phase 0.5 (−0.01 ± 0.04 ppt). The secondary is displaced from phase 0.5 by the eccentric orbit

Orbit	Primary on Gaia's predicted conjunction A: −0.0007 ± 0.0056 in phase (0.1σ); secondary on conjunction B: −0.0042 ± 0.0049 (0.9σ), both at Gaia's epoch with the TESS period. Eclipse spacing 0.61175 against Gaia's predicted 0.6154 ± 0.0092 (0.5σ). e cos ω = 0.1764 ± 0.0001 (Kepler's equation, Gaia's e sin ω held), Gaia 0.183 ± 0.015, implying e = 0.178, ω = 6.4°; the duration ratio gives e sin ω ≈ +0.024 (Gaia +0.020 ± 0.019)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (7 primaries and 7 secondaries: 61.6 and 8.58 ppt; primary mid-times +0.06 ± 0.05 min from QLP's on 4 matched events, secondaries −0.5 ± 1.4 min) and PDCSAP (69.5 and 9.67 ppt, CROWDSAP 0.87–0.89); TARS (65.9 and 9.51 ppt); TGLC (s11). TESS-SPOC TCEs: the primary in s37 (P 18.54296 d, 72.2 ppt, 7.6 h, MES 750, centroid 0.48 ± 2.5″ from the TIC position) and s65 (MES 349, 0.84″), the secondary in s38 (10.4 ppt, MES 80); never flagged as an eclipsing binary
Pixel attribution	TESScut difference images in sectors 37, 65 and 99 (two primaries each): deficit centroid 1.4″, 1.4″ and 1.6″ from the light centroid, the peak deficit on the target pixel, PSF-fit residual 9–22% (crowded field); depth stable across 1×1, 3×3 and 5×5 pixel apertures (61.4, 59.1, 58.5 ppt in s65). A source would need G < 12.78 to make the dip; the brightest within 90″ is G 13.42 at 85.5″, and within 60″ G 13.78 at 28.5″ would need 257%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (one unrelated G 18.5 variable 55″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables (target and sources within 60″), SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, from the Gaia DR3 orbit). The primary eclipse is present in ASAS-3 photometry (2001–2009) at the TESS period, 374 cycles before TESS: +74.0 ± 8.3 mmag in the eclipse core against 75.0 expected, with a trial-period scan peaking at 18.542845 d. TESS-SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4332.154 (2026-10-18 15:44 UTC), 4350.697; secondary: BTJD 4343.497 (2026-10-29 23:59 UTC), 4362.040 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 311112 is a slightly evolved F5 star with a K or early-M dwarf companion of at least 0.48 $M_\odot$ in an 18.54-d orbit with e = 0.18. A near-grazing 67-ppt primary and an 8.7-ppt secondary at phase 0.6118 fall on Gaia's two predicted conjunctions to within 1σ, and the eclipse spacing gives e cos ω = 0.176, within 0.5σ of Gaia's. The deficit lies on the target in three sectors of difference images in a crowded field where no neighbour is bright enough to produce it, and TESS-SPOC and TARS reproduce both eclipses. The eclipse is also recovered at full depth in ASAS-3 photometry two decades old, on the TESS ephemeris carried back 374 cycles; the ASAS-3 data are not part of the ephemeris fit. The QLP depth falls by 12% from sectors 37–65 to 99–100 with the secondary in proportion and the duration unchanged, a change of dilution that a fixed TESScut aperture also shows, though smaller; depths for this star should be quoted with their pipeline.

A92 BD+69 653 = TIC 148877735

TIC 148877735 BD+69 653 P = 3.607450 d

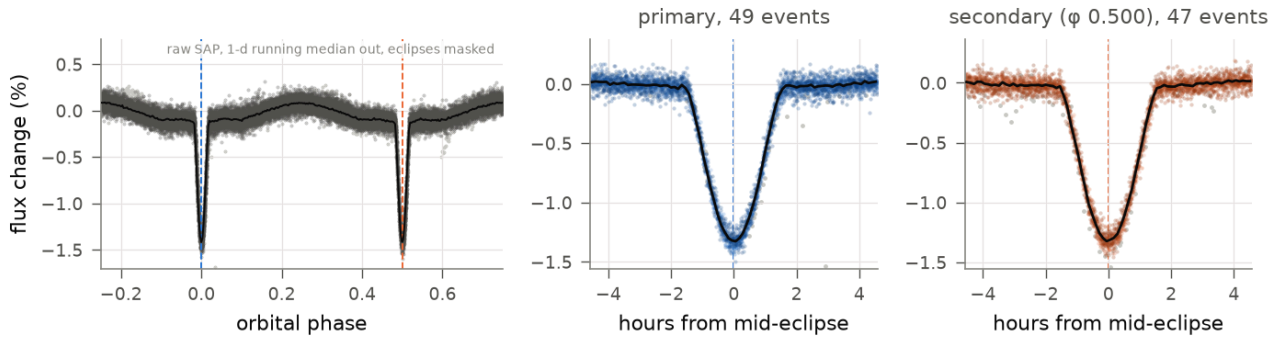


Figure A92. BD+69 653, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 3.607450 d, twice the fold period. Left: the full orbit, with two equal eclipses at phases 0 and 0.500 and an ellipsoidal variation of at least 1–2 ppt peaking at quadrature (the running median removes part of it). Middle and right: the 13.2-ppt, 3.05-h V-shaped primary and the identical secondary (49 and 47 events plotted; 39 and 36 fully covered events fitted individually, 35 and 31 of them in the timing fit).

Identifiers	TIC 148877735; BD+69 653; TYC 4393-1419-1; Gaia DR3 1683351069470284032
Position (J2000)	183.42497, +69.15763 deg (TIC)
Brightness	Tmag 9.33; G 9.66
Stars	F5 (SIMBAD; TIC: T_{eff} 6516 K, 2.54 $R_\odot$ as a single star) at 316 pc; Gaia SB2 orbit (e = 0.002 ± 0.007, K_1 = 89.72 ± 0.61, K_2 = 89.46 ± 0.63 km/s). Twins: q = 1.003, M $\sin^3$i = 1.074 and 1.077 $M_\odot$, about 1.19 $M_\odot$ each at i ≈ 75°, and about 1.8 $R_\odot$ each if they share the light equally: slightly evolved F5 stars. a sin i = 12.8 $R_\odot$
Period	3.6074499 ± 0.0000008 d, timing fit of 35 primaries (31 secondaries give 3.6074495 ± 0.0000009 d) (Gaia: 3.607477 ± 0.000048 d; −0.6σ). Twin eclipses: the period is twice the fold period 1.8037182 d. The two eclipses are identical in depth and duration, so the photometry alone cannot separate P from P/2; the Gaia SB2 orbit, with two equal, anti-phased velocity curves, fixes P
Epochs	Primary BTJD 3110.82736 ± 0.00008 = BJD_{TDB} 2460110.82736; secondary BTJD 3101.80861 ± 0.00009 (phase 0.49997 ± 0.00003). For twins the labels are arbitrary: the primary here is the eclipse at Gaia's conjunction B, of Gaia's secondary component
Primary eclipse	13.21 ± 0.26 ppt, 3.05 ± 0.05 h, V-shaped (flat fraction about 0.2), in raw QLP SAP flux (trapezoid fits to 39 events); 12.4–12.7 ppt with a quadratic baseline in all pipelines
Secondary eclipse	13.16 ± 0.30 ppt, 3.05 h, at phase 0.49997 (36 events)
Orbit	Primary on Gaia's predicted conjunction B: +0.0009 ± 0.0020 in phase (0.5σ); secondary on conjunction A: +0.0011 ± 0.0019 (0.5σ), at Gaia's epoch with the TESS period; Gaia predicts the separation 0.5001 ± 0.0033. e cos ω = −0.0001 ± 0.0001 from the eclipse spacing (Gaia +0.0001 ± 0.0052); equal durations give e sin ω ≈ 0: circular
Pipelines agreeing	QLP SAP; SPOC 2-min SAP (s41, 13 events: mid-times +0.08 ± 0.30 min, depth ratio 1.03); TESS-SPOC SAP (s41 and s74, mid-times within 0.06 min); TASOC and TARS (s14, s15). In the 30-min sectors 14 and 15 the QLP mid-times are 5.3–6.5 min later than TESS-SPOC, TASOC (including its 2-min pixel light curve) and TARS on the same events; those sectors are excluded from the timing fit. SPOC multi-sector TCE (s14–s50): P 1.8037244 ± 0.0000016 d (half the period), depth 12.76 ppt, 3.10 h, impact parameter 0.99, MES 314, 40 transits, not flagged as an eclipsing binary, centroid 0.45 ± 2.53″ from the TIC position
Pixel attribution	TESScut difference images in sectors 47 (8 primaries) and 74 (7): deficit centroid 0.3″ and 0.6″ from the light centroid, PSF-shaped (residual 5.4% and 2.5%); depth stable across 1×1, 3×3 and 5×5 pixel apertures (11.5, 11.3, 11.2 ppt in s47). The only Gaia sources within 60″ (G 18.56 and 19.74, both at 52″) would need 4787% and 14300%. A companion within about 1″ is not excluded (RUWE 0.99)
Catalogue status	

	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search; no ASAS-3 data at this declination. Known SB (SIMBAD SB*, F5, from the Gaia DR3 orbit). SPOC TCE at the half period as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4330.145 (2026-10-16 15:29 UTC), 4333.753, 4337.360; secondary: BTJD 4328.342 (2026-10-14 20:12 UTC), 4331.949 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+69 653 is a pair of nearly identical, slightly evolved F5 stars (q = 1.003, about 1.2 $M_\odot$ each) in a circular 3.61-d orbit seen at about 75°, so that each star grazes the other at conjunction and the two V-shaped eclipses are equal at 13.2 ppt. The photometry alone admits half the period; Gaia's double-lined orbit fixes the true one, and the two eclipses fall on its two conjunctions to 0.5σ. The deficit lies on the target in difference images, and SPOC's own transit search detected the eclipses at the half period with a centroid on the star (MES 314); because the alternate events are equal at that period, it was not flagged as an eclipsing binary. Ellipsoidal variation of 1–2 ppt is visible out of eclipse; it is measured after a 1-d running median on a 3.6-d orbit, which suppresses part of it, so it is a lower bound. In this star's two 30-min sectors the QLP mid-times are about 6 min later than those of three other pipelines on the same events, so the offset lies in the QLP light curve rather than in the star; whether other stars in those sectors share it has not been tested.

A93 HD 112601 = TIC 175734936

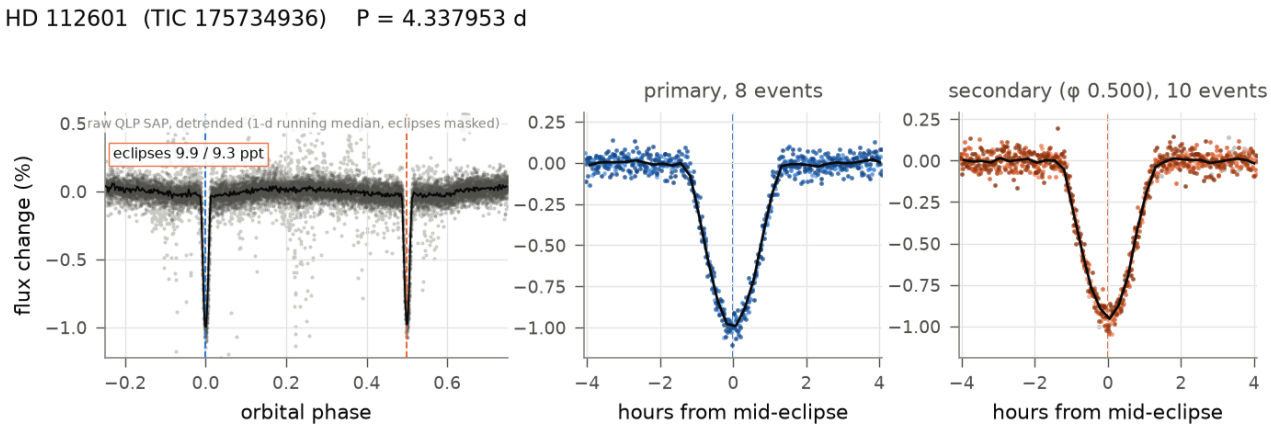


Figure A93. HD 112601, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 4.337953 d, twice the fold period (flagged cadences in light grey). Left: the full orbit, with eclipses at phases 0 and 0.500. Middle: the 9.9-ppt, 2.55-h V-shaped primary (8 events). Right: the 9.3-ppt secondary of the same duration (10 events).

Identifiers	TIC 175734936; HD 112601; BD-01 2746; HIP 63269; TYC 4950-1175-1; 1SWASP J125754.13-020310.2; Gaia DR3 3685481410582132864
Position (J2000)	194.47556, −2.05285 deg (TIC)
Brightness	Tmag 8.95; G 9.34
Stars	F5V (SIMBAD; Gaia GSP-Phot T_{eff} 6194 K, blended light) at 232 pc from the Gaia parallax (TIC 118 pc; RUWE 4.64); Gaia SB2C orbit (circular; K_1 = 80.54 ± 0.64, K_2 = 85.97 km/s, q = 0.94, goodness of fit 5.6). $M_1 \sin^3 i$ = 1.07 and $M_2 \sin^3 i$ = 1.00 $M_\odot$: two F stars of about 1.1 $M_\odot$, a = 14.8 $R_\odot$. A third star is present: one CORALIE spectrum shows three line systems (Schanche et al. 2019), consistent with the high RUWE and the discrepant distances; it dilutes the eclipses
Period	4.337953 ± 0.000001 d, timing fit of 8 primaries and 10 secondaries (Gaia: 4.337861 ± 0.000105 d; +0.9σ). Twin eclipses: the period is twice the fold period 2.168977 d; at the fold period alternate eclipses differ in depth by 0.46 ± 0.08 ppt (5.5σ)
Epochs	Primary BTJD 3380.72231 ± 0.00018 = BJD_{TDB} 2460380.72231; secondary BTJD 3382.89142 (phase 0.50003 ± 0.00006)
Primary eclipse	

	9.89 ± 0.06 ppt, 2.55 h, V-shaped (flat fraction 0.2), in raw QLP SAP flux (trapezoid fits to 8 events); grazing (impact parameter about 0.9)
Secondary eclipse	9.33 ± 0.05 ppt, 2.55 h, at phase 0.50003 ± 0.00006 (10 events); 0.56 ppt shallower than the primary (7σ)
Orbit	Circular orbit (SB2C): no conjunction-phase test. Gaia's reference epoch falls 0.2469 ± 0.0012 in phase (19 min from quadrature) after a TESS primary, as expected if it marks a velocity extremum; the secondary at phase 0.50003 agrees with a circular orbit
Pipelines agreeing	QLP SAP (s46, s91); SPOC 2-min SAP (s46: 9.69 and 9.28 ppt, event-to-event scatter 0.19 and 0.21 ppt, standard errors 0.08 and 0.09 ppt, so the difference is 3.4σ) and PDCSAP (9.65 and 9.25 ppt); QLP mid-times later than SPOC's by +0.39 ± 0.22 min (5 primaries) and +0.19 ± 0.17 min (6 secondaries; standard errors); TESS-SPOC SAP (s46: 9.60 and 9.20 ppt). SPOC and TESS-SPOC TCEs at half the period (2.16895 and 2.16896 d, 9.4 ppt, 2.61–2.63 h, impact parameter 0.98, MES 181 and 199), not flagged as an eclipsing binary, centroids 0.27–0.62 ± 2.50″ from the TIC position
Pixel attribution	TESScut difference images in sectors 46 and 91, primaries and secondaries separately (3–6 events each): deficit centroid 0.4–0.7″ from the light centroid, PSF-shaped (residual 7–9%); depth stable across 1×1, 3×3 and 5×5 pixel apertures (8.1, 8.3, 8.3 ppt for the s46 primaries, a core average). No Gaia source within 60″; the G 12.05 star at 80.5″ would need 12% and would displace the centroid by pixels. A source within about 1″ is not excluded, and the spectroscopic third star is one
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI). Known SB (SIMBAD SB*, F5V). Previously catalogued: SIMBAD lists the type EB* among its object types, from the SuperWASP disposition catalogue (Schanche et al. 2019), which lists 1SWASP J125754.13-020310.2 as an eclipsing binary with follow-up flag “blended eclipsing binary” and three line systems in one CORALIE spectrum, with no period. ASAS-3 shows the eclipse marginally at the TESS ephemeris (+9 mmag, 2.7σ, against about 10 mmag expected). SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4330.734 (2026-10-17 05:43 UTC), 4335.072, 4339.410; secondary: BTJD 4328.565 (2026-10-15 01:40 UTC), 4332.903 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 112601 is a hierarchical triple whose inner pair, two F stars of about 1.1 $M_\odot$ with q = 0.94, eclipses every 4.338 d in a circular orbit. The two eclipses are grazing and nearly equal (9.9 and 9.3 ppt), and their 7σ depth difference is reproduced in SPOC 2-min photometry (3.4σ in one sector); Gaia's double-lined period agrees to 0.9σ. Both eclipse kinds lie on the target in difference images. The SuperWASP disposition catalogue lists it as an eclipsing binary, with a follow-up disposition of blended eclipsing binary and a note that one spectrum showed three line systems; no period or eclipse parameters were published. The ephemeris, depths and near-twin nature given here are new. Because SIMBAD carries that flag as an eclipsing-binary type, the system counts as previously catalogued under the definition of Section 1 and is not among the new systems counted in this paper. The third star dilutes the eclipses and accounts for the large RUWE and the discrepant distances; spectra resolving all three components would give its light fraction and, with the eclipses, masses for the inner pair.

A94 HD 114997 = TIC 245194662

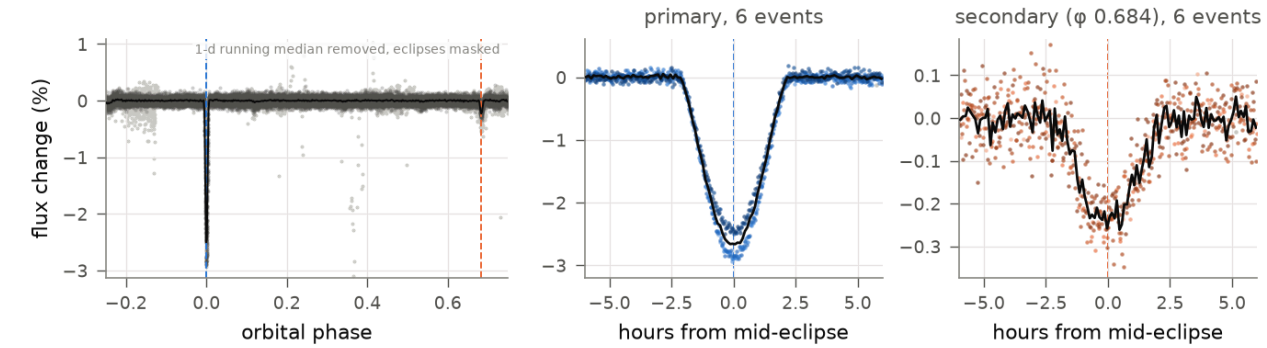


Figure A94. HD 114997, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 17.279547 d (flagged cadences in light grey). Left: the full orbit, with the secondary at phase 0.684. Middle: the six 28-ppt, 4.2-h V-shaped primaries (the shallower points are the diluted sector-100 events). Right: the six 2.5-ppt, 4.0-h secondaries.

Identifiers	TIC 245194662; HD 114997; CPD-53 5525; TYC 8649-1616-1; Gaia DR3 6068115377041311360
Position (J2000)	198.79864, −53.67158 deg (TIC)
Brightness	Tmag 8.92; G 9.35
Stars	G3V (SIMBAD; TIC: T_{eff} 6096 K, 1.38 $R_\odot$, 1.14 $M_\odot$) at 121 pc; Gaia SB1 orbit (e = 0.300 ± 0.009, ω = 358.1 ± 1.6°, K_1 = 23.87 ± 0.20 km/s). f(M) = 0.0211 $M_\odot$, so m_2 ≈ 0.36 $M_\odot$ at i = 90° (0.29–0.43 $M_\odot$): an M dwarf, radius ratio ≥ 0.17; a = 32.2 $R_\odot$. Crowded field (78 Gaia sources within 60″; TIC contamination ratio 0.13)
Period	17.2795467 ± 0.0000052 d, timing fit of 6 primaries; the 6 secondaries give 17.279539 ± 0.000033 d (Gaia: 17.28057 ± 0.00138 d; −0.7σ)
Epochs	Primary BTJD 3353.15051 ± 0.00022 = BJD_{TDB} 2460353.15051; secondary BTJD 3364.96274 (phase 0.68360 ± 0.0001)
Primary eclipse	28 ppt, 4.1–4.2 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 6 events): 27.7–28.5 ppt in sectors 11, 37 and 64 and 24.0 ppt in sector 100, most likely diluted by the red semi-regular variable 50″ away; TESS-SPOC PDCSAP 28.4–28.7 ppt. A grazing eclipse (impact parameter about 0.84; SPOC 0.96)
Secondary eclipse	2.5 ppt (sd 0.3), about 4.0 h, at phase 0.68360 (6 events); recovered at 2.1–3.6 ppt in every pipeline. Displaced from phase 0.5 by the eccentric orbit; equal durations give e sin ω ≈ 0 (Gaia −0.010)
Orbit	Primary on Gaia's predicted conjunction A: +0.0039 ± 0.0028 in phase (1.4σ); secondary on conjunction B: −0.0006 ± 0.0037 (0.2σ), at Gaia's epoch with the TESS period (1.3σ and 0.7σ when conditioned on the TESS period through Gaia's full covariance). Eclipse spacing 0.6836 against Gaia's predicted 0.6880 (0.6823–0.6938). e cos ω = +0.2926 ± 0.0001 from the eclipse spacing, Gaia +0.2999 ± 0.0094 (0.8σ)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s37, s64: 3 primaries, 26.2 and 28.6 ppt, mid-times −0.3 min from QLP's); TARS (4 primaries, 27.6 ppt, +0.01 min); eleanor-lite (s11, 26.9 ppt, +0.8 min). TGLC (s11) agrees in time (+0.7 min) but not in depth (39.5 ppt). TESS-SPOC TCE in s64: P 17.27980 ± 0.00027 d, 29.1 ppt, 4.40 h, impact parameter 0.96, MES 320, centroid 0.18 ± 2.5″ from the TIC position; a TCE in s37 (MES 105) combined primary and secondary at 11.81 d
Pixel attribution	TESScut difference images in sectors 64 and 100 (two primaries each): deficit centroid 2.8″ and 2.1″ from the catalogue position, with the peak deficit on the target pixel; the light centroid is pulled 10–11″ east by neighbours. On the target's 3×3 pixels the deficit is PSF-shaped to 14–19%; the depth falls in larger apertures (23.3, 22.4, 18.6 ppt in 1×1, 3×3, 5×5 in s64) as neighbours' light enters. Two Gaia sources within 60″ are bright enough: the red semi-regular variable (G 11.83, BP−RP 4.7, 50.3″; about 13% needed in the TESS band) and G 12.13 at 52.3″ (39%); both lie 2.4 px from the deficit centroid and lose only 0.8–8.4 ppt at their own pixels, the target's wing, against about 300 ppt if either were the source. All others would need more than 200%

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search. Known SB (SIMBAD SB*, G3V). Listed in Kostov et al. (2025) table 2. The only catalogued variable within 60″ is the G 11.83 neighbour at 48.7″, ASASSN-V J131506.19-534018.9 (VSX type SR, P = 513 d; a Gaia DR3 long-period variable); a long-period variable cannot produce a 4-h eclipse. The primary is present in ASAS-3 photometry (2000–2009) at the TESS ephemeris: +26.5 mmag (4.9σ) in the eclipse core, holding across the TESS period ± 8σ and absent at Gaia's period. TESS-SPOC TCEs as above, not promoted to a TOI
Next eclipses	Primary: BTJD 4338.085 (2026-10-24 14:06 UTC), 4355.364; secondary: BTJD 4332.617 (2026-10-19 02:53 UTC), 4349.897 (allow ±2 min; single secondaries scatter by 4.6 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 114997 is a G3 dwarf with an M-dwarf companion of about 0.36 $M_\odot$ in a 17.28-d orbit with e = 0.30, seen nearly edge-on so that both conjunctions are grazed. The 28-ppt primary and the 2.5-ppt secondary fall on Gaia's two predicted conjunctions, and the secondary's displacement to phase 0.684 gives e cos ω = 0.293, within 0.8σ of the spectroscopic value. That offset is fixed by the orbit and could not be reproduced by a blended neighbour. In this crowded field the difference images place the deficit on the target pixel, away from the two bright neighbours, five pipelines recover the eclipses at the same times (TGLC with a deeper 39.5 ppt), and the TESS-SPOC transit search detects them with a centroid on the target (MES 320). The primary is also present in ASAS-3 photometry at the TESS ephemeris (six in-eclipse points, 4.9σ), which confirms that the ephemeris holds about 20 years earlier; the ASAS-3 data are not part of the timing fit. The QLP depth in sector 100 is diluted, so depths should be quoted with their pipeline and sector.

A95 HD 234179 = TIC 154346187

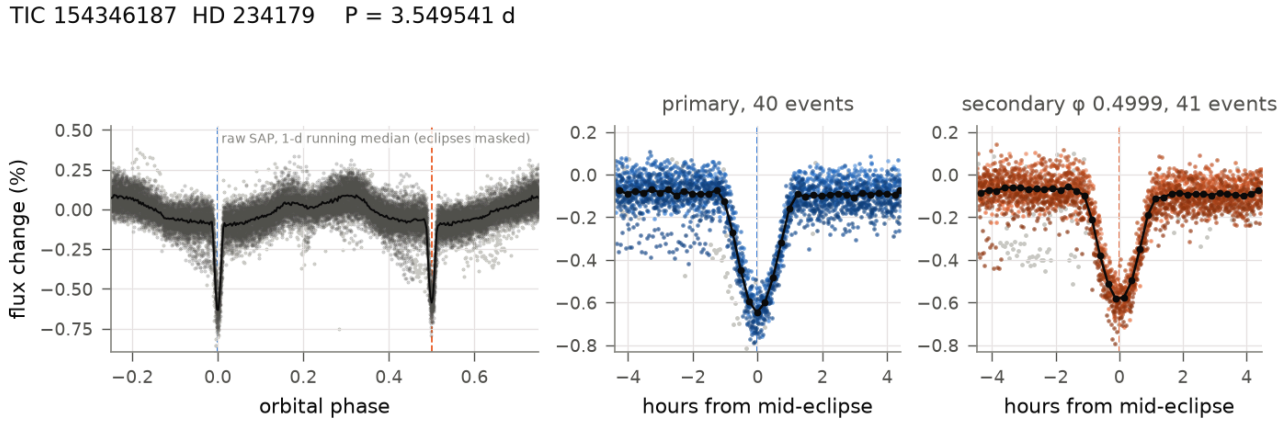


Figure A95. HD 234179, QLP SAP flux with a 1-d running median removed (eclipses masked), folded on the fitted P = 3.549541 d, twice the fold period. Left: the full orbit, with eclipses at phases 0 and 0.4999 and a 1–2 ppt non-sinusoidal modulation out of eclipse. Middle: the 5.6-ppt, 2.2-h V-shaped primary. Right: the 5.1-ppt secondary (40 and 41 events plotted; 36 and 31 fully covered events in the timing fit).

Identifiers	TIC 154346187; HD 234179; BD+53 1736; TYC 3860-1655-1; Gaia DR3 1605802239486264448
Position (J2000)	222.01636, +53.16917 deg (TIC)
Brightness	Tmag 8.98; G 9.38
Stars	G0 (SIMBAD; TIC: T_{eff} 6119 K, 2.65 $R_\odot$ from the combined light) at 256 pc; RUWE 5.5. Gaia DR3 gives two orbits for this source: an SB2C (circular; P = 3.549504 ± 0.000043 d, K_1 = 104.83 ± 0.56, K_2 = 77.41 km/s, goodness of fit 23.1) and an astrometric orbit (P = 541.2 ± 2.2 d, e = 0.113 ± 0.022, photocentre semi-major axis 1.66 mas = 0.40 AU). Inner pair: M $\sin^3 i$ = 0.95 and 1.28 $M_\odot$, about 1.05 and 1.4 $M_\odot$ at i ≈ 75–78°, a sin i = 12.8 $R_\odot$. Hierarchical triple: Gaia's astrometric orbit is the outer orbit of the eclipsing pair about an unresolved third body; the timing amplitude requires $M_3 \gtrsim 0.66\ M_\odot$

Period	3.549541 ± 0.000002 d, timing fit of 36 primaries (Gaia SB2C: 3.549504 ± 0.000043 d; +0.9σ). Twin eclipses: the period is twice the fold period 1.77478 d; at the fold period alternate eclipses differ in depth (5.85 ± 0.09 and 5.18 ± 0.08 ppt, 5.6σ)
Epochs	Primary BTJD 2489.65835 ± 0.00031 = BJD_{TDB} 2459489.65835; secondary BTJD 2491.43279 ± 0.00035 (phase 0.49991 ± 0.00010)
Primary eclipse	5.59 ppt, 2.20 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 36 events); 5.5–5.9 ppt in every sector; grazing (impact parameter about 0.9)
Secondary eclipse	5.09 ppt, 2.24 h, at phase 0.49991 ± 0.00010 (31 events); about 10% shallower than the primary in every pipeline
Orbit	Circular orbit (SB2C): no formal conjunction-phase test. If DR3's circular solutions fix ω = 0 so that the periastron epoch is the ascending node, the primary falls −0.0007 ± 0.0009 in phase (−3.8 ± 4.8 min, −0.8σ) from conjunction A. Outer orbit: the 67 eclipse times (four seasons) are consistent with the light-travel-time curve whose shape is fixed by Gaia's astrometric orbit. With the period, eccentricity and phase fixed and the amplitude, its sign and a linear ephemeris free, K = −3.39 ± 0.32 min (error scaled by √1.6 for χ^2/dof = 102.4/63, down from 391.5/64 without the term; O−C rms 1.39 min, from 2.71), against a minimum of 3.25 min predicted from Gaia's photocentre orbit. About 11% of randomly chosen outer phases fit as well as Gaia's, so the chance of such a match is about 0.1. A free sinusoid search (50–3000 d) finds 528 d, but 318 and 1463 d fit almost as well ($\Delta\chi^2 \approx 5$)
Pipelines agreeing	QLP SAP (s49, s50, s76, s77: 5.75 and 5.01 ppt); TESS-SPOC SAP and PDCSAP (5.44 and 4.93 ppt), mid-times −0.15 ± 0.25 min (17 primaries) and −0.11 ± 0.29 min (17 secondaries) from QLP's; TARS (5.50 and 4.92 ppt). TESS-SPOC single-sector TCEs in s49, s50, s76 and s77 at half the period (1.7747–1.7749 d, 5.1–5.3 ppt, 2.2 h, MES 71–91), not flagged as an eclipsing binary, centroids 0.49–0.99″ from the TIC position
Pixel attribution	TESScut difference images in sectors 49 (6 primaries) and 76 (5): deficit centroid 0.9″ and 1.3″ from the light centroid, PSF-shaped (residual 6–9%), on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (4.35, 4.39, 4.41 ppt in s49, a core average). The only Gaia sources within 60″ (G 17.17 at 42.4″, G 18.69 at 44.0″) would need 740% and 3000%. The third body (about 8 mas) is unresolved and its light is part of the target's
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, Gaia DR3 variability tables, SB9, WDS, ExoFOP (no TOI or CTOI), web search; no ASAS-3 data. Known SB (SIMBAD SB*, G0); SIMBAD references are catalogues only. TESS-SPOC TCEs at half the period as above, not promoted to a TOI
Open issues	The outer period is sampled sparsely by seven sectors in four seasons, so the timing does not by itself fix it. The predicted minimum amplitude assumes that the third body's fraction of the light is below its fraction of the mass. The mass budget does not close: K = 3.39 min needs $M_3 \approx 0.69\ M_\odot$, and a main-sequence star of that mass would shrink the photocentre orbit to about 0.34 AU, against Gaia's 0.40 AU, so the third body would have to be under-luminous for its mass. The out-of-eclipse modulation of 1–2 ppt, probably spots on a synchronised star, is not analysed. Eclipse times may depart from the linear ephemeris by up to ±3.4 min of light-travel time
Next eclipses	Primary: BTJD 4328.321 (2026-10-14 19:44 UTC), 4331.870, 4335.420; secondary: BTJD 4330.095 (2026-10-16 14:19 UTC), 4333.645 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 234179 is a hierarchical triple. Its inner pair, of about 1.05 and 1.4 $M_\odot$ in Gaia's circular 3.55-d double-lined orbit, shows two grazing eclipses of 5.6 and 5.1 ppt, unequal in every pipeline, so the true period is twice the fold period. Gaia's separate 541-d astrometric orbit for this source, which is the outer orbit of the eclipsing pair about a third body, predicts a light-travel-time variation of at least 3.25 min if the third body supplies a smaller fraction of the light than of the mass. The eclipse times are consistent with it: a light-travel-time variation is present (QLP and TESS-SPOC times agree to under 0.4 min, so it is not a pipeline effect), and fitting only its amplitude, with the period, eccentricity and phase fixed by Gaia, gives 3.39 ± 0.32 min and removes three-quarters of the timing χ^2. The test is not strong: 67 timings in four seasons, three free parameters plus the sign of K, a chance coincidence of about 0.1, and outer periods of 318 and 1463 d not excluded. The amplitude also implies a third body of about 0.69 $M_\odot$

that is fainter than a main-sequence star of that mass, which is unexplained. The deficit is on the target in difference images and TESS-SPOC reproduces both eclipses. Eclipse timing over a full outer orbit, and radial velocities of the third star, would test the outer orbit and give the outer mass ratio and the mutual inclination of the two orbits.

A96 HD 140021 = TIC 426877400

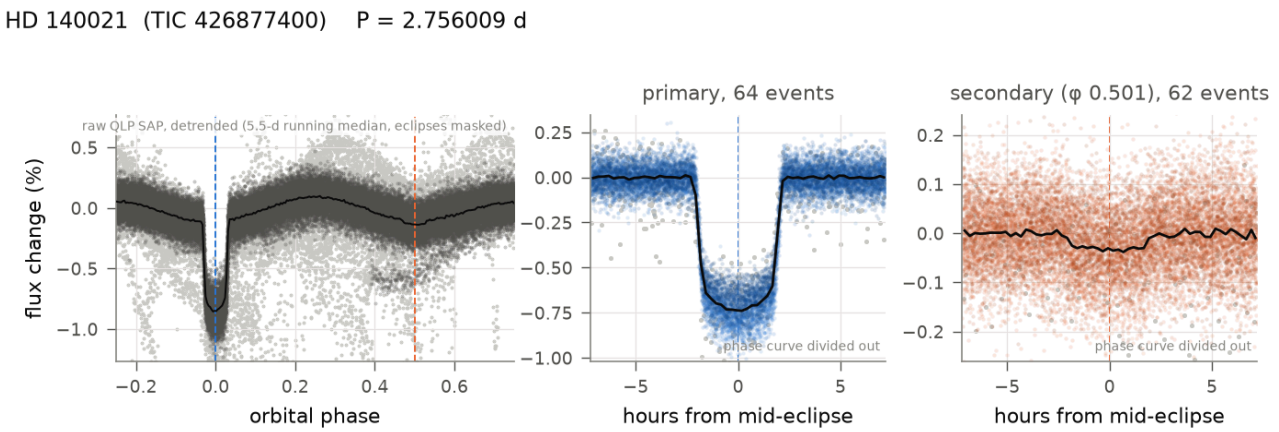


Figure A96. HD 140021, raw QLP SAP flux (sector medians removed, flagged cadences light grey), detrended by a 5.5-d running median with the eclipses masked and folded on P = 2.7560089 d. Left: the full orbit, with the 0.9-ppt ellipsoidal variation; the points near −0.5% at phases 0.38–0.48 are the sector-99 start ramp. Middle: the 7.6-ppt, 4.5-h flat-bottomed primary; right: the 0.31-ppt secondary near phase 0.50 (both with the phase curve divided out).

Identifiers	TIC 426877400; HD 140021; CD−71 1198; CPD−71 1917; TYC 9268-1801-1; Gaia DR3 5794909926060103424
Position (J2000)	236.72191, −71.67431 deg (TIC)
Brightness	Tmag 9.62; G 10.00
Stars	F6/7IV (SIMBAD; TIC: T_{eff} 6215 K, 2.12 $R_\odot$) at 271 pc; Gaia SB1 orbit (e = 0.079 ± 0.030, ω = 8 ± 15°, K_1 16.8 km/s). The mass function (0.00134 $M_\odot$) gives $m_2 \geq 0.13\ M_\odot$ and the transit depth $R_2 \approx 0.18\ R_\odot$: an M-dwarf companion. Ellipsoidal variation of 0.9 ppt (cos 4πφ amplitude) and a 0.25-ppt sin 2πφ term, possibly Doppler beaming
Period	2.7560089 ± 0.0000006 d, timing fit of 61 primaries (Gaia: 2.756044 ± 0.000090 d; −0.4σ). Odd and even primaries agree (7.42 ± 0.11 and 7.37 ± 0.11 ppt)
Epochs	Primary BTJD 3574.15160 ± 0.00014 = BJD_{TDB} 2460574.15160 (errors scaled by 2.1 for χ^2/dof = 4.4); secondary BTJD 3575.533 (phase 0.501 ± 0.010, from the fold; the error is the spread between detrending choices and sector subsets, which give 0.5085–0.521)
Primary eclipse	7.6 ppt (median of 61 events), 4.5 h, flat-bottomed, in raw QLP SAP flux (trapezoid fits with a local quadratic baseline); 7.7–8.1 ppt in sectors 12–93 and 6.4–7.1 ppt in sectors 99–102, a QLP aperture effect (the difference images give 7.9–8.1 ppt in both sectors 65 and 100); TESS-SPOC SAP 7.3 ppt, 4.3 h
Secondary eclipse	0.31 ± 0.02 ppt (17σ), 4.7 h, at phase 0.501 ± 0.010, measured in the fold against a model of the phase curve; single events are below the noise
Orbit	Primary 0.028 ± 0.010 in phase after Gaia's conjunction A (2.8σ), inside the eclipse half-width of 0.036; secondary −0.019 ± 0.013 from conjunction B (1.4σ); evaluated at the eclipse nearest Gaia's epoch, and the same with Gaia's or the TESS period. Conjunction B is excluded for the primary (55σ). e cos ω = +0.002 ± 0.015 from the secondary phase, against Gaia's +0.075 ± 0.028
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s12, s39, s65, s66; 27 events, 7.3 and 7.9 ppt, 4.3 h; mid-times of 25 matched events within −0.6 ± 1.9 min of QLP's); TARS (43 events, 7.6 ppt); eleanor-lite (s12, 7 events, 7.4 ppt). TESS-SPOC TCEs in s39 (P 2.75605 d, MES 110), s65 (MES 97), s66 (MES 62) and s56–s69 (P 2.755995 ± 0.000026 d, 8.15 ppt, MES 100), none flagged as an eclipsing binary; multi-sector centroid 0.45 ± 2.54″ from the out-of-transit centroid
Pixel attribution	

	TESScut difference images in s65 (9 events) and s100 (8 events): deficit centroid 1.2″ and 2.2″ from the light centroid, on the target's pixel, PSF-shaped (residual 10–14%); depth 8.1/8.1/7.8 and 7.5/7.9/7.3 ppt in 1×1/3×3/5×5 pixel apertures. Three Gaia sources within 60″ are bright enough in principle: G 13.55 at 54.7″ (would need 20%), G 14.94 at 35.7″ (69%) and G 15.03 at 26.3″ (75%); each would move the deficit centroid by more than 1 px, and the G 13.55 star's own pixel loses only 0.5–3.3 ppt. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX (its only entry within 60″ is a rotational variable 52″ away), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists, and the SuperWASP false-positive catalogue (Schanche et al. 2019); no SIMBAD reference or ADS full-text record mentions variability. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	The secondary phase depends on the detrending at the 0.01 level, so the eclipses do not yet test Gaia's eccentricity; a radial-velocity orbit would. The primary O−C shows structure of ±1–2 min between sectors (rms 1.7 min)
Next eclipses	Primary: BTJD 4329.298 (2026-10-15 19:11 UTC), 4332.054, 4334.810; secondary: BTJD 4327.923 (2026-10-14 10:10 UTC), 4330.679 (allow ±2 min, plus event-to-event scatter of 1.7 min rms, for the primary; the secondary times are uncertain by about 40 min, from the phase error)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 140021 is an F6/7 subgiant eclipsed every 2.756 d by a low-mass companion. The flat-bottomed 7.6-ppt transit, the 0.31-ppt occultation near phase 0.50 and the 0.9-ppt ellipsoidal variation fit one geometry: a companion of about 0.18 $R_\odot$ and at least 0.13 $M_\odot$, an M dwarf, whose occultation depth matches its expected TESS-band surface brightness. The primary falls 0.028 ± 0.010 in phase after Gaia's conjunction A (2.8σ), inside the eclipse, and the secondary 0.019 ± 0.013 before conjunction B. The shallow secondary's phase depends on the detrending at the 0.01 level, so the eclipses give only $e \cos \omega = 0.002 \pm 0.015$, consistent both with a circular orbit, as expected for a 2.76-d F+M binary, and with Gaia's $e = 0.079 \pm 0.030$. Radial velocities would measure the eccentricity and give the M dwarf's mass and, with the transit, its radius.

A97 TYC 1529-387-1 = TIC 351595900

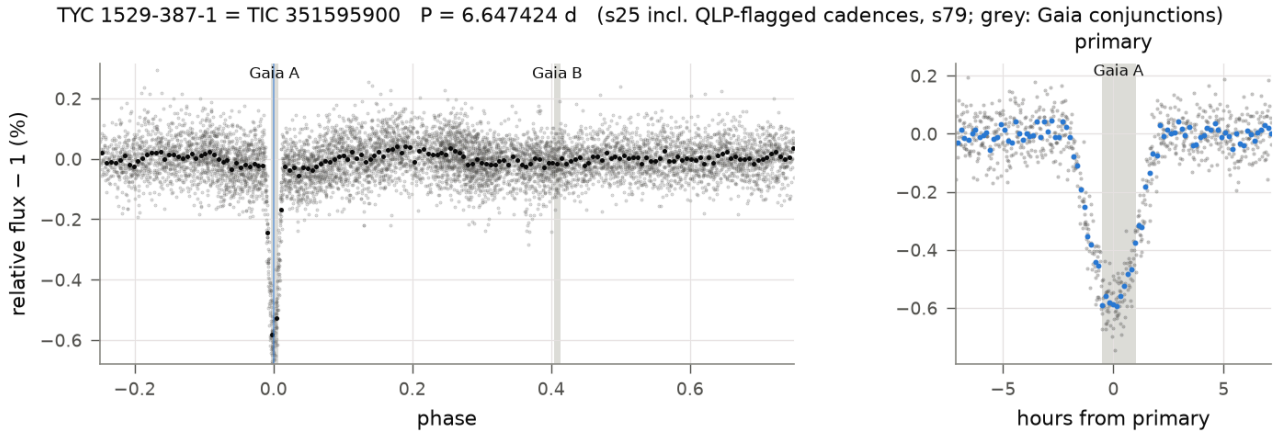


Figure A97. TYC 1529-387-1, raw QLP SAP flux of sectors 25 (including the QLP-flagged cadences, which cover three of the four sector-25 eclipses) and 79, sector medians removed, detrended by a 0.75-d running median with the eclipses masked and folded on $P = 6.647424$ d; grey bands mark Gaia's predicted conjunctions A and B. Left: the full orbit, with nothing at conjunction B (phase 0.41). Right: the 5.9-ppt, 4.1-h V-shaped primary, centred on conjunction A.

Identifiers	TIC 351595900; TYC 1529-387-1; Gaia DR3 4561305128630471168
Position (J2000)	253.94271, +19.00811 deg (TIC)
Brightness	Tmag 9.32; G 9.73
Stars	

	Evolved F star (TIC: T_{eff} 6198 K, 3.12 $R_\odot$, log g 3.53) at 339 pc; Gaia SB1 orbit (e = 0.147 ± 0.013, ω = 172.7 ± 3.5°, K_1 31.5 km/s). With the inclination from the eclipse ($i \approx 79°$) the companion is about 0.38 $M_\odot$, an M dwarf
Period	6.647424 ± 0.000005 d, timing fit of 7 primaries in sectors 25 and 79, flagged cadences included; an event is used when it is fully covered and its trapezoid fit converges inside the search window, which excludes the sector-79 event at BTJD 3457.65, whose cadences are all flagged and carry artefacts of 26–32 ppt; the 219 cycles between the sectors are unambiguous (Gaia: 6.647769 ± 0.000184 d; −1.9σ). Odd and even primaries agree (6.01 ± 0.10 and 6.03 ± 0.15 ppt)
Epochs	Primary BTJD 2826.14200 ± 0.00052 = BJD_{TDB} 2459826.14200
Primary eclipse	5.9 ± 0.2 ppt, 4.1 h, V-shaped (grazing), in raw QLP SAP flux (trapezoid fits to 7 events, O−C rms 1.8 min); TESS-SPOC SAP 5.7 ppt, 4.1 h
Secondary eclipse	Not detected: < 0.2 ppt (3σ) at Gaia's conjunction B (phase 0.409), −0.06 ± 0.05 ppt at phase 0.5. The occultation of an M dwarf by an F subgiant is expected at about 0.1 ppt
Orbit	Primary on Gaia's conjunction A: Δφ = −0.0015 ± 0.0048 (0.3σ, −0.25 h), with Gaia's orbit carried by the TESS period (−0.025 ± 0.013, 1.9σ, with Gaia's own period); conjunction B is 0.42 in phase away. No secondary, so no e cos ω test
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (all 7 primaries, 5.7 ppt, 4.1 h, median O−C −0.1 and −1.2 min; the sector-25 cadences flagged by QLP are unflagged in TESS-SPOC; one matched event 0.5 min apart); TARS (5 events, 6.1 ppt). TESS-SPOC TCE in s79: P 6.6486 ± 0.0008 d, 5.77 ppt, 4.14 h, b 0.99, MES 38, not flagged as an eclipsing binary; centroid 1.39 ± 2.51″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in s79 (3 events) and s25 (2 events): deficit centroid 1.5″ and 2.8″ from the light centroid, on the target's pixel, PSF-shaped (residual 13% and 23%, at low signal-to-noise); depth 5.4/5.2/5.1 ppt in 1×1/3×3/5×5 pixel apertures (s79). The brightest Gaia neighbour that could matter, G 15.46 at 25.7″, would need a 118% eclipse, and its own pixel loses only the target's wings; all 21 sources within 90″ are excluded. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all 11 sources within 60″), SB9, WDS, the ExoFOP TOI and CTOI lists; ADS full text has nothing on the star. Known SB (SIMBAD SB*, from Gaia DR3, its only reference)
Next eclipses	Primary: BTJD 4328.460 (2026-10-14 23:05 UTC), 4335.107, 4341.755 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

A grazing eclipse of an evolved F star by an M-dwarf companion of about 0.38 $M_\odot$. The single eclipse kind is the companion's transit at conjunction A, on Gaia's predicted phase to 0.3σ; the occultation at conjunction B, which Gaia's eccentric orbit places at phase 0.41, is expected at about 0.1 ppt and lies below the 0.2-ppt limit. The 7 events span two sectors 219 cycles apart, and three of the four sector-25 eclipses fall entirely in cadences that QLP flags but TESS-SPOC does not (the fourth, at BTJD 2008.51, is unflagged); with the flagged cadences excluded, the period from sector 79 alone (6.64776 ± 0.00075 d) agrees. Radial velocities over one orbit, with the transit, would give the companion's mass and radius; more TESS sectors would tighten the period, now 1.9σ below Gaia's.

A98 BD+45 2574 A = TIC 157469799

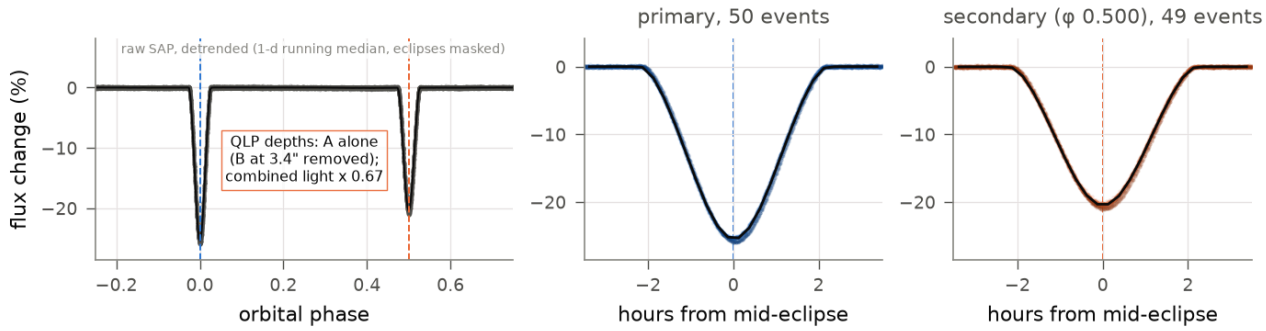


Figure A98. BD+45 2574 A, raw QLP SAP flux (sector medians removed), detrended by a 1-d running median with the eclipses masked and folded on P = 3.5063244 d, twice the fold period. Left: the full orbit, with two unequal eclipses half an orbit apart. Middle: the 245-ppt, 3.85-h primary; right: the 196-ppt, 3.85-h secondary at phase 0.500 (50 and 49 events plotted; 41 and 44 fully covered events in the timing fit). QLP's flux is corrected for the light of the 3.4″ companion B, so these are depths in A alone; in the combined light they are 0.67 times as deep.

Identifiers	TIC 157469799; BD+45 2574 A; TYC 3095-174-1; Gaia DR3 1345750246115251840; WDS J17399+4119 A (A 695 A)
Position (J2000)	264.96617, +41.31447 deg (TIC)
Brightness	Tmag 8.95; G 9.33
Stars	TIC: T_{eff} 6171 K, 1.52 $R_\odot$, at 147 pc; Gaia double-lined circular orbit (SB2C; K_1 83.5, K_2 91.2 km/s, q = 0.92), $M_1 \sin^3 i$ = 1.01 and $M_2 \sin^3 i$ = 0.93 $M_\odot$: two late-F/G dwarfs, $i \approx 87°$. Component A of the visual pair A 695 (WDS 17399+4119, 3.3–3.4″, first measured 1893); B (Gaia DR3 1345750216051287040, G 10.07) shares A's parallax (6.73 and 6.76 mas) and proper motion, and Gaia measured its velocity as constant to 3.6 km/s
Period	3.5063244 ± 0.0000002 d, timing fit of 41 primaries (and 3.5063237 ± 0.0000002 d from 44 secondaries) (Gaia: 3.506308 ± 0.000145 d; +0.1σ). The two eclipses differ in depth by 25% (about 60σ): the period is twice the fold period, 1.753154 d
Epochs	Primary BTJD 3172.71372 ± 0.00002 = BJD_{TDB} 2460172.71372; secondary BTJD 3016.68227 ± 0.00003 (phase 0.49999). Both fits with the QLP sector-80 times shifted by −2.70 min (see Pipelines agreeing); O−C rms 0.24 and 0.45 min
Primary eclipse	245.1 ± 2.6 ppt in QLP SAP flux, which carries the TIC dilution correction (contamination ratio 0.495), i.e. the depth in A alone; 166 ppt in the combined light of A and B; 3.85 h (trapezoid fits to 41 events); odd and even 245.8 and 245.5 ppt
Secondary eclipse	196.4 ± 4.3 ppt in A alone (131 ppt in combined light), 3.85 h, at phase 0.49999 (44 events)
Orbit	Circular orbit (SB2C): no conjunction-phase test. If Gaia's reference epoch for a circular orbit is the time of the primary's maximum velocity, the deeper eclipse falls 0.0025 ± 0.0027 in phase from the predicted conjunction (0.9σ) and is the eclipse of the more massive star; this depends on that convention and is not used for the attribution
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s25, s26; mid-times −0.05 and +0.27 min from QLP's; PDCSAP 253 and 199 ppt, SAP 167 and 131 ppt); TESS-SPOC (all seven sectors, mid-times within ±0.5 min of QLP's in s25–s79); TARS. In sector 80 the QLP times are 2.7 min late against TESS-SPOC (−2.69 ± 0.05 and −2.72 ± 0.06 min for the two eclipses) and TARS, the same offset as for A11. SPOC TCEs at the half period in s25, s26 and s14–s26 (1.7531525 d, MES 4313), flagged as suspected eclipsing binaries; difference-image centroid 1.23 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in s40, s53, s79 and s80 (6–8 events per stack, both eclipses): the deficit is a copy of the blended image (residual 6–7%), 1.0–1.2″ from the light centroid, with depth stable across 1×1/3×3/5×5 pixel apertures (133–142 ppt). Companion B, 3.4″ away, is unresolved, but the deficit centroid moves −1.03 ± 0.03″ along the A→B line in all eight stacks, against −0.94″ predicted for eclipses on A and +1.85″ for eclipses on B; B is excluded. No other Gaia source within 60″ is brighter than G 17.17

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2 and 3, Gaia DR3 variability tables, SB9, the ExoFOP TOI and CTOI lists (no ASAS-3 coverage); SIMBAD's references for the pair are visual double-star measurements, and ADS full text has nothing on eclipses. Known SB (SIMBAD SB* for A, from Gaia DR3)
Next eclipses	Primary: BTJD 4329.801 (2026-10-16 07:13 UTC), 4333.307, 4336.813; secondary: BTJD 4328.048 (2026-10-14 13:09 UTC), 4331.554 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+45 2574 A is a double-lined pair of late-F/G dwarfs (q = 0.92) in a 3.5-d circular orbit, the brighter member of the visual binary A 695. The two eclipses, 25% and 20% of A's light, are what a slightly unequal twin produces at $i \approx 87°$. Because the 3.4″ companion B is unresolved in TESS, the attribution rests on the motion of the deficit centroid along the A–B line, which matches an eclipse on A and excludes B, and on B's constant Gaia velocity; the period agrees with A's double-lined orbit to 0.1σ. QLP's flux for this star is already corrected for B's light, so its depths exceed those in SAP flux from the other pipelines by the factor 1.495. B shares A's parallax and proper motion, so the system is a triple, with B at a projected 500 au. A joint fit of the eclipses with Gaia DR4 epoch velocities, or new spectra, would give both eclipsing stars' masses and radii.

A99 TYC 3925-1852-1 = TIC 352236055

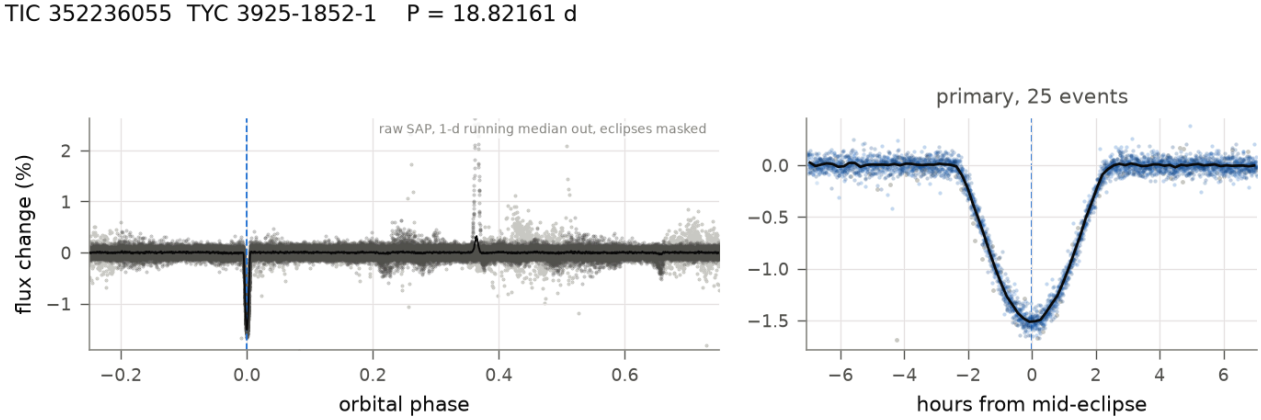


Figure A99. TYC 3925-1852-1, raw QLP SAP flux (sector medians removed, flagged cadences light grey), detrended by a 1-d running median with the eclipses masked and folded on P = 18.821605 d. Left: the full orbit, with one eclipse kind; the brightening at phase 0.365 is a single sector-82 QLP event absent from TESS-SPOC and TARS, and the shallow dip near 0.655 comes from one sector-73 orbit. Right: the 14.9-ppt, 4.7-h V-shaped eclipse, every event normalised by its own out-of-eclipse median.

Identifiers	TIC 352236055; TYC 3925-1852-1; Gaia DR3 2141529070981241088
Position (J2000)	292.15643, +55.31563 deg (TIC)
Brightness	Tmag 9.28; G 9.70
Stars	Subgiant (TIC: T_{eff} 6047 K, 2.53 $R_\odot$) at 271 pc; Gaia SB1 orbit (e = 0.398 ± 0.004, ω = 230.8 ± 0.6°, K_1 40.8 km/s). The mass function (0.102 $M_\odot$) gives $m_2 \geq 0.70\ M_\odot$, a K dwarf; $i \approx 83°$ from the eclipse duration
Period	18.821605 ± 0.000006 d, timing fit of 23 events, with the two sector-80 times (O–C +2.7 min each) corrected by −2.70 min for the QLP sector-80 timestamp offset measured in A98; dropping them instead gives 18.821603 d (Gaia: 18.821345 ± 0.00093 d; +0.3σ). Odd and even events agree (15.00 and 14.83 ppt), and the 2P fold shows equal depths at phases 0 and 0.5
Epochs	Primary BTJD 3160.83009 ± 0.00017 = BJD_{TDB} 2460160.83009 (errors scaled by 1.7; O–C rms 1.8 min)
Primary eclipse	

	14.9 ± 0.4 ppt, 4.7 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 23 fully covered events in 16 of the 18 QLP sectors); TESS-SPOC SAP 13.6 ppt and PDCSAP 14.4 ppt on the same five events. It falls at conjunction B, where the visible star is in front: a grazing occultation of the companion
Secondary eclipse	Not detected: < 0.13 ppt (3σ) at Gaia's conjunction A (phase 0.6665 ± 0.008), and nothing above S/N 6.3 at any phase. At conjunction A the stars are 44 $R_\odot$ apart against 23 $R_\odot$ at B, and the inclination that gives the observed chord at B gives an impact parameter of about 1.6 at A, so no eclipse is expected there
Orbit	Eclipse on Gaia's conjunction B: $\Delta\varphi = +0.0010 \pm 0.0008$ (1.2σ, +0.4 h), with the TESS ephemeris carried to the orbit nearest Gaia's periastron epoch (147 cycles) and Gaia's elements drawn with their full covariance; conjunction A is excluded (+0.3345, 148σ). No secondary, so no $e \cos \omega$ test (Gaia: -0.252 ± 0.004)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s14, s41, s82; 5 events, 13.6 and 14.4 ppt, 4.6 h; mid-times within 0.0–0.5 min of QLP's at 200-s and 600-s cadence, 3.6 min in the 30-min s14); TARS (5 events, 14.5 ppt); TASOC (s14, 14.0 ppt). No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images in s41 and s82 (2 events each): deficit centroid 1.1″ and 0.4″ from the light centroid, PSF-shaped (residual 7–10%), depth 12.4/12.7/12.2 and 12.4/12.3/11.7 ppt in 1×1/3×3/5×5 pixel apertures. Only two Gaia sources within 60″ are bright enough: G 13.18 at 42.5″ (would need 38%) and G 14.05 at 42.9″ (84%); either would put the deficit about 2 px away, and their own pixels lose less than the target's. A companion within about 1″ is not excluded (RUWE 1.10)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists (no ASAS-3 coverage); ADS full text has nothing on the star. Known SB (SIMBAD SB*, from Gaia DR3, its only reference)
Next eclipses	Primary: BTJD 4327.770 (2026-10-14 06:25 UTC), 4346.591 (allow ±2 min, plus event-to-event scatter of 1.8 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 3925-1852-1 is a subgiant with a K-dwarf companion in an 18.8-d orbit of $e = 0.40$. The one eclipse it shows falls on Gaia's conjunction B, where the stars are 23 $R_\odot$ apart and the visible star passes in front: it is a grazing occultation of the companion, and its depth measures the companion's light. At conjunction A, near apastron, the separation is nearly twice as large and the same inclination misses, which accounts for the empty conjunction A (< 0.13 ppt). A 15-ppt occultation requires most of the companion's disc to be covered, somewhat more than the impact parameter from the duration allows with the TIC radius, but within the uncertainties of that radius and of the primary's mass. With 23 events over five years, it is a well-sampled eccentric binary that eclipses at only one conjunction. Radial velocities of the companion, or a spectrum showing its lines, would fix its mass and the inclination.

A100 HD 353655 = TIC 425304484

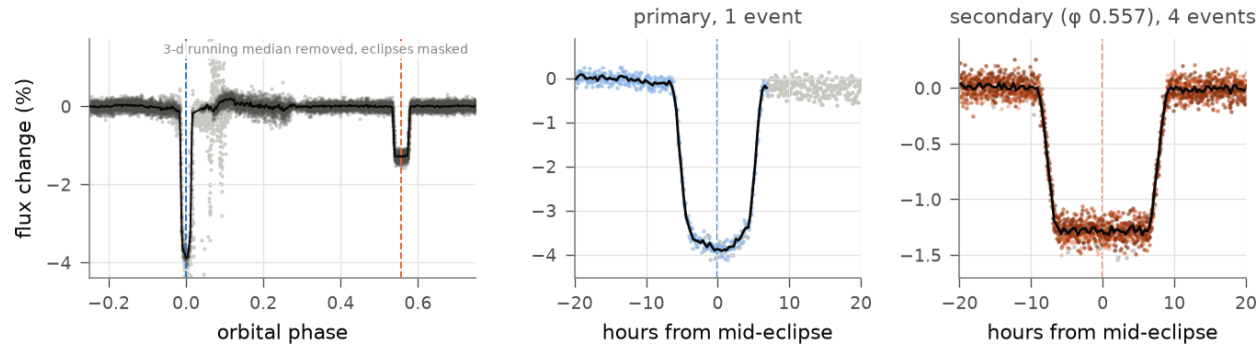


Figure A100. HD 353655, raw QLP SAP flux of sectors 54 and 81 (sector medians removed, flagged cadences light grey), detrended by a 3-d running median with the eclipses masked and folded on P = 16.260285 d. Left: the full orbit, with the deep eclipse at phase 0 and the shallow one at phase 0.557. Middle: the single 35–38-ppt, 12-h primary (sector 81; the grey points after egress are flagged scattered-light cadences). Right: the four 13-ppt, 18-h flat-bottomed secondaries. Zoomed events are normalised by their own out-of-eclipse median.

Identifiers	TIC 425304484; HD 353655; TYC 1615-291-1; Gaia DR3 4317746535565107840; WDS J19475+1512AB (TDT 1753)
Position (J2000)	296.88746, +15.20305 deg (TIC)
Brightness	Tmag 9.46; G 9.94
Stars	G0 (SIMBAD; TIC: T_{eff} 5955 K, 4.14 $R_\odot$, log g 3.24, a subgiant) at 440 pc; Gaia SB1 orbit (e = 0.208 ± 0.009, ω = 64.4 ± 1.7°, K_1 35.4 km/s). The mass function (0.0698 $M_\odot$) gives m_2 = 0.58–0.67 $M_\odot$ for M_1 = 1.1–1.4 $M_\odot$, and the occultation depth a companion near 4500 K: a K dwarf. The WDS companion at 2.6″ is G 19.0
Period	16.260285 ± 0.000024 d, timing fit of 4 secondaries in sectors 54 and 81, 45 cycles apart; Gaia's period excludes 44 or 46 cycles (Gaia: 16.25919 ± 0.00100 d; +1.1σ). The two eclipses differ in depth and duration, so P is the orbital period
Epochs	Primary BTJD 3531.06071 ± 0.00039 = BJD_{TDB} 2460531.06071 (one event); secondary BTJD 3231.16518 ± 0.00053 (phase 0.55656 ± 0.00004)
Primary eclipse	35–38 ppt, 12.1 h, flat-bottomed, from the one sector-81 event at BTJD 3531.06 (the other predicted events fall in data gaps or flagged cadences): 35.0 ppt in raw QLP SAP flux (trapezoid fit) and 35.1 ppt in TESS-SPOC SAP, against 38.0 ppt in TESS-SPOC PDCSAP, 37.5 ppt in TARS and 37.9 ppt in the 1×1-pixel difference image; the end of egress and the baseline after it fall in flagged scattered-light cadences, which limits the baseline fit. ASAS-3 V (2000–2009) folded on this ephemeris shows it at +48.5 mmag (11.5σ; eight of nine in-eclipse points at +44 to +53 mmag); the signal vanishes at Gaia's period
Secondary eclipse	12.9–13.5 ppt (13.0), 18.2 h, flat-bottomed (total), at phase 0.55656 (4 events); ASAS-3 +15 mmag (3.9σ). It falls 0.557 in phase after the primary because the orbit is eccentric; the duration ratio, 18.2/12.1, gives e sin ω ≈ +0.20
Orbit	Primary 0.0002 ± 0.0021 in phase from Gaia's conjunction A (0.1σ), secondary −0.0014 ± 0.0035 from conjunction B (0.4σ), with the TESS period and Gaia's (e, ω, T_p) and their covariance at Gaia's epoch. e cos ω = 0.08897 ± 0.00006 from the eclipse spacing against Gaia's 0.0897 ± 0.0083 (−0.1σ); e sin ω ≈ 0.20 from the durations against Gaia's 0.187
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (primary at BTJD 3531.06093, 0.3 min from QLP's; secondaries 12.6–13.7 ppt, within 1–3 min of QLP's); TARS (primary +0.8 min, 37.5 ppt; two secondaries, 13.7–14.3 ppt). TESS-SPOC TCE in s54: P 16.2598 ± 0.0015 d, 14.5 ppt, 18.3 h, MES 181, centroid 1.3 ± 2.5″ from the out-of-transit centroid; in s81 a TCE at 23.50 d (MES 231) pairs a secondary with the primary. ASAS-3 detects both eclipses
Pixel attribution	

	TESScut difference images: for the primary (s81, one event) the deficit centroid is 2.5″ from the light centroid, PSF-shaped (residual 2.7% in 3×3), depth 37.9/36.9/34.9 ppt in 1×1/3×3/5×5 pixel apertures; for the secondary (s54 and s81, two events each) 2.0″ and 2.7″ from the light centroid (residual 12–23%, at lower signal-to-noise under scattered light). In this crowded field (260 Gaia sources within 60″) none is bright enough to make the primary: the brightest, G 14.15 at 54.5″, would need 174% (65% for the secondary), and its own pixel loses only 0.8–5.9 ppt. A companion within about 1″ is not excluded; the 2.6″ WDS companion is 9 mag too faint
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 3 and 4, Gaia DR3 variability tables, SB9, the ExoFOP TOI and CTOI lists; ADS full text has nothing on the star. Listed in Kostov et al. (2025) table 2. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	TESS has observed the primary once; its timing rests on that event and on ASAS-3, and its depth is uncertain at the 35–38 ppt level. The companion radius implied by the transit is large for its mass unless the subgiant's radius is about 20% below the TIC value
Next eclipses	Primary: BTJD 4327.815 (2026-10-14 07:30 UTC), 4344.075; secondary: BTJD 4336.865 (2026-10-23 08:43 UTC), 4353.125 (allow ±5 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 353655 is an evolved G star with a K-dwarf companion in a 16.26-d orbit of e = 0.21. At conjunction A, near periastron (separation 0.81a), the companion transits the subgiant in a 35–38-ppt, 12-h eclipse; at conjunction B (1.18a) the subgiant totally occults the companion in a flat-bottomed 13-ppt eclipse half again as long. Both eclipses fall on Gaia's predicted conjunctions, and the two orbital quantities the eclipses measure, $e\cos\omega$ from their spacing and $e\sin\omega$ from their duration ratio, both match the spectroscopic orbit. TESS observed the deeper eclipse only once, but ASAS-3 photometry two decades earlier recovers it at 11.5σ on the TESS ephemeris. The transit depth requires a radius ratio of 0.19–0.20 and so, with the TIC radius of the subgiant, a companion of at least 0.77–0.81 $R_\odot$, larger than the 0.6–0.7 $R_\odot$ expected for a 0.6–0.7 $M_\odot$ dwarf; the two agree only if the subgiant is about 20% smaller than the TIC gives. A radial-velocity orbit and ground-based photometry of a primary eclipse would determine the radii.

A101 HD 192721 = TIC 14596588

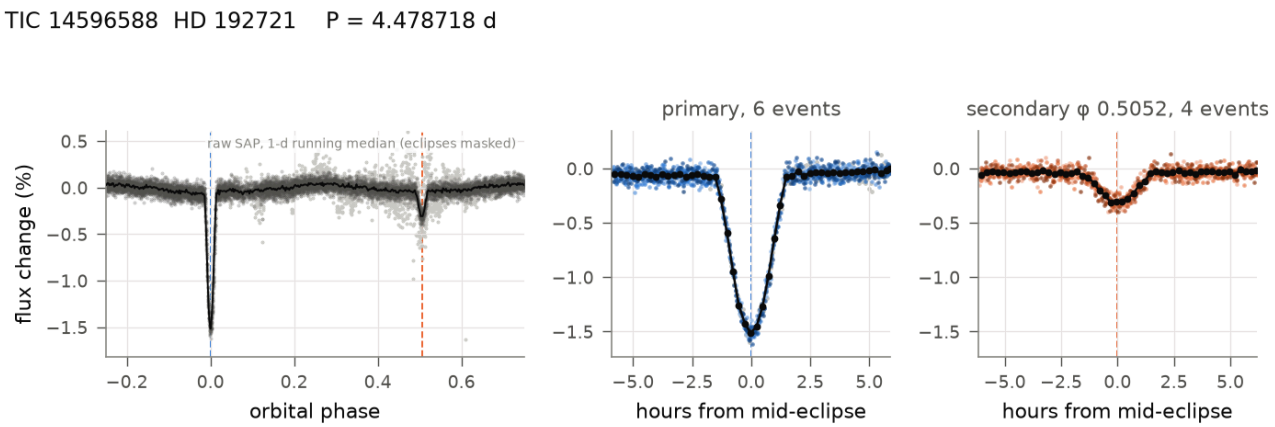


Figure A101. HD 192721, raw QLP SAP flux of sector 92 (the only sector; sector median removed, flagged cadences light grey), detrended by a 1-d running median with the eclipses masked and folded on P = 4.478718 d. Left: the full orbit. Middle: the 14.4-ppt, 3.0-h primary; right: the 2.8-ppt, 3.1-h secondary at phase 0.5052, displaced from 0.5 by a small eccentricity.

Identifiers	TIC 14596588; HD 192721; BD–14 5697; TYC 5752-100-1; Gaia DR3 6876279994232577536
Position (J2000)	304.21183, –14.19728 deg (TIC)
Brightness	Tmag 8.89; G 9.12
Stars	

	A1mA8-F2, an Am star (SIMBAD; Renson 53787; TIC: T_{eff} 7530 K, 2.31 $R_\odot$) at 276 pc; Gaia SB1 orbit (e = 0.009 ± 0.011, K_1 51.4 km/s). The mass function (0.063 $M_\odot$) gives $m_2 \geq 0.73\ M_\odot$, and the secondary-to-primary depth ratio (0.19) a companion near 4500 K: a K dwarf. A co-moving G 14.54 star lies 3.9″ away (parallax 3.63 against 3.57 mas)
Period	4.478718 ± 0.000065 d, timing fit of 5 primaries in one sector (Gaia: 4.478818 ± 0.000142 d; −0.6σ). Odd and even primaries agree (14.40 and 14.37 ppt in the 2P fold)
Epochs	Primary BTJD 3818.15412 ± 0.00013 = BJD_{TDB} 2460818.15412; secondary BTJD 3820.41690 ± 0.00060 (phase 0.50523 ± 0.00013)
Primary eclipse	14.4 ppt, 2.95 h (ingress 1.16 h), V-shaped (grazing, b ≈ 0.9), in raw QLP SAP flux (trapezoid fits to 5 events, O−C rms 0.24 min); TARS 13.6 ppt; photometry from SPOC-calibrated TESScut images 14.3 ppt
Secondary eclipse	2.8 ppt, 3.1 h, at phase 0.50523 ± 0.00013 (4 events); the offset from 0.5 (34 min) is a small eccentricity, the light-travel delay accounting for under 0.1 min of it
Orbit	Primary −0.0034 ± 0.0168 in phase from Gaia's conjunction A (−0.20σ), secondary +0.0028 ± 0.0168 from conjunction B (+0.17σ), with the TESS period and Gaia's (e, ω, T_p) and their covariance at Gaia's epoch (−0.70σ and −0.55σ with Gaia's period carried 766 cycles to the TESS epochs). $e \cos \omega$ = +0.0082 ± 0.0002 from the secondary phase against Gaia's −0.0009 (+0.0061/−0.0101), +1.1σ
Pipelines agreeing	QLP SAP; TARS (5 primaries and 4 secondaries, 13.6 and 2.7 ppt; primary mid-times −0.25 ± 0.36 min from QLP's); aperture photometry on SPOC-calibrated TESScut images (14.3 and 2.8 ppt; primary mid-times −0.01 ± 0.20 min from the QLP ephemeris). No SPOC 2-min or TESS-SPOC light curve exists for this sector, and so no TCE
Pixel attribution	TESScut difference images in s92: the primary deficit (5 events) is 0.3″ from the light centroid, PSF-shaped (residual 3.9%), depth 12.8/12.3/12.1 ppt in 1×1/3×3/5×5 pixel apertures; the secondary deficit (4 events) is 1.3″ from the light centroid. The 3.9″ companion (G 14.54) would need a 214% primary, and the deficit-centroid shift along the line to it excludes it at 11.8 standard deviations for the primary and 3.0 for the secondary (where it would need 41%). The brightest other Gaia source within 60″, G 14.44 at 50.8″, would need 196%. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables, SB9, WDS, the ExoFOP TOI and CTOI lists; Paunzen et al. (2013) list it as constant or undetermined in STEREO photometry of chemically peculiar stars; ADS full text has nothing on eclipses. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	One sector only, so the period is known to 6.5×10^{-5} d. ASAS-3 (463 points, 2000–2009) shows no dip at this period carried 1622 cycles back (+1.6 ± 6.0 mmag); a scan of trial periods gives a marginal dip of the expected size (+16.4 ± 4.1 mmag, S/N 4.0) at 4.479112 d, 6σ from the TESS period among about 100 independent trials, which is not adopted
Next eclipses	Primary: BTJD 4328.728 (2026-10-15 05:25 UTC), 4333.207, 4337.685; secondary: BTJD 4330.991 (2026-10-17 11:44 UTC), 4335.469 (allow ±32 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 192721 is a metallic-line A star with a K-dwarf companion in a 4.48-d orbit, seen in grazing eclipse. Its Am classification fits a short-period binary, in which tidal synchronisation slows the star's rotation. Both eclipses fall on Gaia's predicted conjunctions, and the secondary's displacement to phase 0.5052 measures a small eccentricity ($e \cos \omega$ = 0.008) that Gaia's orbit, consistent with zero, does not resolve. The data are a single sector, with no SPOC product; the period comes from five primaries and its uncertainty grows to about half an hour by the next eclipses. Ground-based timing of one or two eclipses would fix the period and test the marginal ASAS-3 alternative at 4.47911 d.

A102 HD 194269 = TIC 72160748

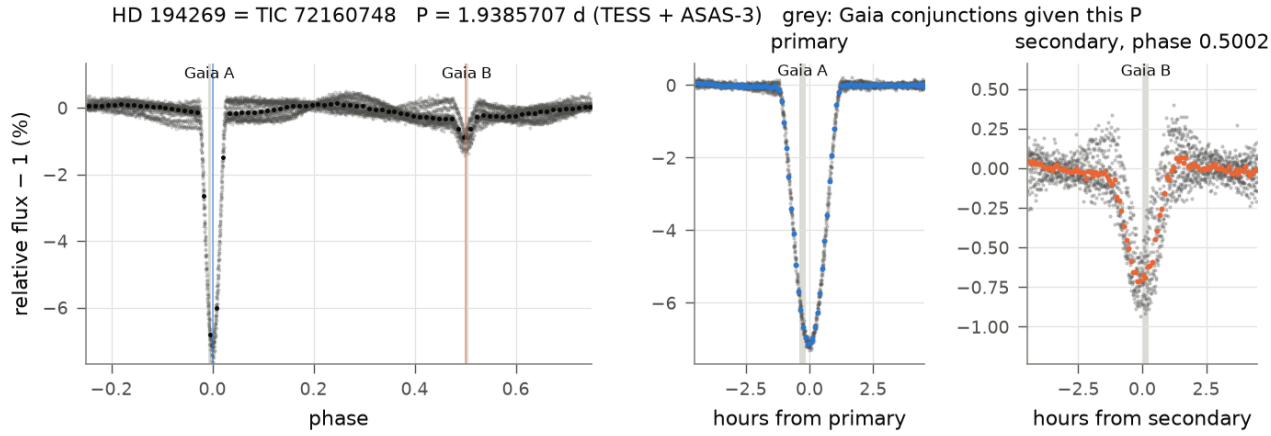


Figure A102. HD 194269, raw QLP SAP flux of sector 92 (the only sector; sector median removed), detrended by a 0.75-d running median with the eclipses masked and folded on P = 1.9385707 d, the period from TESS and ASAS-3; grey bands mark Gaia's conjunctions A and B recomputed with this period. Left: the full orbit; the residual out-of-eclipse structure is spot modulation not at the orbital period. Middle: the 69-ppt, 2.3-h V-shaped primary; right: the 7.1-ppt secondary at phase 0.5002.

Identifiers	TIC 72160748; HD 194269; BD−21 5717; CPD−21 7674; TYC 6341-1343-1; Gaia DR3 6859061882657414144
Position (J2000)	306.37372, −21.48726 deg (TIC)
Brightness	Tmag 8.65; G 9.04
Stars	F8V (SIMBAD; TIC: T_{eff} 6174 K, 1.26 $R_\odot$) at 106 pc; Gaia SB1 orbit (e = 0.070 ± 0.012, ω = 318 ± 5°, K_1 58.9 km/s). The mass function (0.0408 $M_\odot$) gives $m_2 \approx 0.49\ M_\odot$ at $i \approx 81.5°$, and the depth ratio (0.10) a companion near 3800 K: an early M dwarf. The primary is active: about 1% spot modulation within the sector and a ROSAT X-ray association
Period	1.9385707 ± 0.0000005 d from ASAS-3 V (520 points, 2000–2009) folded on the TESS epoch over a scan of trial periods (eclipse 64 mmag deep, S/N 14.8; next-best period S/N 3.9); TESS alone: 1.938562 ± 0.000029 d, timing fit of 11 primaries. The TESS period carried back 3700 cycles is uncertain by 0.11 d, 0.055 in phase, so the ASAS-3 cycle count is unambiguous (Gaia: 1.938490 ± 0.000016 d; +5.0σ)
Epochs	Primary BTJD 3815.72949 ± 0.00012 = BJD_{TDB} 2460815.72949; secondary BTJD 3814.76054 (phase 0.50017 ± 0.00016)
Primary eclipse	69.0 ppt, 2.3 h, V-shaped, in raw QLP SAP flux (trapezoid fits to 11 events with a quadratic baseline; 70.2 ± 1.5 ppt per event with linear baselines, drifting from 68.8 to 73.5 ppt in the second orbit as the scatter rises); SPOC 2-min SAP 69.7 ppt. Odd and even primaries agree (69.8 ± 0.8 and 70.5 ± 0.6 ppt)
Secondary eclipse	7.1 ppt, 2.3 h, at phase 0.50017 ± 0.00016 (11 events); SPOC 7.09 ppt
Orbit	With Gaia's published orbit the primary falls +0.096 ± 0.019 in phase from conjunction A (5.2σ) and the secondary +0.063 ± 0.013 from conjunction B (4.7σ). Gaia's period is 5.0σ short and correlated with its eccentricity (ρ = −0.84); conditioning Gaia's elements on the measured period gives e = 0.021 ± 0.006 (still 3.5σ from circular) and offsets of +0.0057 ± 0.0025 (2.3σ, +0.26 h) for the primary and −0.0032 ± 0.0012 (2.7σ) for the secondary. $e \cos \omega$ = +0.0003 ± 0.0003 from the secondary phase against Gaia's +0.052 ± 0.011 (−4.9σ; +0.014 ± 0.005 after the conditioning, 2.9σ)
Pipelines agreeing	QLP SAP; SPOC 2-min SAP and PDCSAP (s92; 11 primaries, 69.7 and 70.8 ppt, 2.27 h, median O−C +0.05 min against QLP's +0.09 min); TARS (69.1 ppt); ASAS-3 V (64 mmag). SPOC TCEs: the primary (P 1.9385709 ± 0.0000051 d, 72.8 ppt, 2.42 h, b 0.90, MES 721) and the secondary (7.09 ppt, 2.33 h, phase 0.49999, MES 77), neither flagged as an eclipsing binary; primary centroid 0.12 ± 2.50″ from the out-of-transit centroid
Pixel attribution	

	Difference images of 11 primaries in s92 in both the TESScut cutout and the SPOC target-pixel file: deficit centroid 0.2″ from the light centroid, PSF-shaped (residual 2.0% and 1.9%), depth 62.8/62.5/62.1 ppt in 1×1/3×3/5×5 pixel apertures; the secondary deficit (12 events) is 2.3″ (0.1 px) from the light centroid. The brightest Gaia source within 90″, G 13.89 at 49.4″, would need a 616% primary (63% for the secondary, while its own pixel loses only the target's wings). A companion within about 1″ is not excluded (RUWE 0.92)
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all 22 sources within 60″), SB9, WDS, the ExoFOP TOI and CTOI lists; no SIMBAD reference or ADS full-text record mentions variability. Known SB (SIMBAD SB*, from Gaia DR3)
Open issues	Even with the measured period, Gaia's orbit differs from the eclipses at 2.3–3.5σ (conjunction offsets 2.3 and 2.7σ, e cos ω 2.9σ, *e* 3.5σ from circular). The update is linear, 5σ from Gaia's published values, not a refit of the epoch velocities, which are not public
Next eclipses	Primary: BTJD 4327.512 (2026-10-14 00:14 UTC), 4329.451, 4331.389; secondary: BTJD 4328.482 (2026-10-14 23:30 UTC), 4330.420 (allow ±2 min; single secondaries scatter by 1.6 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 194269 is an active F8 dwarf with an early-M companion of about 0.5 $M_\odot$ in a 1.94-d orbit. The secondary at phase 0.5002 shows the orbit to be circular, as expected at this period, and the spot modulation and X-ray emission fit a primary kept magnetically active by the close orbit. With Gaia's published elements the eclipses fall 5σ from the predicted conjunctions. ASAS-3 photometry 3700 cycles before TESS fixes the period at 1.9385707 d: the single-sector TESS period, carried back, is uncertain by only 0.055 in phase, so the cycle count is secure. Gaia's period is 5.0σ short of this. Because Gaia's period and eccentricity are strongly anti-correlated, correcting the period reduces the tension in Gaia's e = 0.07 without removing it: the conditioned orbit has e = 0.021 ± 0.006, still 3.5σ from circular, e cos ω 2.9σ from the eclipses' value, and conjunctions 2.3 and 2.7σ from the eclipses. The system shows how a Gaia period error of a few σ can propagate into both the conjunction phase and the eccentricity, and Gaia DR4 epoch velocities would test this directly.

A103 BD+41 3794 = TIC 188258837

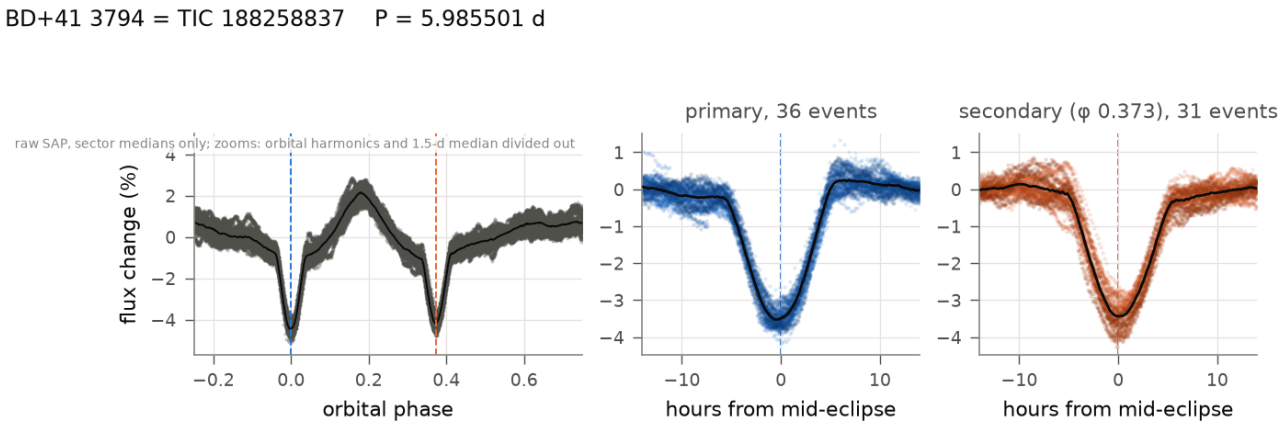


Figure A103. BD+41 3794, raw QLP SAP flux with only the sector medians removed, folded on the fitted P = 5.985501 d. Left: the full orbit, with the two eclipses (dashed lines at phases 0 and 0.373) and the 2.5% periastron brightening peaking at phase 0.18. Middle and right: the primary (36 ppt, 11.0 h) and the secondary (32 ppt, 10.45 h), with the orbital harmonics and a 1.5-d running median divided out (36 and 31 events plotted; 31 and 29 fully covered events in the timing fit).

Identifiers	TIC 188258837; BD+41 3794; ALS 11405; LS III +42 16; TYC 3161-510-1; Gaia DR3 2068151478876679168
Position (J2000)	308.00918, +42.20725 deg (TIC)

Brightness	Tmag 8.92; G 9.75
Stars	B0.2 III (SIMBAD) at 1390 pc, in the Cygnus OB region at b = +1.5°; the TIC T_{eff} of 9072 K is an effect of reddening (E(B−V) = 1.24). Gaia SB1 orbit (e = 0.198 ± 0.048, ω = 345.0 ± 15.6°, K_1 = 135.5 km/s). The mass function, 1.45 $M_\odot$, gives a companion of at least 10.6 $M_\odot$ (a B1–B2 dwarf) for an 18 $M_\odot$ primary; a ≈ 42 $R_\odot$. Stochastic low-frequency variability of 2–3 ppt rms, and a 2.5% periastron brightening at phase 0.18, the same in all eight sectors
Period	5.9855010 ± 0.0000159 d, timing fit of 31 primaries; the 29 secondaries give 5.9854465 ± 0.0000256 d (1.8σ apart). Gaia: 5.986001 ± 0.001368 d (−0.4σ)
Epochs	Primary BTJD 3144.94717 ± 0.00108 = BJD_{TDB} 2460144.94717; secondary BTJD 3135.20653 ± 0.00175 (phase 0.3726)
Primary eclipse	36.2 ± 2.3 ppt, 11.0 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 31 events); 32.5 ppt in TESS-SPOC SAP, 34.0 ppt in PDCSAP. O−C rms 8.4 min
Secondary eclipse	31.8 ± 3.7 ppt, 10.45 h (29 events), at phase 0.3726 after the primary, displaced from 0.5 by the eccentric orbit. O−C rms 14.3 min. The separation is constant to ±0.0008 in phase across eight sectors over five years: no apsidal motion is detected (< 0.0003 per year)
Orbit	Primary at Gaia's conjunction B: −0.0031 ± 0.0166 in phase (0.2σ); secondary at conjunction A: −0.0133 ± 0.0193 (0.7σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The swapped assignment is off by 0.386 ± 0.019. The eclipse spacing gives e cos ω = +0.201 (Gaia +0.184 ± 0.049, 0.35σ); the duration ratio, 1.05 ± 0.05, gives e sin ω ≈ −0.02 ± 0.03 (Gaia −0.048 ± 0.052). Gaia's periastron falls at phase 0.15. The primary is the eclipse of the companion by the B0.2 giant, at the conjunction where the stars are closer (0.914a against 1.013a), which makes it the deeper of two grazing eclipses (b ≈ 0.75–0.8)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s56, s75, s76, s82): on 17 matched primaries and 17 secondaries the mid-times differ from QLP's by +0.15 and +0.17 min (median), 0.6–0.8 min rms, and the depth ratios are 0.97 (SAP) and 1.00 (PDCSAP). The 8–14 min O−C scatter is therefore intrinsic to the star. No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of five primaries each in s56 and s82 and four secondaries in s56: deficit centroid 0.5–0.9″ from the light centroid, peaked on the target pixel and a scaled copy of the target's image over the central 5×5 pixels (residual 8–18%); depth 36–39 ppt in 1×1 and 3×3 pixel apertures, slightly diluted in 5×5 by a neighbour. The brightest Gaia source within 60″ (G 14.41, 47″) would need a 266% eclipse. Three brighter stars at 64–89″ (G 9.98, 11.85, 11.92) would need 8–30% eclipses and lose ≤ 5 ppt at their own pixels. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI); no ASAS-3 data. SIMBAD type SB*. Its eleven SIMBAD references are OB-star catalogues, photometry and Galactic-structure papers; none reports eclipses or an orbit. Two neighbours 47″ away appear in Kostov et al. (2025) table 2, most likely through this star's eclipses in their apertures. Known SB
Open issues	The eclipse times scatter by 8–14 min about the linear ephemeris because of the star's own red noise, which limits the timing precision
Next eclipses	Primary: BTJD 4330.076 (2026-10-16 13:46 UTC), 4336.062, 4342.047; secondary: BTJD 4332.307 (2026-10-18 19:18 UTC), 4338.292 (allow ±14 min, plus event-to-event scatter of 8 min rms for the primary and 14 min for the secondary)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+41 3794 (ALS 11405) is an early-B giant in the Cygnus OB region in an eccentric (e ≈ 0.2) 5.99-d orbit with a companion of at least 10.6 $M_\odot$. Both eclipses are grazing and of similar depth, and they fall on Gaia's two conjunctions with the TESS period; the spacing gives e cos ω in agreement with Gaia's. A 2.5% brightening peaks near Gaia's periastron phase and repeats unchanged over five years; it is the tidal and irradiation signature of periastron passage in a massive eccentric pair. The giant's stochastic low-frequency variability shifts individual eclipse times by several minutes, which TESS-SPOC reproduces event by event. A double-lined spectroscopic orbit would give the masses of a young massive binary directly, and continued timing would measure apsidal motion, which is not yet detected.

A104 BD+42 3880 = TIC 353453805

BD+42 3880 = TIC 353453805 P = 14.350949 d

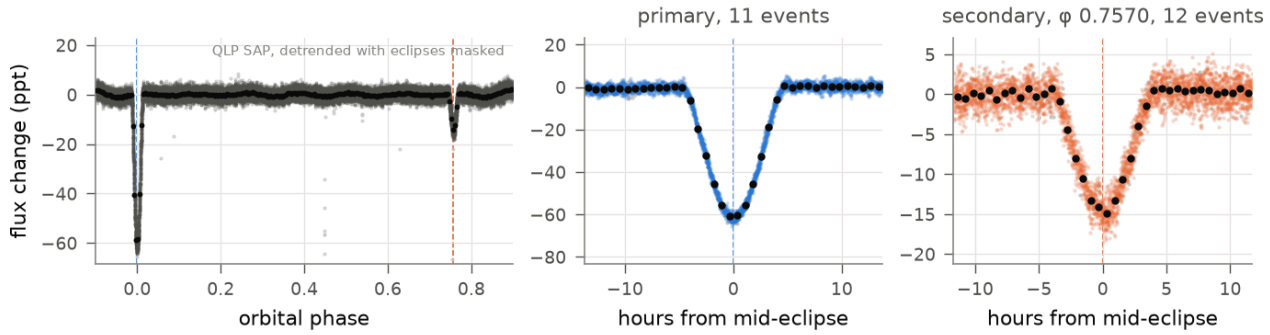


Figure A104. BD+42 3880, QLP SAP flux with sector medians removed, detrended with the eclipses masked, folded on the fitted P = 14.350949 d. Left: the full orbit, with the primary at phase 0 and the secondary of the eccentric orbit at phase 0.757. Middle: the 11 primaries (61 ppt, 8.6 h, nearly V-shaped). Right: the 12 secondaries (14.6 ppt, 7.3 h).

Identifiers	TIC 353453805; BD+42 3880; TYC 3174-31-1; Gaia DR3 2066657521752860544
Position (J2000)	312.47466, +42.83717 deg (TIC)
Brightness	Tmag 9.50; G 9.75
Stars	A2 (SIMBAD; TIC: T_{eff} 9825 K, 4.3 $R_\odot$ for the combined light) at 658 pc; Gaia SB2 orbit (e = 0.4754 ± 0.0084, ω = 24.06 ± 0.91°, K_1 = 86.6 and K_2 = 75.6 km/s, goodness of fit 17.1). With the inclination from the eclipse durations (82.9°) the masses are 2.06 and 2.36 $M_\odot$, two late-B or early-A stars with $R_1 + R_2$ = 5.4 $R_\odot$; a ≈ 40.7 $R_\odot$
Period	14.3509487 ± 0.0000051 d, timing fit of 11 primaries (O−C rms 0.23 min); the 12 secondaries give 14.350951 ± 0.000017 d. Gaia: 14.352119 ± 0.000459 d (−2.55σ)
Epochs	Primary BTJD 3309.02608 ± 0.00010 = BJD_{TDB} 2460309.02608; secondary BTJD 3319.88908 (phase 0.75695 ± 0.00005)
Primary eclipse	60.9 ppt, 8.6 h in raw QLP SAP flux (trapezoid fits to 11 events; 59.7–61.6 ppt per event), nearly V-shaped; 55.5 ppt in TESS-SPOC SAP, 56.5 ppt in PDCSAP
Secondary eclipse	14.6 ppt, 7.3 h (12 events), at phase 0.757: the orbit is eccentric, and no dip is present at phase 0.5. The secondary is shallower mainly because it is more grazing (impact parameter 0.89 against 0.60 at the primary)
Orbit	Primary at Gaia's conjunction A, secondary at conjunction B (the swapped assignment is off by 0.23 in phase). With Gaia's conjunctions carried to the TESS epochs with the TESS period: secondary +0.0027 ± 0.0034 (0.8σ), primary +0.0180 ± 0.0022 (**8.2σ**). The eclipses are 0.75695 apart in phase against Gaia's predicted 0.7722 ± 0.0054 (2.8σ); at Gaia's e sin ω = 0.194 the spacing requires e cos ω = 0.408 (Gaia 0.434 ± 0.009), that is e = 0.452 and ω = 25.4°. The longer eclipse falls at the closer conjunction (0.648a against 0.960a), where the chord is more central
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (all seven sectors): matched mid-times differ from QLP's by −0.02 ± 0.13 min (11 primaries) and −0.18 ± 1.04 min (secondaries); depths 8% (SAP) and 6% (PDCSAP) shallower for both eclipses alike, a common scale difference between the pipelines. TESS-SPOC TCEs in every sector: primary P 14.350–14.352 d, MES 172–304; the secondary as a second TCE (MES 17–75); centroid offsets 0.2–2.0 ± 2.5″ from the TIC position
Pixel attribution	TESScut difference images of two primaries each in s56 and s75: deficit centroid 0.2″ and 1.5″ from the light centroid, a scaled copy of the star's image (residual 3.6–4.7%), core depth 48–49 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 49 Gaia sources within 60″, the brightest (G 15.95, 58″) would need an 1850% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025, ExoFOP (no TOI or CTOI). SIMBAD type SB*; its references are Vilnius photometry, extinction studies and diameter catalogues. Known SB
Open issues	

	The primary lies 0.018 in phase (6 h) from Gaia's conjunction A, 8σ on Gaia's formal errors. This is the same discrepancy as the eclipse spacing (2.8σ) and the period (2.55σ), and Gaia's goodness of fit of 17 suggests that its formal errors are underestimated. A ~2-ppt brightening near Gaia's periastron phase (−0.07) is suggested but not established
Next eclipses	Primary: BTJD 4327.943 (2026-10-14 10:34 UTC), 4342.294; secondary: BTJD 4338.806 (2026-10-25 07:17 UTC), 4353.157 (allow ±2 min, plus event-to-event scatter of 1.9 min rms for the secondary)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+42 3880 is a pair of late-B or early-A stars of about 2.1 and 2.4 $M_\odot$ in a 14.35-d orbit with e ≈ 0.45, seen at about 83°. Both eclipses are partial; the deeper and longer one falls near periastron, where the stars are closest and the chord is most central, and the secondary falls at phase 0.757. The attribution rests on the pixels and on TESS-SPOC, which reproduces both eclipses with the same timings and its own threshold-crossing events. The eclipse spacing, e cos ω = 0.408, differs from Gaia's 0.434 ± 0.009 by 2.8σ, and Gaia's period is 2.55σ from the TESS period. Because both tensions come from a solution with a goodness of fit of 17, we attribute them to underestimated Gaia errors; the TESS values should supersede them. Gaia DR4 epoch radial velocities fitted jointly with the eclipse timings would give both masses and radii.

A105 BD+67 1313 = TIC 321718006

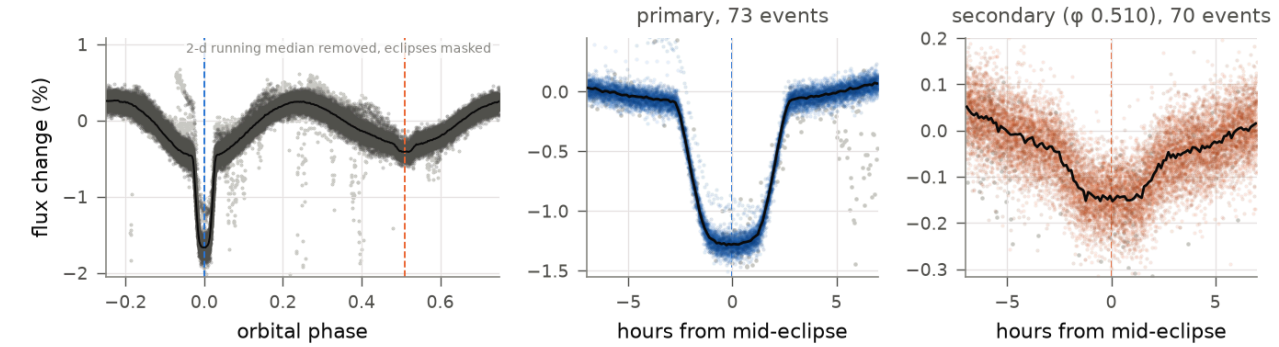


Figure A105. BD+67 1313, QLP SAP flux with sector medians removed and a 2-d running median removed with the eclipses masked, folded on the fitted P = 3.925329 d; flagged cadences in grey. Left: the full orbit, with the ellipsoidal modulation (about 0.6% peak to peak) and the eclipses at phases 0 and 0.510. Middle: the grazing primary (11.8 ppt, 5.6 h). Right: the secondary (0.9 ppt, 5.3 h); each event is normalised by its own out-of-eclipse median, so the zoom keeps part of the ellipsoidal curvature.

Identifiers	TIC 321718006; BD+67 1313; SAO 19407; TYC 4461-1835-1; Gaia DR3 2222179382751488384
Position (J2000)	321.50404, +67.71901 deg (TIC)
Brightness	Tmag 8.84; G 9.21
Stars	F8 (SIMBAD; TIC: T_{eff} 6455 K, 3.56 $R_\odot$, log *g* 3.45) at 336 pc; Gaia SB1 orbit (e = 0.015 ± 0.011, K_1 = 29.03 ± 0.47 km/s). The mass function, 0.00995 $M_\odot$, gives a companion of about 0.30 $M_\odot$ (0.24–0.35) for the TIC mass of 1.31 $M_\odot$: an M dwarf grazing an evolved F star. Ellipsoidal variation of about 0.6% peak to peak
Period	3.9253291 ± 0.0000007 d, timing fit of 68 primaries (O−C rms 0.98 min). Gaia: 3.925365 ± 0.000079 d (−0.45σ)
Epochs	Primary BTJD 3144.07847 ± 0.00008 = BJD_{TDB} 2460144.07847; secondary BTJD 3146.08177 (phase 0.51035 ± 0.00015)
Primary eclipse	11.8 ppt, 5.6 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 68 events; 11.7–11.9 ppt per event); 11.2–12.9 ppt depending on the baseline and pipeline. Odd and even depths agree to 1.0σ

Secondary eclipse	0.9 ppt, 5.3 h (63 events), at phase 0.51035. The depth is uncertain by a factor of about two (0.7–1.7 ppt) because it depends on the baseline model of the ellipsoidal variation. The depth ratio, 0.075, matches the TESS-band surface-brightness ratio of a 3300–3500 K dwarf and a 6300 K primary
Orbit	Primary at Gaia's conjunction A: +0.0003 ± 0.0034 in phase (0.1σ); secondary at conjunction B: +0.0037 ± 0.0033 (1.1σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipse spacing gives e cos ω = +0.0163 ± 0.0002 (Gaia +0.0104, +0.0105/−0.0084; 0.6σ). At conjunction A the F star is behind, so the deeper eclipse is of the F star
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP: 19 primaries, mid-times within +0.00 and +0.06 min of QLP's (median), 0.22 and 0.33 min rms; depth 11.2–11.3 ppt against 11.7 ppt. TESS-SPOC multi-sector TCE (s56–s69) at 3.925314 ± 0.000022 d, MES 186, 11.5 ppt, 5.57 h, impact parameter 0.84, centroid 0.45 ± 2.5″ from the TIC position; the secondary as a second TCE (MES 12.4, 0.90 ppt)
Pixel attribution	TESScut difference images of six primaries each in s56 and s83: deficit centroid 0.4″ and 0.6″ from the light centroid, PSF-shaped (residual 2.2–3.4%), depth 12.1–12.6 ppt in 1×1, 3×3 and 5×5 pixel apertures. Of 22 Gaia sources within 60″, the brightest (G 16.31, 44.5″) would need an 822% eclipse. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 3 and 4, ExoFOP (no TOI or CTOI). Listed in Kostov et al. (2025) table 2. SIMBAD type SB*, with four references (diameter catalogues, the TESS asteroseismic target list and Gaia DR3). Known SB
Next eclipses	Primary: BTJD 4329.528 (2026-10-16 00:36 UTC), 4333.453, 4337.379; secondary: BTJD 4327.606 (2026-10-14 02:28 UTC), 4331.531 (allow ±2 min; single secondaries scatter by 12 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+67 1313 is a slightly evolved F8 star (log g 3.45) with an M-dwarf companion of about 0.3 $M_\odot$ in a 3.93-d, nearly circular orbit. The 11.8-ppt primary is a grazing eclipse (impact parameter about 0.82), and the 0.9-ppt secondary at phase 0.510 is the occultation of the M dwarf. Both lie on Gaia's conjunctions to within 1.1σ, and the secondary's phase gives a small, significant eccentricity (e cos ω = 0.016) consistent with Gaia's. The eclipses sit in the minima of a 0.6% ellipsoidal modulation, which leaves the secondary depth uncertain by a factor of two. The system was listed only as an unvetted machine-learning candidate and as a TESS-SPOC threshold-crossing event that did not become a TOI. A spectroscopic mass and radius for the F star, combined with a fit to the grazing eclipse, would fix the radius of the M dwarf.

A106 HD 205273 = TIC 352935659

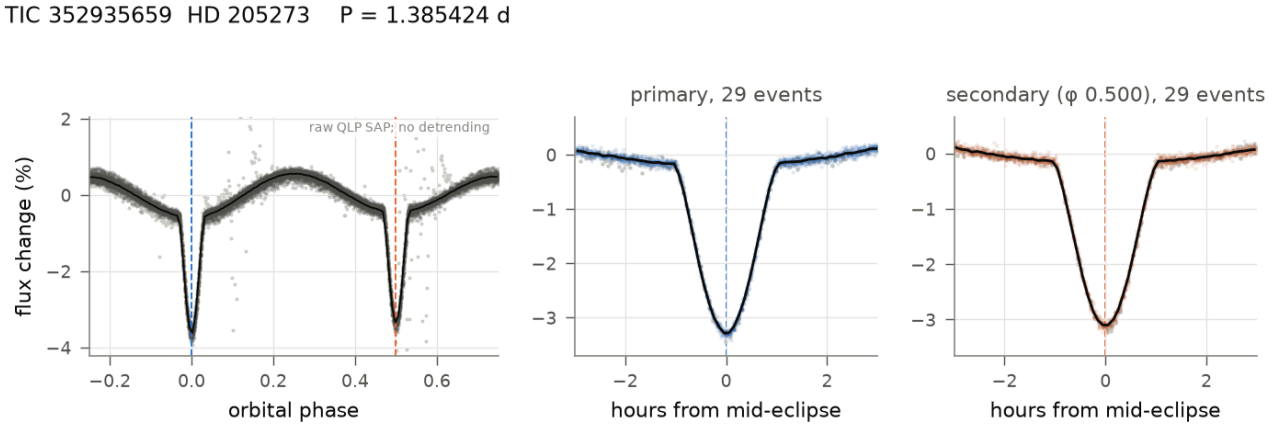


Figure A106. HD 205273, raw QLP SAP flux with sector medians removed and no detrending, folded on the fitted P = 1.385424 d; flagged cadences in grey. Left: the full orbit, with the ellipsoidal modulation of about 1.1% peak to peak and two grazing, V-shaped eclipses at phases 0 and 0.500. Middle and right: the primary (30.6 ppt) and the secondary (29.2 ppt), both 1.95 h long, each event normalised by its own out-of-eclipse median (29 events of each kind plotted; 27 and 26 fully covered events in the timing fit).

Identifiers	TIC 352935659; HD 205273; BD+04 4700; SAO 126895; TYC 550-1303-1; Gaia DR3 2699348496989329536
Position (J2000)	323.51547, +4.92603 deg (TIC)
Brightness	Tmag 8.71; G 9.05
Stars	F5 V (SIMBAD; TIC: T_{eff} 6456 K, 1.80 $R_\odot$ for the combined light) at 165 pc; Gaia SB2 orbit (e = 0.011 ± 0.004, K_1 = 122.77 ± 0.58 and K_2 = 127.64 km/s, q = 0.962). With the inclination from the eclipse durations (about 71°) the masses are about 1.37 and 1.32 $M_\odot$: two F5 dwarfs of about 1.3 $R_\odot$ each; a sin i = 6.9 $R_\odot$, so a ≈ 7.3 $R_\odot$. The ROSAT source 1RXS J213405.3+045548, 27.6″ away and within its positional error, is probably this binary
Period	1.3854241 ± 0.0000004 d, timing fit of 27 primaries (O–C rms 0.32 min) and 26 secondaries (0.24 min). Twin eclipses: the period is twice the fold period 0.692713 d. Gaia: 1.3854194 ± 0.0000044 d (+1.1σ)
Epochs	Primary BTJD 3402.27646 ± 0.00009 = BJD_{TDB} 2460402.27646; secondary BTJD 3402.96917 (phase 0.50000 ± 0.00009)
Primary eclipse	30.6 ppt, 1.95 h in raw QLP SAP flux (trapezoid fits with a local quadratic baseline to 27 events); 28.9 ppt in TESS-SPOC SAP
Secondary eclipse	29.2 ppt (sd 0.46), 1.95 h (26 events), at phase 0.50000; 27.6 ppt in TESS-SPOC SAP. The 1.3-ppt (4%) difference between the two eclipses is the same in QLP and TESS-SPOC and is significant at 11.6σ in QLP
Orbit	Primary at Gaia's conjunction A: +0.0021 ± 0.0014 in phase (1.5σ); secondary at conjunction B: −0.0048 ± 0.0018 (2.6σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipses are exactly half a period apart, e cos ω = 0.0000 ± 0.0002, against Gaia's +0.0107 ± 0.0040 (2.7σ); the secondary's offset is this difference. The deeper eclipse, at A, is of Gaia's star 1, the slightly more massive star
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP and TARS (s55, s82): matched mid-times within −0.01 and −0.08 min of QLP's for primaries and secondaries (median; rms 0.05 and 0.15 min), with the same unequal depths; TARS within +0.36 and −0.15 min. TESS-SPOC TCEs at half the period, 0.6927155 d (MES 308 in s55, 626 in s82, 29.0 ppt, impact parameter 0.97), centroids 2.1 and 1.1 ± 2.5″ from the TIC position. ASAS-3 photometry shows the eclipses about 8,500 cycles before TESS: the 20 points in the predicted eclipse windows are 34 mmag fainter than the rest (10σ)
Pixel attribution	TESScut difference images stacking both eclipses, 27 events in s55 and 31 in s82: deficit centroid 0.6″ and 0.1″ from the light centroid, PSF-shaped (3×3 residual 1.1–1.5%), depth 26.3–26.9 ppt in 1×1, 3×3 and 5×5 pixel apertures; the per-event pixel depths alternate between the two eclipses. The brightest of the 7 Gaia sources within 60″ (G 18.62, 22″) would need a 20,000% eclipse. A companion within about 1″ is not excluded

Catalogue status	Not catalogued as eclipsing in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI). SIMBAD type SB*, with six references (spectral classification, photometric and diameter catalogues, a study of young stars); none reports variability. Known SB
Next eclipses	Primary: BTJD 4327.740 (2026-10-14 05:39 UTC), 4329.125, 4330.511; secondary: BTJD 4328.432 (2026-10-14 22:16 UTC), 4329.818 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

HD 205273 is a detached pair of nearly equal F5 dwarfs (q = 0.96) in a 1.385-d orbit, seen at about 71°, so that both eclipses are grazing and V-shaped. The two eclipses differ in depth by 4%, the same in two pipelines and in the alternating depths of the difference images, which identifies the true period as Gaia's SB2 period. The eclipses are exactly half a period apart, as expected for a tidally circularised orbit of this period; Gaia's small eccentricity (0.011 ± 0.004) is most likely the bias that affects fitted eccentricities of orbits with tidally distorted components, and it accounts for the 2.6σ offset of the secondary. The eclipses are also present in ASAS-3 photometry, so the ephemeris holds across about 8,500 cycles. A light-curve and radial-velocity solution would give masses and radii for two main-sequence F stars.

A107 BD+32 4452 = TIC 397680054

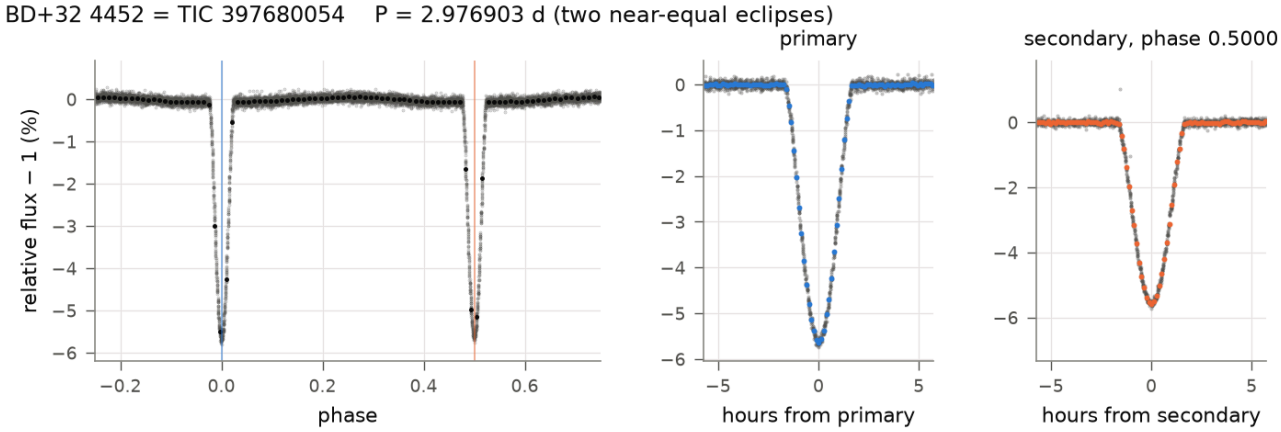


Figure A107. BD+32 4452, QLP SAP flux with sector medians removed and a running median removed with the eclipses masked, folded on the fitted P = 2.976903 d, twice the fold period. Left: the full orbit, with two near-equal eclipses at phases 0 and 0.5000. Middle and right: the primary (55.7 ppt) and the secondary (55.2 ppt), both V-shaped and 3.15 h long.

Identifiers	TIC 397680054; BD+32 4452; SAO 72451; TYC 2739-1543-1; Gaia DR3 1901472839026009344
Position (J2000)	338.24512, +33.00083 deg (TIC)
Brightness	Tmag 9.49; G 9.82
Stars	F2 (SIMBAD; TIC: T_{eff} 6629 K, 2.04 $R_\odot$ for the combined light) at 276 pc; Gaia SB2C orbit (circular; K_1 = 103.1 ± 0.7 and K_2 = 100.7 ± 29.4 km/s). M $\sin^3 i$ = 1.29 and 1.32 $M_\odot$, q ≈ 1; with the inclination from the eclipse duration (about 79°), about 1.37 $M_\odot$ each, uncertain by more than ±50% because K_2 is uncertain by ±29%: twin F stars, a sin i = 12.0 $R_\odot$
Period	2.976903 ± 0.000001 d, timing fit of 14 primaries (O–C rms 0.49 min); the 12 secondaries give 2.976902 ± 0.000001 d. Twin eclipses: the period is twice the fold period 1.48845 d. Gaia: 2.976915 ± 0.000039 d (−0.3σ)
Epochs	Primary BTJD 2681.70466 ± 0.00012 = BJD_{TDB} 2459681.70466; secondary BTJD 2739.75430 (phase 0.50001 ± 0.00006)
Primary eclipse	55.7 ppt, 3.15 h in raw QLP SAP flux (trapezoid fits to 14 events), V-shaped (flat fraction 0.2); 53.9 ppt in TESS-SPOC SAP, 55.2 ppt in PDCSAP

Secondary eclipse	55.2 ppt, 3.15 h (12 events), at phase 0.50001. The primary is deeper by 0.46 ± 0.13 ppt (3.6σ; 5.4σ in s56 alone), and TESS-SPOC SAP and PDCSAP show the same 0.45–0.55-ppt difference. Odd and even depths within each eclipse agree
Orbit	Circular orbit (SB2C): no conjunction-phase test. The secondary at phase 0.50001 ± 0.00006 gives e cos ω = 0.0000 ± 0.0001
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (s16, s56; 15 primaries and 15 secondaries) and TARS: the same T14 (3.16–3.17 h), median O−C within 0.2 min of the ephemeris in every product, depths 53.5–55.2 ppt against QLP's 54.3–55.0 in the same per-event fits. TESS-SPOC TCE in s56 at half the period, 1.4884527 d, MES 1077, 56.0 ppt, impact parameter 0.95; centroid 0.59 ± 2.50″ from the TIC position and 0.09 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images of nine primaries and nine secondaries in s56 and six primaries in s16: deficit centroid 0.1–0.2″ from the light centroid, on the target pixel, PSF-shaped (residual 1.6–2.2%), depth 46.7–48.0 ppt in 1×1, 3×3 and 5×5 pixel apertures. The brightest Gaia source within 60″ (G 15.37, 28.9″) would need a 929% eclipse; all 43 sources within 90″ are excluded. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing in: VSX, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables (the target and all 24 sources within 60″), Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2–4, ExoFOP (no TOI or CTOI); no ASAS-3 data. SIMBAD type SB*, with three references (diameter catalogues and Gaia DR3). Known SB
Next eclipses	Primary: BTJD 4327.932 (2026-10-14 10:15 UTC), 4330.909, 4333.886; secondary: BTJD 4329.421 (2026-10-15 21:58 UTC), 4332.397 (allow ±2 min)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+32 4452 is a pair of nearly identical F2 stars, of about 1.4 $M_\odot$ each (uncertain by more than ±50%, since Gaia's K_2 is uncertain by ±29%), in a circular 2.98-d orbit, seen at about 79°, so that each star grazes the other and the two V-shaped eclipses are 56 ppt deep. The period is Gaia's double-lined circular period; the 0.5-ppt depth difference between the eclipses, seen in three products, shows that the stars differ slightly in surface brightness and fixes which eclipse is which. A circular orbit has no predicted conjunction phase to test, so the attribution rests on the pixels, where both eclipses are centred on the target in two sectors, and on TESS-SPOC and its own threshold-crossing event. A double-lined radial-velocity curve with a better K_2 than Gaia's (±29%) would give precise masses for two F stars.

A108 TYC 2747-436-1 = TIC 239649487

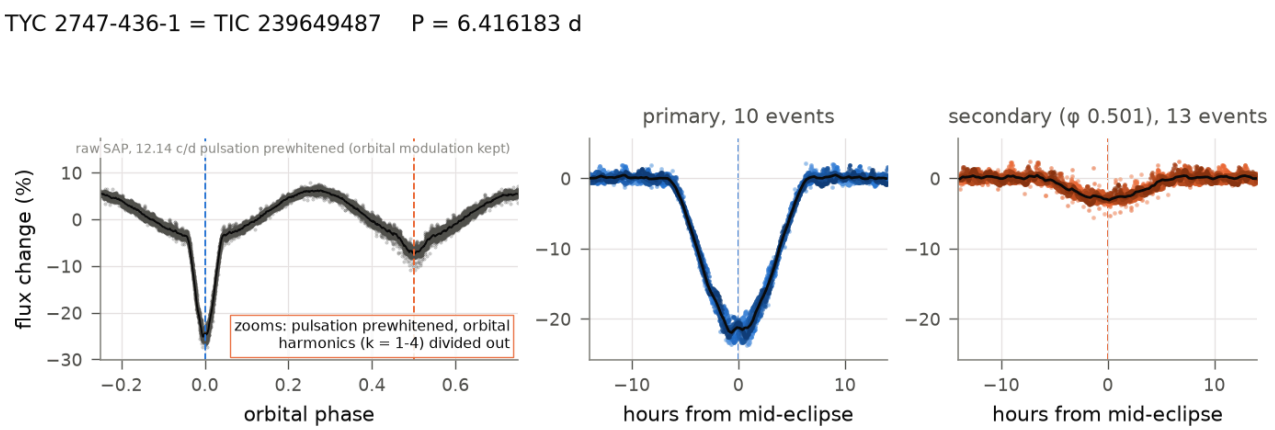


Figure A108. TYC 2747-436-1, raw QLP SAP flux folded on the fitted P = 6.416183 d. Left: with the 12.14 c/d δ Scuti pulsation prewhitened and the orbital modulation kept, showing the ellipsoidal variation with the primary at phase 0 and the secondary at phase 0.501 in the deeper ellipsoidal minimum. Middle and right: with the pulsation prewhitened and orbital harmonics (k = 1–4) divided out, the primary (214 ppt, 12.6 h) and the secondary (about 28 ppt, about 11.7 h; 13 secondaries plotted, 11 fully covered and fitted).

Identifiers	TIC 239649487; TYC 2747-436-1; Gaia DR3 1904322292128388096

Position (J2000)	338.96658, +35.84712 deg (TIC)
Brightness	Tmag 9.13; G 9.53
Stars	T_{eff} 7236 K (Gaia DR3; TIC: T_{eff} 6182 K, 3.82 $R_\odot$, a single-star estimate for the blended light) at 389 pc; Gaia SB2 orbit (e = 0.0096 ± 0.0095, K_1 = 30.91 and K_2 = 111.34 km/s, q = 0.278). Masses about 1.55–1.65 and 0.43–0.46 $M_\odot$ for i ≥ 75°; a ≈ 18 $R_\odot$. The more massive star is a high-amplitude δ Scuti pulsator: 12.1381 c/d (1.977 h, semi-amplitude 50 ppt) with harmonics at 24.276 and 36.414 c/d, an independent mode at 15.6913 c/d (17 ppt) and combination frequencies. An ellipsoidal modulation of semi-amplitude 48–53 ppt, phase-coherent over five years, suggests that the low-mass secondary is close to filling its Roche lobe (about 5.0 $R_\odot$); a semi-detached Algol with a pulsating mass gainer is the working hypothesis, pending a light-curve model
Period	6.4161826 ± 0.0000024 d, timing fit of 10 primaries (O–C rms 1.44 min); the 11 secondaries give 6.416179 ± 0.000040 d. Gaia: 6.416343 ± 0.000171 d (−0.94σ)
Epochs	Primary BTJD 3233.51625 ± 0.00014 = BJD_{TDB} 2460233.51625; secondary BTJD 3236.72891 (phase 0.50071 ± 0.00037)
Primary eclipse	214 ± 2 ppt, 12.6 h in QLP SAP flux with the pulsation prewhitened and the orbital harmonics divided out (trapezoid fits to 10 events), flat fraction 0.2; in the raw pulsating flux, 214 ppt in QLP against 203 ppt in TESS-SPOC SAP, 206 ppt in PDCSAP and 202 ppt in TARS. The pulsation amplitude, in units of the out-of-eclipse flux, falls in the primary-eclipse cores (39.4 against 50.5 ppt) and not at the secondary, so the pulsator is the star eclipsed at primary
Secondary eclipse	About 28 ppt (26–37 ppt per event), about 11.7 h (11 events), at phase 0.50071 ± 0.00037. Its core is W-shaped in s56 and s83, probably from residual pulsation or an imperfect ellipsoidal model; the pulsation amplitude does not drop in this eclipse
Orbit	Primary at Gaia's conjunction A: −0.0018 ± 0.0028 in phase (0.6σ); secondary at conjunction B: −0.0062 ± 0.0023 (2.7σ), with Gaia's conjunctions carried to the TESS epochs with the TESS period. The eclipse spacing gives e cos ω = +0.0011 ± 0.0006 (Gaia +0.0078 ± 0.0074, 0.9σ): the orbit is circular to 0.001, and Gaia's small, upward-biased eccentricity moves its conjunction B to phase 0.505, which accounts for the secondary's offset. The primary is the eclipse of the more massive star (K_1 = 30.9 km/s)
Pipelines agreeing	QLP SAP; TESS-SPOC SAP (s16, s56, s83): the ten primary mid-times agree with QLP's to −0.10 min median, 0.61 min rms, depth ratio 0.94; TARS +1.7 min median, 5.0 min rms, depth ratio 0.92. The secondary is too shallow against the ±5% pulsation for an event-by-event comparison in raw flux. No SPOC 2-min data and no TCE
Pixel attribution	TESScut difference images of four primaries each in s56 and s83: deficit centroid 1.0″ and 0.4″ from the light centroid, on the target pixel, a scaled copy of the target's image (residual 3.7–4.0%), core depth 187–193 ppt in 1×1, 3×3 and 5×5 pixel apertures. All 16 Gaia sources within 60″ are G ≥ 15.15; the brightest (56.6″) would need a 3800% eclipse, and none could produce the 10% peak-to-peak pulsation. A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing or pulsating in: VSX, GCVS, ASAS-SN variables, SB9, WDS, Gaia DR3 variability tables, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. 2025 tables 2 and 3, ExoFOP (no TOI or CTOI); no ASAS-3 data. The only VSX object within 60″ is the neighbour Gaia DR3 1904322257768650624 (rotational variable, G 15.15, 56.6″). SIMBAD type SB*; its only reference is Gaia DR3. Known SB
Open issues	The light shares are not consistent: the 22% drop in pulsation amplitude would imply that the pulsator supplies about 95% of the light, while a 48–53 ppt ellipsoidal semi-amplitude from a lobe-filling 0.45 $M_\odot$ star needs that star to supply about half; a light-curve model is needed to resolve this and to test the semi-detached configuration. The W-shaped core of the secondary is unexplained, and its event times scatter by 17 min rms. QLP's primary is about 6% deeper than TESS-SPOC's
Next eclipses	Primary: BTJD 4330.683 (2026-10-17 04:17 UTC), 4337.100, 4343.516; secondary: BTJD 4333.896 (2026-10-20 09:23 UTC), 4340.312 (allow ±2 min; single secondaries scatter by 17 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

TYC 2747-436-1 is a double-lined eclipsing binary of about 1.6 and 0.45 $M_\odot$ in a circular 6.42-d orbit, whose more massive star is a high-amplitude δ Scuti pulsator with a dominant period of 1.98 h. The 21% primary eclipse is of the pulsator: the pulsation amplitude falls during the primary eclipse and not

during the secondary. The large ellipsoidal modulation suggests that the 0.45 $M_\odot$ companion is close to filling its Roche lobe, which would make this an oEA system, an Algol with a pulsating mass gainer; this remains a hypothesis, because the size of the amplitude drop and the ellipsoidal amplitude imply different light shares for the two stars, and only a light-curve model can reconcile them. Both eclipses lie on Gaia's conjunctions; the secondary's 2.7σ offset follows from Gaia's small and probably spurious eccentricity. Neither the pulsation nor the eclipses was previously catalogued. Pulsation timing around the orbit would measure the pulsator's light-travel-time orbit independently of Gaia, and eclipse mapping of the pulsation would constrain its mode geometry.

A109 BD+43 1585 = TIC 274331697

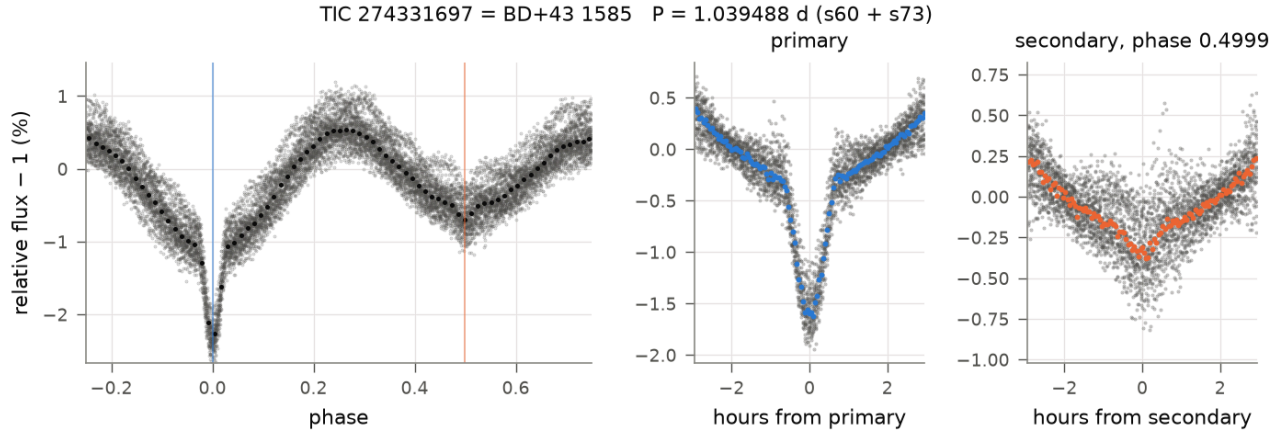


Figure A109. BD+43 1585, raw QLP SAP flux (sector medians removed; a running median with the eclipses masked, long enough to keep the orbital wave) folded on the sector 60/73 ephemeris, P = 1.0394878 d; the 2.5-d^{-1} pulsation is not removed and accounts for most of the scatter. Left: the full orbit, with the 6-ppt ellipsoidal double wave and the measured eclipse phases 0 and 0.4999 marked. Middle: the 1.3-h V-shaped primary. Right: the 1.4–1.6-ppt secondary. In both zooms the eclipses sit on the curvature of the wave, so the apparent depths there exceed the fitted ones.

Identifiers	TIC 274331697; BD+43 1585; TYC 2949-1328-1; Gaia DR3 963883711690164992
Position (J2000)	100.50093, +42.89616 deg (TIC)
Brightness	Tmag 9.62; G 9.95
Stars	F2 (SIMBAD; TIC: T_{eff} 6812 K, 1.59 $R_\odot$) at 246 pc; Gaia SB1 orbit (e = 0.019 ± 0.018, ω = 148 ± 36°, K_1 = 72.8 ± 1.4 km/s); RUWE 3.11 with no sign of a resolved double (ipd_frac_multi_peak 0). The mass function (0.0416 $M_\odot$) gives $m_2 \approx 0.6\ M_\odot$ at the inclination implied by the eclipse duration ($i \approx 69$–$70°$, impact parameter about 0.9), and the depth ratio (0.12) a companion near 3800–3900 K: a late-K or early-M dwarf. A pulsation at 2.503 d^{-1} (with its harmonic and close peaks at 2.48–2.58 d^{-1}), 1.2–3.2 ppt and changing from sector to sector, is in this star's light: a γ Doradus-type pulsation of the F2 primary or variability of an unresolved third star
Period	1.0394878 ± 0.0000005 d, timing fit of 34 primaries in sectors 60 and 73 (Gaia: 1.0394549 ± 0.0000076 d; +4.3σ). The period is not constant: the sector 20 and 60 epochs give 1.0395047 d (+6.5σ from Gaia's), 16 ppm longer, and no linear ephemeris fits the three sector epochs (χ^2 3678 for 1 degree of freedom)
Epochs	Primary BTJD 3134.11958 ± 0.00009 = BJD_{TDB} 2460134.11958; secondary BTJD 3073.30944 ± 0.00052 (phase 0.49991 ± 0.00050). Sector 20 epoch BTJD 1854.49221 ± 0.00013, 25.8 min early on this ephemeris
Primary eclipse	12.6 ppt (per-event scatter 0.4 ppt), 1.28 h, V-shaped (flat fraction 0.2), in QLP SAP flux after a model of the orbital harmonics and the pulsation (fitted with the eclipses masked) is divided out (trapezoid fits to 34 events); 12.3 and 12.7 ppt in sectors 60 and 73 in raw-flux fits. Odd and even primaries agree (12.55 ± 0.08 and 12.46 ± 0.11 ppt). Grazing
Secondary eclipse	1.44 ± 0.03 ppt in the fold (1.6 ppt median per event), 1.3 h, at phase 0.49991 ± 0.00050 (34 events)
Orbit	

	The result depends on how the epoch is carried to Gaia's, because the period changes. From the sector 20 epoch, 1409 cycles after Gaia's reference epoch, with the sector 20–60 period: primary −0.0105 ± 0.0067 in phase from conjunction A (1.6σ), or −0.011 ± 0.024 (0.5σ) when the observed 16-ppm period change is carried as a systematic error; secondary +0.041 ± 0.042 from conjunction B (1.0σ). From the sector 60/73 epoch with the sector 60–73 period: +0.027 ± 0.006 (4.2σ) on the formal error, +0.027 ± 0.043 (0.6σ) with the systematic. With Gaia's own period: +0.048 ± 0.010 (4.7σ) from sector 20 and +0.104 ± 0.019 (5.6σ) from sectors 60/73. The period that would put the sector 20 primary exactly on conjunction A, 1.039490 ± 0.000008 d, lies within the range of the TESS periods. $e \cos \omega = -0.0001 \pm 0.0008$ from the secondary phase against Gaia's −0.015 ± 0.014 (1.1σ): circular
Pipelines agreeing	QLP SAP; TESS-SPOC SAP and PDCSAP (11.7–12.7 ppt, median O−C −0.26 to +0.35 min against the QLP ephemeris in sectors 60 and 73); TARS (11.9 and 12.3 ppt, +0.21 and −0.39 min). All four products put the sector 20 eclipses 25–30 min early, and the sector 20 secondaries shift with the primaries (−22 to −28 min). TESS-SPOC TCEs in sector 60 (P 1.0394974 ± 0.0000150 d, 12.09 ppt, 1.33 h, b 0.99, MES 76) and sector 73 (P 1.0395090 ± 0.0000134 d, 12.35 ppt, MES 65), with the secondary as a second TCE in each (1.71 and 1.15 ppt, phase 0.5007); none flagged as an eclipsing binary; centroids 0.37 and 0.28 ± 2.50″ from the out-of-transit centroid
Pixel attribution	TESScut difference images in sectors 20, 60 and 73 (11, 16 and 19 primaries): deficit centroid 0.8″, 0.9″ and 0.6″ from the light centroid, PSF-shaped (residual 5.5%, 3.6%, 3.7%), on the target pixel; depth stable across 1×1, 3×3 and 5×5 pixel apertures (12.24, 12.29, 12.17 ppt in sector 60). The secondary stacks are on the target as well. The only neighbour bright enough, Gaia DR3 963883608610951552 (G 13.67 at 40.1″, classified δ Scuti/γ Doradus/SX Phoenicis by Gaia and in VSX), would need a 40% eclipse, and its own pixel loses only the target's wings. Per-pixel amplitude maps place both the ellipsoidal wave and the 2.503-d^{-1} signal on the target (0.06–0.20 px), not on the neighbour (1.7–1.9 px). A companion within about 1″ is not excluded
Catalogue status	Not catalogued as eclipsing or variable in: VSX (only the neighbour above), ASAS-SN, Prša et al. 2022, IJspeert et al. 2024, Kostov et al. (2025) tables 2–4, Gaia DR3 variability tables (target and all sources within 60″ except the neighbour), SB9, WDS, ExoFOP (no TOI or CTOI), web search; SIMBAD references are catalogues only. Known SB (SIMBAD SB*, from Gaia DR3). TESS-SPOC TCEs as above, not promoted to a TOI
Open issues	The eclipse period differs by 16 ppm between the sector 20–60 and 60–73 baselines, and both eclipses shift together, as light-travel time in a wider orbit would produce; three epochs cannot fix the outer orbit. Gaia's period is 43–48 ppm shorter than every TESS value (4.3–6.5σ on the formal errors); with the observed 16-ppm change added as a systematic error the difference is 1.8–2.7σ, so it may reflect the same variation over Gaia's 2014–2017 baseline. The ellipsoidal semi-amplitude (6.3 ppt) is below the roughly 11 ppt expected, within the factor-of-two uncertainty of the stellar radius and the gravity-darkening coefficient. Whether the pulsation belongs to the primary or to a third star is not determined
Next eclipses	Primary: BTJD 4328.491 (2026-10-14 23:44 UTC), 4329.531, 4330.570; secondary: BTJD 4327.971 (2026-10-14 11:15 UTC), 4329.011 (allow ±30 min: the formal ephemeris error is 0.8 min, but the sector 20 epoch lies 25.8 min from this ephemeris, and a timing variation of that size may be present; single secondaries scatter by 4.2 min rms)
Method	Gaia-orbit period fold, G 9–10 subsample (Section 2.4); confirmed by independent re-analysis (Section 2.5)

BD+43 1585 is an F2 star with a late-K or early-M companion of about 0.6 $M_\odot$ in a circular 1.04-d orbit, seen at about 70°, so that both eclipses are grazing: a 12.6-ppt primary and a 1.4–1.6-ppt secondary on a 6-ppt ellipsoidal wave. The deficit is on the target in difference images in three sectors, TESS-SPOC and TARS reproduce both eclipses, and the δ Scuti-type neighbour 40″ away is excluded both as the eclipse source and as the source of the 2.5-d^{-1} pulsation, which lies in the target's own light. The eclipse times are not on one linear ephemeris: sector 20 is 25.8 min early against sectors 60 and 73 in every pipeline, and the secondary shifts with the primary, as light-travel time would produce and apsidal motion in a circular orbit would not. With RUWE 3.1, this points to a third body. The changing period also governs the comparison with Gaia. Carried with the TESS period from the epoch nearest Gaia's, the primary is consistent with conjunction A (1.6σ on the formal error), whereas Gaia's own period, 43–48 ppm

shorter, places it 4.7σ away, so the comparison has no single significance. Eclipse timing over several seasons would give the outer orbit, and Gaia DR4 epoch astrometry and velocities would test whether Gaia's short period reflects the same variation.

S3.4 Eclipsing systems from the M-dwarf transit searches (A1–A4)

A1 TIC 311958594 = UCAC4 083-054218

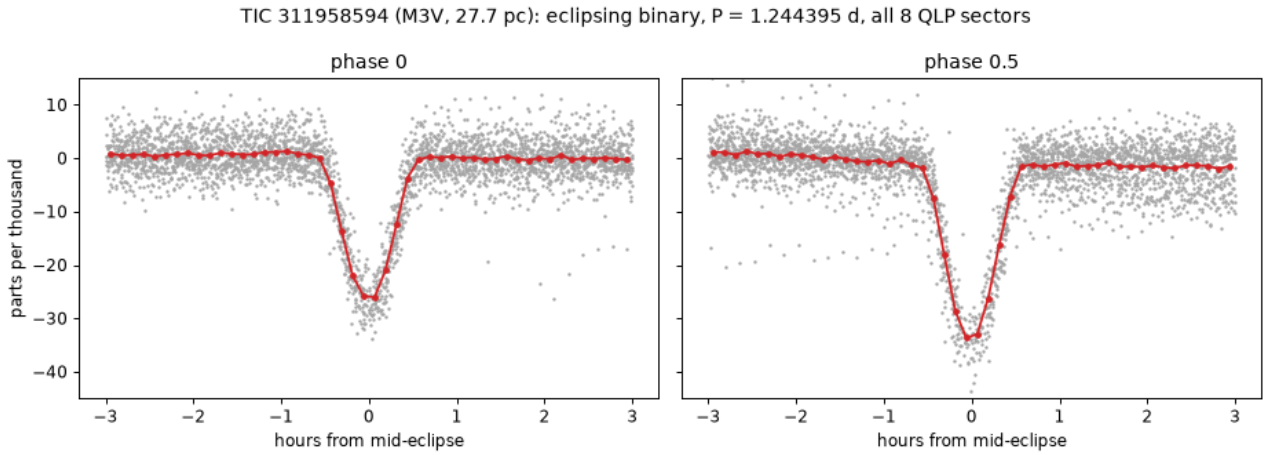


Figure A1. TIC 311958594, all 8 QLP sectors folded on P = 1.2443947 d: the two unequal V-shaped eclipses, 18.5 and 22.8 ppt (1.85% and 2.28%) deep in the QLP aperture, half an orbit apart, at phase 0 (left) and phase 0.5 (right).

Identifiers	TIC 311958594; Gaia DR3 5803288101584023040; 2MASS J17342735-7326519; UCAC4 083-054218
Position (J2000)	17:34:27.3370 −73:26:51.896 (263.613904, −73.447749 deg; TIC/Gaia-propagated)
Brightness	Tmag 9.82; G 11.0004; V 12.235
Stars	M3V (0.56 $R_\odot$ in TIC) + M dwarf, near-equal, unresolved; at 27.7 pc. Co-moving with the K7 dwarf TIC 373015604, 98″ away (~2,700 AU): a hierarchical triple
Period	1.2443947 ± 0.0000016 d (no Gaia orbit)
Epoch	BTJD 1629.45849 = BJD_{TDB} 2458629.45849 (the shallower eclipse)
Eclipses	2.28% and 1.85% (QLP box depths), V-shaped, ~0.7 h, half an orbit apart; grazing (a central eclipse of two 0.5 $R_\odot$ stars at this period would last ~2 h). Depth ratio B/A = 1.20–1.33 in the six sectors at 10-min cadence or finer, each at +4.7 to +8.9σ
Out of eclipse	Brightening at quadrature (ellipsoidal) and a ±5–10 ppt spot wave locked to the orbit, whose phase drifts from sector to sector
Pixel attribution	Deficit centroid 3–5″ from this star and 13–15″ from the giant HD 158062 (TIC 311958593) in S093 and S100, two field orientations
Catalogue status	Not in any catalogue searched (Section 2.7) except Kostov et al. (2025), table 2. SIMBAD lists it as a proper-motion star. Not in Gaia DR3 `vari_eclipsing_binary` or NSS; RUWE 1.09. Gaia photometry puts it ~1 mag above a single M3V (M_G 8.79 at BP−RP 2.48), as a near-equal unresolved pair should be
Open issues	Type EW or EB uncertain; the detached geometry has not been fitted
Method	M-dwarf transit search (Section 2.3); confirmed by independent re-analysis (Section 2.5)

The signal was first detected as a 0.622-d, 240-ppm transit-like signal on the K7 dwarf TIC 373015604. Its depth depended on the aperture (~450 ppm in sectors 39, 66 and 93, ~0 in sectors 100, 102 and 103), and the per-pixel deficit lies ~95″ away. In the M dwarf's own QLP light curve the signal is two unequal eclipses at 1.2443947 d. After a 1-d running-median detrending, the inequality appears insignificant in two sectors (S066 at +1.1σ, S039 at −2.0σ). The detrending is the cause: beneath the drifting spot wave, it tilts the baseline differently under each eclipse. Measured in a 43-min core against a local linear baseline, the eclipses are unequal in every sector at 10-min cadence or finer. In sectors 12

and 13, at 30-min cadence, the ratio is 1.11, because the cadence smears a 0.7-h eclipse. Parallaxes of 36.12 and 36.01 mas and proper motions of (24.1, −58.6) and (23.1, −54.3) mas yr^{-1} tie the M dwarf to TIC 373015604. A spotted, tidally locked M+M pair is common at this period.

A2 TIC 183494856 = UCAC4 070-000968

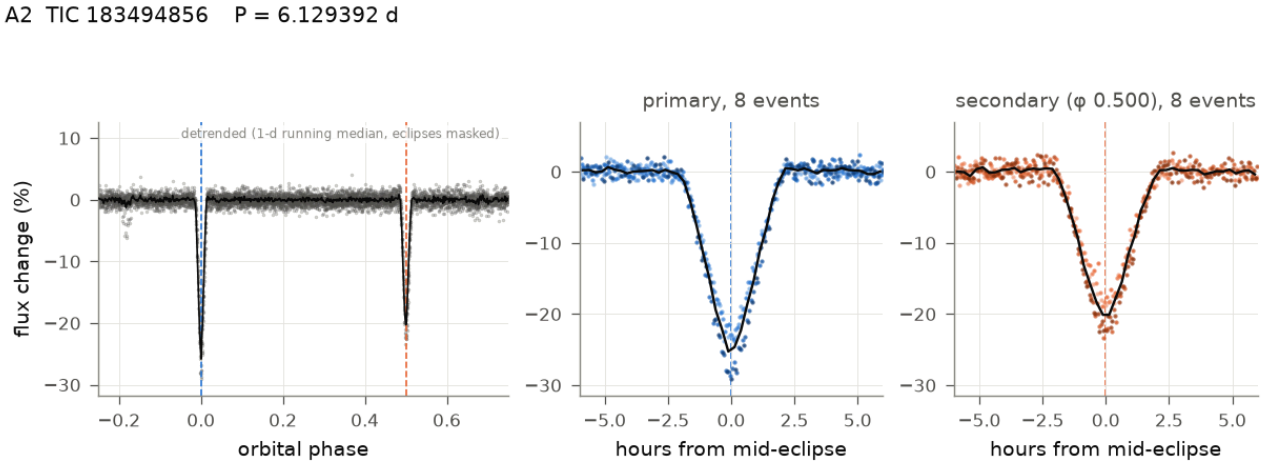


Figure A2. TIC 183494856 folded on P = 6.129392 d with the deeper eclipse at phase 0. The shallower eclipse falls at phase 0.50010. Absolute depths depend on the aperture; their ratio, 0.79, does not.

Identifiers	TIC 183494856; Gaia DR3 4636365763800590208; 2MASS J01134830-7602338; UCAC4 070-000968
Position (J2000)	01:13:48.3096 −76:02:33.825 (18.451290, −76.042729 deg)
Brightness	Tmag 14.81; G 15.2985; V 15.48
Stars	Solar-type primary (TIC: T_{eff} 5601 K, 0.979 $R_\odot$, 0.99 $M_\odot$, ρ 1.055 $\rho_\odot$) at 1,150 pc (parallax 0.840 ± 0.034 mas); RUWE 0.955
Period	6.129392 ± 0.000006 d (no Gaia orbit)
Epoch	BTJD 3.74608 = BJD_{TDB} 2457003.74608 (deeper eclipse)
Primary eclipse	21–29% (own 3×3-pixel aperture, by sector), 29.1% (GSFC-ELEANOR-LITE S001), 44.9% (TGLC S001); V-shaped, partial
Secondary eclipse	17–23% (own aperture), at phase 0.50010 (+0.9 min): consistent with a circular orbit. Shallow/deep ratio 0.79 in all four reductions
Timing	rms 2.1 min over 27 of 31 eclipse times, 757-d baseline, 124 cycles (sectors 1, 13, 27, 28)
Pipelines agreeing	Own aperture photometry, GSFC-ELEANOR-LITE (S001, S013), TGLC (S001). TASOC's corrected flux has 88% scatter and is unusable; too faint for QLP
Pixel attribution	Deficit centroid 1.1″ (S27) and 1.3″ (S28) from this star, 70.5″ and 71.2″ from the M dwarf on which it was found
Catalogue status	Not listed in any of: VSX, the ASAS-SN variable-star catalogue (II/366; Jayasinghe et al. 2018), Gaia DR3 `vari_eclipsing_binary`, `vclassre` and NSS, Prša et al. (2022), ATLAS variables, Kepler/K2 EBs, OGLE-IV, WISE variables (Chen et al. 2018), Gaia DR2 variables, SIMBAD (nearest entry a galaxy 10.7″ away). Listed only in Kostov et al. (2025), table 2. No TOI or CTOI
Open issues	Type EA or EB uncertain; no morphology fit. Sectors 94 and 95 put the star in the inter-CCD gap, so no longer baseline exists
Method	M-dwarf transit search (Section 2.3)

The signal was first detected as a 1.9 $R_\oplus$ planet candidate on the 0.217 $R_\odot$ M dwarf TIC 183494855 (P 3.06465 d, 6,400 ppm, 24.5σ). Its half-depth width of 1.77 h exceeded the 1.00-h maximum possible transit on that star, and the alternating events differed at 3.4σ when folded at twice the period. The centroid moved the signal 70″ to this star. Here the eclipses are 20–29% deep, and the 1.77-h duration fits a star of 1.06 $\rho_\odot$ at impact parameter ~0.7. At Tmag 14.81 the star is below QLP's ~13.5 limit, which explains its absence from full-frame-image pipeline products.

A3 HIP 83929 = TIC 85379511 = G 181-29

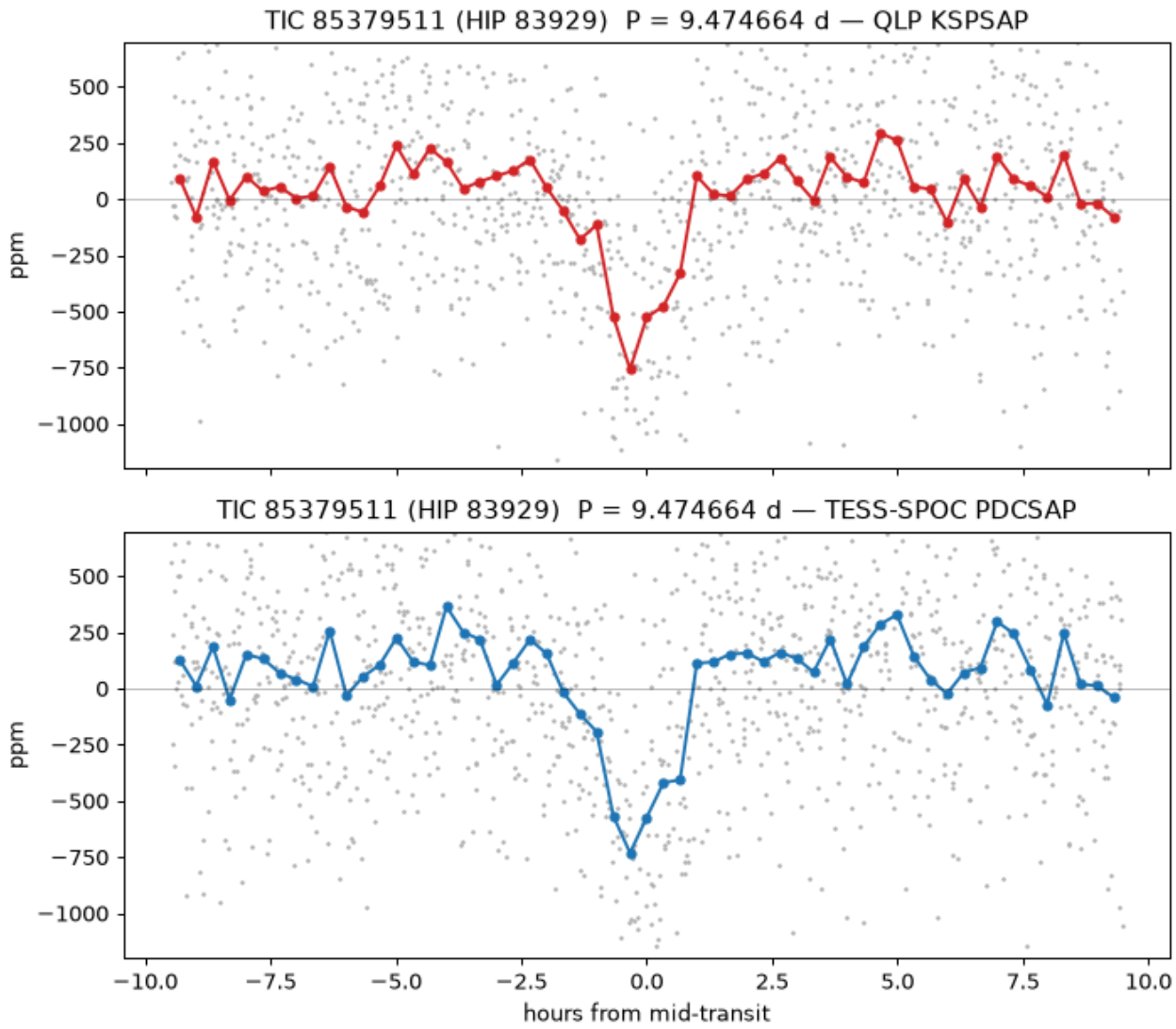


Figure A3. HIP 83929 folded on P = 9.474664 d: the 515–581 ppm, 1.58-h dip in QLP KSPSAP (top) and TESS-SPOC PDCSAP (bottom) flux. Phase 0.5, where no dip is found, is not shown.

Identifiers	HIP 83929; TIC 85379511; Gaia DR3 1354026652389939968; G 181-29; 2MASS J17091965+4126095
Position (J2000)	17:09:19.6205 +41:26:09.856 (257.331752, +41.436071 deg)
Brightness	Tmag 9.66; G 10.4472; V 11.02
Stars	K4V, 0.67 $R_\odot$, at 39.1 pc, with an unseen stellar companion (Gaia DR3 RUWE 3.87; robust RV amplitude 73.2 km s^{-1} over 24 epochs, χ^2 p-value 0.0). If the RV period equals the eclipse period, the companion is ~0.37 $M_\odot$ (0.365 $M_\odot$; an estimate, not a lower limit)
Period	9.474664 d (photometric; Gaia publishes no orbit for this star)
Epoch	BTJD 1993.379 = BJD_{TDB} 2458993.379
Primary eclipse	515 ± 40 ppm (QLP), 581 ± 39 ppm (TESS-SPOC); 1.58 h, against a 3.0-h maximum central duration: grazing
Secondary eclipse	None: 15 ± 42 ppm at phase 0.5 (predicted $\lesssim$ 25 ppm for an M-dwarf companion)
Pipelines agreeing	QLP (7 sectors, 12/13 events positive) and TESS-SPOC (7 sectors, 13/14 events positive). The QLP odd/even difference (600 vs 435 ppm) reverses in SPOC (500 vs 647), so it is not an alternating-depth signal
Pixel attribution	

	On target in five sectors (deficit centroid 0.5–3.2σ from the star); the only star inside one pixel, TIC 1270686103 (Tmag 16.37, 19″ N, would need a 25% eclipse), is rejected in three of the five sectors (at 5.4σ in S52) while S53 and S79 leave it level with the target; brighter neighbours excluded at ≥ 9.5σ. In a cyclic-shift null test, 0 of 100 trials reach the observed coherent SNR of 12.6
Catalogue status	Not listed as a variable of any eclipsing type. VSX: HAT-151-0000522, type “VAR”, P 12.895517 d, amplitude 0.012 V (a rotational signal 20× the eclipse depth at a different period; SIMBAD types the star Ro*); kept under Section 2.7, and the eclipses are new. In Hipparcos since 1991; 21 SIMBAD references, none about binarity; no TOI or CTOI; ExoFOP empty. A TESS-SPOC threshold-crossing event at 9.4748 d, the eclipse period (632 ppm, MES 8), exists in the multi-sector data-validation products and was never promoted (Section 2.7)
Open issues	Gaia DR3 publishes the RV amplitude, not the RV period: a planet is strongly disfavoured but not formally excluded. One night of spectroscopy or Gaia DR4 epoch RVs would resolve it. Sectors 78 and 79 were originally measured shallower in both pipelines (SPOC 219/233 ppm against ~700); an independent re-reduction does not reproduce this (TESS-SPOC 473 and 607 ppm, QLP not shallower), so no trend is claimed
Method	M-dwarf transit search (Section 2.3), coherent search; the sole survivor of the vetting of 40 peaks; arithmetic independently cross-checked
Status	Probable. The dip is real (three reductions, 6–7σ, 0 of 400 random-phase boxes reach it) and on or near the target, but the companion's radial-velocity period is unknown, so its mass and even its stellar nature rest on the assumption that the RV period equals the eclipse period; the one-pixel neighbour is rejected in only three of five sectors; no timing fit, so no next eclipse is predicted

The signal resembles a 1.7 $R_\oplus$ super-Earth with an equilibrium temperature of ~590 K on a bright K dwarf. Every photometric test passed: seven sectors over four years with no single sector carrying it, a duration well inside the star's maximum, pixels on target, a second pipeline, and 0 of 100 trials in a null test using the same grid and filter as the observation. Gaia DR3 radial velocities strongly disfavour a planet. An RV amplitude of 73.2 km s^{-1} indicates a stellar companion. At the photometric period the mass function gives ~0.37 $M_\odot$, and the primary's photocentre wobble would be ~0.71 mas at 39.1 pc, the right size for RUWE 3.87. A 9.47-d planet inside such a binary would also be dynamically difficult to retain. One alternative, a ~1 $M_\odot$ white dwarf on a 60–66 d orbit, is disfavoured because its wobble (~6 mas) would be an order of magnitude more than the RUWE reflects. The dip is most plausibly a grazing stellar eclipse. Its nature is undetermined until the companion's period is measured.

A4 TIC 70665829 = UCAC4 537-050325

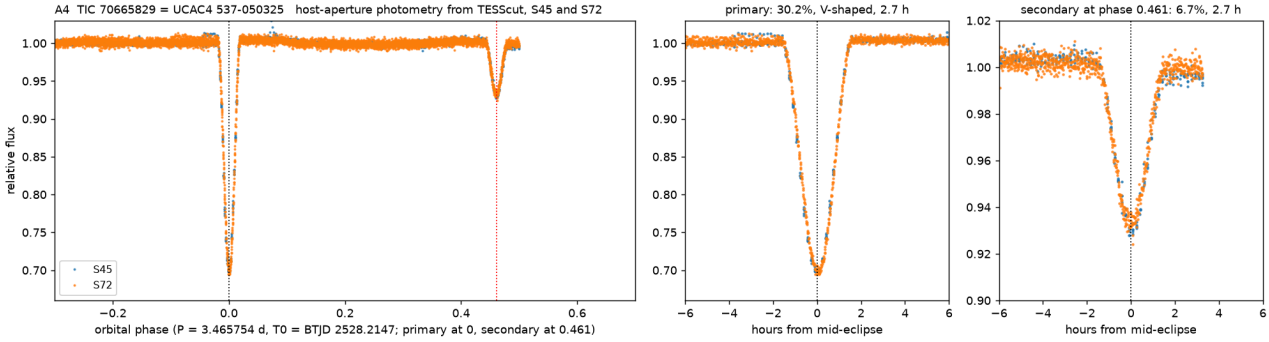


Figure A4. TIC 70665829, aperture photometry extracted at the host in the TESScut cutouts of sectors 45 and 72, folded on P = 3.465754 d with the 30% primary at phase 0 and the 6.7% secondary at phase 0.461.

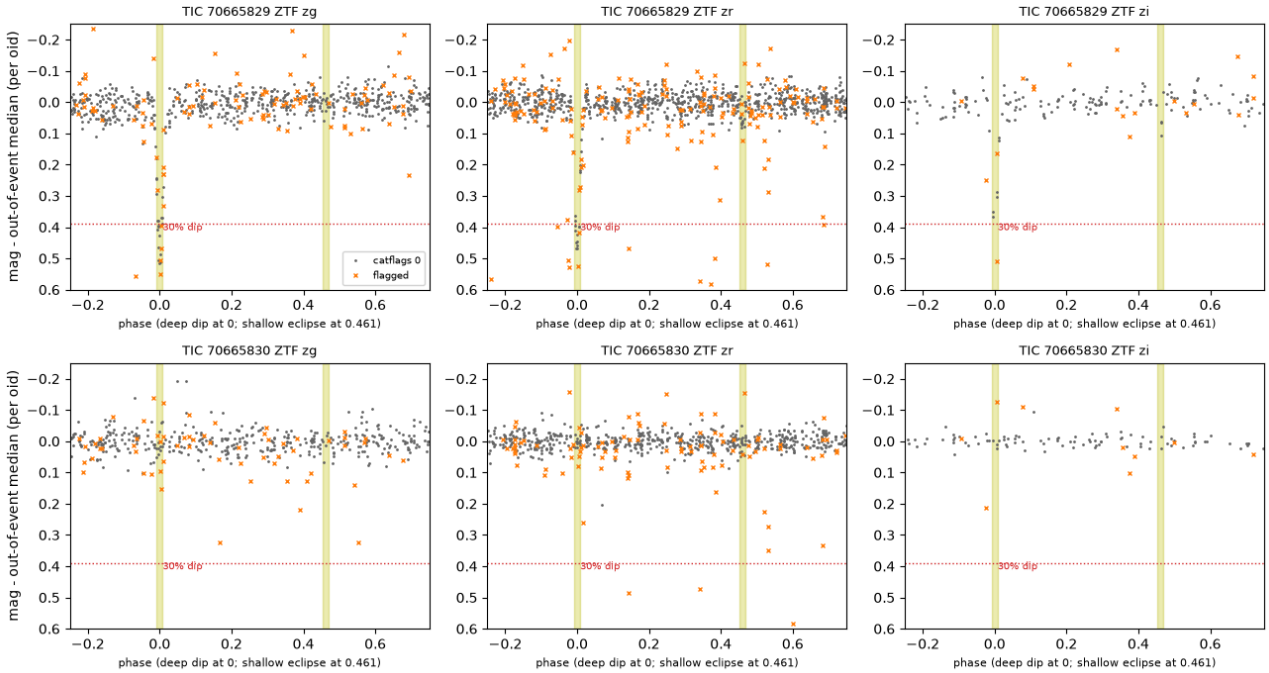


Figure A4b. ZTF *g*, *r* and *i* photometry (2018–2025) of TIC 70665829 (top) and of the M dwarf TIC 70665830 at 60″ (bottom), which ZTF resolves, folded on the TESS ephemeris with the primary at phase 0 (yellow bands: the predicted eclipse windows). Every ZTF epoch inside a primary window is faint on TIC 70665829, by 0.40 mag at the median; the same windows on the M dwarf show nothing.

Identifiers	TIC 70665829; Gaia DR3 3890379452571731456; UCAC4 537-050325
Position (J2000)	155.384788, +17.203515 deg
Brightness	Tmag 12.29; V 13.12; range at primary minimum T 12.294–12.684, at secondary minimum 12.370
Stars	K dwarf (TIC: T_{eff} 5080 K, 0.76 $R_\odot$, 0.85 $M_\odot$) at 240 pc + an M dwarf of ~0.54 $M_\odot$ (from the mass function; the two-sphere fit gives R_2 0.42–0.54 $R_\odot$, T_2 ≈ 3600–3700 K, L_2/L_1 0.07–0.11); *i* 87–89°, both eclipses partial (impact parameter 0.36–0.56 $R_\odot$)
Period	3.4657540 ± 0.0000093 d (joint timing fit of 11 primary and 10 secondary eclipses; O–C rms 0.3 min over 218 cycles). Gaia NSS SB1: 3.465888 ± 0.000285 d, agreement at 0.5σ; the primary and secondary periods agree to 1.1 ± 1.6 s
Epochs	Primary T_0 = BTJD 2528.2147 ± 0.002 = BJD_{TDB} 2459528.2147; last observed primary BTJD 3283.7496. Secondary at phase 0.4607 (BTJD 2529.8114)
Primary eclipse	30.2% (29.8–30.5% over the 11 events; 30.7–31.1% after removing the ~3% of foreign light in the aperture), V-shaped, full width 2.75 h, FWHM 1.56 h
Secondary eclipse	6.73 ± 0.06% at mid-eclipse, V-shaped, full width ~2.7 h, at phase 0.4607: $e \cos \omega = -0.062$, against Gaia's −0.077 ± 0.040 (0.4σ); equal durations give $e \sin \omega \approx 0$. The depth ratio 0.22 is the M/K surface-brightness ratio in the TESS band
Orbit	Gaia SB1: K_1 60.97 ± 1.75 km s^{-1}, *e* 0.077 ± 0.040, ω 175 ± 22°, mass function 0.081 $M_\odot$; RUWE 1.85. With the K dwarf (the RV star) eclipsed at the deep event, Gaia's ω predicts the secondary at phase 0.451 ± 0.025 (observed 0.461, 0.4σ); the other assignment predicts it at 0.451 and the deep eclipse is at 0.539 (3.5σ)
Pixel attribution	Same method for both eclipse kinds, in two field orientations: the 30% eclipse's deficit centroid is 2.7″ (S45) and 2.0″ (S72) from this star, the 6.7% one's 4.4″ and 3.2″, both peaking on the star's pixel; 58–60″ from the M dwarf TIC 70665830 on which the signal was first seen. No star within 90″ could supply the 30%: the only one brighter than Tmag 17, the M dwarf at 60″, puts 1.5–4% of its light in the aperture and would need an intrinsic depth above 1500%
Independent photometry	ZTF resolves the pair: 46 epochs inside predicted primary windows over 2018–2025 are all faint on TIC 70665829, median +0.396 mag (30.6%), while the same windows on TIC 70665830 give +0.006 mag. QLP, TESS-SPOC and TARS light curves of the star show the primary at 27–30% in sectors 45, 46, 48, 72 and a January 2026 sector; the M dwarf's own 2-min SPOC aperture sees it at 1–2%, as contamination
Catalogue status	

	Known SB (SIMBAD SB*, from the Gaia solution); nothing in VSX within 60″; not in the TESS-era eclipsing-binary tables; nothing at the position in the ASAS-SN variables
Next eclipses	Primary: BTJD 4330.4068 (2026-10-16 21:49 UTC), 4333.8725, 4337.3383, 4340.8040, 4344.2698, then every 3.46575 d; secondary 1.597 d after each primary
Method	M-dwarf transit search (Section 2.3), from one of four remaining eclipse-like candidates of the faint late-M search, as a 6.7% eclipse; the 30% primary was identified later in the same photometry and attributed to this star by the tests described below

The faint-M search found a 1.58% dip at 3.46577 d on the M dwarf TIC 70665830, clean on the fold but 1.5 times longer than that star allows. The pixels put it on TIC 70665829, the Tmag 12.29 star three TESS pixels (60″) away, where Gaia already had a single-lined orbit at the same period. The search's fold put the 6.7% eclipse at phase 0 and tested for a secondary only at phase 0.500. On that fold the 30% eclipse falls at phase 0.539, 3.3 h later, so the search detected only the 6.7% eclipse although the 30% eclipse is in the same photometry. Five independent lines of evidence (pixels, dilution, period identity with the Gaia orbit, ZTF, and Gaia's ω) show that the 30% eclipse is this system's primary. A single K + M geometry at $i \approx 88°$ fits both eclipses. Of the four remaining eclipse-like candidates from that search, two were already in VSX as BY Draconis rotators and one (TIC 234495456, Section 3.8) was an uncatalogued rotator whose apparent secondary was the harmonic structure of a sinusoid.

S3.7 Planet candidates (C1–C3)

C1 TIC 4206066 b and c: Rabtsevich's two candidates, recovered in the public data

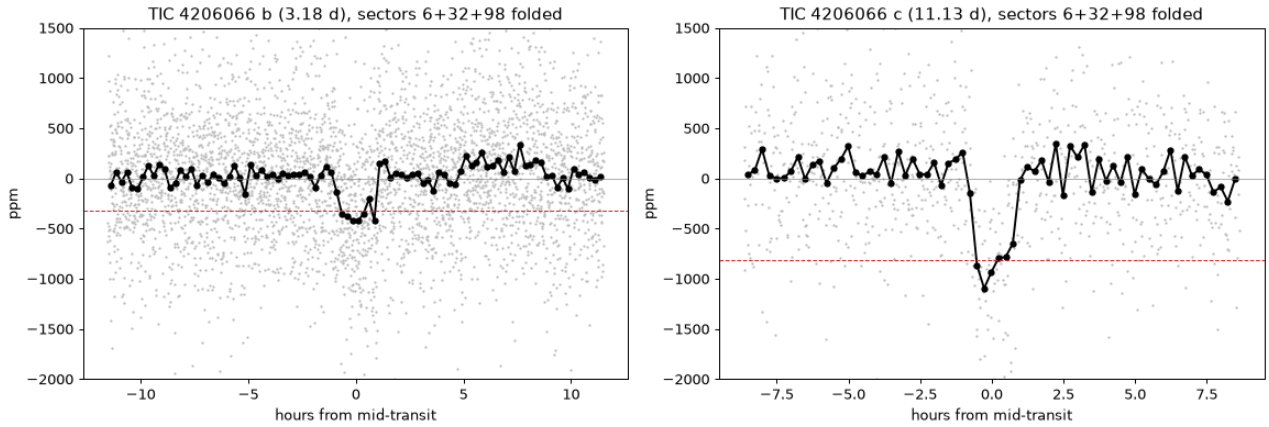


Figure C1. TIC 4206066 folded on 3.182785 d (b) and 11.132690 d (c), QLP sectors 6, 32 and 98.

Star	K dwarf, Tmag 9.68, T_{eff} 4010 K, 0.648 $R_\odot$, 0.63 $M_\odot$, 35.6 pc; contamination ratio 0.0027; RUWE 1.21. No TOI, CTOI or SPOC TCE on 2026 October 9
Data	QLP sectors 6 (30 min), 32 (10 min) and 98 (200 s); SPOC 2-min sector 32 as a cross-check
b	P 3.182785 d, T_0 BTJD 1471.60086; 320 ± 28 ppm (box); 16–17 events with data, all 16 single-event depths positive; odd/even 335 ± 38 vs 305 ± 40 ppm; phase 0.5: −25 ± 25 ppm; 7.3σ; ≈1.3–1.4 $R_\oplus$; SPOC sector 32: 272 ± 43 ppm
c	P 11.132690 d, T_0 BTJD 1473.74286; 811 ± 51 ppm (box); 6 events; 11.6σ; ≈2.0–2.2 $R_\oplus$; ~1.5 h against ~3.1 h for a central transit, so impact parameter ~0.87, near-grazing; SPOC sector 32: 625 ± 97 ppm
Look-elsewhere null	0.5–12 d search, 64 moving-block nulls on a 200,000-point grid. Stage 1: c at SNR 15.75 against a largest null of 8.57; stage 2 (c removed): b at 11.68 against 8.53; 0/64 at both stages. On a refined grid 0/64 still holds (largest refined null 8.62)
Pixels	c: no Gaia-resolved neighbour can produce it (all excluded at ≥ 5.5 sd). b: on target in sector 98, but an RP 18.0 star 12″ away (p = 0.85; would need a ~65% eclipse) is not excluded; one sector-98 event dropped post hoc as a pointing dipole (documented; the event can be restored)
Status	

An independent recovery, from the public data, of the two candidates reported by Rabtsevich (2026). TESS DDT program 100 (2-min cadence) was approved to observe it in November (its listing was verified); nothing has been reported

The search was made blind to the periods published by Rabtsevich (2026), and it recovered both signals. In sector 98, c was the top peak and b the second; in sector 32, b was the top peak. Here c is the stronger detection, whereas Rabtsevich (2026) was less certain of it. The grazing shape of c and its shallower sector-32 events (433 and 481 ppm against 800–1,100 elsewhere, ~2.5σ) are also consistent with a blended eclipsing binary. Pixel analysis, however, shows that c is fixed on the star, while b, the more planet-like signal, could still be produced by a faint background binary. Ground-based photometry with small pixels would resolve b.

The null also yields a general grid criterion: on a log-period grid the worst first-to-last transit-time slip is about B · Δlog P / 2 for baseline B, independent of period. Here a 20,000-point grid gave a 294.8-min slip (5.12 of the shortest boxes) and missed b entirely, while 200,000 points gave 29.5 min.

C2 TOI-5394.01 = HD 89277 = TIC 61109252: an occultation of an M-dwarf companion

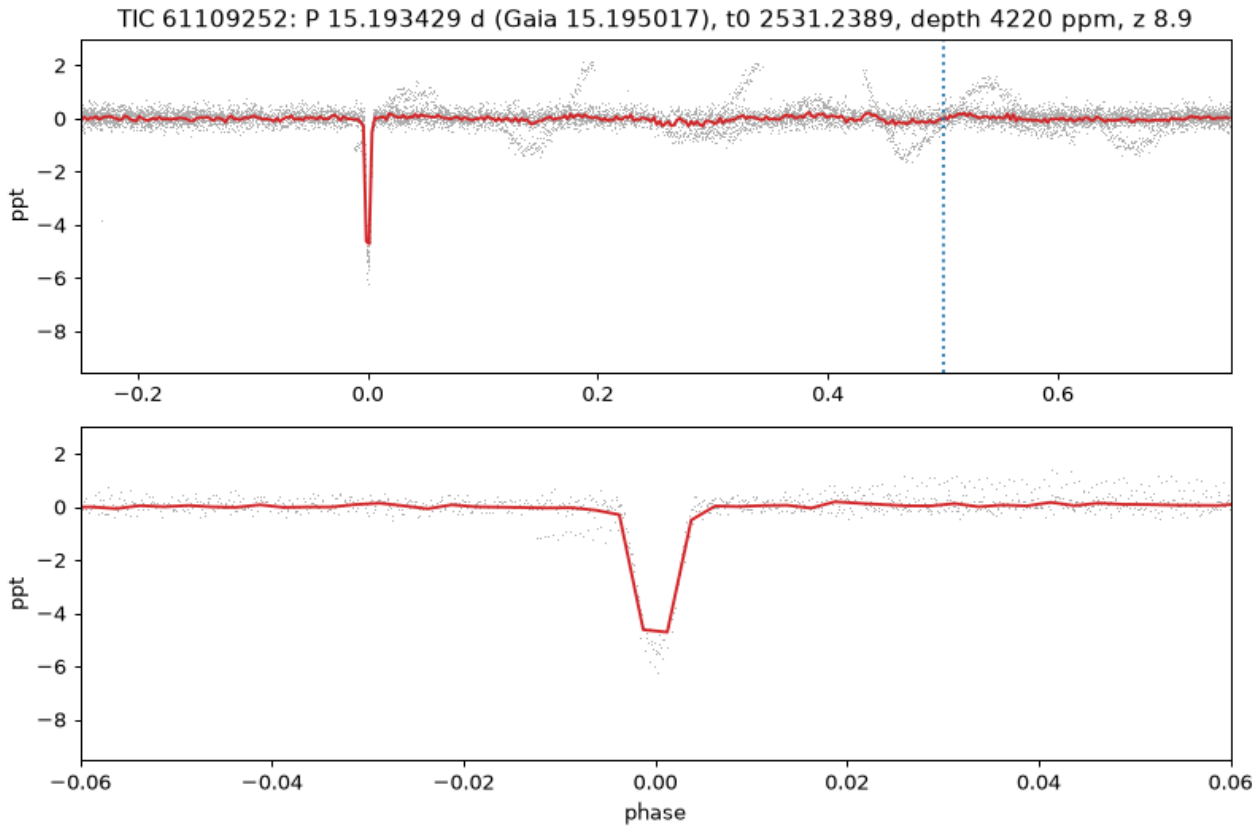


Figure C2. HD 89277 folded on the TOI period, 15.193429 d: the 4.7-ppt V-shaped dip falls at the conjunction with the companion behind the G star.

Identifiers	TIC 61109252; TOI-5394.01; HD 89277; HIP 50469 (TIC cross-match); Gaia DR3 623779716969345920
Star	G0V, T_{eff} 5977 K, 1.07 $R_\odot$, 64 pc, Tmag 8.02
ExoFOP	P 15.1935123 ± 0.0002 d, 6,228 ppm (QLP), disposition APC; photometric BEB; a 2023 note on the Gaia SB1; TRES RVs "out of phase... companion not responsible for the transits"; a chromatic MuSCAT2 event
Gaia DR3 SB1	Significance 131: P 15.195017 ± 0.0014 d (1.0σ from the TOI), e 0.292 ± 0.009, ω 246.3 ± 2.3°, K_1 27.43 ± 0.21 km s^{-1}; f(m) = 0.0284 $M_\odot$, so M_2 ≈ 0.40 $M_\odot$ for M_1 = 1.1 $M_\odot$ and i ≈ 90°
Conjunction	

	The dip falls at the conjunction with the companion behind the G star: $\Delta\varphi = -0.027 \pm 0.017$ (-0.013 ± 0.010 projecting the QLP ephemeris back); the other conjunction is at $+0.396 \pm 0.019$
Geometry	$a = 29.6\ R_\odot$; at $i \approx 87.0°$ (from MuSCAT2's 2.68-h duration) about 39% of the M dwarf is covered, predicting ~5.7 ppt; QLP has 5.7 (4.2–4.7 ppt in our fold). At the other conjunction closest approach is $1.92\ R_\odot$ against $1.46\ R_\odot$ for contact: no eclipse
Independent check	The occultation interpretation holds: TRES RVs 5.3 and 47.1 km s^{-1} against Gaia's predicted 6.7 and 46.2; MuSCAT2 g/r/i/z depths 1.0/2.1/4.7/7.3 ppt, as red as an M-dwarf occultation should be
Status	Not submitted as an ExoFOP community note

All of this evidence was on ExoFOP but had not been combined. Read as a planet transit, the "out-of-phase" RVs argue for a background eclipsing binary. Read as an occultation of the Gaia companion, they are exactly what is expected, so the companion is the cause and the background-binary reading is wrong. The chromatic MuSCAT2 depths, deepest in the reddest band, are what an occultation of a cool M dwarf should give.

C3 TOI-6284 and TOI-1467: independent recovery of published candidates

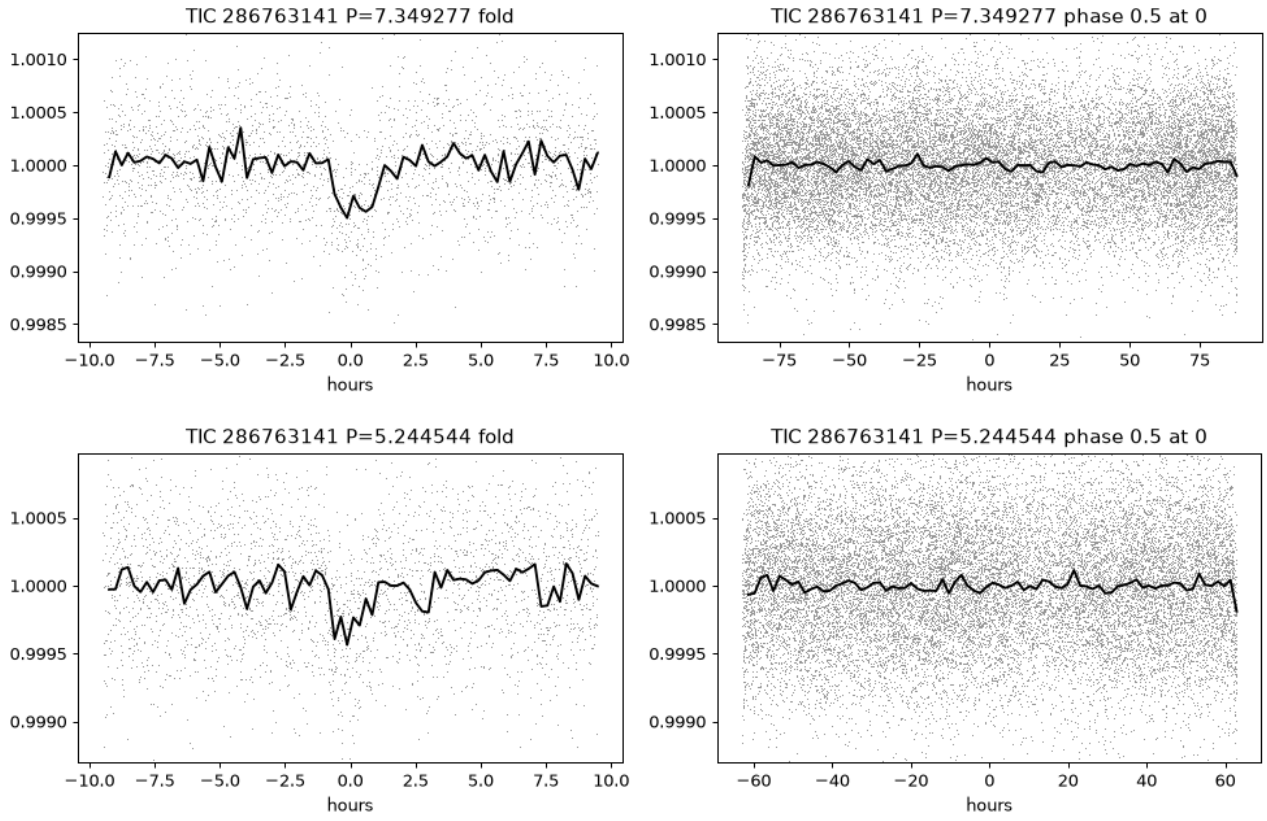


Figure C3a. TOI-6284 (TIC 286763141) with TOI-6284.01 masked, folded on 7.349277 d (top) and on 5.244544 d (bottom), centred on the transit (left) and on phase 0.5 (right).

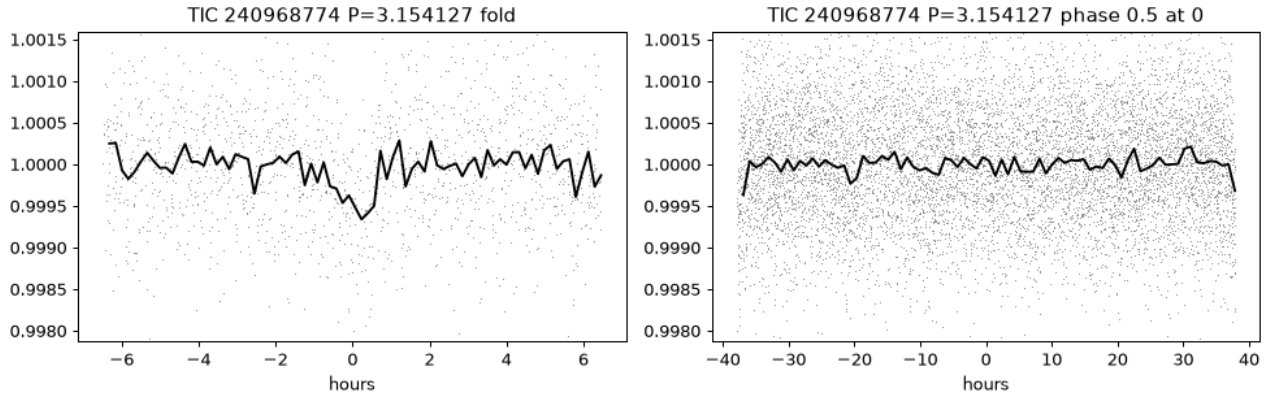


Figure C3b. TOI-1467 (TIC 240968774) with b masked, folded on 3.154127 d, centred on the transit (left) and on phase 0.5 (right).

Search	78 known M-dwarf planet hosts in the parent sample (101 signals), every known signal masked; SPOC 2-min where it exists, QLP otherwise; coherent BLS 0.5–30 d. Calibration: TOI-270 b, TOI-1730 b and L 98-59 b each recovered as the top peak when left unmasked
TOI-6284 (TIC 286763141)	M dwarf, T_{eff} 3548 K, 0.47 $R_{\odot}$, 21.0 pc, Tmag 9.72; TOI-6284.01 at 3.4476 d. New signals at **7.349277 d** (420 ± 32 ppm, SNR 13.3, ~1.05 $R_{\oplus}$, 21 events all positive; SPOC TCE 1 of the S1–S96 run, P 7.349322 d) and **5.244544 d** (326 ± 28 ppm, SNR 11.7, ~0.92 $R_{\oplus}$, 26 events all positive; in no SPOC DV report). Period ratios 1.521 and 1.401, near 3:2 and 7:5. Cyclic-shift null 0/36
TOI-1467 (TIC 240968774)	T_{eff} 3834 K, 0.49 $R_{\odot}$, 37.4 pc, Tmag 10.60; TOI-1467 b at 5.971 d. New signal at **3.154127 d**, 532 ± 59 ppm (SNR 9.0–9.2), ~1.24 $R_{\oplus}$, 21 of 21 events positive; already SPOC TCE 2 (P 3.154093 d, 513 ± 84 ppm). Cyclic-shift null 0/60
Prior report	All three signals are in Tschudi (2026; arXiv:2607.23781): TOI-6284 at 5.244499 d (298 ± 90 ppm) and 7.349312 d (465 ± 125 ppm), and TOI-1467 at 3.154156 d (533 ± 215 ppm)
Status	An independent recovery with different code, not a discovery. None is a TOI or CTOI yet

A literature check made after the search found all three signals in a preprint (arXiv:2607.23781, v3 dated 2026 August 3). This work adds the independent recovery, made with different code and data handling (depths of 326 ± 28 and 420 ± 32 ppm, against the preprint's 298 ± 90 and 465 ± 125). It also adds one fact the preprint does not cite: SPOC flagged the 7.349-d signal as a threshold-crossing event in its sectors 1–96 run. For TOI-6284 the tests do not exclude TIC neighbours inside one pixel at Tmag 17.6–17.9 (13–20″ away, which would need ~60% eclipses), or anything closer than Gaia resolves. Excluding them requires ground-based photometry. The re-reduction was not completed once the preprint was found.

S3.8 Other objects

TYC 3707-251-1 = TIC 316045876: a 5.854-d eclipsing binary, withdrawn as a 197-d system

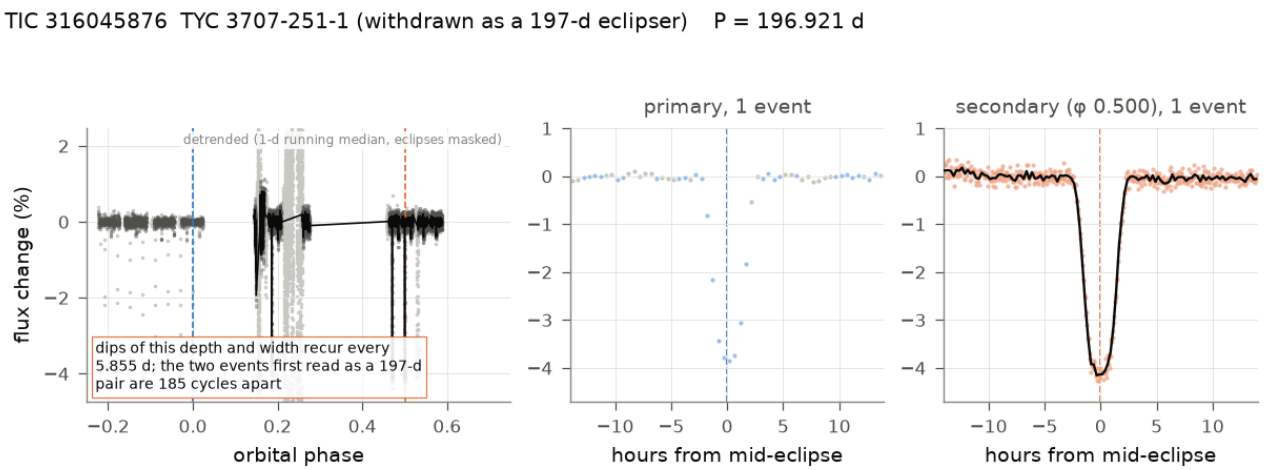


Figure D3. TYC 3707-251-1, QLP SAP flux folded on P = 196.921 d, the period of the earlier 197-d interpretation. Left: the whole orbit, detrended with a 1-d running median with the eclipses masked. Middle and right: the two dips previously read as the primary and the secondary (φ 0.500) eclipses of a 197-d orbit, one event each. Both are members of a series of ~4%, ~3-h dips every 5.855 d.

Withdrawn. The claim of a 197-d eclipsing binary is withdrawn: the two events belong to a 5.855-d series; the star, or another star in its aperture, remains an uncatalogued eclipsing variable at a much shorter period.

Identifiers	TIC 316045876; TYC 3707-251-1; Gaia DR3 448490316232971392
Brightness	Tmag 9.94; G 10.53
Stars	TIC: T_{eff} 5937 K, 3.09 $R_{\odot}$. Gaia pure astrometric orbit: P = 198.4044 ± 0.9570 d, i = 90.532°, e = 0.224
Original claim	Two raw-SAP events at BTJD ~1835.924 (sector 19) and ~2918.987 (sector 59), 36 and 39 ppt (TESS-SPOC 39 and 40 ppt), read as primary and secondary eclipses eleven 98.4603-d half-cycles apart (P_{orb} = 196.921 d), on opposite Gaia conjunctions at joint 2.1σ
What the data show	~4%, ~3-h dips every 5.855 d in the raw QLP SAP flux: sector 18 at BTJD 1794.94, 1800.80, 1806.66 and 1812.51; sector 19 at 1818.36, 1824.21 and 1835.92; sector 59 at 2913.13, 2918.99, 2930.69 and 2936.55 (spacings 5.85 and 11.71 d). Both catalogued events are members of this series. Shallow secondary dips are visible at half phase
Fitted ephemeris	Fit to all cadences: *P* = 5.854343 ± 0.000012 d, T_0 = BTJD 1794.9476, from 12 clean eclipses in sectors 18, 19, 59 and 86 (three more partly in flagged cadences); primary 3.8%, odd and even equal (3.65 vs 3.67%); a 0.34% secondary at exactly phase 0.50 (7× the bin scatter), so the orbit is circular and this is a real eclipsing binary, not a blend of two signals
Relation to Gaia's orbit	The 12 eclipse times are not linear (χ^2 671/10; O−C about −4, +3, −5 min across the three epochs). A light-travel-time delay with the shape fixed by Gaia's 198-d astrometric solution (P, T_{peri}, e, ω) and one free amplitude fits with χ^2 34/9 and K = 5.1 min (4.4–6.6 over the Gaia covariance). This amplitude is above the 2.2-min floor the astrometric photocentre requires. A quadratic ephemeris with the same parameter count gives χ^2 120. Gaia's period with a random phase does as well 10% of the time, and a blind scan aliases at 129, 158, 198 and 276 d, so the timings agree with Gaia's orbit but do not find it on their own
Open issues	Which star in the aperture hosts the eclipses is not pixel-verified, though the timing link to Gaia's orbit argues for the target. Sector 86 has further dips under scattered-light flags
Catalogue status	In no catalogue checked except Kostov et al. (2025), table 2
Method	Edge-on astrometric-orbit search (Section 2.6), as a 197-d candidate; orbit phase checked by independent re-analysis (Section 2.5); the 5.855-d series was identified later, verified in the raw data and refitted with the timing analysis above

The original search and its independent check tested for repeats only near Gaia's 198-d period and half of it, not at unrelated short periods. An independent refit confirms the 5.854-d series and finds a circular orbit with a shallow secondary. The eclipse timings vary in a way consistent with light-travel time on Gaia's 198-d orbit, which would make this a hierarchical triple: the 5.854-d eclipsing pair is itself the 198-d orbiter. That reading is consistent and modestly supported, not proven. The object is a new eclipsing binary at 5.854 d. It is not counted among the 104 new systems and is listed separately, because it entered the catalogue under a wrong period. Its host is not yet pixel-verified.